Now There Was a Lady!

By Darren H. Tanke

Alberta Palaeontological Society

Now There Was A Lady!

Hope Johnson, LL.D.

1916–2010

Darren H. Tanke

Alberta Palaeontological Society

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Alberta Palaeontological Society
P.O. Box 68024 Crowfoot PO
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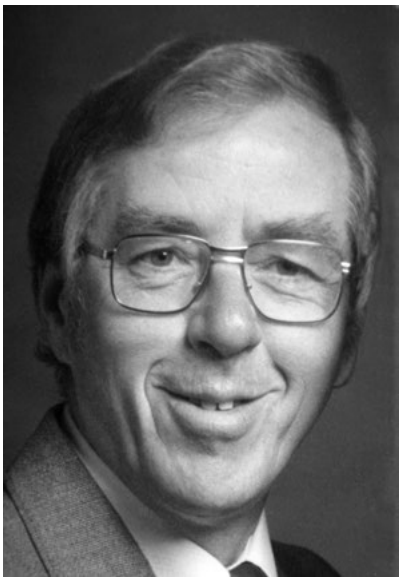

Hank William Tanke, *circa* 1979.

To my father, Hank William Tanke (September 28, 1935, The Hague, Holland – January 31, 2018, Victoria, Canada), who taught me that without a university education I would need dedication, passion, and a lot of hard work in order to achieve my goals and dreams in life. Thanks, Dad, I made it.

He didn't live to see this book's completion, but saw late drafts of it and was very encouraging.

As much as this book is a tribute to Hope Johnson's life story and her many achievements, it is also dedicated to the memory of Vicki Lynn Leuck (March 15, 1961–October 27, 2014), a good friend and fellow amateur fossil collector of Hope's. Vicki, who resided in Medicine Hat with her husband Shane (both of whom Hope described to the author as "very fine people" and in family notes as "excellent people"), passed away suddenly but peacefully in her sleep from heart failure at the age of 53 (Anonymous, 2014b). It was shortly before the author hoped to meet and interview her for her recollections of Hope, for inclusion in this book. Shane and Vicki have been active in sharing their palaeontological hobby with students, and especially the discovery of Late Cretaceous insects, dinosaur and bird feathers in amber recovered from southern Alberta. They have been duly noted or acknowledged in a number of newspaper, magazine and technical publications (Lamolet, 2013; McKellar *et al.*, 2011; Engel *et al.*, 2013; McKellar *et al.*, 2008, 2013; Pelletier, 2012; McKellar and Engel, 2009, 2011a-c, 2014; Revell, 2013).

Vicki Lynn Leuck, 1961–2014.

Vicki and Shane Leuck teaching a school class about vertebrate palaeontology, with some of the Hope Johnson collection. Both images courtesy of Shane Leuck, Medicine Hat.

Contents

iii Dedication

vii Acknowledgements

Forewords
1 Dr. Philip J. Currie
3 Allan C.J. Jensen
5 David A.E. Spalding

10 Introduction

14 Acronyms used in this book

15 Map of Hope Johnson country

17 Hope's family heritage and early life (1916 to 1938)

35 1939 to 1947: World War II, marriage and a new home in Alberta

50 1948 to 1956: Finding her first fossils and Hope's palaeontological self-education

56 1957 to 1969: Hope's amateur palaeontology career takes off

107 The 1970s: Her career continues, development of the first fossil guidebook

135 Slowing down? 1980 to 2000

155 2001 to 2010: Publication of the new fossil guidebook and the closing chapter of a life well lived

165 Overview

195 Afterword

197 Notes

208 Addendum: Hope Johnson's art in colour

236 Portraits of Hope through the years

238 References

276 Index

Acknowledgements

This book began as a paper on March 28, 2009 as a proactive approach to my "Remember Me . . ." series on historically early and lesser-known people on professional crews, or special amateurs, all involved in Alberta vertebrate palaeontology to some degree. Hope Johnson taught me much, especially in my early vertebrate palaeontology career (the first being that the little vertebra I found in Horsethief Canyon near Drumheller was a *Champsosaurus* dorsal and the second that I had interpreted it upside-down!) so it is indeed an honour and privilege to recognize her numerous contributions and to write this biography.

The author could not have done it alone—many people graciously and generously helped, with newspaper clippings, sharing their memories of Hope, suggesting other potential contacts, providing images, proofreading, hosting, and in many other ways. *Thank you all!*

First and foremost, the author thanks the late Hope Johnson, who at my request jotted down numerous detailed notes about her life, knowing this biography would some day meet fruition, though neither of us—I'm sure—had any idea of its eventual size! You were so very busy, Hope!

Of the Johnson children, Lawrence, Margaret Noone, and Barbara Preston (and her husband Ed) provided many details of their mother's life, access to family photographs, notes, artwork, family history, military records, and provided manuscript comments. Hope's granddaughters, Rachel Noone and Julia Strickland also provided images of Hope's artwork. Their openness and willingness to help ensure that the remarkable story of Hope Johnson is more accurate and complete.

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A massive computer crash and dying hard drive—just as the manuscript neared completion—was very alarming, but staff at the Reality Bytes computer shop in Drumheller saved the day. The author thanks Michael Sharpe, Ashley Charles Bancroft-Uwins, and especially Information Technology specialist Krista Cosman.

In a document this detailed, compiled over twelve years, and covering a long historical time span, combined with fading memories of some interviewees, some errors are bound to have crept in. The author has made every effort to fact-check and double-check everything herein, but assumes sole responsibility for any historical or factual errors. A great many people freely shared images of Hope's artwork—so many, in fact, that it was impossible to show them all. An effort has been made to include art objects owned by her family and friends (and some museum pieces), created with different media, subject matter, location, style, and works that she created over the decades.

Finally, a huge thank-you to Alberta Palaeontological Society (APS) editor Howard Allen for his editorial and photo editing skills, the cover design, Figure 127, APS collection database search; and to APS volunteer proofreaders Wayne Braunberger, Mona Marsovsky, and Vaclav Marsovsky. Mona, in particular, took on the job with astonishing thoroughness.

Foreword by Dr. Philip J. Currie

Vertebrate Palaeontologist; University of Alberta, Edmonton

I think it is fair to say that anyone interested in vertebrate fossils in Alberta probably knows who Hope Johnson was. Johnson and Storer (1974) is a much-sought-after classic for the identification of common Late Cretaceous fossils. Hope's specimen illustrations for that book were an efficient way to identify fossils, particularly those of dinosaurs from southern Alberta. The guide, originally published by the Provincial Museum of Alberta (now the Royal Alberta Museum) and fortunately republished as a new, expanded edition by the Alberta Palaeontological Society (APS) in 2009, has also proven immeasurably useful to enthusiasts and professionals far beyond the borders of Alberta. It's a gift I still give to new students when they start working with dinosaurs, and copies are always available in our field camps in Alberta.

I first met Hope Johnson in 1976 when I toured southern Alberta after accomplishing my lifelong goal of becoming a dinosaur palaeontologist. I of course knew of her through "Johnson and Storer," and by reputation from stories told by friends and colleagues like Gilles and Jane Danis, Peter Dodson, Gord Edmund, Dale Russell and John Storer. I also knew that in the early 1970s she was responsible for many of the displays of isolated bones and teeth in what we called the "chicken coops" near the entrance of Dinosaur Provincial Park. The buildings were horrible—the visitor often had a hard time seeing through the glass because of the glare from the sun, and by the late 1970s most of the specimens were covered in dust and dead flies, as the buildings fell into neglect when Hope's employment at the Park ended. But the exhibits themselves were remarkably good, and included all of the common fossils of the area so that an astute tourist could identify anything he or she saw in the surrounding badlands. Organizing these displays, complete with Hope's illustrations and labels, was the genesis of the fossil identification guide that she did with John Storer.

A healthy, robust woman with long braided hair that was usually coiled atop her head, she exuded enthusiasm and a *joie de vivre* that was as large as her laugh. Both this time, and on subsequent visits, she was always a gracious host. She selected specimens for me to see from her vast personal collection of fossils. Sometimes the fossils were particularly well-preserved examples of things well-known, but most of the time they were small, enigmatic things that she had not been able to identify herself. Over the years, I never managed to see the whole collection, and every time I saw her, she showed me new puzzlers that she had selected for my benefit. Hope was still actively collecting fossils, and would periodically arrange to visit me in either our field camps in Dinosaur Park, or at the museums in Edmonton (up to 1982) or Drumheller. Sometimes she would bring friends who had been collecting fossils with her, and had found interesting specimens of their own. She had an uncanny knack of finding new and interesting fossils to show me. After years of revealing a never-ending procession of interesting specimens that I had never seen before, I joked that she had divided up her collections long ago with the

intention of only showing me certain ones when I had reached a certain stage in my own development or interests. They were all meticulously identified and labelled, and she never hesitated to turn over any and all specimens that I considered important to be catalogued into the provincial collections. In fact, she seemed disappointed if she came up with something interesting that I did not want to accession. Over the years, more than a thousand of her specimens were incorporated into the collections of the Provincial Museum/Tyrrell Museum. She instilled in others her sense of responsibility towards the care and curation of fossils, and many a person from Medicine Hat either came to my office with specimens for me to identify or accession, or sent them via Hope.

Although I knew Hope primarily through her fossil collections, I was also very aware of her artistic abilities and always asked to see her latest drawings and paintings when I visited her. Her interests in botany, archaeology and public education were so strong that I am sure there were people in those fields who were never aware of how highly she was regarded by vertebrate palaeontologists (both professionals and amateurs).

It was a great day in 1981 when I learned that Hope had been given a doctorate of laws by the University of Lethbridge—she had all of the qualities that I expect in my own students, and she had done so much for the Province of Alberta. Although none would question that she deserved that honour, she modestly seemed a little embarrassed by it, and I never saw her even mention it to others.

The last time I saw Hope was in her assisted living for seniors apartment in Medicine Hat. By that time, she had not been able to go into the field for several years, and for the first time had no new fossils to show me. But she had lost none of her knowledge, curiosity, enthusiasm or artistic productivity.

Hope Johnson was a remarkable woman for the breadth of her knowledge and interests, and her willingness to share knowledge and specimens with anyone from a rank amateur to a professional palaeontologist. She was a responsible collector who found many specimens that have been catalogued into the collections of the Canadian Museum of Nature, the Royal Alberta Museum, the Royal Tyrrell Museum of Palaeontology, and many other museums (both small and large). Her illustrations and paintings span the range of scientifically accurate to creative, but they were always educational and aesthetically pleasing. When she passed away in 2010, it was the end of an era.

Foreword by Allan C.J. Jensen

Artist/Curator; Medicine Hat, Alberta

To understand art, we must look inside it; see what makes it work, and how it is put together. When we explore these aspects, we are considering art's design. Design is the structure of art. Design is all around us: in nature, in our environment, in works of art. There are more opportunities for training our judgement, stimulating our visual curiosity and adapting creative ideas for our personal use than we could ever exhaust. Nature with its almost unlimited supply of forms is a great source of design. It has always been a primary stimulus for artists. Especially helpful is the way nature offers so many diverse forms and combinations. Many people think of science and art as being at the far ends of the spectrum of human endeavor. Both, however, require keen powers of observation and a deep understanding of process.

For Hope Johnson science and art were closely linked and became an important dimension of her life's work. She was able to blend her art and artist's eye with a fine scientific mind and keen interests in natural history, botany and palaeontology. As well, the southern Alberta landscape begged to be explored and painted. From an early age of three or four years, Hope was drawing and painting. Her interest in visual arts developed as the years progressed. Although largely self-taught, she received some instruction at a private school in the Vancouver area which introduced the Royal Drawing Society methods of Professor Ablett, though beyond that she was self-taught in various artistic methods and was reportedly an open-minded artist and not averse to trying out and learning new techniques. As well, her mother (who was a teacher) started a small private school where Hope received her elementary education.

She enlisted in the Canadian Women's Army Corps in late 1941. During the war she was posted to several locations in Canada, reaching Suffield, Alberta in 1945. There she met her future husband, Olafur Johnson, who was a meteorological officer at Suffield.

While living in Ralston (adjacent to Suffield) with her husband and growing family, Hope's interest in the visual arts was reinvigorated by the local community. It included people interested in various aspects of the arts. She became a founding member of the Ralston Art Club, painted scenery for and acted in productions of the Ralston Drama Club and organized numerous art shows. The members of the Ralston Art Club arranged for monthly classes to be taught by Dr. Harry Wohlfarth of the Department of Extension at the University of Alberta. Because of her earlier training in drawing, Hope was appointed local instructor between his visits. She learned to paint with oils and later began painting scenes in watercolours. Hope became thoroughly familiar with the prairie, river valleys and badlands of southern Alberta. A strong interest in natural history prompted her artistic talents to turn increasingly to the depiction of shells, plants, bones, fossils and detailed landscapes. Her paintings and drawings were shown in the Canadian Nature Art shows sponsored by the Canadian Nature Federation and the

National Museums of Canada. In addition, her work was frequently exhibited in Medicine Hat and area where she continued to be actively involved with the local art community. She became a member of the Hat Art Club and also the Strathcona Centre where she taught oil painting to a group of interested adult students. Hope found that the landscape and the river valleys were interesting from every point of view and she continued her consuming desire to illustrate what she found and saw. A unique characteristic of Hope's illustrations are the pen-and-ink techniques she used to define the fossils, bones, shells and plants she drew. She found a combination of line drawing and refined pointillism very suitable to portray the shapes and surfaces of her subjects. As a result, they take on highly believable aspects of three-dimensional objects.

In 1981 Hope received an honorary Doctor of Laws degree from the University of Lethbridge. In his presentation to the Chancellor requesting the award, Dr. J.-P. Christopher Jackson (Associate Professor, Department of Music, University of Lethbridge) and her fossil collecting friend noted, "Hope Johnson truly exemplifies the all too seldom found 'expert amateur'. " She was eminently an artistic scientist who practiced many skills, most of which involve aesthetic as well as scientific understanding.

In 1999, I had the opportunity to curate an exhibition of Hope Johnson's work which was organized by the Medicine Hat Museum and Art Gallery for the Alberta Foundation for the Arts Travelling Exhibition Programme. Consisting of fifteen works, *From Fossils to Fine Art* was a visual examination of the overlapping roles of Dr. Hope Johnson—naturalist, palaeontologist and artist. Travelling to some two dozen Alberta venues over a two-year period, it sought to show Hope's interest in a wide range of "nature studies," allowing each of her interests to extend and amplify the other. Included were watercolours on paper, oils on canvas, photographs, drawings, pen-and-ink illustrations as well as fossil specimens.

Following the "travelling exhibition" another art show entitled *Hope Johnson: A Retrospective* was mounted for the Medicine Hat Museum and Art Gallery in 2001. The large exhibition of fifty-six works included watercolours, oils, pen-and-ink drawings as well as ink-and-wash works.

Somehow, we frequently consider art and science as having little, if any, relationship with each other. Hope Johnson considered them to be partners in her lifelong examination of the wonders and beauty of the natural world and the interpretation of its secrets. A well-developed sense of observation and a thorough knowledge of process are prime requirements in both fields. An added facet is a curiosity of the microcosm as well as the macrocosm; an examination of minute details always helps us understand the "big picture."

To see the World in a grain of sand,
And a Heaven in a wild flower,
Hold Infinity in the palm of your hand,
And Eternity in an hour.

— William Blake

Foreword by David A.E. Spalding

Geologist, museologist, naturalist, writer; Pender Island, British Columbia

I first met Hope Johnson in Dinosaur Park on May 18, 1968. I was a recent immigrant from Britain and this was my first Alberta spring; my field notes were full of sightings of coyotes and pronghorn, of unfamiliar birds and flowers.

Those who did not know Alberta in those days should not envisage the infrastructure and public stature that Dinosaur Park has today. The road from Brooks was gravel, poorly signed, and the park itself was little developed, with vaguely defined boundaries which overlapped local ranches. Cattle wandered through the campground, the few staff were not knowledgeable about fossil resources, and few tourists were attracted because the park had almost no promotion. There were only token exhibits: two or three fragmentary skeletons housed in little sheds and a few bones. To see any sort of dinosaur models locally you had to go to Drumheller or Calgary Zoo; and to see skeletons probably the only good ones in Alberta were in the Drumheller Fossil Museum and the University of Alberta (UA) Geology Museum—one in each. Apart from a few professors—Charles Stelck and Richard Fox at the UA and Len Hills at the University of Calgary, there was little local expertise. Canadian vertebrate palaeontologists were largely based in Toronto, Ottawa and Montreal, whence came occasional expeditions that checked in with Hope in the summer. In those days, Alberta dinosaurs (and other vertebrate fossils) were only a potential resource, little studied by scientists for decades, largely unrecognized and unfunded by governments at any level, and of little public interest.

Of the meeting my field notes record only that "I met Charles Heath party, and drove into area of 'new' dinosaur." However, my companion and colleague, Don Taylor (Stelck and Burwash, 1983), commented in his notes that the party also included Velma Burton and Hope Johnson, and that he had "demonstrated how to take out a tibia." The next day Don and I stopped off in the Medicine Hat museum where I saw Hope's palaeontology exhibit and noted that her little pamphlet on "Dinosaurs" was on sale (Johnson, 1968a). This casual encounter was to be the first of numerous contacts I had with Hope, mainly in the course of my fifteen years at the Provincial Museum of Alberta (PMA; now the Royal Alberta Museum) and subsequently, through my involvement with Dinosaur Park (where Hope was for a while employed as a naturalist) and the Royal Tyrrell Museum. Though we did not see each other often enough for us to become close friends, I learned to have a real respect for her knowledge and abilities as a botanist, a palaeontologist, an informal educator, and an artist drawing and painting natural objects and landscapes. Ironically—since I was in the museum field—I only occasionally visited local historical museums as they rarely had natural history exhibits, and I was largely unaware of Hope's substantial contribution to local museums until I had a chance to read the draft of this book.

I don't remember Hope talking to me about her past; there were always flowers or fossils or paintings uppermost in her mind. But I was interested in the scientific history of the area and over the years have

explored aspects of the palaeontology and history of the province's dinosaurs in a variety of media. An early insight into the history of vertebrate palaeontology came when, on my arrival in Canada in October 1967, my friend and colleague Bruce McCorquodale (Spalding, 1997) gave me a copy of the report of a 1963 conference on "Vertebrate Paleontology in Alberta" (Folinsbee and Ross, 1965). The thirty-four conference delegates included many notable figures. The U.S. representatives included Edwin Colbert and Alfred Romer. The Canadian elite was represented by half a dozen palaeontologists from Ontario, including Charles M. ("Charlie") Sternberg, Loris Russell, and William Swinton. The UA representatives included Don Taylor (my companion on that Dinosaur Park trip), Ray Harrison (already planning the new Provincial Museum) and Bruce McCorquodale (then at the Saskatchewan Museum), who were now among my new colleagues. There were only three amateurs, and Hope (described only as "Mrs. O. Johnson") was one of only two women attending the conference (other than those accompanying a husband who was a delegate).

The conference reviewed the record of palaeontological discovery in Alberta. The delegates were strongly supportive of a projected Vertebrate Palaeontology program at the UA and a modest exhibit of dinosaurs in the new Provincial Museum, then in planning. In due course both came to pass; thus Hope played a small part in getting the Provincial Museum off the ground. If Hope contributed to the discussions it was not recorded in the published record, but it seems that the group was aware of the quality of her work. There was much concern about a scientific perspective of the "rock hound" fraternity, expressed for the group by British fossil fish specialist T.S. Westoll. There was observation of the worst examples: "those who collect trinkets for sale, gathering bones, stripping the sites and spoiling them for good." But there was also recognition of "the magnificent work done by these non-professional palaeontologists who are doing a splendid job in preparing the way," which can reasonably be seen as a tribute to Hope, the only "non-professional palaeontologist" in the group. She would have greatly widened her network of contacts at the conference, but few who were present would have anticipated the strong role Hope would play in Alberta palaeontology during the succeeding decades. It was from opportunities like this that Hope developed her network of expert advisors, those to whom she could send questions and specimens, and who began to check in with her for new discoveries when they visited the province.

My interest in the proceedings was initially in the attendees I knew or knew about. I had met Swinton as an undergraduate when he came to speak at the University of Sheffield in the late fifties. He had then been England's "Mr. Dinosaur" at London's Natural History Museum—he was the first specialist to write a trade book entirely about dinosaurs (Swinton, 1934)—and he also played an important role in establishing museum training programs in the UK, from which I had benefitted. He preceded me to Canada, eventually becoming director of the Royal Ontario Museum; though we never met again we corresponded in later years. I had also corresponded with fish palaeontologist Westoll, the only British delegate, as part of an unsuccessful attempt to get involved in vertebrate palaeontology at a graduate level in the UK. By the time I arrived in Alberta four years after the conference, Ray Harrison was the Provincial Museum director (and my boss); Bruce McCorquodale became a friend and a valued colleague; and Don Taylor was on my staff.

Over subsequent years I met almost everyone who had been at the conference, as I developed my own network of contacts. As Head Curator of Natural History at the recently opened Provincial Museum and Archives of Alberta, I was building a staff, a collection, and a program largely from scratch at my base in Edmonton, and planning to fill the empty galleries with exhibits reflecting the diversity and beauty of Alberta nature. Almost everything was new to me—Alberta's geography and natural history,

the history and culture of Alberta, the professional and amateur experts in these fields, and even a lot of the vernacular language of my new community.

However, I followed the grand old Canadian tradition of galloping off in all directions and was beginning to make sense of it all. Beyond the museum, I had to get to know the province on the ground. As I travelled I visited parks, museums, university departments, fish and game departments and local naturalists, and started to build a network of contacts. I was pleasantly surprised by the number of professional natural science specialists in universities and the public service, but found the amateur natural history movement less well supported. Outside the biggest cities, a few naturalists were local experts—Kerry Wood in Red Deer, Helen Schuler in Lethbridge—and, yes, Hope Johnson in Ralston—were among the few people who were knowledgeable about the flora, fauna, and sometimes fossils, of their areas. They may have been linked with a local society, but there was no viable provincial naturalists' organization, and no effective national one. To market the museum and build my network I offered free talks to Alberta naturalist groups, so I soon got connected with the naturalists' movement.

Complementary to my museum duties, I found myself with a busy volunteer career. I helped to run the Edmonton Natural History Society, through which I soon participated in creating the Federation of Alberta Naturalists in 1970. I created the Alberta Naturalist magazine in 1971, but by that time I was also involved in the new national organization, the Canadian Nature Federation (CNF), and began to attend its conferences. I served on its board and contributed to the new national magazine, *Nature Canada*. Inside and outside the museum, these developments involved Hope, too.

At the museum I began to hire a team of curators, including in the first few years Julie Hrapko (Botany), Hugh Smith (Zoology), and John Storer (Palaeontology). Hope was active in botany and palaeontology and she soon developed a relationship with the two curators in those fields. I also found myself involved with Hope's art career. One of my museum responsibilities was to supervise production of a series of sixteen habitat groups (dioramas) featuring the major species and ecological regions of the province. Here I was concerned not just with science, but with aesthetics. I had an interest in nature art since childhood contact with the work of Britain's bird illustrators; it was time to expand and develop that interest.

Alberta's leading artist in this field had been Carl Rungius (1869–1959; Whyte and Hart, 1985) and our sister museum in Calgary, the Glenbow Museum, had a substantial collection of his work. We soon set up an exhibit of Rungius' work at the museum, supported by a small publication written by Glenbow's curator of art, Lorne Render, and myself (Render and Spalding, 1969). As we began the diorama program the museum had to acquire artists with the complex skills needed to paint natural backgrounds and blend them with real specimens. Ralph Carson was already on staff, though not painting for the museum until I was able to put him on a project which demonstrated his skills. I later hired Ludo Bogaert as a foreground artist—he later also tackled full-size dioramas (Spalding, 1974). I also reviewed the work of contract painters and we were able to secure artists Hugh Monahan (who unfortunately died before he was able to complete a diorama) and Clarence Tillenius.

My volunteer activities again complemented my museum work. On arrival at the first Canadian Nature Conference in Ottawa in 1971 I was delighted to see the first National Canadian Nature Art exhibit, arranged in conjunction with the conference by Executive Director Ted Mosquin. I don't remember how extensively I was involved in bringing the exhibit to the attention of Alberta artists, but in Ottawa I was delighted to see that Alberta was strongly represented, with 40% of the art works by Alberta artists. Here was an opportunity! Within a few days of getting back to Edmonton I had arranged for all the Alberta works to come to the PMA for exhibition before they went home to their artists or owners.

Alberta hosted the second CNF conference in Edmonton in 1972 and there the museum was the sole host of the second national art show. It included thirty-four works, selected by a jury comprising Terry Fenton (of the Edmonton Art Gallery), Professor Ray Salt (ornithologist) and David Richeson (then Extension Supervisor of the museum). Two works by Hope were included: *Early Summer* and *Wild Dock* (Anonymous, 1972c).

In several subsequent years the National Museum of Natural Sciences (now the Canadian Museum of Nature) organized and circulated national shows across the country in association with the CNF: (*e.g.* those shown at the conferences in Wolfeville, 1973; London, 1974 and Victoria, 1975). My later role was less onerous, though to support Alberta artists in the show I produced and published a list of twenty-four Alberta nature artists in 1973, which was expanded by 1976 to include thirty-seven names (Spalding, 1976b). Hope, of course was included, but I don't remember whether she submitted in a later year when I served on the jury for the national show.

In 1970 I had hired a vertebrate palaeontologist for the museum: John Storer. A student of the Royal Ontario Museum's Loris Russell, John was keenly interested in Cenozoic mammals, but was also ready to tackle dinosaur material. I can't remember which of us first saw Hope's collection of fossil drawings, but I was developing a series of publications by staff and outside experts; with John's input and collaboration, Hope's *A Guide to Alberta Fossils from the Age of Dinosaurs* (Johnson and Storer, 1974) was published by the museum in 1974 as the fourth in the series.

When Philip J. Currie replaced John as Curator of Palaeontology in 1976, I accompanied him on a tour of key sites in the south of the province in July. We visited fossil and interpretive sites in the Drumheller, Dinosaur Park and Milk River areas, checking for fossils and making or visiting contacts. On July 12 we visited the Medicine Hat Museum and its new National Exhibition Centre, then headed for Redcliff to meet with Hope. Among topics discussed were location of collections, a film being done by Hope and Dale Russell of the National Museum, for the BBC, the impact of the new permit legislation, and the possibility of Hope borrowing bird specimens to assist her in doing paintings. Phil and I left Hope at 2:15 p.m., ate supper in Red Deer and finally reached Edmonton that night; my field notes for the trip end with arrival home at 11:00 p.m. (Spalding, various dates, field notes).

In later years I and my colleagues also arranged a number of one- and two-person and thematic nature art shows at the museum, starting with Rungius' works borrowed from the Glenbow and continuing with a show by Charles Lacy and Peter Karsten in 1975 and 1976 (Spalding and Scott, 1975). The first thematic show was *A Parliament of Owls* followed by *Transient Treasures*, a collection of prairie wild flowers by Helene White (White, 1977). Hope was included in this select group in 1978, when she was the subject of a one-person show called *Images from Alberta's Southeast: The Art of Hope Johnson*. In this she exhibited twenty-five pictures or groups in various media, including pencil, oil, pen-and-ink, and watercolour. The work featured (as far as I can remember or tell from the titles in the exhibit list) a self portrait, two portraits of First Nations subjects, geological features (eleven), fossils (four) and flowers (five). Julie Hrapko, Curator of Botany, wrote the information sheet about the exhibit and I believe she also curated it (Hrapko, 1978).

After a year of secondment to undertake interpretive planning for the then-unnamed Tyrrell Museum, and a few other special projects, I left the Provincial Government service in 1982 and began parallel careers as a heritage consultant, musician and writer. My interests in palaeontology did not lapse: during the '80s I published a new edition of Sternberg's *Hunting Dinosaurs* (Sternberg, 1985) and with colleagues undertook a number of interpretive projects for Dinosaur Park and ACCESS TV. In 1990 I moved to British Columbia and after a year or two was able to focus on writing books. Two of these

feature Alberta's dinosaur resources: *Dinosaur Hunters* (Spalding, 1993) from a world-wide perspective, and *Into the Dinosaurs' Graveyard* (Spalding, 1999) which tells Canada's story and mentions Hope several times. I kept in touch with Alberta, bringing in occasional cultural tours and speaking at a Tyrrell Museum symposium on Dinosaur Provincial Park, in 2005 (Spalding, 2005). This was the last time I met Hope—sitting in a wheelchair, but still wearing her long braids and still keenly interested in what was going on.

I am only sorry I had not had the opportunity to know Hope better. However, Darren Tanke's thorough and intriguing work gives us all the opportunity to learn more about this fascinating woman and great Albertan.

Introduction

The August 24, 2010 passing of Hope Johnson (Anonymous, 2010a–c) marked the end of an era for Alberta's amateur and professional vertebrate palaeontology communities. Her death affected other disciplines, too, as she travelled in many circles within the province for 65 years. How many among us can truly say they never knew her personally, or of her art work, or learned to identify Alberta prairie plants, or Late Cretaceous bones and teeth through her excellent vertebrate fossil identification books? A relative late-comer to palaeontology, during much of her middle and later life, and especially during the late 1950s to 1980s, Hope was a well-known and respected powerhouse in the Albertan amateur and professional vertebrate palaeontological communities. She was also heavily involved in the naturalist and visual arts communities as well as charitable organizations, but this book focuses more on her extensive activities mostly in Alberta vertebrate palaeontology; as well as providing examples of some of her fossil and botanical drawings and paintings. We will also illustrate scenes and vistas of Alberta's natural splendour; not as seen through her eyes, but how she interpreted it through her ink pen and paint brush.

From 1970 onward, Hope's name appears occasionally in popular books, bulletins, internal reports, scientific or science-related publications, or professional reports by others dealing with invertebrate or vertebrate palaeontology or their interpretation (*e.g.* Ramanujam, 1970; D.A. Russell, 1972d; Danis, 1973; Storer, 1973a, b, 1974; Moser, 1974; Currie, 1980a-b; Danis and Walper, 1980; Sargeant, 1980; Maier *et al.*, 1981; Noble and Rollans, 1981; Harbron, 1983c; Currie, 1984; Dodson, 1985; Barendregt *et al.*, 1988; Willock, 1990; Currie *et al.*, 1993; Gao and Fox, 1996; Stalker, 1996; Spalding, 1999; Gasser, 2001; Quinsey, 2004 (with a photo of Hope); Humber *et al.*, 2005; Tanke, 2011a, b; McKellar and Engel, 2014; and Tanke, 2015b), possibly because she focused more on collecting isolated bones, microvertebrates and other small fossils, a popular facet of vertebrate palaeontology for amateur collectors due to their frequency of occurrence (*e.g.* Noad, 2015), but important for larger-scale studies (Brinkman *et al.*, 2005). She did, however, appear in a great number of newspaper articles for much of her adult life relating to her activities in the military, performing arts, natural and human history, her paintings, and botany. Middle-aged or older readers involved in Alberta palaeontology certainly knew of her or may have met or dealt with her directly, but the younger crowd mostly knows of her by name only, and are now educated on common vertebrate fossil identifications with her three illustrated fossil guidebooks (Johnson, 1968a; Johnson and Storer, 1974; Alberta Palaeontological Society, 2009), the last two in particular.

The author's earlier articles (Tanke, 2011a,b, 2015b) were smaller papers on Hope, cut and pasted with some rewriting and editing, of an earlier version of the present book. Women in vertebrate palaeontology are often overlooked or underappreciated, but there is a recent and growing trend to recognize them (Turner *et al.*, 2010 and citations therein). Few big dinosaur discoveries can be directly attributed to Hope, but she made major contributions in many other ways not only in palaeontology, but in

botany, natural history, grasslands conservation, education and art. In hindsight, probably her biggest contribution to vertebrate palaeontology was not a single fossil discovery, or even a revolutionary idea or concept, but the numerous technical drawings of her fossil finds, which she shared with many via her fossil guidebooks.

An elderly friend told the author that Hope had missed her calling and should have been a teacher. But the friend completely missed the point—Hope *was* a teacher, who taught many of us how to recognize and identify the sometimes bewildering array of Late Cretaceous vertebrate fossils this province is blessed with. This matriarch of Alberta amateur vertebrate palaeontology should neither be overlooked nor forgotten.

Here is her life story in more or less chronological progression, based on a variety of scattered resources. This consists of the author's personal recollections; discussions and interviews with the Johnson family, friends and amateur fossil collectors (Tanke, 2010b); interviews with members of the public (some of whom knew her as far back as 1945) and professional colleagues; Hope's and others' letters; her military records; miscellaneous publications and other documents, especially newspapers; in-depth Internet research; and Hope's own writings about herself and her family history (many penned by her from May 10 to 12, 2006). We are lucky that she wrote these things down. Several voice recordings exist (Puckett, 1973; J. Danis, 1981; J. Danis *et al.*, 1981) and these provided important clues to her past and her work in DPP.

It became apparent in interviews with Hope's friends that they knew little about her earlier life. One remarked, "She was too busy with her present life to talk about her past life" (P. Dudley-Reese, pers. comm., 2014). Others thought she was more interested in fossils, her family, and friends than in her own heritage. It appears she did not keep a scrapbook of newspaper clippings in which she appeared. Visits to her daughter, Margaret Noone in Medicine Hat, uncovered over 250 family photographs, many with detailed notes written on their backs about who was in the picture, their relation to Hope, date, activity portrayed, *etc.* Interspersed among the pictures were additional detailed notes about family history, genealogy, and short biographies—all welcome grist for the historian.

Hope left behind some 35 mm photographic slides of her earlier artwork and field activities. Many of these are unfortunately scratched or damaged in other ways and appear to be remnants of larger sets. Nevertheless, they provide useful information and some were used herein. They were cleaned and scanned but some damage remains uncorrected. The often vibrant but earthy or natural colours that Hope used in her paintings may have faded on some of these slides. Many bear data penned by Hope and the date of processing printed on the slide mounts; these also provide historical information as to her field activities and that information is included herein, as well as data from her three fossil catalogue books, though the more useful data (places, dates, people involved) became scant in later pages. A visit with Hope's granddaughter, Rachel Noone, produced more examples of Hope's art. Appeals in the Medicine Hat News were published on December 30, 2014 (Anonymous, 2014c) and in the Grasslands Naturalists newsletter (Anonymous, 2014a) in the hopes of finding other people who knew her: a few more responses were received.

Hope touched so many lives and passed away relatively recently, so there is a wealth of memories and recollections from friends and colleagues, unlike other subjects from long ago in my "Remember Me . . ." series. Her three personal fossil catalogue books, numbering 341 pages and with many details minutely recorded, also provided many interesting clues about her amateur palaeontology career and her contacts in the 1950s to the 1980s. The author also encouraged Hope to write details about her life, family roots, and palaeontological activities for what was to be an article (but now this book!) which

she knew would appear after her passing. These were penned in 2009, but the author did not see them until March, 2011, after her death.

Some of Hope's personal papers were found to have inconsistent dates regarding specific events, sometimes differing by as much as one year. The reasons for these discrepancies are unknown, but are likely from her lapses of memory, or just a slip of the pen. In those rare cases, the preponderance of other evidence (such as dated museum catalogue records and field notes) is accepted as more accurate and her biographical chronology was adjusted accordingly. A recent paper on Hope by the author (Tanke, 2011a) appeared in March of that year, and a PowerPoint presentation on her is online (Tanke, 2011b). Soon after that paper was published, the large batch of Hope's writings (dozens of carefully written pages) emerged, some of which contradicted earlier statements (or even Hope's own documentation) that had been made in that paper. Here such discrepancies are also noted. This book will feature some of Hope's artwork, sometimes of a surprising subject or style. The author is not an artist in the classic sense, nor an art reviewer or critic, so will not speak on the virtues of her work. The art pieces will speak for themselves regarding composition, style, method, skill, experience, expertise, and imagination of the artist. Hope contributed so much, mainly to Alberta's palaeontological, natural history, and arts communities. The author can only hope he is equal to the task of telling you her fascinating life story. Here you will learn about her adoptive parents and their backgrounds, Hope's life story, what she accomplished across several natural history disciplines, her artwork and love of plants, volunteer work, awards and recognition she received and some of the amateur and professional people she met along the way, who taught and helped her to achieve greatness.

Talking to Hope was always interesting and on occasion she would go off on a related historical tangent as she told a story and taught you something new and interesting. The author has tried to emulate her detailed talking, writing, and teaching style with little sidebar snippets of information related in some way to her interesting life.

The way this publication is produced, in small print runs and on demand, creates the opportunity for updated versions as new information comes along. While the story herein is believed to be mostly complete, there are still unknowns about Hope's life and activities; thus there is still room for updates. For example, there are very few pictures of Hope in a badlands setting, but surely someone took pictures of her there. Virtually nothing is known of her teaching career at her mother's private school, St. Agatha's, in Victoria, British Columbia. Much of her artwork is scattered across the province, country and some even overseas; it would be nice to see those. Hope's 35 mm photographic slide collection seems to be only the sad remnants of a much larger collection. If you, the reader, have more information or imagery, please contact the author for potential future updates to this book and to Hope's historical record. This book is a special project that had its roots in my "Remember Me . . ." series on non-professionals and lesser known characters involved in Alberta vertebrate palaeontology (Tanke, 2008a, b; 2010a, c, d; and Tanke and Walker, 2011). Nearly a dozen more stories in this series are in progress and hopefully will be published sporadically in future issues of the Alberta Palaeontological Society *Bulletin*.

The author hopes this book will also be inspiring to readers; Hope shows us how an everyday person can achieve greatness if they persevere and put their mind to a lifetime hobby or project (or in Hope's case many hobbies and projects). However, that inspiration should be tempered with responsibility on the part of amateur fossil collectors in the province. Many of the fossil localities mentioned in the text are on private land where permission from the landowner is required for entry, or in Provincial Parks where public access may be restricted. Amateur collectors are reminded to get the proper authorizations from private land owners and to respect rules when visiting Provincial and National Parks and

other natural areas. Hope Johnson was very particular and careful in this regard, seeking landowner permission, and respecting Park or protected areas boundaries. Finally, the author reminds readers of the Historical Resources Act, R.S.A. 2000, c. H-9 (**http://www.canlii.org/ab/laws/sta/h-9/20040623/whole.html**) which requires permits for the excavation of most fossil resources within the province of Alberta.

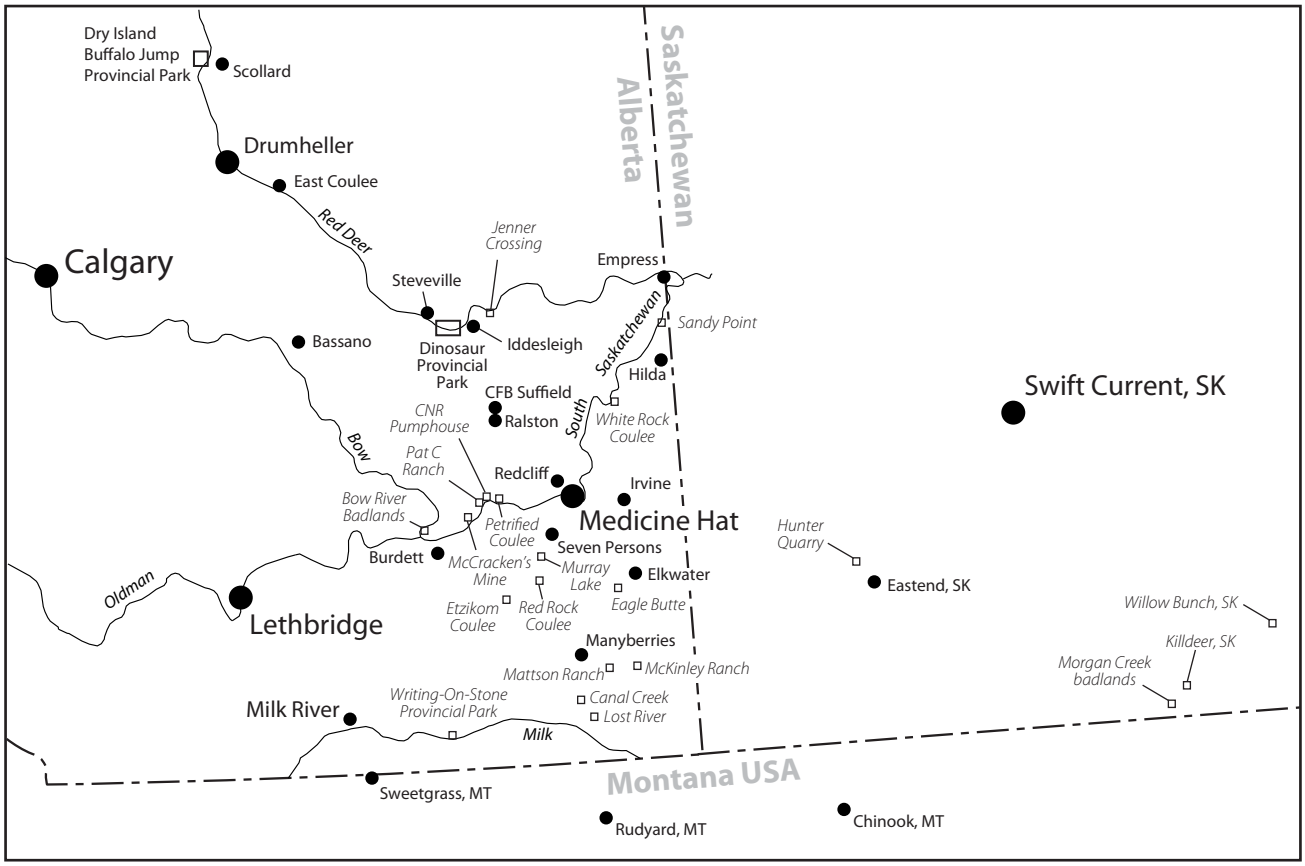
Some comment is necessary regarding name changes to some of the Canadian academic institutions identified herein. Some names have changed significantly over time. For example, the now Canadian Museum of Nature was formerly the National Museum of Canada and before that, the palaeontology section of the Geological Survey of Canada. The Royal Tyrrell Museum of Palaeontology (RTMP) got its start as the Palaeontology Department of the Provincial Museum of Alberta (PMA) and specimens were catalogued under the PMA acronym for several years before the RTMP opened its doors. In the fall of 1982 the Palaeontology Department of the PMA was transferred to Drumheller and became the Tyrrell Museum of Palaeontology. Fossils curated in Edmonton had a PMA number, *e.g.* PMA 67.5.56. After the transfer to Drumheller, the specimen number prefix (only) changed, so the same fossil was now TMP 67.5.56 and later (about 2000) the numbering system was expanded to suit a larger database, so TMP 67.5.56 became TMP 1967.005.0056, *etc.* Such changes are noted in the text. In this sense some fossils donated by Hope initially benefitted the PMA, then years down the road became part of the RTMP collection. A few items she collected stayed behind at the PMA and today are used for educational and comparative purposes.

In the course of researching this book, the author met a lady (L.S.) in her early 90s, at the Valleyview Retirement Community, who was remarkable in her own right—at one point serving as an "ack-ack girl," manning an anti-aircraft battery during the Battle of Britain. L.S. knew Hope the last six years of her life, sharing a table with her at mealtimes and they became friends. There she learned of Hope's zest for life, her encyclopedic knowledge, and was regaled by Hope with many stories from her interesting life. At the conclusion of our discussion about Hope, this woman summed it all up with one remark: "Hope Johnson—Now *there* was a lady!"

Acronyms used in this book

AB	Alberta
AMNH	American Museum of Natural History, New York
AMQUA	American Quaternary Association
APS	Alberta Palaeontological Society, Calgary
ASPG	Alberta Society of Petroleum Geologists, Calgary (CSPG after 1969)
BATUS	British Army Training Unit, Suffield
BBC	British Broadcasting Corporation, United Kingdom
BC	British Columbia
BCWSC	British Columbia Women's Service Corps
CANQUA	Canadian Quaternary Association
CFB	Canadian Forces Base
CMN	Canadian Museum of Nature, Ottawa (NMC prior to 1990)
CNF	Canadian Nature Federation, Ottawa (now Nature Canada)
CPR	Canadian Pacific Railway
CSPG	Canadian Society of Petroleum Geologists (ASPG prior to 1969)
CWAC	Canadian Women's Army Corps
DPP	Dinosaur Provincial Park, Alberta
GSC	Geological Survey of Canada
LL.D.	Doctor of Laws, honorary doctorate degree
MOR	Museum of the Rockies, Bozeman, Montana
NHM, NHMUK	Natural History Museum, London, United Kingdom
NMC	National Museum of Canada, Ottawa (CMN since 1990)
PMA	Provincial Museum of Alberta, Edmonton (RAM since 2005)
RAM	Royal Alberta Museum, Edmonton (PMA prior to 2005)
ROM	Royal Ontario Museum, Toronto
RSM	Royal Saskatchewan Museum, Regina and Eastend
RTMP	Royal Tyrrell Museum of Palaeontology, Drumheller (TMP prior to 1990)
SVP	Society of Vertebrate Paleontology
TMP	Tyrrell Museum of Palaeontology, Drumheller (RTMP since 1990)
UA	University of Alberta, Edmonton
UC	University of Calgary
UK	United Kingdom
UTM	Universal Transverse Mercator, geographical coordinate system
WGS 84	World Geodetic System 84, geographical coordinate standard
WWI, WWII	World War I, World War II

Map of Hope Johnson country



Hope's family heritage and early life (1916 to 1938)

Hope was adopted at the age of four months. Her arrival into this world, gleaned from notes to the family and author, occurred on March 17, 1916 in Vancouver, British Columbia (possibly in a convent or a home for unwed mothers). She was a cherubic, healthy, 9.5 pound (4.3 kg), blue-eyed and fair-haired baby (Figures 1 and 2). The place of birth given here is *contra* Tanke (2011a, b).

Her nineteen-year-old mother, otherwise unidentified, was not ready for the responsibilities of motherhood and gave her up for adoption. Among Hope's notes, curated in the Esplanade Arts and Heritage Centre Archives in Medicine Hat, she states that her mother (identity not given) was American and that her father, also unidentified, was "from the old country," but that both resided in Sardis, now part of southern Chilliwack, British Columbia. Her adoptive father apparently told her these details. Hope thought perhaps her biological father was a young soldier who perished in France during World War I, possibly during the Battle of the Somme, which occurred from July 1 to November 18, 1916. Using these clues, an attempt to find her biological father was made. Almost one hundred men from Chilliwack and area died in WWI (Chilliwack Museum and Archives, 2014) and of these, nine

Figures 1 and 2. Hope at age four months with her adoptive father (left) and at age seventeen months (right). Hope was well known for her long hair most of her life and it is already apparent at a very young age.

had a life or work connection to Sardis; only two died during the time period defining the Battle of the Somme. The first was Lance Corporal James Stronach (Anonymous, 1916a), thirty-five, and originally from Scotland. The other, Private Luke Charles Mahone, age twenty-one, was born in the United States and a member of the Nucc-chah-nulth First Nations from Nitinat (Pitadaht), on the west coast of Vancouver Island, British Columbia. He was a student at the Coqualeetza School in Sardis (Anonymous, 1916b). Neither of these men seems to be a good match, but their information is recorded here just in case.

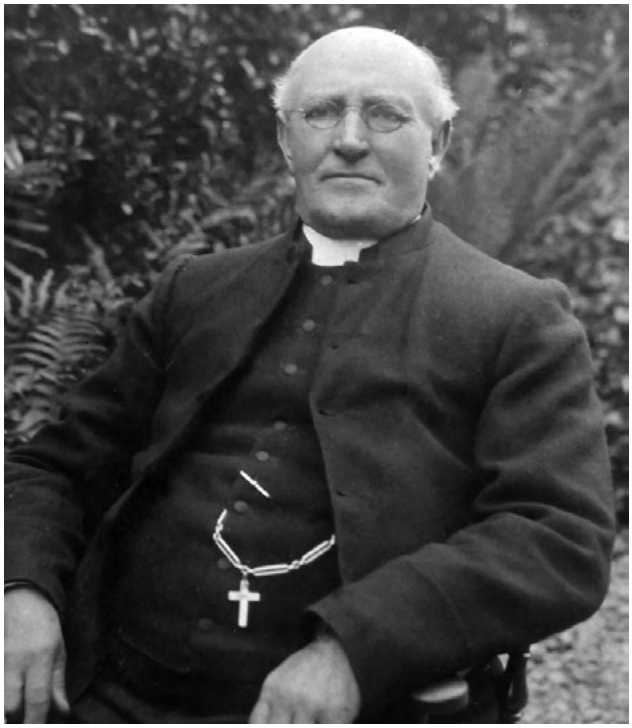


Figure 3. Rev. James Joseph Large, later in life, in an undated picture. Image courtesy of the Johnson family.

Hope's earlier notes to her family stated that she was born in Sardis, but her last penned notes state implicitly that she was born in Vancouver "no matter what my birth certificate says," though the adoption itself was registered in Sardis. Part of the confusion may have arisen because the registrar in Sardis was deaf, according to Hope. In another document, Hope says she was born in Vancouver but her biological parents resided in Sardis. The confusing matter remains unresolved.

As she was adopted, nothing is presently known of her biological parents or any siblings; she was named Barbara Elizabeth Hope Large, though she went by Hope most of her life. The origins of the given name Hope are unclear. Perhaps her parents felt they were giving the waif hope by adopting her. Another possibility is the nearby town of Hope, 28 km to the northeast. Her first name honoured her aunt Barbara, the sister of Hope's adoptive mother. She signed a piece of her artwork at age sixteen, "B.E. Hope Large"; in a 1932 high school class photograph she identified herself as Hope Large; in College she went by Barbara E.H. Large (Victoria Board of School Trustees, 1933); in 1936 she is identified "Miss B.H. Large (Anonymous, 1936c); and on occasion throughout her adult life she signed her name "B.E. Hope Johnson," usually on more formal documents—for example, on her mother's death certificate (1965) and her honorary doctorate paperwork (1981). Her artwork was signed "Hope Johnson," or more typically "H. Johnson," rarely accompanied with a place name, or a date; the latter appear more in her later works.

Hope's writings state that her adoptive middle class parents were "really rather old to have children so I was an only child." They married later in life—too late, perhaps, to have their own biological children, so they adopted. Her parents had an interesting background which certainly moulded Hope's early life and personality, so their lives are detailed here.

Her adoptive father, Herbert "Bert" Ernest Evylyn Large came from Sussex, England. He was not a large man, about 5 ft, 6 in, slight in build, and apparently had a bad temper. When Hope was older and physically big enough, she stood up to his rages, directed toward her mother, to calm him down.

He was born in Withyham, East Sussex, on July 28, 1864[1] to James Joseph Large (1835–1908; Figure 3) and Large's first cousin (such marriages were legal then), Caroline Robinson (*née* Hunt? 1826–1898; Figure 4). He was educated at a good school, St. Swithin's in Winchester, Hampshire.

When he was a boy, his schoolteacher father became an Anglican clergyman and, according to Hope's

notes, was good friends with author, historian, and naturalist Reverend Charles Kingsley (1819–1875; Anonymous, 2014e). Kingsley corresponded with biologist Thomas H. Huxley (1825–1895) for a time, was a tutor to the Prince of Wales, and was also appointed as Queen Victoria's Chaplain in 1859. Kingsley was also good friends with Charles Darwin (1809–1882) and a supporter of Darwin's evolution research, receiving an advance review copy of *On the Origin of Species* and is acknowledged in the second edition of that book. Given this scientific and natural history background it is not surprising to learn that Rev. Large's own interests included natural history, which was shared with son Herbert, who in turn passed it on and encouraged it in a little girl named Hope.

As a young man, Herbert was sent to be with his father, who was giving spiritual guidance to British colonists residing in remote corners of the world. It was an unwise, life-changing move for Herbert, as we shall soon see. His destination was British Guiana (now Guyana). The arrival date is uncertain but Herbert was there by about 1882, appearing in a posed studio portrait (Figure 5). For employment, he was involved in some managerial capacity overseeing labourers. During his stay he contracted malaria from a mosquito bite. He survived, but the illness would seriously afflict his health for the rest of his life.

Figure 4. Caroline Large (*née* Robinson), undated picture. Image courtesy of the Johnson family.

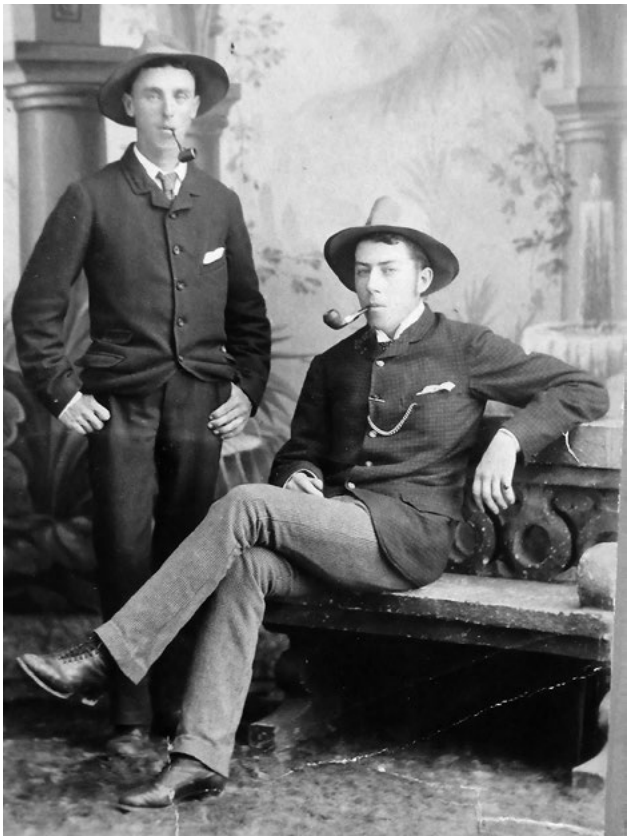


Figure 5. Herbert (left) wrote on the back of this photograph, "Two young Englishmen in British Guiana. Year 1882?" The other man is identified as Bernard Body. Taken in a photographer's studio in Georgetown. Image courtesy of the Johnson family.

The next phases of Herbert's life journey were documented by his future wife, Rachel (Large, 1957a), late in her life, and were also recorded in Hope's notes. The chronology of some of the events is a bit confused. After he recovered from his bout of malaria, Herbert set out for more agreeable surroundings, travelling to Trinidad, Barbados and Texas (the latter he described, according to Hope, as "gun crazy and dangerous"). His work activities and durations of stay in these places are unknown; though he may have been financially supported in the United States, about 1889, by his father (Large, 1957a), buying land (presumably in Texas) but later

Figure 6. Edward Richard Gunson in an undated photograph as a young lawyer in London, England. Image courtesy of the Johnson family.

selling it and finding that "Life was strange and not congenial," (Large, 1957a) perhaps in reference to the gunplay noted above.

Herbert wanted to be back on British Dominion soil and to make a personal fortune, so he moved north to Canada in 1892 (1893 according to Hope). But arriving by port that summer in Victoria, British Columbia, he found the city in the throes of a serious smallpox epidemic (Begbie and Praeger, 1893; Johnson, 2013). Rather than flee, he bravely took it upon himself to help. He volunteered at a quarantine hospital—built in just three days—for treating the growing number of patients. Having once been seriously ill with malaria and cared for by strangers, perhaps he felt obligated to pay the favour forward. In any case, his staying in a dangerous situation speaks volumes about his good character as a younger man. In Victoria he made contact with Dr. John S. Helmcken (1825–1920; Anonymous, 1920a, 2015b) and evidently became friends. Hope's mother wrote a glowing account of the doctor in a newspaper article (Large, 1957b) which was later used in a Royal Jubilee Hospital centennial history (Murphy, 1958, p. 81).

Once the smallpox crisis was over, Herbert set out on foot for regular work and became a helper at a farm (Large, 1957a) owned by Solomon Dean (1839–1903), an early pioneer in the Victoria district (Anonymous, 1933a). Large (1957a) states that the farm was located sixteen miles (26 km) north of Victoria, though period voter listings put Dean in South Saanich, closer to Victoria. Herbert stayed for about a year, working hard and carefully saving his earnings. From there he moved to the mainland and settled in New Westminster, British Columbia where he made friends and through them found a job as a Clerk Steward at the "Sons of England" Men's Club (actually a benevolent society). A 1901 census (Library and Archives Canada, 2015) lists him at age 36, a boarder with the Broad family, employed full time in New Westminster as a club steward, making $500 per year.

By being thrifty he saved some of his income for future plans. Eventually he had enough money to buy a farm, a goal he long coveted. He found his way to Sardis (population about 500), where a sister lived, a Mrs. Walter George Edwards (Ada Elisabeth Large, 1860–?). Herbert lived on Prest Road (Anonymous, 1928), just east of Sardis, and tried farming there for a while (Figure 11). However, his education, physique, disposition and especially his malarial relapses evidently did not lend well to this line of work. His physical illness may have affected his personality: Hope described him as a "tyrant" and she had several serious confrontations with him (B. Preston, pers. comm., 2014).

The fact that his illness affected his ability to work and perform his duties as breadwinner for his family must have been frustrating and demeaning. Large (1957a) states that he settled for good in Victoria and that his latter days were spent "tranquilly and happily there" (Figure 43).

As already noted, Herbert was greatly interested in nature, which obviously rubbed off on Hope. Despite his serious malarial symptoms, he lived a relatively long life, dying at home in Victoria on August 25, 1949, aged 85 (Anonymous, 1949b), of kidney failure (according to Hope), though his death certificate implies arteriosclerotic cardiovascular disease with which he was afflicted in his final two years. He is buried in the Hatley Memorial Gardens in Colwood, Vancouver Island (Figure 18)—coincidentally, less than 500 m from where the author's parents retired.

Hope's adoptive mother, Rachel "Dollie" Emily Gunson (December 21, 1875–June 10, 1965) was born

to Edward "Teddy" Richard Gunson (May 25, 1842 or 1843?–October 3, 1909; Figure 6) and his first cousin Caroline Ada Hunt (1844–1920; Figure 7) in Walthamstow, Essex, then just northeast of London, but now part of Greater London. The family lived on Albany Road Leyton and Prospect Hill in Walthamstow for about ten years. Rachel was their fourth daughter and fifth child. Hope described her mother as "very small (4 ft, 11 in), capable, energetic and bright."

Figure 7. Caroline Ada Hunt. Image courtesy of the Johnson family.

According to Hope, Rachel came from a well-to-do family of five or more children (Figure 8) and was home schooled, at first by a governess, then at a private school for several years. She was cultured and multilingual; among her possessions when she came to Canada was a two-volume set of French plays by J.-B. Poquelin ("Molière,"1622–1673; Poquelin, 1893). Her father was a Notary Public (business lawyer) who suffered from agoraphobia[2] so little Rachel accompanied him on his daily rounds for emotional support. She learned much about life from him as they travelled about London. When the need arose for her father to deal privately with a client, she would be given some money and left alone at a local art gallery or museum where, largely on her own, she learned much about culture and the world through paintings and gallery exhibits. Although apparently not artistic herself, she did appreciate art and years later strongly encouraged Hope's early interest in drawing and painting.

At some point Rachel received teacher training in Cardiff, Wales. It must have been early in her adult life, for in 1901 at age twenty-five, Rachel was a lodger and employed as a Daily Governess for a family living in Rena Cottage on Church Road in Rustington, England (Anonymous, 2011). A 1911 census showed Rachel residing in Littlehampton, West Sussex, employed as a schoolteacher (England and Wales census, 1911). No formal portrait pictures of her as a young woman are known to exist. She appears in a group photograph (Figure 9) at a fancy social garden party.

Figure 8. Some of the Gunson children in 1885. Hope's adoptive mother Rachel is at upper right, age 9. Image courtesy of the Johnson family.

Rachel later decided to emigrate to Canada. Her teaching credentials were at first not recognized here, but this hindrance was eventually overlooked due to a shortage of teachers in British Columbia. She got to western Canada by advertising her interest in helping convalescents.

Details of her first work activities in Canada are a bit confusing, with family and other histories showing her employed at numerous jobs over a very short period of time; perhaps too many jobs, but the details are recorded here for completeness.

Rachel arrived in Canada, alone, at the age of thirty-four, departing from Liverpool and arriving in Quebec on the Canadian Pacific transport and passenger ship *SS Lake Manitoba* on August 13, 1911. At immigration, she gave her former occupation as "teacher" and, curiously, her destination as Calgary, where she planned to work as a "domestic." She headed west by train but did not stop in Calgary. She continued west, where she worked for the Cooke-Hearle (Hurle?) family, who apparently had paid her passage, on Kootenay Lake, British Columbia.

Figure 9. Rachel, standing in the centre foreground wearing a white blouse and dark skirt. The picture was taken at "Rosemead" Garden Party, Littlehampton, Sussex in 1910. Image courtesy of the Johnson family.

Johnson family history has Rachel joining her sister Barbara Avice Gunson (June 8, 1885–March, 1965, Worthing, England) already living in Sardis. However, Anonymous (1939d) has Barbara moving to Sardis following Rachel, in 1913. Barbara (Figure 10) married Wilson Stacey (died September 1949, aged 86, in Worthing, West Sussex), a pharmacist twenty-three years her elder. They lived on Prest Road in Sardis, not far from the Larges. Later, Barbara disapproved of the adoption of Hope, telling Rachel "there is bad blood in her." Well, we all know how wrong she was! Barbara was never kind to Hope as a child. A Johnson family story relates Hope visiting the Stacey home to play with her cousin Christopher; he was given milk to drink—Hope got water.

Barbara moved back to England in 1933 (Anonymous, 1939d); her relationship with Hope in later years is unrecorded.

In 1911 Rachel began teaching at the Coqualeetza Institute (Large, *c.* 1960; United Church of Canada Archives, 2014), one of the now controversial residential schools set up across Canada for First Nations children (Anonymous, 2015a and citations therein). She was noted as being a teacher there during two school inspections conducted in April and September of 1912 (Ditchburn, 1913). The school had about ninety children. She found parts of the work quite troubling:

> I was rather terrified of the huge Indian youths who were difficult to control in class, and even more hard to manage at the breakfast table over which it was my duty to superintend . . . According to rule, Grace must be said before they sat down, but to silence them first was far from easy, and all through the meal they would be hurling cherry stones at the opposite table [where the girls sat]. There was always plenty of such ammunition for the tables were laden with fruit and the children generally, fed like fighting cocks. (United Church of Canada Archives, 2014).

Meting out harsh discipline to disobedient students was also an issue for her: "The size of those great lads whom I was supposed to chastise with a whip—but never could, intimidated me." (United Church of Canada Archives, 2014). Disliking the mistreatment of the children at these schools, she questioned a Baptist Principal at one, "what he felt the main aim of that school was," and according to Hope's notes was told, "to teach the pupils 'to flee from the wrath to come!'" This was

Figure 10. Barbara Avice Stacey, Hope's aunt with her son Christopher Cyril, seen here about 1921. Hope was quite fond of Christopher but lost track of him much to her disappointment. He went on to become a reporter with the Worthing Gazette in Sussex, England. Image courtesy of the Johnson family.

evidently too much for her and she quit soon after. The ugly legacy of these schools is still with us today, leading to the *Truth and Reconciliation Commission Report,* issued in 2015 and the healing that now must occur.

The circumstances under which Herbert and Rachel met is unknown. Family history says they met in Sardis and were married in the Holy Trinity Cathedral in New Westminster on November 30, 1912 (Figure 12). Hope described this union as a marriage of convenience; he 48 and she 35. Asked to explain this comment, Hope's daughter Margaret stated that Rachel, being a spinster in her early thirties, was kept afloat during financial difficulties, along with her other penniless sisters, by an older brother. She did not want to continue being a financial burden on him, so set out for Canada to start a new life and decided to marry the first good man that came along (M. Noone, pers. comm., 2015).

Rachel was involved in the Women's Auxiliary and is credited for her prominent involvement in the training of local children for Christmas 1913 tableaux, including "Fairy Queen's Birthday," "Nursery Rhymes" and others (Anonymous, 1913). This was perhaps the first of a number of mostly festive-season performing arts productions she would be involved with. In 1915, under the auspices of the Sardis Hospital Auxiliary, both Herbert and Rachel presented an "amusing comedy sketch" (Anonymous, 1915).

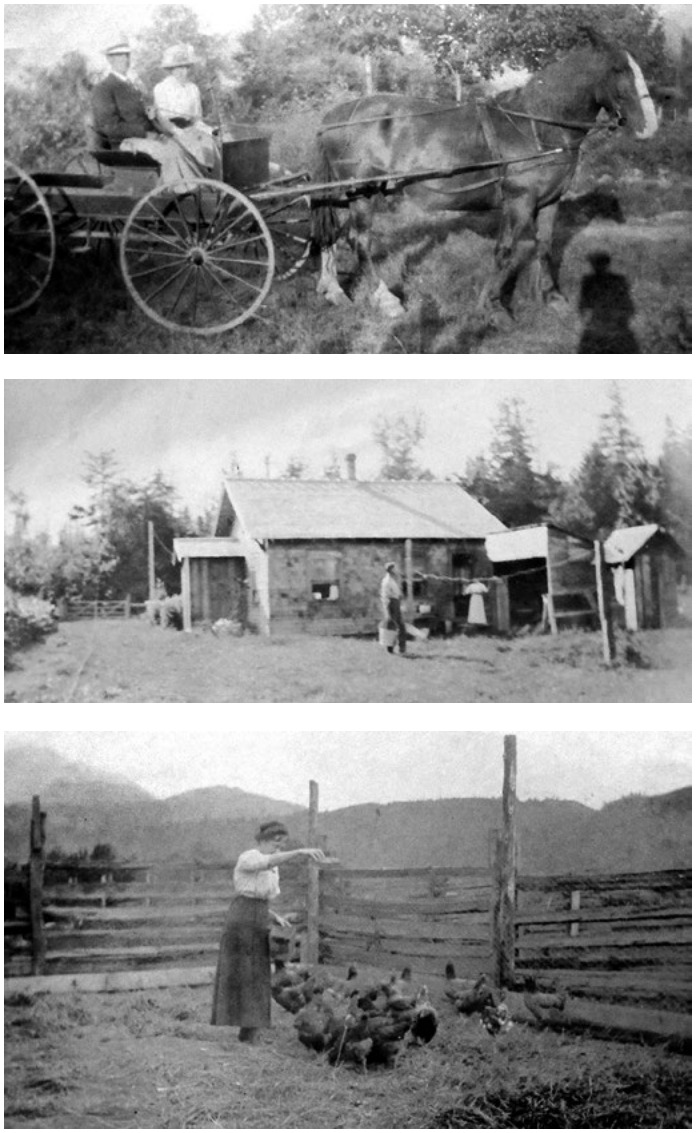


Figure 11. Life on the Larges' farm in Sardis. Top: Herbert and Rachel with "Laddie" the horse. Middle: the farmhouse. Bottom: Rachel feeds the chickens. Left: Herbert and Rachel with "Max" the dog. Images courtesy of the Johnson family.

Advertisements in the *Chilliwack Progress* newspaper for 1915–1916 (not cited here) show Herbert selling farming implements; a sow pig complete with her litter; Rhode Island Red chickens and fertile eggs; cattle; and Flemish Giant and Belgian hares. Despite Herbert and Rachel's domestic post-Edwardian bliss (Figure

13), something was missing from their lives, and that was a child.

It is uncertain how soon after her birth that Herbert and Rachel adopted Hope; the earliest known picture of her was when she was about four months old (Figure 1) and it is thought that she was adopted then (M. Noone, pers. comm., 2015). There is excellent digitized newspaper coverage for the area where the Larges lived during this period, but they must have lived a relatively low key life as they only occasionally appeared in the local paper (Anonymous, 1913, 1915, 1918a, b, 1919, 1921, 1928), and then usually only in the briefest of terms. In a newspaper with regular birth announcements, surprisingly there is no mention at all of Hope's adoption.

The family was one of many in the Sardis area who donated apples to soldiers fighting overseas (Anonymous, 1918a). Herbert stuck it out in the farming business for about five years after he married Rachel, but ultimately his malarial symptoms weakened him too much for that type of work so he sold out. The *Wrigley's Directory* from two years after Hope's adoption lists the family residing in Sardis and Herbert's occupation as "dairying" (Wrigley's, 1918, p. 405). The family moved away from their farming homestead in Sardis in the late summer or fall of 1918 and went to Victoria on Vancouver Island, according to Hope's notes. But they did not stay there long—in fact, Hope may have erred on this account. Anonymous (1918b; Figure 14) announces an auction sale of Herbert's farm and other possessions: "Mr. H.E. Large of the Prest Road will sell his dairy stock, pigs, poultry, implements, feed, furniture by public auction on Tuesday, September 10 at 1:30 p.m."

Figure 12. Rachel and Herbert Large; possibly in a wedding photograph, the style of clothing is suggestive of about 1912. Image courtesy of the Johnson family.

Figure 13. Herbert and Rachel at home in Sardis, 1913. Courtesy of the Johnson family.

If the family was planning on moving to Victoria, the sale of their furniture is inexplicable. Anonymous (1918b) shows that they sold everything except the land itself. But more than a year after the auction, Anonymous

AUCTION SALE

Under instructions from H. E. Large, of Chilliwack, B.C., I will sell by Public Auction, on

TUESDAY, SEPT. 10, 1918.

Commencing at 1:30 p.m., sharp, at his place situated on the Prest road and first place north of John McIntyre's, the following live stock, etc.:

Cattle—One Ayrshire heifer, 3 years old, full milk and bred. One Holstein and Jersey grade, 4 years old, bred May 26th. One Ayrshire grade, 3 years old, bred July 14. One Jersey, 7 years old, full milk and bred. One heifer 2 1-2 years and bred. Two heifer calves.

Pigs—Two pigs.

Poultry—8 hens, 8 pullets, 9 bantams.

Implements—Top buggy, plow, harrows, 1 Planet Jr. seeder and cultivator, lawn mower, set driving harness, set work harness, garden tools, forks, poultry netting, etc.

Feed—Several tons of oat and and clover hay.

Furniture — Bedstead, dresser, washstand, chiffonier, Morris chair, rocker, office chair, davenport in green velour, hand sewing machine, quartered writing desk, dining table, chairs, Perfect pantry, high chair, 1 cook stove, heater, reversible carpet square, crockery, kitchen utensils, etc.

Contributed by Others—Dresser, washstand, iron bedstead and springs, two seagrass chairs, 1 rocker, 3 chairs, lounge, two tables, Bell Oak stove, and other articles.

Terms Cash.

WILLIAM ATKINSON, Auctioneer.

Box 34 Chilliwack. Phone 141

Figure 14. Newspaper advertisement announcing the late summer, 1918 sale of Herbert Large's farming operation. Note how even Hope's high chair was on the auction block. From Anonymous, 1918b.

(1919) notes their overseas departure ("Mr. and Mrs. Large and family") from Sardis, in early December of 1919; so they must have stayed in town, no longer farming. What they did during this period for work is uncertain; an immigration form for an overseas trip in 1921 gives Herbert's occupation as "gardener." Apparently they were planning to leave Canada and return to the land of their birth—and to do so permanently. This was the first of three events beyond Hope's control that almost prevented her ever living in Alberta—and ultimately pursuing all those things she did in and for this province.

Immigration records on the *Ancestry.com* website show that the trio left Saint John, New Brunswick on December 14, 1919 and arrived in Liverpool. What Herbert did for work there is unknown. A photograph in one of Hope's albums (Figure 15) shows her in Rustington, West Sussex (southwest of London, on the English Channel) during this time. Rachel had been a governess in this area in 1901.

About a year and a half after leaving Canada the Larges realized that their move had been a mistake: both Herbert and Rachel were disenchanted by the ways things had changed and with life in general in post-WWI England (M. Noone, pers. comm., 2014). The three returned (in fact they were "glad to return": Large, 1957a) to Canada via Liverpool on October 22, 1921 (October 29 was suggested in Anonymous, 1921), travelling third-class on the White Star passenger liner *SS Megantic,* arriving in Montreal.

Other details on the immigration forms record that Herbert was fifty-seven and a gardener, Rachel was forty-four and Hope was five-and-a-half years old. On Hope's form is the question: "Can you read?" to which "Yes" is written in, suggesting this bright child was already reading. Her place of birth is given as Vancouver, BC. The form records that the family's last address in England was "Warbeck," Spencer's Wood, near Reading, in Wokingham, not far west of London. "Warbeck" (actually Warbrook) is the name of the house the Larges occupied on Basingstoke Road (Figure 16). Their activities in Spencer's Wood are unknown (J. Brice, pers. comm., 2016). The family's religion is stated as Anglican and their home address 1336 Johnson Street in Victoria. Years later, Hope recalled celebrating her fifth birthday on board a ship (Forbes, 1998), but this appears to be in error, or possibly it occurred on another trip or was another birthday.

There is some confusion about Rachel's work activities upon returning to Canada. It is known that she started a small private school called St. Agatha's or St. Agatha's Junior School, which a later newspaper advertisement (Anonymous, 1937b; Figure 17) states was founded in 1924. A timeline pieced together from sketchy information suggests that at the beginning of Hope's elementary schooling the Larges lived in Victoria, where Hope was probably home-schooled by her mother for two years, starting at age six (in 1922) until Rachel erected St.

Figure 15. Hope (left) and "Thomas" in Rustington, West Sussex, England, 1920. Image courtesy of the Johnson family.

Agatha's in 1924. The balance of Hope's elementary schooling, to June 1930, was under the St. Agatha's banner. It appears that the family (and with it St. Agatha's) relocated to the Burnaby-New Westminster district between the summer of 1927 (Anonymous, 1927) and some time in 1930, when Herbert is again recorded as living in Victoria (Vancouver Public Library, 2014; see Page 33). In Hope's military records she stated that her elementary schooling took place in New Westminster, which would refer to her location when she graduated (Grade 8) in September 1930. St. Agatha's activities ceased when Hope entered high school back in Victoria.

Figure 16. The "Warbrook" house, now renamed Woodvale, in Spencer's Wood, England. The Larges lived or stayed here in the early 1920s. Image courtesy of J. Brice, Spencer's Wood.

It seems that all lessons at St. Agatha's were taught in the Large family home. The family rented one floor of a house to live in and dedicated a room or two to school lessons. Rachel also taught shut-ins. A classified ad in 1930 stated that she also taught male mentally handicapped pupils or those needing "special tuition" (Anonymous, 1930b). Other newspaper advertisements in the 1930s (Anonymous, 1937b and 1937c) indicated that the school also specialized in teaching children with poor or defective eyesight and a nursery was added for children aged two to five. She did a good job at this and was paid by the British Columbia and—inexplicably—the Alberta Government, according to Hope's notes, though the author could find no record of her having taught in Alberta.

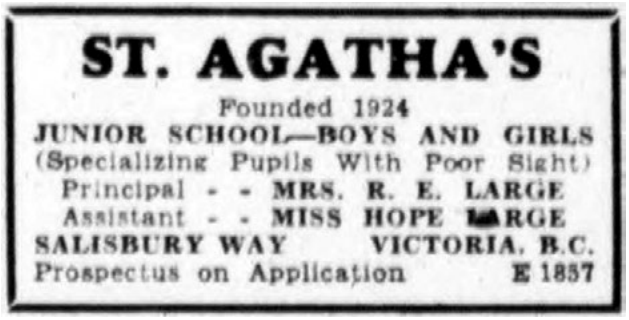

ST. AGATHA'S
Founded 1924
JUNIOR SCHOOL—BOYS AND GIRLS
(Specializing Pupils With Poor Sight)
Principal - - MRS. R. E. LARGE
Assistant - - MISS HOPE LARGE
SALISBURY WAY VICTORIA, B.C.
Prospectus on Application E 1857

Figure 17. Newspaper advertisement for St. Agatha's School, from the Victoria *Daily Colonist*, August 8, 1937, p. 11.

Following a hiatus of five or six years, from the time Hope finished her elementary schooling in 1930, the St. Agatha's name reappears, starting in 1936, when Rachel is recorded (Vancouver Public Library, 2014; see Page 33) as its proprietor, in Victoria. Apparently Hope, by this time twenty years old, assisted her mother with lessons.

The existence of newspaper advertisements for the school suggest that hers was truly a private school and she was paid solely by the students' parents. She taught until at least 1946, though perhaps on a smaller scale and not under the St. Agatha's School banner. While Rachel did teach private lessons, from Hope's last penned notes about her life she thought her mother did not have her own private school in Canada (*contra* Tanke, 2011a, b; but Hope was certainly mistaken: see Anonymous, 1936b, 1936c; Forbes, 1998 and see below). Little is known of these small private schools (Barman, 1980). In a detailed account of them in the Larges' area (Barman, 1998), Rachel's school is not included, nor did the author of that paper or her interested colleagues have any knowledge of it. It must have been quite a small school with few students; Anonymous (1936c) lists eleven: six boys and five girls, participating in a St. Agatha's Junior School Christmas concert.[3] Besides teaching, Rachel was also interested in women's social issues, chairing a meeting of "the newly-organized Women's Movement" in 1933 (Anonymous, 1933b).

Due to husband Herbert's chronic illnesses, Rachel was the breadwinner of the family (M. Noone,

pers. comm., 2014); Hope's notes say that Herbert was not a "good earner." Later in life Rachel penned a number of stories that were published in supplemental issues for the *Colonist Islander* newspaper (Hallmark Society, 2011; Large, 1956, 1957a, 1957b, 1957c). She died in Victoria on June 10, 1965, aged 89. Curiously, her death certificate lists her prior occupation as "home duties," completely overlooking her dedicated teaching past, yet husband Herbert's death certificate gave his occupation as "teacher and farmer"; there is no evidence that he had formal training as a teacher. Rachel was cremated, her ashes interred with Herbert's casket (Figure 18).

Figure 18. Rachel and Herbert's headstone at the Colwood Cemetery, Victoria. UTM coordinates are 10U 463726 5365301 (WGS 84). Image by the author.

Growing up, Hope admitted in her notes that she was a "tomboy" and liked to "play at Indians, pirates, shipwrecks, *etc.*!!" She was intensely curious about almost anything and this trait would be with her for her entire life. She was responsible for the gardening work around the family home when she was young. In her youth she was constantly surrounded by rich flora, nature, and domesticated animals, be they cats, pigs, dogs, horses, chickens, hares, or goats (Figure 19).

She appears in several photographs at age eight (Figures 19 and 20) with her hair down, wild and free, and another with it tied back; it is really hard to see that they are in fact the same girl. Pictures of her in her mid- to late teens and as a young woman are equally different in mood and appearance (Figure 21 a–c). It appears that like many young women, Hope experimented with, and changed her "look," making it difficult at times to identify her in a photo. However, a distinctive "widow's peak" can be seen in her forehead hairline in some pictures, aiding identification.

Figure 19. Hope, age eight (1924) with "Susie," a Toggenburg doe at their homestead, 3711 Old Saanich Road, north of Victoria. Image courtesy of the Johnson family.

Hope did not have any other adoptive or known biological siblings. She learned at age fourteen that she was adopted, but this status never really bothered her; she just accepted it and moved on with her life. That her parents were British was revealed in Hope's soft-spoken voice which had an unmistakable touch of a British accent, especially noticeable when she was younger; it may have been reinforced during her stay in England when she was of pre-school age.

Hope was a bright and gifted child, developing interests in art and drawing at age three (Caldwell, 1985b), even drawing a bird with the leg articulations, confusing to many (*i.e.* the typical "knee pointing-backwards" confusion), but in her drawing it was done correctly (Forbes, 1998). By age four she was being taught French and a bit of Latin by her mother, noting years later that she even developed a proper French accent. She was already developing a strong interest in plants, rocks and animals (Figure 20). At a young age she was exhibiting a desire to teach others—she tried to teach the family dog to read (Forbes, 1998). In her first of many newspaper appearances during her long life, Hope wowed a

reporter with a poetry recital she gave at her mother's private school during a Christmas concert in 1924. The reporter noted (Anonymous, 1924): "Perhaps the outstanding feature of the older girls' part was the recitation of *Hiawatha's Departure* by Hope Large—a difficult poem for a child of eight." The reporter was accurate in assessing the poem's difficulty: a W.W. Longfellow opus, 184 lines in length! Hope also earned an award for her efforts in nature studies and needlework. She is noted in Anonymous (1926a) for reciting a French fable in front of others and participating in several folk dances with other girls, both at her mother's school (Anonymous, 1926b). In her notes she mentions that she was in the Girl Guides in her early teens, but nothing more is known of this. Her first love was plants, surrounded as she was by many native and domesticated varieties in her childhood, and being caretaker of the family garden. She became strongly interested in botany during her teenage years (Caldwell, 1985b).

Figure 20. Hope again at age eight, now looking more prim and proper, with her adoptive cousin, Christopher Stacey (Figure 10). Image courtesy of the Johnson family.

Figure 21 a, b, c. (L–R): Pictures of Hope as a teenager. **a.** Undated, but believed to be age 14. **b.** Age 16. This photograph appeared in her 1931–1932 high school yearbook; see Anonymous, 1932, p. 20. **c.** Carefree and happy; an undated picture, possibly taken in her later teens. Images a and c courtesy of the Johnson family.

Early in her life Hope received some professional art training at a private school near Vancouver that taught the Royal Drawing Society method of Professor Ablett (Anonymous, 2009). The style was introduced by Thomas L. Ablett (1849–1945), founder of the Royal Drawing Society in England in 1888. It promoted the teaching of art to children, since they often draw before learning to write. The technique encouraged them to draw from memory and to try imaginative drawing.

From Grade 9 (September, 1930) until matriculation (University entrance certificate) in June, 1932 she attended nearby Victoria High School (Figure 22), whose school motto, *Palma non sine pulvere* (No reward without effort) Hope apparently took to heart her whole life. She did quite well there, taking Latin and science as specializing options, though mandatory algebra may not have been her forte. Her chemistry and biology classes were liked and well taught and were, according to Hope, "very useful to me my entire life." There she was a three-year member of the girls' Portia Debating Society. Her name could not be found in the 1931–1932 yearbooks to indicate she was involved in any other clubs or team sports. Her nickname in school was an unkind "Hopeless" (M. Noone, pers. comm., 2015). Some details

Figure 22. Hope stands on the far right in this female-dominated 1932 Victoria High School class photograph. Image courtesy of the Johnson family.

of Hope's school year were recorded in her high school yearbook by her chums (Anonymous, 1932, p. 22): "Hope Large: Has lately become one of Portia's leading orators. She 'speaks as one fed on poetry' with a faint touch of classic humor—in Algebra? No hope!"

In her mid-teens she was involved with the Victoria Boy Scouts for about six years, much of it as an assistant or leader, or "Akela." Nothing more is known of this volunteer work, except that she learned St. John's Ambulance First Aid. Her leadership skills and first aid would play an important role later in life. Also, sometime in her youth she took lessons in German (passing with first class honours, according to her military records) and singing and piano for several years; but the latter two she evidently did not pursue into adulthood. By age sixteen the long braided hair that so distinguished Hope most of her life was apparent (Figure 21 a, b) though she did let her hair hang free on occasion (Figure 21 c).

These were tough financial times for the Large family and they were "on relief" during Hope's high school years. Her mother apparently closed her private school once Hope reached high school age. Through ages fifteen to seventeen Hope was able to find summer work as a fruit picker north of Victoria. She picked strawberries, and a *c.* 1930 picture of her with others her age shows them about to go out into the orchards to pick cherries (Figure 23). Efficient fruit pickers could earn up to $1.80 per day, about $27.30 in today's dollars (Bank of Canada, 2018).

Figure 23. Hope (sitting on fender) with fellow fruit pickers. The scene dates to about 1930. Courtesy of the Johnson family.

Hope's oldest surviving art piece comes from about 1932, a portrait of the family's horse, "Laddie" (Figure 25). Nothing else is known about her art activities during this time.

After high school graduation, Hope completed one year (1932; Anonymous, 2014r) at nearby Victoria College of Arts and Science, a subsidiary of the University of British Columbia. She attended from September, 1932 until April, 1933 and appears in a class graduation photograph (Figures 24, 27). The College was then based in a rather swanky setting: the Craigdarroch Castle[4] (Anonymous, 2014g; Reksten, 1987), a four storey granite, oak and stained-glass, thirty-nine room mansion built in the 1890s in the Scottish baronial style (Figure 26). The castle was built for Robert Dunsmuir (1825–1889), a local coal baron (Bowen, 1999). He died prior to its completion and the surviving family lived in it for about seventeen years. From 1919 onward it served a variety of functions but was a college from 1921 to 1946 (Craigdarroch Castle Historical Museum Society, 2014; Reksten, 1987).

In this opulent setting Hope studied chemistry, biology, English, Latin and Homeric Greek. The latter she found very useful in her later geological, palaeontological and botanical studies. Over the years students carved their names into the ornate, dark brown wood panelling in several rooms; autographs included Canada's later historian and author, Pierre Berton.

Figure 24. Hope is seated in the front row on the left (white arrow) with the Victoria College of Arts and Science class of 1932–1933. See Figure 27. Image courtesy of Karen Humble and Tessa Hawkins, Craigdarroch Castle (Victoria).

Figure 25. "Laddie" the family horse, the oldest known piece of artwork done by Hope, *c*. 1932, when Hope was 16 or 17. Image courtesy of the Johnson family. See also the colour version, Figure 161.

Figure 26. Craigdarroch Castle in 2015, Victoria. Photograph by the author.

Figure 27. Detail of Figure 24.

The author and his wife, Patty Ralrick, were given the opportunity to look for Hope's name among the hundreds of signatures, but none was found. Hope wished to pursue a full university education but like many students in the Great Depression she simply ran out of funds and had to drop out of school after the first year. If she had completed her two-year stint, the 1933–1934 term would have cost $108.00 (Victoria Board of School Trustees, 1933, p. 8), or about $2,030 in 2018 dollars. Years later she revealed her regret at missing a university education. But she felt she made up for it over the years by "reading voraciously" (Segal, 1994) and through her regular interactions and numerous other contacts in the professional vertebrate palaeontology and other communities.

We don't know what Hope had planned on doing with her life, though her courses suggest something arts and/or science-based. Her activities in the years between leaving college in 1933 and 1938 are incompletely known. She did not detail them in notes to the author and her family. There are existing interviews with her during her military service and from 1976 in Ralston, Alberta (Smith, 1976). The ongoing digitization of period Victoria newspapers and the posting of these online at the *Internet Archive* (**archive.org**) website and other Internet sources provide some clues to her comings and goings during this time. In Smith (1976, p. 3) Hope vaguely states that for work during this period she "did little bits of this and that." She became involved with the Boy Scout movement in Victoria in 1934 (Anonymous, 1962r). She was "elected to membership" of the Victoria chapter of the

Royal Canadian Astronomical Society on October 2, 1935 (Shaw, 1935). Her subsequent activities with that organization are unknown.

Finding work during the Great Depression was difficult for many Canadians, though Hope was a teacher at her mother's private school around 1936–1937 (identified as an "assistant" in Anonymous, 1936b and 1937b and even as "Miss HB Large, school principal"(!) in Anonymous, 1936c). Her military records disclosed that she did unspecified "odd jobs" during 1937 and 1938. They further revealed that from February to October 1938 she attended business college (possibly Pitman College in Vancouver) and received a "Pitman Prof.[essional] Cert.[ificate]." There she trained as a stenographer and typist, though in a late 2006 letter to a professional colleague and friend she revealed: "Although I was once a stenographer, I never liked typing" and in Smith (1976, p. 3) she states that she "didn't care for" that course. But she made a go of it: her military records disclose that on enlistment, she could write 120 words per minute in shorthand and transcribe shorthand notes at 45–50 words per minute, adding "The department in which I worked naturally demanded a high standard of accuracy in spelling and typing."

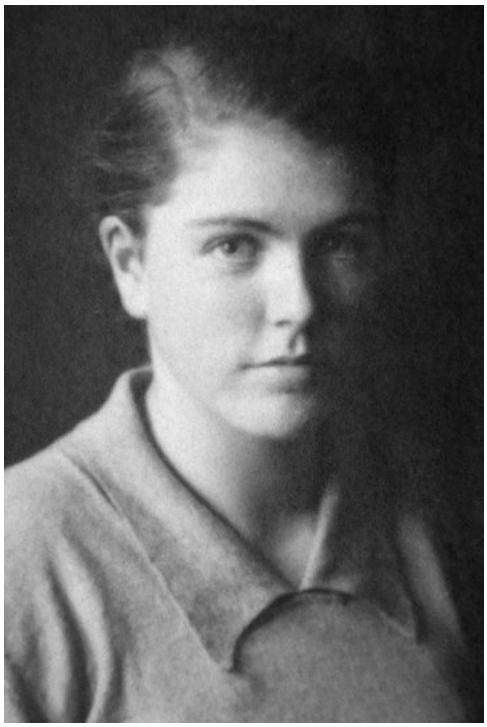


Figure 28. Hope as a young woman, 1930s. Image courtesy of the Johnson Family.

In her early twenties Hope (Figure 29) was employed as a stenographer and ledger keeper clerk at the High School Correspondence Branch of the British Columbia Government's Department of Education, based in the Parliament buildings in downtown Victoria. It is not known when she started there but presumably it was during or after the fall of 1938. By the fall of 1941 she was being paid $70.00 a month, or about $1,073.00 in 2018 dollars. She left her job there in late September or early October, 1941. Anonymous (1941k) records that she left the High School Correspondence Branch to join the Canadian Women's Army Corps (CWAC) and was given a departing gift of a "handsome leather travelling bag and pocketbook" by her Education Department colleagues.

Hope helped with arrangements for a big banquet held in late February, 1937 as part of a Boy Scouts celebration in Victoria (Anonymous, 1937a). Hope was one of two women who made special costumes and did scenic effects for an opera in Victoria in June, 1937 (Anonymous, 1937f). Just before Christmas that same year, Hope's mother Rachel had a play she authored, *The Children go to Bethlehem,* presented at a local church (Anonymous, 1937e, 1937f). It is not known if Hope helped with this play, but she did assist in another production of it three years later. Early in 1938, Hope's mother hosted a party at the Large residence that included a tea and a musical program by members of the "Bach Group String Quartette." Given her love of music, Hope was likely an attendee. Even palm-reading was part of the event's entertainment (Anonymous, 1938a). In late 1938 Hope participated in a two-day religious conference for 225 young people (the "Lower Island Young People's United Executive Officers Congress") where she and several others were committee chairs for the "worship and personal religious living" and welcoming committee of the meeting (Anonymous, 1938c).

The Vancouver Public Library website (Vancouver Public Library, 2014) gives clues to the Large family's activities and whereabouts during this time. The family moved a lot, seemingly ten times in seventeen years (1923–1940), possibly due to their being poor and on relief. The frequent moves must have

had an effect on Hope—she would make new friends only to lose them soon after. This could explain her notable independence as an adult. Most of their homes were located in well-to-do neighborhoods but the Larges, being constrained financially, apparently always rented a floor of the house.

In 1923 Herbert is listed as the proprietor of Dominion Bazaar, residing at 665 Pine Street in Victoria. In 1924 he's listed as residing at 3711 Saanich Road and employed as a salesman for the Fuller Brush Company. An entry for 1925 shows Herbert as a school teacher (training for which he definitely did not have) at his home, again at 3711 Saanich Road.

An H. Large is listed in 1926 as residing in New Westminster, but no other details are given. In 1927 Herbert is listed as the principal of St. Agatha's School, still at 3711 Saanich Road in Victoria; Rachel is listed as a teacher there. In 1928, Herbert is listed as the proprietor of St. Agatha's School at 765 Rayside Avenue in Burnaby. This information conflicts with family history, yet Anonymous (1928) notes him visiting his sister in Sardis and residing in Burquitlam, on the outskirts of Burnaby. There is no mention of Rachel or Hope. This was within the period in which the Large family lived on the mainland (see Page 26).

For 1929 Herbert is recorded as a teacher at St. Agatha's School which was also the Large family home, now at 131 3rd Street, New Westminster. The 1930 entry lists Herbert as a gardener, residing at 1029 Burdett Avenue, back in Victoria. In 1931 Herbert is listed as living at 1526 Fort Street, Victoria. Interestingly, in the same year a "U. Large" is listed as a gardener at Cadboro Bay on the oceanfront in Saanich, north of Victoria: Herbert, it seems, was soon destined for the same place. For 1932 (a 1933 directory was not found online) and 1934, Herbert's occupation is given as a gardener, the address being Cadboro Bay Post Office, though he was recorded on his death certificate as being retired in 1934.

Figure 29. Hope in a 1936 portrait taken in Victoria, age 20 or 21. Image Mo9476, taken by photographer Kenneth McAllister; used with permission of the Victoria Archives.

In 1935 Hope is listed as living at 1526 Fort Street, Victoria, with her parents. In 1936–1937 the family lived at 3367 Salsbury Way, Victoria. Herbert is listed as retired and Rachel as the proprietor of St. Agatha's Junior School at the same (home) address. Surprisingly, Hope is listed as a school teacher there, though what official training she had for this occupation is unknown. A portrait of her was taken at this time (Figure 29). In 1938 she lived with her parents at 1474 Fort Street, but there is no mention of a school being run by her parents or of Hope teaching there. In 1939 her parents moved again, to 1210 Fort Street, Victoria. Presumably Hope (Figure 30) lived there too but is not listed. With yet another move, to 1242 Richardson Street, Hope is confirmed as a stenographer for the British Columbia government from 1940 to 1941. Her military records state that she taught German classes for the Victoria Canadian Legion Services branch and that she took private German lessons, ". . . passing exams with first class honors." It is not clearly stated when these activities occurred, but may have been about 1937–1938.

Both her parents are listed as retired from 1938 onward. At the end of Herbert's life he and Rachel

resided at 1446 Gladstone Avenue. Sometime after his death, Rachel must have moved one final time, as her death certificate states she was "formerly of the Ritz Hotel," in Room 710.

Whatever long-term schooling or life plans Hope had were forever changed when ominous war clouds gathered overseas. Though the circumstances would at first seem unlikely, as we shall see it is certain that Adolph Hitler's failed experiment at National Socialism and his crazy ambitions to conquer Europe, Russia and other countries fortuitously steered Hope toward her interest in palaeontology. How is this possible? Because of World War II, she joined a women's paramilitary group, was sent to eastern Canada for training and service, and was ultimately reposted to Alberta and its fossil riches. Had the war never happened, Hope may have stayed in Victoria with her aging parents, or ended up somewhere else, doing something quite different with her life. This global conflict was the second of several significant events beyond Hope's control that caused her to end up in Alberta.

Figure 30. Hope dressed in chic clothing, possibly her British Columbia Women's Service Corps uniform, in the spring of 1939; WWII was not far off. Image courtesy of the Johnson family.

1939 to 1947: World War II, marriage and a new home in Alberta

Digitized online Victoria newspapers again provide us with glimpses of Hope's activities during this time. In December 1939 another church event was held and Hope is mentioned as participating in the "Metropolitan Young People's Penny Fair" (Anonymous, 1939e).

Hope and another woman are credited with employing their artistic talents by helping with stage sets and arrangements for a musical comedy concert, *A Night of Love* and assisting with set and special costumes for light and grand operas (Anonymous, 1940a, 1940b, 1940k).

Her mother Rachel hosted a number of social teas at their home and Hope was involved in some of these. Rachel hosted a church "Guild of Friendship"[5] tea at the Large residence in mid-November, 1940 (Anonymous, 1940i). In early December, 1940, Hope was an actor in an unspecified role in her mother's nativity play, *The Children go to Bethlehem*. This production was put on by the Guild of Friendship group and was a benefit concert for the homeless and for men fighting overseas. Later that month Hope and her mother hosted a Christmas party at their home for members of the Guild of Friendship group (Anonymous, 1940m). Another December activity was a program of varied songs and plays (Anonymous, 1940l). In that production, Hope and another woman did scenery props for the play and a "patriotic tableau" featured "Britannia" posed by Hope and assisted by Boy Scouts, Sea Cadets, Girl Guides and representatives of the three military branches.

World War II and Hope's involvement with the Canadian Women's Army Corps (CWAC) over roughly five years (fifty-four months) had such a big impact on her life that this period is examined more closely here. Not many of her friends and associates were aware of her extensive wartime service. She was among the first to join the CWACs and among the last to be discharged.

With the outbreak of World War II in early September, 1939, like many other young people in warring nations, a patriotic Hope began to think about how she could best serve her country. As early as 1938, forward-thinking women with historical knowledge and political savvy knew that a war was coming and began to organize themselves into private, unarmed militias. Many had family ties to England. Their homeland was being menaced by rising Nazism and Fascism and accompanying massive military growth, development and expansions into neighboring lands by Germany and Italy. A seemingly inevitable global conflict was again looming and they felt they had to do something to be ready for that eventuality. Hope was one of these women. She joined a group that rapidly grew, evolved, was renamed and restructured, during a time of deteriorating global political and military situations, especially in Europe and east Asia: war was coming and people had to get ready.

Hope already had a good job as a stenographer and clerk for the British Columbia Provincial Government, but she signed up first for civic service, enlisting in the Red Cross Corps (Anonymous, 1942d), into which the British Columbia Women's Service Corps or BCWSC was eventually, but briefly, folded

(Anonymous, 1941g; see also Anonymous, 1940e) and later administrated to a degree by the Canadian Women's Army Corps (CWAC or "Quack" for short).[6] The group later regained its independence as the British Columbia Women's Service Corps (BCWSC).

It was informally organized on October 5, 1938, soon after the Munich Crisis by a Mrs. C. Hugo Raymond, daughter of Brigadier-General J.S. Dunbar (Anonymous, 1940h) and started with only five members (Conrod, 1983, p. 313), though Anonymous (1939b) called the original group the "Victoria Women's Service Club." A formal organization meeting was held on January 17, 1939 (Anonymous, 1939a). At this point ten members are identified and about one hundred women came to the meeting; a few new members joined up. The group enjoyed steady growth. There were 1,600 members in British Columbia by 1940 (Anonymous, 1942e). At their own expense, the BCWSC members designed and made their own uniforms and studied a variety of useful subjects such as first aid, etiquette, map reading, signalling, code-ciphering, anti-gas decontamination, regular infantry military (foot) drill, wireless, army-scale cooking, and light vehicle maintenance. These courses were quite intensive. Anonymous (1940j) provides details on the vehicle course. Seven classes, each with different instructors, covered the following: lubrication—chassis, motor, transmission; steering and brake system; chassis and frame; motor disassembly and operation; cooling system; clutch, transmission, drive shaft; carburetor, ignition, fuel pump and electrical system; springs, shackles, shock absorbers, differential; and tractor and truck operation. The young ladies were taught by lucky volunteer soldiers from the Canadian Army, who at first thought they were going to be teaching middle-aged women. The BCWSC also assisted with army recruiting drives (Anonymous, 1941e). This initial activity centred in Victoria where Hope and her friends assisted in any way they could (mostly as drivers and clerks) at the Esquimalt naval base, home of Military District 11 (British Columbia) headquarters. Similar groups elsewhere in the province soon sprouted up and were later amalgamated under the BCWSC banner. Anonymous (1941a) describes the BCWSC uniform Hope would have worn:

> . . . smart navy blue tunic and skirt with the insignia of the BCWSC, a gold leaf [Figure 31] on the right sleeve. A navy blue beret with the gold leaf insignia on the right side is also worn. The buttons on the uniform are blue leather. Gold stripes, below the insignia on the sleeves indicate the rank of the officer.

For supplementary insignia, the author has also seen for sale online a lapel or similar-style "BCWSC" pin, made by Birks. It measured 0.75 in × 0.8 in (19 mm × 20 mm). Anonymous (1941a) notes the uniform consisted of a navy blue skirt, white shirt, blue sweater and beret.

Hope's military records indicate that she served with the BCWSC for two years and attained the rank of Sargeant. Similar women's volunteer groups were created across Canada, all wanting to serve in a military but non-combative capacity (Anonymous, 1940d, 1940g). On June 23, 1940 the Victoria BCWSC (including Hope) presented a drill review for the public at the Esquimalt military base where they conducted ". . . demonstrations given of the various phases of work undertaken by the women." Anonymous (1942i) records that Hope gave lectures on gas and classes in German grammar and conversation for members of the services. Her military records state that she did this for two years under the aus-

Figure 31. British Columbia Women's Service Corps badge, measuring 6.7 cm tall and 7.6 in wide. With her artistic talents and long service in the group, did Hope design this badge? Image courtesy of Bill Alexander, North Bay, Ontario.

Figure 32. Hope at dockside in Vancouver, 1941. Image courtesy of the Johnson family.

pices of the Victoria Canadian Legion Services. The BCWSC joined a massive Victory Loan military parade in Victoria on June 7, 1941 (Anonymous, 1941c). A photograph of four BCWSC groups from different British Columbia cities and towns marching together is known (Anonymous, 1941d). As noted earlier, the final evolution for all these groups was the CWAC. She served at Work Point Barracks in Esquimalt, British Columbia, which opened August 29, 1941. At the age of twenty-five, Hope joined up for formal military service with the CWACs on September 8 and filled out a variety of forms documenting her personal data, past life, work and possible career intentions. Her attestation paperwork was filled out by a "JB Kennedy Jun. Cmdr."; this is none other than Joan Kennedy (1908–1956), the brains and driving force behind the entire CWAC organization (Conrod, 1983; Allinson, 2011).

The CWAC attestation papers provide some of Hope's personal information. They state that her religious affiliation was with the United Church of Canada. She stood 5 ft, 5 in tall and weighed 135 pounds. She had never had to make a court appearance. She noted she could speak and write in English and French fluently, and knew some German. Surprisingly, when asked if she had anyone entirely dependent on her, she indicated her mother. She stated that she would be available to start with the CWACs on October 1, 1941 and this was believed to have been her official start date, though another form suggests November 3, 1941. It's not known when she left for eastern Canada and her CWAC training. She was in Victoria doing some unspecified recitations at a silver tea put on by her mother on January 24, 1942 (Anonymous, 1942b).

Whatever the actual date, she joined up at a particularly bad time for the Allies. WWII had by then been raging for two years and the Axis forces seemed unstoppable nearly everywhere. Germany had annexed or conquered most of Europe and was attacking and making gains into Russia. Britain was largely isolated but still hanging on after winning the Battle of Britain a year earlier, where a young woman also serving in the military would, sixty years later, become a retirement friend of Hope's. Fighting raged across North Africa and German U-boats were sinking millions of tons of Allied military and commercial shipping. Japan was running roughshod into China and was secretly assembling its powerful naval fleet that soon would launch a surprise attack on Pearl Harbor and the Philippines, drawing the United States into the war.

Hope received her regimental number soon after signing up with the CWACs (Anonymous, 1941h). Asked what jobs she thought she could best do with them, Hope indicated her stenography background and other skills in her civilian life, but also suggested she would be good as an assistant librarian, commenting:

> With reference to my second choice, I have good knowledge of four languages—*i.e.* English, Latin, French and German. I have some knowledge of Greek (passed University Greek A, Univ. of BC). I have also considerable knowledge of this type of work through my work in the Education Department. I am very interested in languages, and am always studying something.

The CWACs had concern for what their members would do for employment when the war was over. Hope had a guarantee of re-employment as a stenographer at her old job upon finishing her military service, but stated on a March 11, 1942 form that she had no desire to return to that line of work. She may have seen the war and the CWACs as an opportunity to leave home, mature, learn new things, and make a life for herself once the war was over. On the same form, where asked what she would like to do for work after the war, she stated that she would like to try mixed farming and raising livestock, a job for which she claimed she had four years experience. Other possible jobs were language study or something artistic like carving or painting.

The CWACs (and other all-women military units) were an important contribution to Canada's war effort during World War II. When one woman was asked why she joined the CWACs, she stated "There was a war on. And I didn't want to sit and knit" (Brennan, 1997). Canadian politicians and military leaders were at first strongly opposed to women serving in the military (Allinson, 2001): war was seen as a man's endeavour, despite other warring nations of the time having active women's components. The CWACs even went to Ottawa to consolidate their efforts and try to get some recognition (Anonymous, 1940f). In this they eventually succeeded. An early supporter of the CWAC concept was the Honourable J.L. Ralston, the Canadian Minister of Defence and for whom the town of Ralston, Alberta was named (eventually to be Hope's home for some three decades). Based on Ralston's support and other factors, the federal government eventually agreed to utilize women in a military role—in fact, Canada was the last to do so.

This was the first time women were officially inducted into the Canadian military. Each woman could replace a man who could then be sent into an active war zone (Anonymous, 1941j, 1942f; Dean, 1942; Martin, 1942). CWACs signed on for the duration of the war plus a period of time (usually about one year) after cessation of hostilities. Applicants had to be eighteen to forty-five years of age and in good health. Selected women had six classifications (Keshen, 2004) and many job types to choose from (Anonymous, 1941i; Table 1). Skills learned in the army could be applied to postwar careers.

Draftswomen	Cooks/Dietitians	Dental/Operating room Assistant
Car/Truck driver	Light Vehicle Mechanic	Laundress
Parachute Packer	Stenographer	Typist
Medic	Instructor	Orderly
Clerk (Cipher, pay, records, general duty)	Canteen Helpers/Waitress	Messengers
Radiographers	Bookkeepers	Cleaners
Lab Assistant	Switchboard Operators	Trainers
Administrators	Quartermasters	Mechanics (vehicle, radio, typewriter)
Accountants	Teletypist	X-ray/lab technician

Table 1. Some of the job opportunities available to CWACs during World War II.

Recruitment drives and advertising were active through much of the war (Anonymous, 1942h). Upon acceptance into the CWAC, all women were issued a smartly tailored, brown-skirted uniform (Figure 35). The snappy uniform was popular with CWAC members, the Canadian public (Anonymous, 1942c, 1944a; Canadian Institute of Public Opinion, 1944; Dundas, 2001) and with women serving in other Allied war branches.

In Paulter (2005), Hope discloses that her military serial or regimental number was w.11002 and states that she was the third woman to join the Canadian army. However, this is not entirely true. Her serial number, broken down, translates thus: "w" stands for Woman. The "11" represents Military District 11, which was all of British Columbia and Yukon. The next set of numbers (starting at "000") represented the recruit herself. This indicates Hope was not the third woman to join the Canadian Army, but was the third woman to sign up with the CWACs in Military District 11, the third of an eventual 2529 women from that district (Conrod, 1983, p. 244). So she signed up with the CWACs in her area early—very early, in fact. This, plus the fact that the CWACs started in her home city of Victoria leads the author to speculate if Hope may have had some inside knowledge about the formation of the group, perhaps even playing a role in its early development (though her name does not appear in Anonymous, 1947a) and once it launched was able to be among the very first of an eventual 22,600+ women to join. A fascinating clue is in her notes to her family where she makes the following statement, which seems to suggest she was in fact involved in the early development of the CWAC prior to her own joining them:

"After the War of 1939 broke out women began thinking about training in the Defence Forces. I joined a group in Victoria, several of whom became the 1st HQ (headquarter) staff of the CAN WOMEN'S ARMY CORPS."

But note that this was actually Military District No. 11 headquarters (Ward, 1996, p. 20), not 1st Headquarters, as in Hope's account. She was in No. 1 Platoon of "M" Company. That she signed up with Joan Kennedy, as noted earlier, almost assuredly confirms that Hope must have been involved in the early development of the CWACs.

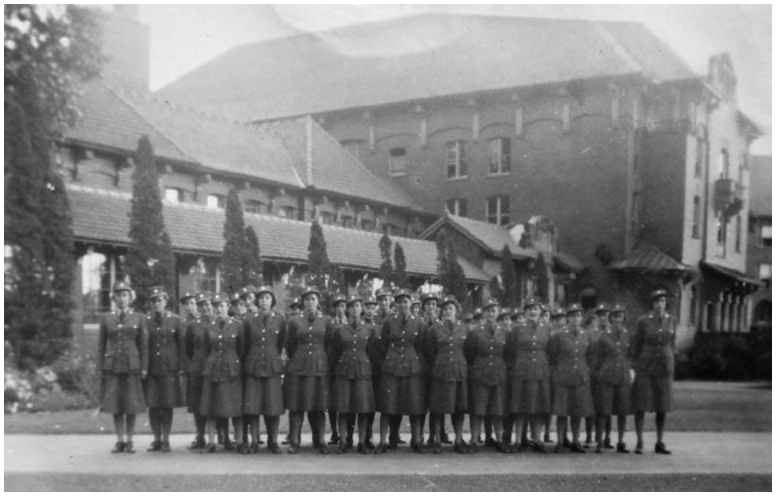


Figure 33. Hope wrote on the back of this photograph: "One of my platoons at Ste. Anne de Bellevue 1942." Image courtesy of the Johnson family.

Like many women in war, she served on the "Home Front," with her military duties landing her at bases across central and western Canada. Life and training for a CWAC recruit was the same Canada-wide, so comments in Bowman (1975), Conrod (1983), Ward (1996) and others would apply equally to Hope. Initially she would have undergone the PULHEMS physical examinations test: P= general physical condition; U= upper body; L= lower body; H= heart; E= eyes and ears; M= mentality; and S= stability. Recruits also had to provide a urine sample; inoculations were given; then she was measured for her uniform, which arrived several weeks later. Hair length regulations stipulated that it had to end one inch above the collar; so if Hope had long hair at this point it would have been cut and she'd be identified as "Volunteer Private H. Large w.11002."

She headed east on a long trip by train and passed through—perhaps even stopped at—stations in Suffield and/or Medicine Hat, Alberta, the district where she would, after just a few years, spend the rest of her life (Figure 38). The trip would have taken days to complete and she would have been transported with a growing number of other CWAC recruits in sleeper cars for the long trip east.

Many of the girls and younger women were children of the Depression and had experienced shared hardships. Now they were going on to a new life, a good paying job, health care, free room and board, and a big adventure.

Hope's first military posting for basic training and other instruction was the large CWAC base in Sainte-Anne-de-Bellevue, near Montreal, for ten months, based at the Macdonald Agricultural College (Anonymous, 1942a, 2014l; Figure 33). Earlier in 1942 the camp was turning out 500 trained CWACs every month but by the summer efforts were being made to increase this number to 1,000 (Anonymous, 1942h). Hope's notes state that the college was practically empty on her arrival because of the Depression. CWACs there had to share accommodation with McGill University students (Ward, 1996).

Once settled, the women would do basic training and courses comprising four or more weeks of parade drill, platoon drill without weapons, salute drill, personal hygiene, military law, military organization, morale, map reading, continuous kit and uniform inspections, drill square, physical education and exercising, the dreaded social diseases film and lecture, gas mask use and fire drills. This training was very real and tough and was a shock to many, their lives now regimented and without privacy. It was physically and psychologically demanding. Many were homesick. They were all on probation and held the lowly rank of "Volunteer." But if they passed muster, they would be assigned a higher rank and go on to more specialized training as required for their service to Canada. Group training built cohesiveness, camaraderie, and team spirit.

Hope's service is noted in a newspaper article (Figure 34). By mid-summer 1942 the already-promoted Sgt. Hope Large guided a Canadian Press reporter (Ecker, 1942) and a new recruit on a tour of the base. The reporter commented that the new girl would transform with training and a uniform, and "in a very few weeks, like Sgt. Large, she'll stand tall and proud in her smart CWAC uniform. She'll walk with poise and look people she meets straight in the eye. She'll have found her job in the army and she'll be doing it well."

SGT. INST. HOPE LARGE, C.W.A.C.

A VICTORIA girl who recently was made sergeant instructor and is stationed at Macdonald College, Ste Anne de Bellvue, Que., is the daughter of Mr. and Mrs. Herbert Large, Richardson Street. Prior to joining the C.W.A.C. in October, 1941, Sgt. Large was employed in the correspondence department of the Department of Education in the Parliament Buildings here. Before leaving for Quebec she was stationed at Work Point, holding the rank of corporal. While at Work Point she gave lectures on gas, and held classes in German grammar and conversation for members of the services. Sergeant Large has lived mostly in Victoria, coming here at the age of two years. She received her early schooling at St. Agatha's School, of which her mother, Mrs. H. Large, was principal. Later she attended Victoria High School and Victoria College. Her work at Macdonald College includes instruction in first aid and physical training.

Figure 34. Newspaper article from the Victoria *Daily Colonist*, August 22, 1942.

Hope earned a short leave this summer and visited her parents in Victoria. Her mother saw her off in Vancouver (Anonymous, 1942g). It appears Hope was in charge of some women in Quebec (Figure 35). A limited number of recruits with maturity, demonstrated responsibility, aptitude for team leadership, or with valuable prior experience and job skills in their civilian life were considered for further promotion and training. They were sent to officer schools and Hope was one of them. Part of the CWAC job was marching in public for recruitment, morale and celebratory purposes. Most women on the Sainte-Anne-de-Bellevue base (Hope included?) marched in a large parade on September 2, 1942 commemorating the third year of Canada's entry into the war (Anonymous, 1942j). They marched alongside men whose units fought at Dieppe.

She was interviewed by a senior CWAC woman from the "Personnel Selection Board" on January 12, 1943. The following details about Hope were recorded:

> Height 5′5″, age 26. Weight 156 pounds. Health "good." She is a well-developed, sturdily built young woman with short-cropped brown hair, direct blue eyes and short features. Expresses herself well and readily.
>
> She has kept herself busy with a wide range of hobbies and outside interests. Has never been active in competitive sports, but enjoys swimming and cycling. Has a variety of reading preferences—biography, history, English classics, and Russian and German authors. Appreciates Dorothy Parker [satirist, writer, critic, poet; 1893–1967] humor. Has marked artistic ability, prefers broad canvasses and paints in water color and postal [*sic:* pastel?] paint. Is fond of producing unusual color combinations and effects. Sews, does fancy work, and tries

her hands at off bits of wood-carving, preferably complicated and delicate design. Taught German classes under auspices of the Victoria Legion Services—2 years. Had a Wolf Cub pack for 6 years and held rank of Sargeant in B.C. Women's Service Corps, 2 years. Studied piano and voice for several years but now restricts her musical interests to collections of classical recordings. Is single. Has considerable experience dealing with others and good ability as a teacher. She is original in thought, is extremely versatile, with superior intellectual endowment and good educational background. She has kept up her study of German, and has a well-rounded general knowledge in the fields of her interests. At times she is somewhat lacking in social poise and mature judgment. Is an excellent conversationalist, a quick thinker and is capable of handling responsibility, but is probably at her best in work requiring independent action.

General comments:

Throughout this interview Cadet Large was cooperative and discussed her interests, military activities, and ambitions freely, although at times manifesting some restlessness and apparent insecurity. She has a very keen mind, is able to see relationships quickly, has for the most part a good sense of values and is well informed generally. She seems reasonably well poised but at times seems to lack finesse. She is not interested in duties of a routine nature but prefers work that affords an opportunity of initiative and originality.

Figure 35. Hope is front-and-centre in this detail of a CWAC group photo. The undated image is stamped on the back, "Milk & Sullivan 486 Kent St." traced to Ottawa, only two blocks from the CMN. This may be a graduation photo from when she trained women for Army clerical work in Ottawa. If true, a 1943–1944 date is suggested. Image courtesy of the Johnson family.

Hope took a cadet course and received her officer's commission on April 2, 1943 after spending time at a training centre in Kitchener, Ontario. The graduation ceremony is noted in Anonymous, 1943b. She received her "pips" (officer's rank insignia) at a Sherbrooke, Quebec training centre from Major-General Thomas Louis Tremblay (1886–1951), the Inspector General for eastern Canada (see also Anonymous, 1943a, 1943c). As a young troop leader at these bases her Boy Scouts First Aid training came into play: she was an instructor to more junior-ranking CWACs in First Aid. The CWAC training period in First Aid was extremely rapid for her. She was surprised to learn she would train what would become other First Aid instructors. She protested that she had not enough time to take the full course given by the military, but she was kept in the instructor role regardless, citing her full First Aid training in Boy Scouts years earlier. Anonymous (1942i) records that in addition to First Aid, she also gave instruction in physical training.

She continued training others in First Aid and other duties and was then posted to Number 40 Company at Glebe Barracks (the former Ottawa Ladies College) in Ottawa, where she was an army instructor and a Platoon Officer in charge of a large group of women (Figure 35), being trained for Army clerical work at the Ottawa High School of Commerce (Vocational Training School No. 3), with her stenography experience coming into use. Most of her war service was spent in this city.

Figure 36. An originally hand-tinted portrait of Hope taken in 1943, age 26–27. The picture was taken at the New Paramount Studio on 104 Sparks Street in Ottawa. Many servicemen and women had their portraits taken there during WWII. Image courtesy of the Johnson family.

She had a formal portrait in her uniform taken this year; Hope had developed into a handsome young woman (Figure 36). In the summer of 1943 she earned some extended leave and travelled all the way west to Victoria to be with her parents (Anonymous, 1943d).

Life on the base was not all hard work. The ladies had a beauty parlour in their barracks, the first for CWACs across Canada (Anonymous, 1944b). As someone already interested in natural history, perhaps she found time on leave to visit the natural history exhibits at the Victoria Memorial Museum (now the Canadian Museum of Nature, or CMN), a fifteen minute walk from the School of Commerce and a mere two blocks from a shop where it is known she had at least one photograph developed (Figure 36). If so, she would have seen the impressive Alberta dinosaur skeletons and skull exhibits there, the results of the "Great Canadian Dinosaur Rush" (Figure 37); it would have been her first exposure to Alberta vertebrate palaeontology.

On August 11, 1944 her work performance in the CWACs was reviewed and discussed in a military document. Three of the commanding officers under whom she served (from early December, 1943 until the summer of 1944) assessed her abilities and stated:

> 1. Proving to be a good administrative officer.
>
> 2. As Company Commander has administered CWAC personnel attending Ottawa School of Commerce in an excellent manner.
>
> 3. This officer has mostly handled VTS (Vocational Training School) personnel attached to #40 Admin Unit for instruction. She is keen, enthusiastic and energetic. She is a naturally shy individual and to overcome this, adopts a "hail fellow well met" attitude which irritates some people. It is felt with more varied army experience she will become a good Company officer and her gutsy approach will be less apparent as she gains confidence in her own ability.

It was further stated:

> Lieut. Large is on strength of the No. 3 V.T.S. (Vocational Training School) and as such handles mostly V.T.S. personnel. She is extremely conscientious about the welfare of her personnel, is clever and has become essential to V.T.S. However, it is proposed to attempt to have her taken on strength of CWAC and thereby placed in the promotion stream. She ought to be a good Company officer. She is rather self-conscious, to overcome which she adopts a breezy attitude which is a contradiction to her lack of self-confidence.

Regarding her future in the Canadian military:

> Wants to remain in the service and continue her present work or would take an appointment where her knowledge of language is utilized.

Medically, this report indicated she was now suffering from "flat feet."

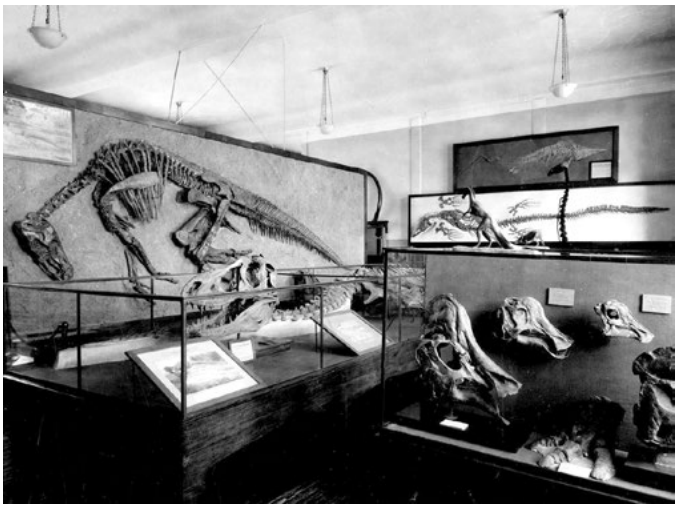


Figure 37. Some of the mostly Alberta dinosaur mounts on exhibit at the now-named Canadian Museum of Nature (CMN) in Ottawa, Ontario, seen here in 1935, just seven or eight years before Hope was posted to Ottawa. No major gallery renovations are known to have occurred during the 1935–1943 period (because of the Great Depression and war), so this would be how Hope saw them if she visited. Image by Dr. Kindle, © CMN; used with permission.

Hope also served in Peterborough and Brockville, Ontario, where in the latter she took an army administration course. Her last military posting was in early 1945, to Suffield, Alberta at the new, isolated, Anglo-Canadian military base. During the war that corner of the province became militarized with the development of various air training bases for aircrews under the British Commonwealth Air Training Plan (Conrad, 1989; Carter, 2001), including: a top secret radar testing station located near Jenner, Alberta (Hope, in Smith, 1976 called this the RWPU or Radio Wave Propagation Unit); Camp 132, one of the largest Prisoners of War camps (1943–1946) in Canada, capable of holding more than 12,000 German prisoners of war; and the sprawling Suffield base.

She was transferred there to assist another CWAC, a Captain "Ting" Tingley who was in charge of all the CWACs there. In the absence of another woman, a Dorothy A. Robinson, Hope assumed command of #34 Administration Unit, overseeing three platoons of women (Smith, 1976, p. 3). So on occasion, Hope was second in command of the CWACs at Suffield (R. Johnston, pers. comm., 2014).

She arrived on January 21, 1945, finding a flat, treeless and frozen terrain surrounding the large base (Figure 38); she later stated that "what looked like a frozen ocean was a bit baffling." It had been cool and snowy at her military postings back east; but in Alberta she said the "windswept wintry plains were a new experience." In a 1973 interview (Puckett, 1973; Dryden, 1981) Hope commented on her love of plants in British Columbia and she still retained this interest upon arriving in Suffield, but "botany began to shrink before my eyes as it were" upon seeing the stark landscape for the first time. Another CWAC officer, arriving the year before, described the bleak scene even better (Ward, 1996, p. 132):

> As I gazed out the hut windows, the sand heaped on the sills and was being pounded against the less than tight window casings, by the constant winds. Three scrawny Hereford [cattle] loomed over the horizon, the wind at their backs and with not a blade of anything in sight for them to eat. What a sad, lonely landscape! No wonder the Department of National Defence should designate the vast area as a training ground for chemical warfare. Not much vegetation for flame throwers to blast to cinders either.

Asked in Smith (1976), what Hope thought upon arriving at Suffield and how it felt knowing she would possibly be staying there for years, she commented: "I never had any particular ideas about staying here for any particular length of time or anything like that. I'm not a person that ever plans ahead—I think that leads to an awful lot of disappointments."

But despite its stark appearance, Hope would come to love this land, with its diverse natural history (Stevens, 1972) and scenery; in (Smith, 1976) she stated, "I tend to like desert regions. I'm intrigued by the struggle for survival." A former CWAC and friend recalled Hope's days on base. While Hope was a commanding officer and greatly respected, she was different from other officers in not being of a stiff regimental "military type" and when the weather warmed up with the arrival of spring, she was soon

spotted planting flowers in front of the officers' quarters, hoping they would grow (F. Bickerton, pers. comm., 2014).

Life in the early days on the Suffield base was a bit rough on the "bald prairie" and is recounted in Various authors (1961). There were poor roads that were impassable when wet, no stores for shopping; there was "no nothing" (F. Bickerton, pers. comm., 2014). Medicine Hat was 56 km away by rail or the Trans-Canada highway, then a rough gravel road. And there was mud. Lots of it. Hope recalled years later (Johnson, 2003) that during the early years at Ralston, from 1949 to 1955, the family had to deal with mud: ". . . we had mud roads, mud school yards, etc., etc." In the same letter she noted that she also suffered from bad allergies during her days in southeastern Alberta: "Ralston was very windy at times, and I suffered for many years from very severe 'hay fever' —read weed fever Russian thistle, pig weeds, and sage—all wind-pollinated species. I lived on anti-histamines. Ug. Took shots too—for years, June to 20th October, until I was about 48, no more shots, just pills. I'll be 87 in March, so it doesn't seem to have shortened my life. I should really have liked to live in a teepee (part time) . . ."

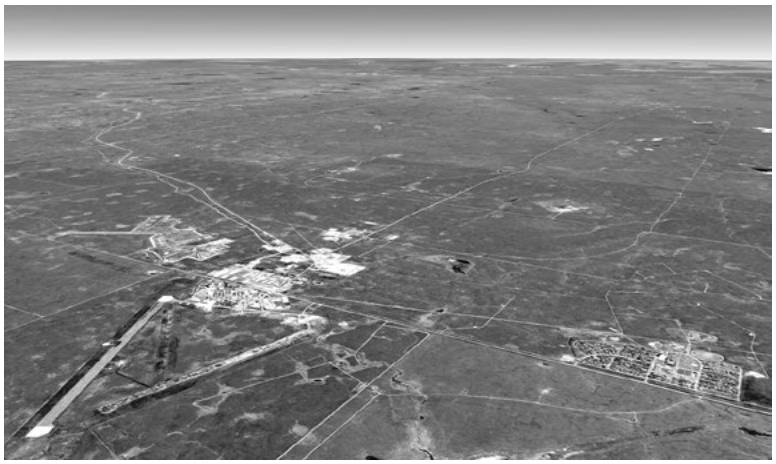


Figure 38. The Suffield base seen in perspective view to the northeast. The total lack of trees is evident. The town of Ralston is in the lower right corner. Imagery from Google Earth, reproduced under terms of use. Image © 2018 DigitalGlobe.

The base where Hope worked was secret and called the Suffield Experimental Station, though all the locals knew there was something more than agricultural research going on there, given all the tight security. In fact, the base was first used for top secret chemical and biological warfare training and testing, as well as WWII smoke and flame weapons experimentation (Anonymous, 1949a, 1951a; 1961e; 1974b; Morgan, 1967; Chown, 1989; Avery, 1998; Laforce, 2006) and became a leading facility in chemical and biological warfare research.

It was also the scene of some spectacular experimental detonation of TNT, up to 500 t (*e.g.* Hopkins, 1959, 1961; Anonymous, 1960e; Shiels, 1960a, b; Anonymous, 1961b; Jones, 1963, 1970, 1976, 1977 (among others); MacKenzie, 1983; see National Film Board, 2015, for film footage of a 1962 test), to simulate atomic blasts, but without the negative effects of radiation fallout. Hope witnessed some of these huge blasts firsthand.

The British had lost their chemical warfare testing area in Algeria when France surrendered to Germany during WWII. So a large area of land was appropriated and a new testing base was built from scratch west of Medicine Hat (Goodspeed, 1958; MacKenzie, 1983; BATUS, 1987; Boulet, 1994). Today it is used by the British Army for heavy artillery practice, war games and simulated tank battles (Anonymous, undated; Prin, 1971; Helm, 1988; Edwards, 1998; Anonymous, 2014h; Warner, 2018). More recently it has been a venue for terrorism attack mitigation experiments. Today, when prospecting for fossils in the Hilda, Sandy Point and White Rock Coulee areas one cannot help but notice the distant thump and rumble of exploding shells.

Ward (1996, p. 130–131) notes that the CWAC presence at the base was greatly reduced in 1944 with the intention of abolishing all CWACs there. But they returned and Hope spent about a year and a half supervising CWAC staff who worked in the laboratories (chemistry technicians), the photographic sec-

tion and as secretaries and radar observers. By the end of the war 584 personnel were present on the base, though the number of women could not be established at the time of writing.

In her first of what would be many appearances in the local newspaper over her lifetime, it is recorded (Anonymous, 1945a) that Hope donated blood for the seventh time at a Red Cross Blood Donor Clinic. She was already an accomplished artist by this point, having had two of her paintings shown at the Armed Forces Art Exhibition in Ottawa in 1944 (Delday, 1977). Details of women's participation in this show are given in Brandon (2009), yet Hope's name does not appear as a participant. She was surprised when one of her exhibited pieces at the Armed Forces exhibit sold for $7.00 (Delday, 1977), but in the decades to follow, many more sales would occur.

Hope's arrival at Suffield marked her first experience with the flat, dry, and expansive Alberta short-grass prairies which would be her home for the next sixty-seven years. Her main natural history interest at this point was still botany, but she found the Alberta prairies "somewhat uninspiring" (Bush, 1969, p. 10). This was hardly surprising, having been raised in damp and lush coastal British Columbia and having just arrived from the forested lands of central Canada. But on arriving she was quickly inspired by someone else: Olafur "Ollie" Johnson (Figure 40). They met the day she arrived at Suffield and, coincidently, on his birthday. She was curious about the absence of trees and had a query about the local climate. The two met this way as he was a civilian meteorologist on base.

Olafur was born on January 21, 1909, the fifth of nine children born to Jon Jónsson (1864–1925; surname later changed to Johnson) and Guthrun Thorvaldsdottir (Thorvaldson, 1984), poor Icelandic immigrant farmers from the north coast of Iceland. Repeated volcanic eruptions had ruined their crops and livestock so they set out for Canada and a new life. They settled in Arnes, near Gimli, Manitoba where many Icelanders had already immigrated due to similar and other natural disasters (Courtright, 1984).

It was a hard life for many of the new settlers in an often unforgiving land (Gimli Women's Institute, 1975). The Johnsons were not spared the hardship. Life became even tougher for the poor family when Olafur's father died young, leaving his mother to raise the five remaining children.

Young Olafur was smart and attended post-secondary schools in Regina and Toronto, ultimately earning his Master of Science degree (with Honours) in Math and Physics from the University of Manitoba in 1932 (Figure 39). During the war

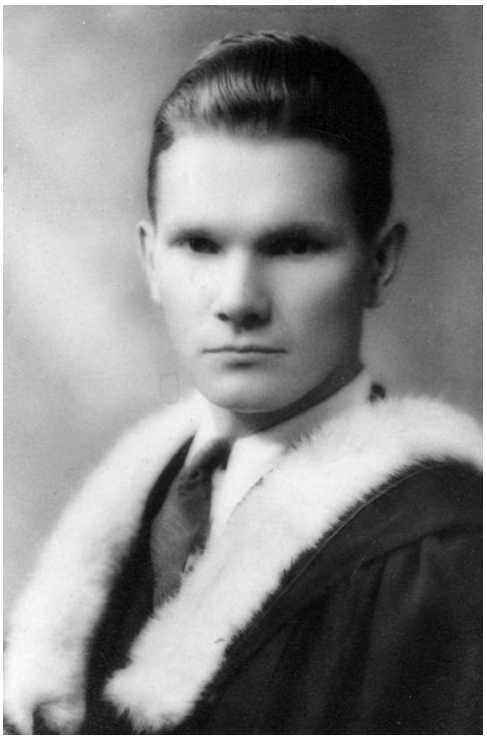


Figure 39. An intense-looking Olafur Johnson in his 1932 University of Manitoba graduation photo, captioned: "A serious and high-minded young man whose ability and determination will no doubt ensure his success in the higher realms of Physics and Mathematics." (University of Manitoba, 2014, p. 155). Image courtesy of the Johnson family.

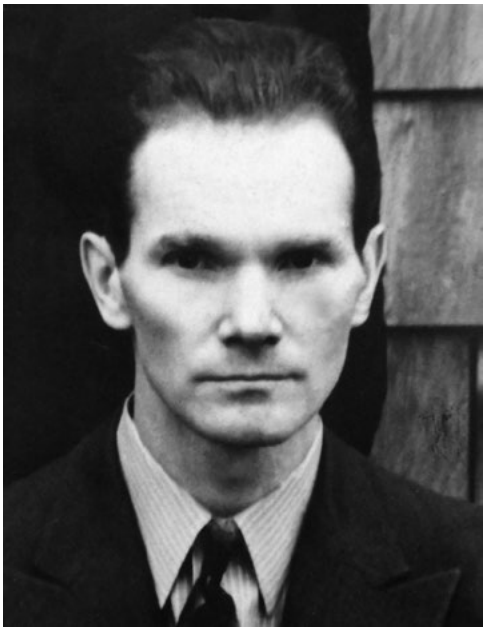


Figure 40. Olafur in 1939, age thirty, when he served at the RAF station in Debert, Nova Scotia. Image courtesy of the Johnson family.

he was seconded (Arnes Book History Committee, 1990) to a Royal Air Force (British) mine-sweeping unit in Debert, Nova Scotia.[7]

He was later a civilian meteorological officer seconded by the Canadian Government (Department of Transportation) to the Suffield base in Alberta. While riding the CPR passenger train to an unknown destination, he received a telegram *en route* asking him get off at the Suffield station to fill in for a few weeks for someone who hadn't shown up for a work assignment. That person never did appear and Olafur did such a good job that he ended up staying for some thirty years!

Here, then, is the third big quirk of fate that resulted in Hope staying in Alberta. If that person had shown up for work at Suffield, Ollie would have continued on to his unknown destination. Hope would never have met him and after the war she would likely have moved on to pursue other things, perhaps back in British Columbia, to be close to her aging parents. What a loss to Alberta palaeontology and other disciplines that would have been!

Figure 41. Olafur and Hope in Winnipeg, 1945. Such candid street photographs were popular in North American and Europe in the 1940s (Kidd, 2013). Image courtesy of the Johnson family.

At Suffield, Ollie was one of three meteorological officers and Hope wrote that he had an "excellent success rating" in the science of weather prediction, which was critical given some of the top secret blasting and testing of toxic aerosols and gases there. Precise meteorological forecasting ensured the best experimental results and no one wanted those lethal toxins and carcinogens drifting over the base or populated civilian areas. He wrote at least one scientific paper (Johnson, 1959), found among Hope's papers after her passing.

Hope visited Ollie's parents in the summer of 1945 and must have had a good reception. In a freelance street photographer image, she is seen beaming and confidently striding down a Winnipeg street with him (Figure 41). She spent part of her leave this summer with her parents (Anonymous, 1945b) on Vancouver Island and again at Christmas with her now-fiancé, Olafur (Anonymous, 1945d). Hope's mother did not care much for Olafur because she found him quiet and shy, and "too foreign-looking." Regardless, history shows that he and Hope would go on to have a good life together.

With the war over and the CWACs being disbanded, the need for Hope's services gradually diminished. Her military records for the early summer of 1946 show her using up much accrued vacation time and generally closing out her military career. Hope "retired" (her words) from the CWAC in July, 1946 with the rank of 2nd Lieutenant (her notes say 1st Lieutenant but her military records state otherwise). She served in the CWAC longer than the woman who started the group, Joan Kennedy (Anonymous, 1945c). Her military records report that she was "struck off the strength" on July 15 and awarded the Canadian Volunteer Service Medal and the War Medal, 1939–1945.

Ollie and Hope visited her parents in Victoria and were married there in a small ceremony at the Metropolitan United Church on September 5, 1946 (Anonymous, 1946c; Figure 42). Details of the wedding, subsequent celebratory events and bridal fashion may be of interest to some readers and can be found in Anonymous (1946a). They honeymooned on the island somewhere north of Victoria.

In an interesting aside, Anonymous (1946b) notes that Hope's mother Rachel left Victoria at the

Figure 42. Hope and Olafur's wedding celebrations. Image courtesy of the Johnson family.

end of September, 1946 to take a teaching position at Canal Flats, British Columbia as part of the Windermere School District in the east Kootenays. How long she stayed in Canal Flats is unknown; the author could find no record of her there.

In a military exit interview document from July 13, 1946, Hope declared her future plans were "Establishing a home in Medicine Hat, Alta." Under a section titled "Medical Officer's statement of Physical Limitations," the doctor stated that her "Feet may be troublesome where long standing or walking is required" which is ironic given all the decades and many miles of strenuous hiking she would soon be doing in her quest for fossils and other natural history pursuits!

With the war concluded, the need for CWACs was over and the organization disbanded. It was resurrected during the Korean War (Hope, by then a mother, did not rejoin), later to be absorbed into the regular Canadian military.

Even after leaving her military service, the Suffield base would continue to play an important role through most of Hope's later life. She was occasionally allowed to prospect for fossils on base property and would have a large fossil exhibit in the officers' mess or club for about eight years. She was allowed to watch some of the experimental large-scale blasting done there. On base, she met Earth scientists who became her friends, her teachers of geology, and some became fellow fossil collectors. Over the years she would teach many military staff how to paint. Her daughter Barbara would marry a base scientist and her son Lawrence, who went on to become a nuclear scientist, had a summer job there (Anonymous, 2014d, 2014k).

In the absence of housing for married couples in Ralston, the Johnsons resided for a year and a half in Medicine Hat (Smith, 1976, p. 3). When proper quarters were built in 1948 they moved back west to 13 Kingston Avenue in Ralston (then known as the Crown Village; Figure 44 a–c; Anonymous, 2014n), southeast of and adjacent to the Suffield base.

The Johnson family would reside there for twenty-six years, the first time Hope had a steady long-term household, having moved frequently as a child and while serving in the military. At first it was tough for Hope to live

Figure 43. Hope's parents, later in life, perhaps taken during Hope's 1946 wedding in Victoria. Image courtesy of the Johnson family.

Figure 44 a-c. The Johnsons' Ralston home as it appeared in 1949 (a), about 1965 (b), and October 2015 (c). In this simple home, Hope learned about vertebrate palaeontology and completed hundreds of fossil drawings and badlands paintings, among others. Many Canadian palaeontologists were hosted at this home. Images a and b courtesy of the Johnson family. Image c by the author.

there as a civilian on the windy, bald prairie (Figure 45), with few amenities and as a new mother with a young son to care for and another baby on the way (F. Bickerton, pers. comm., 2014)—in fact, unexpected twin girls (M. Noone, pers. comm., 2015), who Hope jokingly called "double trouble" in a family photograph album (Figures 46, 47).

Today a quaint town, when Ralston was first built there was much bulldozing and grading activity, strong winds blowing the loose topsoil, and no lawns: windblown dirt was everywhere. It was not until 1953 that the major construction was completed. When it rained, all the soil turned into a quagmire.

There was no store in town so getting groceries was often problematic. The Johnsons had no car and had to shop in Medicine Hat, reached via the Trans-Canada highway which was often in the most deplorable condition. Hope (in Smith, 1976), described it as narrow ". . . frequently under water," with ". . . enormous puddles and great rocks." One could order groceries from the Station Supply Depot on base but orders were often received incomplete or contained the wrong items (Anonymous, 1969h). As a domestic, these and other privations caused Hope much stress (Smith, 1976). She chummed around with a few other wives in town who helped get her through the then dreary conditions there.

Moving back to Ralston allowed Olafur to be close to his job at the renamed DRES (Defence Research Establishment Suffield) and they now had the space to plant a lawn, a big garden, trees, shrubs and flowers. Olafur pursued his love of tennis, was President of the Tennis Club for a time and for about three decades taught indoor and outdoor lessons. In Ralston he did well at his meteorology job,

Figure 45. Hope and Ollie with young son Robert during their early days in Ralston. The background confirms Hope's comments that Ralston was at first a rather desolate place to live. Image courtesy of the Johnson family.

Figure 46. The Johnson family in May, 1949 with (left to right): Margaret, Robert, and Barbara. The boy standing on the left is a neighbour, identified as "Terry." No lawns, gardens, or trees yet. Image courtesy of the Johnson family.

providing weather data and forecasts for the military and civilians (Anonymous, 1971b), eventually becoming Head of the Meteorological Department at Suffield until his retirement in 1976, after more than thirty years of what was sometimes a stressful job.

Hope became involved in local clubs and activities, though when these occurred is lost to history. For about four years she was a member of the Ralston Drama Club where she painted posters and sets for plays held at several officer clubs at the Suffield base and for the Ralston School's concerts. She even briefly tried her hand at acting, portraying a Scottish nun in *High Ground* and a maid in *The Lady's Not for Burning*. Hope had arrived in Alberta during an important time in the provincial government's support for the arts. On March 27, 1946, the government introduced the "Cultural Development Act," supporting the arts—the first government anywhere in North America to formally do so (see Schindeler, 1998; and Townshend, 2005, p. 211–212). The Visual Arts Board was the last to form in 1948. The government provided financial assistance so that Albertans who wanted to could enjoy various aspects of the arts. For visual artists like Hope these were exciting and encouraging times.

Hope and Olafur were soon raising four children born in rapid succession: Robert Leslie (born September 4, 1947); twins Margaret Rachel (Noone) and Barbara Gudrun (Preston), born October 9, 1948; and Lawrence Haraldur (born August 24, 1950); Figure 47.

Figure 47. The children of Olafur and Hope Johnson in April, 1951. L–R: Margaret, Lawrence, Robert and Barbara. Image courtesy of the Johnson family.

1948 to 1956: Finding her first fossils and Hope's palaeontological self-education

While many involved in vertebrate palaeontology can trace their interest in fossils and prehistoric life back to a young and impressionable childhood, Hope's involvement in the field began comparatively late in her life.

There is no record of her collecting rocks or fossils on Vancouver Island, even though fossils have been collected there for over 170 years (*e.g.* Ludvigsen and Beard, 1994). Certainly she was aware fossils existed; and as a youth she had seen some invertebrate fossils on display at the Provincial Museum in Victoria that were derived from marine shales collected north of Nanaimo. Reviews of historical newspapers from this time reveal articles about dinosaur discoveries and other fossil finds of which Hope must have been aware. However, it was not until the age of thirty-one, in 1947, that Hope became directly involved with and interested in fossils (J. Danis, 1981).

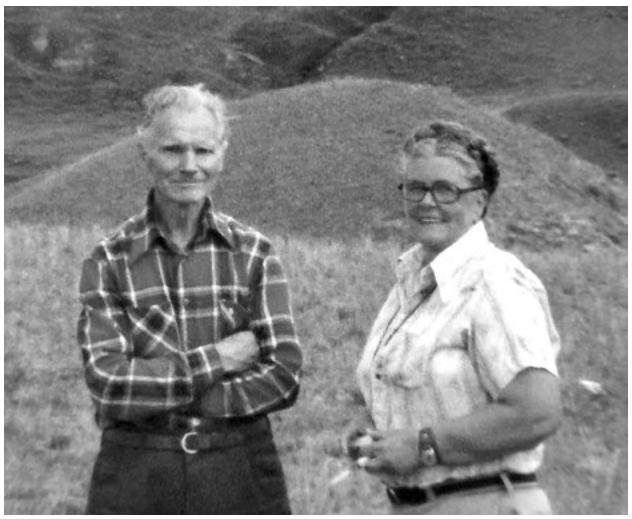


Figure 48. A rare image showing Ollie and Hope together in the field, in this case Petrified Coulee, in 1979. Image courtesy of Bill Lawson, Redcliff.

To begin with, she was more interested in rocks and especially botany. In Puckett (1973) she states that a person (unidentified) alerted her to fossils in the Ralston region. Settled into their new home, she began to explore the countryside on foot. When Ollie came home from work he would take care of the twin girls while Hope and son Robert explored the prairie underfoot. Olafur did not really care for the dirty, hot work of exploring, though later he did join her on occasion, if the dinosaur bones she collected were big and he would help her with difficult extractions (Bush, 1969; Figure 48).

Hope would put her two-year-old son Robert in a child's wagon (Figure 52) and tow him around on cross-country explorations nearby (Smith, 1976, p. 4), each foray usually under an hour. With hard rubber tires rolling on a stony surface it must have been a rough ride for the boy.

She relates one of her first trips onto the prairie to see what natural history treasures it had to offer: "I went to see what was on the prairies. It was stones." And lots of them. She wanted to see flora, especially emerging spring plants, but the local climate of low precipitation, long, hot summers, intense sunlight, and drying winds meant many of the hardy, native plants were small and had few or small flowers. A five-year drought had reduced plant numbers and she found only a few drought-resistant species such

as the prairie parsley, *Lomatium foenicalaceum* (Figure 160) of which Hope, writing to a friend many years later, noted: "It may be small, but certainly has a long name," and the buffalo or golden bean, *Thermopsis rhombifolia* (Figure 49). These and the other plants did not much interest her, at least not at first. No popular books had been written on the local flora, but she would eventually change that. Possibly all she had to go on at first was a 1949 mimeographed copy of *Flora of the Farming and Ranching Areas of the Canadian Prairies* by Archibald Budd. It is said that mimeographs of this document were popular with both professional and amateur botanists at the time (Best, in Budd and Best, 1964).

Figure 49. Technical pen-and-ink illustration of Buffalo Bean. Undated, unpublished image courtesy of the Johnson family.

Unimpressed at first with the local vegetation, she turned her attention to things geological, another of her many interests. Countless rounded rocks, large and small, littered the prairie landscape around her home (Johnson, 1961l), but were completely unfamiliar to her as they were derived from glacial till—deposits of a type Hope had never seen. Long despised by farmers, these field stones were of great interest to her. She was intrigued by the colourful granites, limestones, gneisses, mica schists with pretty red garnets embedded within, and quartzites, all scraped off the Canadian Shield far to the north (Shetsen, 1984); and brown to honey-coloured quartzite cobbles, black chert, agatized wood and other rocks deposited in and eroded off of the upper Eocene to Miocene Cypress Hills Formation. The cobbles themselves originated from outcrops in the mountains of northwestern Montana, several hundred kilometres away (*e.g.* Leckie and Cheel, 1989). Such rocks would eventually become the main subjects for some of her paintings (Figures 183 and 184) and many years later she had a display of them in her retirement apartment.

She was also interested in crystals and after her death a number of crystals of local Bearpaw Formation gypsum and varieties of quartz such as those found inside septarian nodules, convoluted chalcedonies, and brilliant "Herkimer diamonds" were found among her collection.

Among this menagerie of geological material were some sedimentary rocks sourced from a gravel pit across the road from her home. In these Hope noticed that some contained marine invertebrate fossils. In Smith (1976, p. 4) she states: "I remembered a colleague of Ollie's saying 'I believe this is a great place for fossils,' so I thought to myself, 'Well if it's a great place for fossils, obviously some of these things I'm seeing are probably fossils and I should look harder at them.'" In a letter to a friend some fifty-five years later she recalled there were "lots" of fossils there. The gravel pit was located on the west side of the north-south road (the Jenner Road) near Ralston and the Suffield base; it has been abandoned and now largely reclaimed by nature (Figure 50).

Figure 50. The north end of an abandoned and overgrown gravel pit that matches Hope's location descriptions, visited by the author on October 20, 2015. Straight line distance from the centre-north part of the gravel pit to the former Johnson residence in Ralston is just 800 m (0.5 mile). Coordinates of the gravel pit are UTM 12U 487577; 5565019 (WGS 84). Photo by the author.

Visits to other gravel pits in the area

for more fossils were also rewarding, with discoveries of the odd marine nodule containing fossils or even isolated fossil bones of Late Cretaceous marine plesiosaurs. Some of the older glacial till rocks contained Palaeozoic invertebrates (Figure 51) such as Devonian and Ordovician corals of different types (solitary/horn; *Faberophyllum*), tabulate (*Favosites*), and rugose (*Lithostrotion*); nautiloid cephalopods (*Michelinoceras*); crinoid stems; and brachiopods (large and small species of *Atrypa*). She found other rocks among the gravels that contained Late Cretaceous fossils. She described these rocks as "dull, grey ones, rounded and sometimes elongated, with shell fragments visible in them." She initially had trouble identifying them, lamenting in a fossil catalogue notebook that the identifications are "the best I could do, but there must be many errors as I had little help."

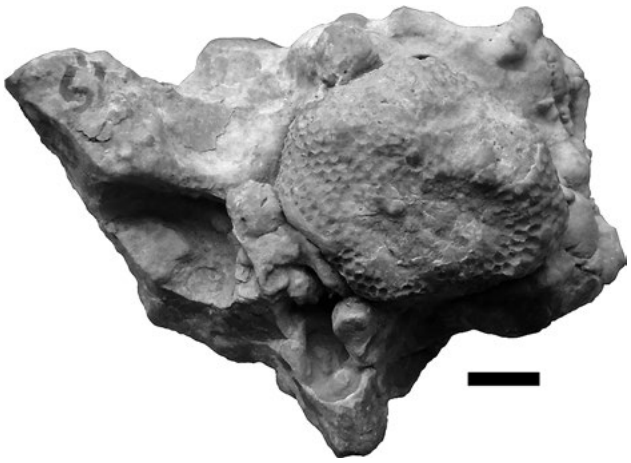


Figure 51. An interesting Palaeozoic fossil found in Hope's collection (her catalogue #52) after her passing, recovered by a friend from glacial till near Manyberries, Alberta. It may be referable to the receptaculitid alga *Cyclocrinites*. Image by the author. Scale bar = 1 cm.

Most glacial till in southern Alberta is derived from points far away, but stones containing Late Cretaceous marine fossils came from the local Bearpaw Formation which often produces rounded sandstone or mudstone concretions in the nearshore or offshore Bearpaw facies. These rounded or disc-shaped concretions, when found broken apart naturally or are smashed open with a hammer, sometimes contain a nucleus formed by biological objects such as a crab claw, wood, shell, fish vertebra, or sometimes entire specimens. Larger fossil inclusions are often seen protruding on the edges of the concretions. It is likely one of these that drew Hope's eye. Apparently her first fossil finds were pieces of the large Late Cretaceous marine clam, *Inoceramus* (L. Johnson, written comm., 2010). Hope's notes record that this was a rare genus in the Ralston gravel deposits, but given their size and thickened shell, would be easy to see protruding from a concretion. Later she found several specimens of the rare conus (= guard or gladius of some authors) of a coleoid squid (*Actinosepia canadensis, sensu* Waage, 1965). One went to the University of Alberta (UA; Johnson, 1992a) and one was later donated to the Royal Tyrrell Museum of Palaeontology (RTMP). Construction of an airfield at the Suffield base meant more rocks from the gravel pits were brought in for her to examine. The child's wagon then became useful not only for transporting her son (Figure 52), but for hauling some of her geological and palaeontological finds home for cleaning and further study.

She sent some of her fossil finds along with her supportive husband Ollie,[8] who visited the Physics Department at the UA in Edmonton (Forbes, 1998) to attend a meteorology conference (Johnson, 2003). From there the fossils went to the Geology Department (Sanderson, 1999) where they were examined. The specimens were identified and Hope received encouragement from Professor Percival S. Warren[9] (Figure 54 a; Stelck, 1970).

In later correspondence Warren took the time to carefully identify her finds using the Latin binomials of the Linnean System (with which she was familiar, thanks to her Greek and Latin language training) and he suggested further sources of published technical information on the fossils. Late in life, in notes to the author and her family, Hope said she still treasured Warren's first letter (it was not found among her papers after her passing). Hope's notes record that she had a "short but interesting" mail correspondence with Warren over a four to five year period; her fossil catalogue (Johnson, undated 1),

suggests this exchange took place in the late 1950s. Dr. Charles Stelck (1917–2016) from the UA and Dr. Loris S. Russell (1904–1998) of the Royal Ontario Museum (ROM) and National Museum of Canada (NMC; see Seymour, 2004) also had some contact with Hope and identified fossils for her (*e.g.* Johnson, 1961a). Hope may have had some contact with the former Head of the UA's Geology Department, Professor John A. Allan (1884–1955), though by that time he had retired.[10]

Hope's first catalogued fossil finds are reworked Late Cretaceous marine invertebrate fossils recovered from Pleistocene gravel pits, such as the ammonites *Placenticeras*, *Rhaeboceras*, *Scaphites* and *Baculites* and the clam *Inoceramus*, among others. The specimens date back to May 1952, and July 1953 (Johnson, undated 1). She began to study the local and regional geology using the 1940 Geological Survey of Canada Memoir 221, *Geology of the Southern Alberta Plains* (Landes, 1940; Russell, 1940), which she borrowed so often from the Suffield base library that they eventually gave it to her. This book not only identified many Late Cretaceous invertebrate fossils she found, but also gave locality information and maps for other sites she could explore at will. A set of coloured geological maps accompanying the book helped her date the bedrock she was exploring. She continued to visit local gravel pits in July and September 1956 and recovered a variety of invertebrate fossils.

In the early to mid-1950s, Hope, along with three others, started the Ralston Art Club. Two were medical doctors at the Suffield base: Dr. Kenneth "Ken" R. Clementi (1915–2003), a British-born health officer and Dr. William C. Stewart (1919–2007), a physiologist (Delday, 1977). Hope's notes also record Mrs. Joan Thain (1925–), wife of the local school principal, being involved with the club. Clementi arrived on the base in 1951 (Stouffer, 1971) so the Art Club started sometime after that; Hope's notes suggest about 1954.

Figure 52. Hope's first child Robert, with the wagon they both used to explore the prairie and active gravel pits close to home for fossils. Image courtesy of the Johnson family.

All four founding members were skilled artists. Arrangements were eventually made for their club to become part of the UA's Art Extension Department (Caldwell, 1985a; Clark, 1985; see also Wohlfarth, 1978). Professor Harry Wohlfarth (1921–1996) of that department (Cormack, 1981) became a regular visitor to Ralston, once a month for ten months each year.

At first the club had a steady membership of twenty-five adult members and even more junior members, Ralston being a family community. Later, there was a high membership turnover rate because many of the art club members were military personnel from the local BATUS (British Army Training Unit Suffield) base who were stationed for two years or less before being rotated back to the British Isles (Caldwell, 1985a).

Workshops were conducted on weekends, except in winter (because of the potential for bad highway conditions). Longer workshops were held in the summer. Paintings were made of still life objects, others from the artist's imagination, and even clothed live models were employed, the art club members taking turns in that capacity. Sketching classes were taught in the mid-1960s by Allan C.J. Jensen (1941–) of Medicine Hat. It is widely believed that the Ralston Art Club's workshops were held only in town, but one of Hope's paintings—an uncommon mountain scene (Figure 165)—is annotated on the back: "When I was living at Ralston & a member of

the Art Club we visited the Banff area . . . Painted at the site." So the group must have taken outdoor field trips from time to time. Some interviewees thought Hope may have taken art lessons at the Banff School of Fine Arts (Cameron, 1954), but other than possibly visiting it on her own or with the Ralston Art Club, the author could find no evidence that she took classes there. Painting colleague Evelyn Kleis (pers. comm., 2016) recalled that they all went to Red Rock Coulee on a number of occasions.

Hope later described instructor Wohlfarth as someone who "fired people up." He was an interesting character: an artist (painting/sculpting), scientist and adventurer (Anonymous, 1961a, 1974a, 1986; Pasnak, 1967; Ross, 1996). Alberta author Robert Kroetsch described Wohlfarth as having an "eccentric and antic carnival spirit" (Kroetsch, 1993, p 224). As a scientist, he became a leading authority on colour kinetics and "colour psychodynamics," or the effect of light and especially colours on people's physiology, behaviour and achievement (Cobb, 1970; Zilm, 1970; Anonymous, 1984b). He did not have a driver's license, so he regularly travelled to many rural communities across Alberta by Greyhound Bus, promoting art and teaching classes (Anonymous, 1960j). It is said that he travelled more than two million kilometres by city and rural bus this way (Lammi, 2002).

Ralston and area residents were encouraged to come and view the Club members' artwork and offer their comments (Anonymous, 1960d). According to Hope's own notes, Wohlfarth, recognizing her aptitude as an artist and illustrator, made her the "Local Instructor" of the club, even though at this point she had no experience working with oil paints. She acted on his behalf whenever he could not attend (Forbes, 1998). She fulfilled this role from 1954 to about 1975 and was a Life Member of the Club. She was never paid for any of this work. Under Wohlfarth's tutelage, students like Hope were able to show their art work at an (unidentified) exhibit in British Columbia (Anonymous, 1958a).

It is not known for certain when Hope first exhibited her paintings. Anonymous (1951b) relates how the Alberta Government's Visual Arts Board of the Cultural Activities Branch conducted its fifth annual travelling arts display across the province and that in 1951 fourteen of thirty-three artists in the show lived in smaller communities (double the number from previous shows). Many of them were trained in community art classes by the Extension Department of the UA. It is therefore possible that Hope's artwork was among those exhibited. The Ralston Art Club presented several of its own shows over the years and Hope had no trouble finding buyers for her work. At an early stage, she would frame each piece for $2.85 and sell it for $25.00.

The Steveville badlands northeast of Brooks, Alberta were made into a provincial park (first named Steveville Dinosaur Provincial Park, later shortened to Dinosaur Provincial Park [DPP]) in 1955 (Jensen, 1971). It, along with other parks in existence or in development, were part of a bigger plan by the provincial government to increase tourism as a major revenue source (Anonymous, 1956a). Former Park employee (from 1974 to 1978) Ron Chamney recalled that his father, T. Potter Chamney, was involved in some of the planning for the Park and he thought Hope might have been involved in some way, too; perhaps in the late 1950s and/or the early 1960s (R. Chamney, pers. comm., 2015). Certainly by 1955 she was collecting dinosaur and other vertebrate fossils in Petrified Coulee: her notes and RTMP fossil catalogue records confirm this. Also in 1955 the Alberta Government was encouraging amateur and professional artists to explore, paint and draw the natural splendour of our province (Anonymous, 1955a). It is unknown if Hope was already painting the badlands at this point.

Hope's earlier activities in the Scouting movement came to the fore again in the late 1950s. A Suffield base geophysicist friend named Gareth Jones (see next chapter) led several Boy Scout troops in his native Wales and had one in the Medicine Hat area; he would ultimately be involved in the Scouting movement for some sixty years (Ashwick, 2000). No doubt his involvement with that organization inspired Hope to rejoin, too—probably strengthened by the fact that her two boys were now old enough

to join. She became a Cub Scout leader, first as an assistant and then as a full leader, from 1956 (Anonymous, 1962r) until 1970. People recalling this time said she was very good with the boys. Their activities included long and rigorous wilderness hikes, up to 10 miles (16 km) in length, with a strong natural history education component.

Ollie helped with the Scouting movement too, helping to raise funds by recycling bottles and papers, the money raised to buy the boys badges and other items. These were happy times for the Johnsons, but also worrying times during a tense phase of the Cold War. One of Olafur's grimmer tasks in November, 1961 was reassuring worried Medicine Hat area people that radioactive fallout from a Soviet 50–58 megaton hydrogen bomb test (the largest man-made blast of all time) would pass high overhead and not harm them (Anonymous, 1961c). Such dangerous times may have inspired the subject matter in one of Hope's more unusual art pieces, in 1964.

1957 to 1969: Hope's amateur palaeontology career takes off

Hope continued visiting the gravel pits that had rewarded her earlier searches. Frequent revisits were worth the effort. The uncemented and unstable vertical walls of gravel pits meant that countless rocks on the exposed faces were always dropping out and rolling to the bottom of the pit where experienced eyes could examine them for fossils. Her fossil catalogue books record visits to local and regional gravel pits in the Ralston area every year from 1955 through 1961—and even in the winter, on February 6, 1962. During this period she collected a plesiosaur centrum in one of the gravel pits. Olafur joined her for a fossil hunt in 1956 in the badlands of the Jenner area.

In 1957 a newly-hired civilian member of the scientific staff at the Suffield base was Gareth H.S. Jones[11] (Figure 54 b), a colleague of Ollie's. He started as a summer student but was later hired on as a permanent staff member, which was a stroke of good luck for Hope. Jones' formal education background included engineering, geophysics and geology, and he would soon pursue and earn his Ph.D., so he was a valuable contact for her. She had immense respect for him given his background, intellect, and Earth sciences knowledge, which he freely shared. He was interested in palaeontology, British iguanodont dinosaurs, and had professional contacts at the Royal Ontario Museum (ROM) in Toronto.

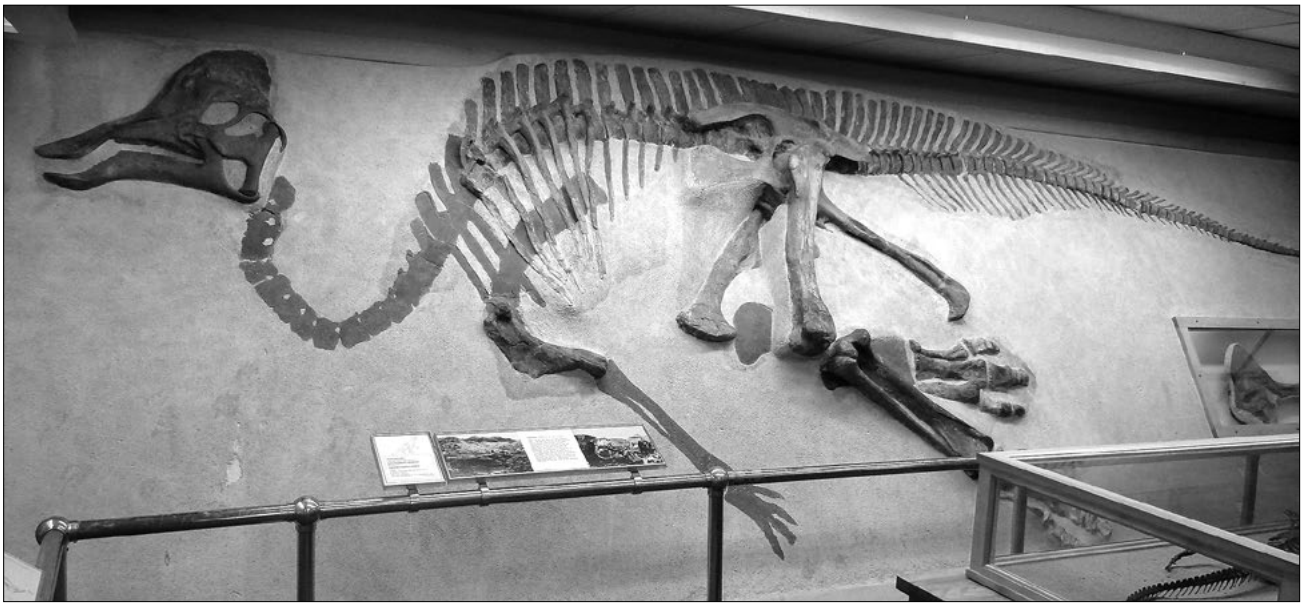


Figure 53. Composite mount of *Corythosaurus* (made from three partial skeletons) on display at the Palaeontology Museum on the University of Alberta campus. Mounted by George F. Sternberg and son in 1935. Photographed by the author in the fall of 2015.

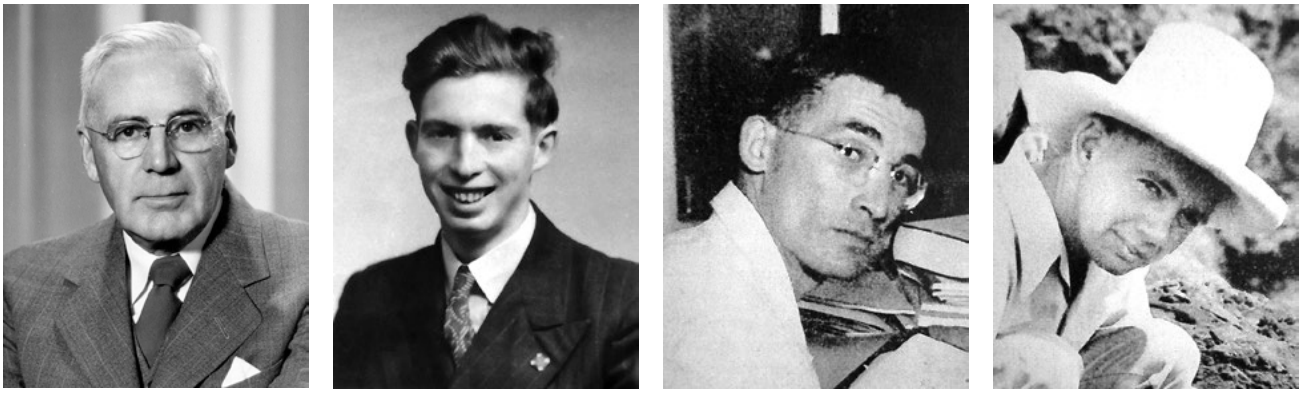


Figure 54 a-d. (L–R) Four Earth scientists who mentored Hope. a) Professor Percival S. Warren (1890–1970), geologist at the University of Alberta, Edmonton, undated. b) Dr. Gareth Hubert Stanley Jones (1924–1997), geophysicist at the Suffield base, about 1957. c) Dr. Roy Lemon (1927–2001) ROM invertebrate palaeontologist, about 1959. d) Dr. Gordon Edmund (1924–2004) ROM vertebrate palaeontologist, about 1959. Figure 54 a courtesy of Department of Earth and Atmospheric Sciences, University of Alberta; Figure 54 b courtesy of Brian Ashwick, Medicine Hat; Figure 54 c and d modified from Bush, 1960.

Hope was a natural fit—she had access to the abundant fossil remains and those of iguanodont descendants—the hadrosaurs. Her ROM contacts through Gareth Jones were soon to teach her much about vertebrate palaeontology. The two were good friends over the next forty years and he would be a frequent fossil collecting companion during the late 1950s and early 1960s. He showed her how to use Russell's 1940 paper and interpret the geological maps, and they explored some locations described therein. In 1957 she took Jones to a locality called "Petrified Coulee"[12] on George Lapp's Ranch and other local Foremost Formation outcrops. Together they found many fossils and Hope taught him much—as revealed in a letter from the ROM's Gord Edmund to the NMC's Wann Langston: Edmund writes that Jones was "quite an advanced amateur in vertebrate palaeontology" (Edmund, 1959).

Because Hope was intrigued by small vertebrate fossils ("microfossils") and lived in an area rich with them, she and her friends made many discoveries there. Petrified Coulee became one of her favourite field localities. Even after having been carefully picked over, a few rainfalls would cause the poorly cemented sediments to erode down quickly by several millimetres (*e.g.* Bayrock and Broscoe, 1972; Barendregt and Ongley, 1979; author, pers. observation), exposing a new crop of small fossils to await discovery. Hope and her friends did much prospecting for fossils on weekends, often with select eager Cub Scouts from her troop in tow.

Jones travelled to visit colleagues every few months at the University of Alberta (UA) in Edmonton and Hope would sometimes accompany him with fossils to have identified there. This would give her a chance to visit the small and little-known palaeontology museum on campus (Anonymous, 1935a, 1935b, 1935c, 1990c) which at the time included a nearly complete and open-mounted trionychid turtle, a panel mount of an adult *Corythosaurus* composite skeleton (Figure 53), mounted lower jaws and hind limb of a ceratopsian, a mounted skull of *Gorgosaurus* (Figure 55), and the (still) only-known complete skull and partial skeleton of the pachycephalosaurid *Stegoceras*.

Hope was now hooked on palaeontology and quickly developed a consuming interest in the abundant fossils near her home, figuring out their identities and geological ages. Long ago, a well-known vertebrate palaeontologist told the author he was of the opinion that museum fossil preparators and scientific illustrators knew the anatomy of fossil animals better than scientists because they spent much more time handling the fossils in the museum and often spent more time than scientists in the field. Hope adopted both strategies but began to focus on illustration, drafting mostly simple but very effective stippled pen-and-ink drawings of material she found. Years later, in a dialogue about the interpre-

tation and identification of one of her bone drawings, she wrote the author (Johnson, 1996) stating: "I am not imaginative in my drawings." No embellishments were ever added, just the anatomy she saw.

It is not known when she first did a drawing of a fossil as she rarely put a date on her drawings or paintings; nor do we know for certain when she began drawing local plants.

This new interest in fossils was heady stuff for Hope, but due to her age and other circumstances, she had essentially "missed the boat" with respect to Alberta's "Great Dinosaur Rush" (see Russell, 1966b; Spalding, 1999). By the late 1940s the field was pretty much dead in Alberta. The romantic days of competing institutions collecting hundreds of major dinosaur specimens in the province were over, largely curtailed by the Great Depression and then WWII. After the war, Charles M. Sternberg (1885–1981; Russell, 1982a–c, 1990) resumed some small-scale fieldwork but after the 1947 field season his days in Alberta on behalf of the Geological Survey of Canada were over as he approached mandatory retirement age in 1950. He would be contracted later to do work in Alberta (Anonymous, 1961d, 1962a).

In the late 1940s the UA in Edmonton had the aforementioned small palaeontological museum, the Late Cretaceous component of which was built in large part by George F. Sternberg in 1935 (Anonymous, 1935b, 1935c), though upgraded several times since (Anonymous, 1990c). The Calgary Zoo had their Prehistoric Park of concrete dinosaurs, which opened in 1937. It underwent rapid and continuous growth for several decades (Van Wagner, 1956; Anonymous, 1957a, 1959b; Gibson, 1973; Debus, 2006). Exhibits included two walk-through fossil display houses featuring Alberta dinosaur material, the first with a *Corythosaurus* skeleton acquired from the Geological Survey of Canada (Anonymous, 1938b). A couple of years later the second fossil house featured isolated material collected just after WWI, mostly by William E. Cutler (1878–1925; Anonymous, 1950).

Figure 55. UALVP 10, *Gorgosaurus* skull, displayed at the Palaeontology Museum on the University of Alberta campus, photographed by the author in the fall of 2015.

Alberta-based vertebrate palaeontology courses and field programs were still in the future with the UA and the Provincial Museum of Alberta (PMA) beginning their programs and dedicated fieldwork in the mid-1960s. In 1948 John E. Storer (1944–), who would eventually help Hope publish her first fossil guide (Johnson and Storer, 1974) was only four years old and Philip J. Currie, who would also be a valuable contact in the future, would not be born for another year. Another possible supporter, Professor Allan at the UA, died of a heart attack in 1955. So Hope was pretty much on her own at first when it came to learning about animal anatomy and fossil identification, though she did receive some support from ROM staffer A. Gordon Edmund (Seymour, 2005) starting in 1957 or 1958.

Unlike today, popular books on dinosaurs and vertebrate fossils were comparatively rare in the 1950s and 1960s. Fortunately many of the Alberta dinosaurs and other Late Cretaceous vertebrates had already been named as well as described and illustrated by the likes of Barnum Brown, Charles W. Gilmore, Lawrence M. Lambe, Richard S. Lull, William A. Parks, Loris S. Russell, Charles M. Sternberg, and Nelda E. Wright, so there was an excellent knowledge-base and scientific publications on which to begin her self-training. A few technical anatomy textbooks such as Romer's classics, *Osteology of the Reptiles* (1956) and *Vertebrate Paleontology* (1966), were also consulted. Hope's son Lawrence remembers anatomy textbooks in the Johnson home (L. Johnson, pers. comm., 2010). She wrote to some of these

dinosaur researchers and got reprints of their scientific articles. A 1964 letter in Canadian Museum of Nature (CMN) files, written by Hope to C.M. Sternberg, documents her request for several reprints of his papers and states that she already had reprints of many of his earlier works.[13]

A good number of the lower vertebrates such as fish, frogs, turtles and crocodiles have changed little through their evolutionary history. With living representatives available for study, she was able to understand their anatomically similar or nearly identical osteology with less difficulty than the more mysterious fossil forms presented. It was these smaller types of animals and their fossil remains, known as "microvertebrates" and smaller bones of larger animals that Hope would search for and specialize in.

Microvertebrate material from Alberta was among the first Late Cretaceous vertebrate fossils noted or described (Panton, 1884, 1885; Lambe, 1902; see also Brinkman *et al*., 2005), but it was largely overlooked for the next fifty to sixty years as American and Canadian museums pursued large dinosaur skeletons for their research, collections and public galleries. Hope not only searched for microvertebrate remains, but also for the locations of microvertebrate-rich fossil sites or "microsites" that would be useful for bulk wet-sieving techniques then just beginning to be applied in western North America (McKenna *et al*., 1994). Though she was well aware of the microvertebrate screening technique, there's no evidence that she tried it herself, preferring the traditional method of getting down on one's hands and knees, with nose to the ground, finding the tiny fossils visually. She did this a lot over her long life, which no doubt led to a comment in her advanced years (Johnson, 2009) that palaeontology was "hard on the knees."

Late Cretaceous geology of the southern Alberta plains had also been basically roughed out by this time (L.S. Russell, 1940 and citations therein). So with this published literature, Hope began the challenging task of teaching herself about geology and especially vertebrate palaeontology. Palaeontology textbooks existed at this time (Williston, 1925, would have been useful), but there were few popular palaeontology books or magazine articles on extinct life and specifically dinosaurs (*e.g.* von Zittel, 1925; Reed, 1930; Lull, 1935; Fenton, 1941; Colbert, 1945, 1947; Shimer, 1933; Swinton, 1948; Simpson, 1953; Swinnerton, 1950; Raymond, 1958). These were good general introductions, but they dealt little with Late Cretaceous vertebrates, so she had to turn to the technical scientific descriptions.

Even with the technical literature to back her up, many bones were unillustrated. Bones she found were in varying states of completeness, from animals of varying ages, or from animals only partially known or completely unknown to science and undescribed. Canadian palaeontologists before Hope dealt with the same issues. Lawrence Lambe (1863–1919) was confused during his first forays into Alberta, finding it difficult to know what isolated or fragmented bone pieces came from what types of animals. He had problems correctly reassociating incomplete specimens and accurately identifying them (Spalding, 1999, p. 41).

But Hope persevered in her studies and steadily acquired what became an encyclopedic knowledge of animal skeletal anatomy both past and present, which amazed many who met her. One gruesome but nevertheless valuable way to learn comparative osteology is collecting and studying the bones of modern animals. Road-killed remains of many terrestrial and flying species abound on Alberta's highways, especially in the late spring and summer. Such carcasses, providing they are not too damaged or decayed, can be skinned, boiled and the flesh removed, to yield good bone specimens for study. At her 2010 memorial service, son Lawrence related that when he was a boy, one day his mother had something in the pressure cooker on the stove. When asked what was cooking, Hope replied, "rattlesnake stew!" Of course it was not actually the family's supper—Hope was rendering the carcass down to retrieve the bones and gain a better understanding of fossil snake vertebrae, skull bones, and teeth. Her notes also mention having a pickled specimen of a painted turtle (*Chrysemys picta*) gifted to her from

the UA, that she boiled up to use the bones for comparative purposes. In an early 1969 letter to a colleague she commented that she had "a number of collections of animal bones which I've accumulated over quite some time."

The late T. Potter Chamney,[14] who knew and worked with some of the early dinosaur collectors, had the highest respect for Hope, admiring her dedication to vertebrate palaeontology and her excellent field identification skills (R. Chamney, pers. comm., 2010). Hope's fossil catalogue books (Johnson, undated 1–3) contain comments about fossil identifications and other contact she had with professionals in the palaeontological community. These names include David Bardack (UA, 1932–); Philip J. Currie (PMA, RTMP, UA, 1949–); Richard C. Fox (UA, 1933–); Len V. Hills (UC, 1933–2013); Dale A. Russell (NMC, 1937–); Loris S. Russell (NMC, ROM); Charles M. Sternberg (GSC, NMC); palaeobotanist Wilson N. Stewart (UA, 1917–2004); John E. Storer (PMA); and Percival S. Warren (UA). Any or all of these individuals could have or did help Hope learn about fossil vertebrate material. Others that can be added to this list are Rufus Churcher (1928–), Gord Edmund and Archie M. Stalker. Another contact was Canadian ichthyologist Ed J. Crossman (1929–2003) who, according to Hope's notes, helped her identify some of the gar and sturgeon material she collected. Bison expert Michael C. Wilson (1948–; Figure 66) was also an occasional visitor to her home (M. Wilson, pers. comm., 2015).

Hope soon began to share her knowledge publicly and to promote her rewarding palaeontology hobby. One wonders if this educational component of her persona was due to her own mother's teaching background, Hope assisting her mother with lessons, and Hope's debating skills learned in high school. She also gave talks to schoolchildren about geology and especially palaeontology but virtually nothing is known about the duration of these activities, at least early on.

Vertebrate palaeontology in Alberta continued to be fairly low-key during the late 1950s and early 1960s. Some professional fieldwork did occur with the ROM and NMC conducting smaller scale work: prospecting, working low-diversity bonebeds, or collecting single dinosaur skeletons. Most of this work occurred away from the Medicine Hat area, so it is unlikely that Hope encountered those crews. One crew she might have visited was that of Dr. Wann Langston (1921–2013; Bell *et al*., 2013) who in 1958 led the excavation of a horned dinosaur skull and skeleton of what became the type (CMN 41357) of the ceratopsian *Vagaceratops* (formerly *Chasmosaurus*) *irvinensis* (Anonymous, 1958d-e; Sackman, 1958). But big changes were coming, some that Hope would help with. A friend of Gareth Jones, Roy Lemon[15] (Figure 54 c), from the ROM, was doing some fieldwork in Alberta for the GSC in July of 1962 (Johnson, 1967a) and suggested that Hope and Gareth contact the ROM's vertebrate palaeontology department about their fossil bone finds. Hope would later send the ROM some invertebrate fossils at Lemon's request.

Through Lemon she met Dr. A. Gordon "Gord" Edmund (Figure 54 d; Seymour, 2005), Curator of Vertebrate Palaeontology at the ROM. When this meeting first occurred is uncertain; Edmund's name first appears in Hope's fossil catalogue books in late August 1958, putting them both in today's Dinosaur Provincial Park (DPP). It is possible they first met in 1957, when Edmund was visiting Alberta on a long orientation tour of western Canadian field localities with Levi Sternberg as a guide (Edmund, 1957). Gord and Hope evidently got along quite well and thus began a good relationship between the amateur and professional, each teaching the other. He suggested to Hope the names of other professionals she could contact for scientific information (J. Danis, 1981). Later on, Edmund said he didn't know whether to categorize Hope as an "amateur professional" or a "professional amateur" (Bush, 1969).

Apparently it was Edmund who first encouraged Hope to properly document and catalogue her fossil finds; she writes (Johnson, undated 1, p. 19): "Dr. A.G. Edmund started me off on this catalog. He knew about catalogs, but not about dinosaurs etc" (see Figure 56). This comment seems a bit imperti-

nent, but while Edmund was the professional and Hope the amateur, she must have felt the roles were somewhat reversed, for she writes (Johnson, undated 1, p. 64) in late 1961: "AGE wasn't (isn't) very 'up' on dinosaurs, very 'guessy'." Years later she commented that while she often sent fossils to the ROM for identification, "In the past I had numerous misidentifications from them. A pity." (Johnson, 1976b). She may have been right about Edmund—though he had full academic training, Hope had more field and relevant anatomical experience in Alberta and she felt she had the upper hand when it came to bone

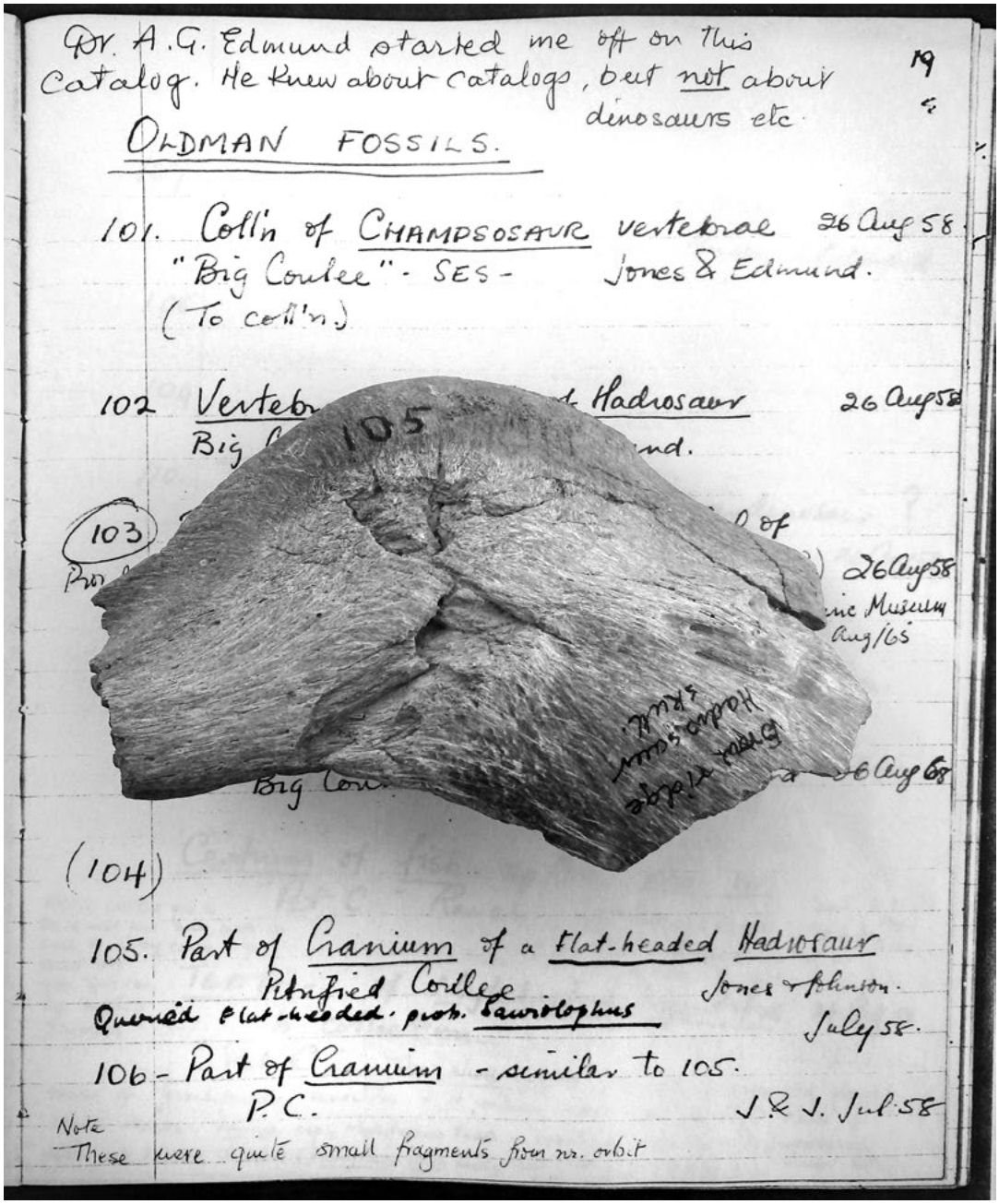


Figure 56. A page of Hope's fossil catalogue book from July, 1958, with her comment about Edmund seen at the top. Also seen here is her specimen #105 which she identifies as a "Part of Cranium of a Flat-headed Hadrosaur," and later added "…. flat-headed. prob. [ably] *Saurolophus*," collected by Gareth Jones and her from Petrified Coulee. It is re-identified here as a large hadrosaur right surangular of unknown species, in right lateral view. Image by the author.

and teeth identifications. It is doubtful there was any animosity between the two as they had contact over many subsequent field seasons, Edmund requesting some specimens Hope had found for the ROM collections, recognizing her with a nomination into the Society of Vertebrate Paleontology about ten years later and arranging for her to loan fossil horse material to an exhibit in Medicine Hat in 1979.

1958

Hope was angling for professional illustrator work late in 1958 with the now-named Canadian Museum of Nature (CMN). She had been drawing technical illustrations of local plants for some time. A mid-December 1958 letter to her from Loris S. Russell acknowledges receipt of some of her original artwork of native Alberta plants. He showed them to the museum's Chief Botanist, Dr. Alf E. Porsild (1901–1977) who commented that her drawings were "very good botanically and show not only skill in drawing but good understanding of form and gross morphology."

Russell could offer Hope no employment opportunities but said he would keep her in mind or recommend her work to others. He also commented that he hoped a crew from the museum could visit her next year and "do anything that we can to help you in improving your collections." Pencilled on the edge of the Russell letter is a humorous comment by Hope: "I like 'gross morphology' don't you?" Despite this rejection, some of her plant drawings were used by another museum for illustration purposes (see 1962 section, Page 68).

Through connections made during her CWAC days or through Olafur, who still worked at the Suffield base, Hope was able to conduct fossil hunting and collecting on the military base itself this year, something that even to this day is only granted to professionals in extremely rare cases.[16] Unexploded "dud" ammunition litters the firing range and only rare and carefully escorted trips into the area are approved. Gord Edmund, Gareth Jones, and Hope were permitted to have a look there in August.

Her notes also record an August 26 foray with Gareth Jones into a marine-influenced microsite yielding lamnoid shark, *Myledaphus* ray fish and crocodile teeth, *Champsosaurus* vertebrae and then-unreported sawfish teeth (*Ischyrhiza*) from a site on the South Saskatchewan River south of Suffield. The sawfish teeth would later be described in the scientific literature (Storer and Johnson, 1974). The next day the pair was joined by Gord Edmund and they visited the southern part of what is now DPP, in the Princess area.

Petrified Coulee was prospected numerous times this year. Hope's fossil catalogue records nine trips there from June through November. A complete lambeosaurine ischium was recovered in July and outcrops upstream were visited a month later.

A November visit with Gareth Jones was made to badlands that are now incorporated into DPP. Hope's notes make frequent reference to a company called "Oakalta" that had oil or gas wells and access roads south of today's DPP core area, an unknown distance west of Wolf Coulee.

1959

In 1959 Hope painted a large and detailed mural (Figure 167, measuring 6.5 m long × 1.2 m tall) for the Ralston Public Library that depicted the Ralston area's geographical, social, palaeontological and geological histories. Late that year, the Minister of National Defence visited the Suffield base. On a side trip to the Library he formally unveiled Hope's mural (Bridge, 1959c).

Many murals worldwide are painted directly onto concrete or cinder block walls; fortunately, Hope worked on wooden panels so that when the library moved into a new building in 2014 the mural was moved as well (Sanchez, 2014). The mural is very good for its time and reflects her ample knowledge of geology and palaeontology. It also reveals minor gaps in her knowledge at this point—for example, an ammonite is shown swimming with its shell upside-down (Figure 57). Her mural includes a Late Cretaceous marine turtle; at the time, marine turtles were unknown from Alberta's fossil record and would remain so for decades.

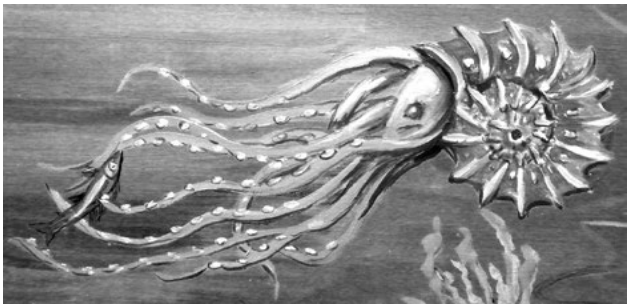


Figure 57. Detail of the upside-down ammonite from Hope's 1959 Ralston Public Library mural.

Hope left a detailed pamphlet explaining the mural and this text is repeated here, verbatim:

The Mural
by Hope Johnson

This mural may be roughly divided into three parts:

The Left Hand Section

This represents the succession of geological (and biological) events which have taken place in this area.

(1) This dark blue part is a rather conventionalized portrayal of primitive life in the sea. In this painting it may be considered as representing the whole period until the Upper Cretaceous Age. (Until about 100–90 million years ago.)

(2) This part shows the kind of life flourishing about 90–70 million years ago, at which time a sea stretched through the central part of North America, from the polar regions to the Gulf of Mexico. TYLOSAURUS is shown at the left—a large sea-serpent like reptile, and below it a CHAMPSOSAUR, a relatively small crocodile like reptile which did not become extinct until well after the end of the Cretaceous period. There is also a marine turtle, a large gar pike (LEPIDOSTEUS) with hard enameled scales, an ammonite swimming and a nautiloid on the sea bottom.

As we move toward Part (3), we must imagine that we are in brackish water (estuarial) representing a change in the land-water proportion at this time, *i.e.*, about 70 million years ago. The rocks and shales of this time are called "Foremost" and "Oldman" formations—the Foremost being below the Oldman, and showing a more marine type of life than the Oldman, but quite near to shore, and probably part salt–part fresh water. The sea retreated gradually south and east, and a low bayou country stretched north and south along the western shores of the sea, covering the part of the continent we now live in. In the marshier parts, and in the water itself lived crocodiles, Champsosaurs, and many kinds of duck-billed dinosaurs. Sharks and ray-like fish (MYLEDAPHUS) and the gar pike LEPIDOSTEUS lurked offshore. The duck-billed dinosaurs, which were all vegetarians, are called, as a class, HADROSAURS. The large yellowish beast with the high crown is a typical crested Hadrosaur of a type called CORYTHOSAURUS. About ten types of this dinosaur have been found near here, mainly in the Red Deer badlands to the north of us. The one pictured would be about 30 feet long, weighed probably 20 thousand pounds, and was mainly aquatic. The curious high-crested grayish dinosaurs to the right of the Corythosaurus are called PARASAUROLOPHUS. They were a smaller and lighter type of Hadrosaur than Corythosaurus. In all the crested types the nostrils ran up into the high head bones or crest, and down back into the windpipe, forming an airlock, so that the animal could eat efficiently under water.

(3) Part 3 shows three types of Horned Dinosaur ("CERATOPSIA" = "HORNED FACES"). These are from left to right STYRACOSAURUS, MONOCLONIUS, and TRICERATOPS. The total length of these was about 30 feet, the height at the hip about 9 feet, and the length of the head and crest, about 9 feet. In spite of their formidable appearance

they were plant eaters. To the right of these monsters are 2 ostrich-mimic dinosaurs of a type called STRUTHIOMIMUS (ostrich mimic). They are fleeing from a GORGOSAUR. Gorgosaurs were carnivorous, about 30 feet long, like a small edition of TYRANNOSAURUS REX. They are thought to be scavengers mainly but had formidable teeth, a hugh [sic] gape, and powerful 3-clawed feet. Their fore-limbs appear to have been virtually useless.

(4). Between (3) and (4) we see dark clouds, because the next period is quite obscure to us as far as this area is concerned. The dinosaurs all died out about 60 million years ago. A recent suggestion by a well-known scientist is that unfavourable mutations caused by an unusually high level of cosmic radiation may have destroyed them. Mammals of many types replaced them = BRONTOPS, many Rhinoceros-like grazers, deer, camellines, and mammoths. The land surface was much higher than it is now—as high as the top of the Cypress Hills (4800 feet) and the subsequent Ice Age eroded it away.

(5) The coming of the Ice Age is represented by snow and ice,

(6) The extreme upper left-hand corner shows a typical bit of the "badland" country on the South Saskatchewan River some 14 miles away. This is the "Oldman Formation," the hardened sand and mud of the time of the dinosaurs, and there their bones lie today, washed out by the rains and bared by the wind.

The Central Section
This represents the area today—the Central Laboratory, the school, *etc.*, and an aerial view of the Station.

The Right Hand Section
This is a schematic representation of the time scale, and shows many of the fossils found locally, as well as the source of the natural gas.

Most early amateur collectors simply surface-collected or chunked fossils out of the ground without the consideration or benefit of consolidating glue, but not Hope. By the early summer of 1959, like a professional museum field worker, she was using the scientific standard shellac mixed with methyl hydrate, as a fossil preservative (Eastcott, 1965) and making standard plaster and burlap field jackets to collect larger specimens, using toilet paper (which she called "palaeontological tissue") as separator (B. Ashwick, pers. comm., 2014). These were half-jackets only. Jackets she made in DPP some ten years later were rather rough examples with unsmoothed and untrimmed plaster exteriors and edges, but effective nonetheless.

At this time Hope and Gareth were also employing hospital-grade "Gypsona"-brand plaster bandages used by hospitals for setting broken limbs. Although in common use by the Royal Tyrrell Museum of Palaeontology (RTMP) since 1982 (Shultis, 1983), Hope's use of this type of plaster bandage appears to be one of the first documented palaeontological applications for them in Alberta. Another surprise is the use of explosives by Gareth and Hope to remove overburden covering moderate-sized and isolated specimens. A witness to one of these blasts said that primer cord (in those days it was possible to buy this from a hardware store) was used as the main explosive, with a blasting cap and fuse cord to set it off. The explosion was not large—more like "ditching dynamite" (B. Ashwick, pers. comm., 2014) and broke the rock up in place rather than blasting pieces aside. If the 1959 date is correct, Hope's use of explosives marks their last known use in a palaeontological context in the province; until now, the last known use of explosives was in 1956 when Wann Langston used them north of Drumheller to collect an *Edmontosaurus* skeleton for the CMN.

In a letter to the editor of the *Medicine Hat News* (Bridge, 1959b), Hope reveals that she and a few others were collecting fossils at nearby Petrified Coulee (she made visits April 19, September 20, October 3 and November 1, 1959) and sending important fossils to Canadian museums. In the letter she also pushes for tourism infrastructure upgrades at the site such as a lookout, signage, hand railings and road improvements.

Her specimen number 367, found at Petrified Coulee in 1959, was a complete hadrosaur sacrum with nine fused vertebrae. It was later donated to a Drumheller museum (see 1964 section, Page 76).

Hope hunted for fossils at a place she called "Hand Coulee" (its actual location is currently unknown) on April 5. On July 1, Hope and Gareth Jones visited an Oakalta badlands property on the Sand (= Little Sandhill) Creek and near Wolf Coulee in today's DPP. Her notes place her in Milk River in July, but it is unclear if this was the town or the river. She also collected near Sweetgrass, Montana on July 5.

During the 1958–1959 period Hope was a panel member of the Ralston Home and School Association (Anonymous, 1959c). How long she served in this function is unknown. It is recorded that she and another woman were trying to have kindergarten teachers included in the Alberta Teachers' Association pension plan (Anonymous, 1958b).

1960

It is unfortunate that Hope rarely dated her artwork, but in a rare instance we see a series of beautiful and delicately-rendered watercolours all done on the same size and type of card, one bearing a 1960 date. With visiting Professor Wohlfarth and the Ralston Art Club she was involved in a sketching and painting project, themed "Between Darkness and Dawn" (Anonymous, 1960a). From June 9–12, Wohlfarth led a three-day short course on outdoor sketching (Anonymous, 1960c).

Hope visited Petrified Coulee eight times, from April through October. She also travelled to Eastend, Saskatchewan in September. Outcrops on the south bank of the South Saskatchewan River south of Suffield had long tempted her, so on September 18 she and a friend waded across the river and secured some fossils among the steep outcrops there. Also in the fall, she visited outcrops below the brickworks at Redcliff and found some unusual fossil wood. A piece was sent to an unidentified American palaeobotanist who thought it was palm wood, but a UA palaeobotanist and Ph.D. student named Ramchandra (possibly Reddy P. Ramchandra) thought it was a rare specimen of bark from a taxodiaceous tree. It was later described and published as the new genus *Cupressinocortex albertaense* by Ramanujam (1970) who also acknowledged Hope as the donor of the fossil. A fossil wood specimen matching this description is shown on the UA (2016) Paleobotanical Collection website.

On June 1 and 11 she visited a site on the South Saskatchewan River called "Shaman's Head"; the location of this site is not well documented, but apparently it was not very rich in fossils. Outcrops in today's DPP on the Oakalta property were examined on August 18. Additionally, she made a three-day visit to the badlands with her Cub pack (Anonymous, 1960b). During this time and well into the 1960s, the local newspaper makes reference to the Ralston Cubs and Scouts making forays into the badlands for fossil collecting. There is little doubt that Hope led these trips but she is not specifically identified so the articles are not cited herein. White Rock Coulee was visited in July and October, producing a typical variety of Late Cretaceous vertebrate fossils. At this stage Hope was regularly mailing fossils to the ROM for identification. Anything they wanted they kept; the rest were returned.

In the course of research for this book, a surprising find was a 1960 letter to the editor of the Ralston newspaper written by a clearly frustrated Hope, complaining bitterly about all the work and cost involved for the sake of unhealthy Halloween treats and for mothers, expected to design competitive costumes for the children attending the school party and trick-or-treating later that night (Johnson, 1960). She exclaims (her capitals), "IT SEEMS TO ME, THIS WHOLE BUSINESS IS BECOMING A TIRESOME NUISANCE." She goes on to suggest a group costume party or a bonfire or fireworks. It appears that no one responded to her tirade. Her comments seem unusual, for later in life she loved the Halloween parties held at her retirement home and was an active participant (see Page 178).

1961

The summer of 1961 produced an exciting surprise with the discovery of eastern short-horned lizards, *Phrynosoma brevirostris,* discovered at Petrified Coulee, resulting in a revision in the scientific knowledge of their northernmost occurrence. Hope was involved in this event, sending specimens (and later, scorpions) to the Zoology Department at the UA in Edmonton (Bridge, 1961a). Cottonwood Consultants (1986, p. 39) had her collecting *Phrynosoma* specimens about a dozen miles south of her Ralston home in June and July. The find also resulted in her appearance on a local television program with some of these curious reptiles (Bridge, 1961b). Thirty years later she would be involved in research on them (Powell and Russell, 1991).

Beginning October 30, 1961 the *Medicine Hat News* posted a series of twelve articles by Hope on the local geology and palaeontological history (Johnson, 1961b-m; Figure 58). Anonymous (1961k) congratulated Hope on this series and wrote, "if the *News* is requested by a sufficient number of readers, this material may be produced in a more permanent form." In fact, these articles were indeed so popular that the *Medicine Hat News* later combined them into a tabloid form (see 1963 entry, Page 75). The editors of the local paper, *The Ralston Co-Operator* were "gratified to know this" (Anonymous, 1961m).

Hope was in the field as early as March 19, collecting shark teeth on the "Pat C Ranch." In a letter to Loris S. Russell, dated April 13, she commented that she had not done much collecting so far, visiting local gravel pits and a shell bed that yielded shark teeth (Johnson, 1961a).

July 29 saw her near Bassano, exploring Horseshoe Canyon Formation outcrops. She didn't report finding anything, though she made a painting of the outcrops there (figured in Bush, 1969, p. 13). Petrified Coulee was visited numerous times this year between June 15 and November 12, including a field trip with the Bassano Cub troop on July 9. A 35 mm slide with a November 1961 processing date shows Gareth Jones exploring a place identified as "Mountains of the Moon Coulee" (Figure 59). Cottonwood trees with green leaves—indicating a river—are visible in the background. The location

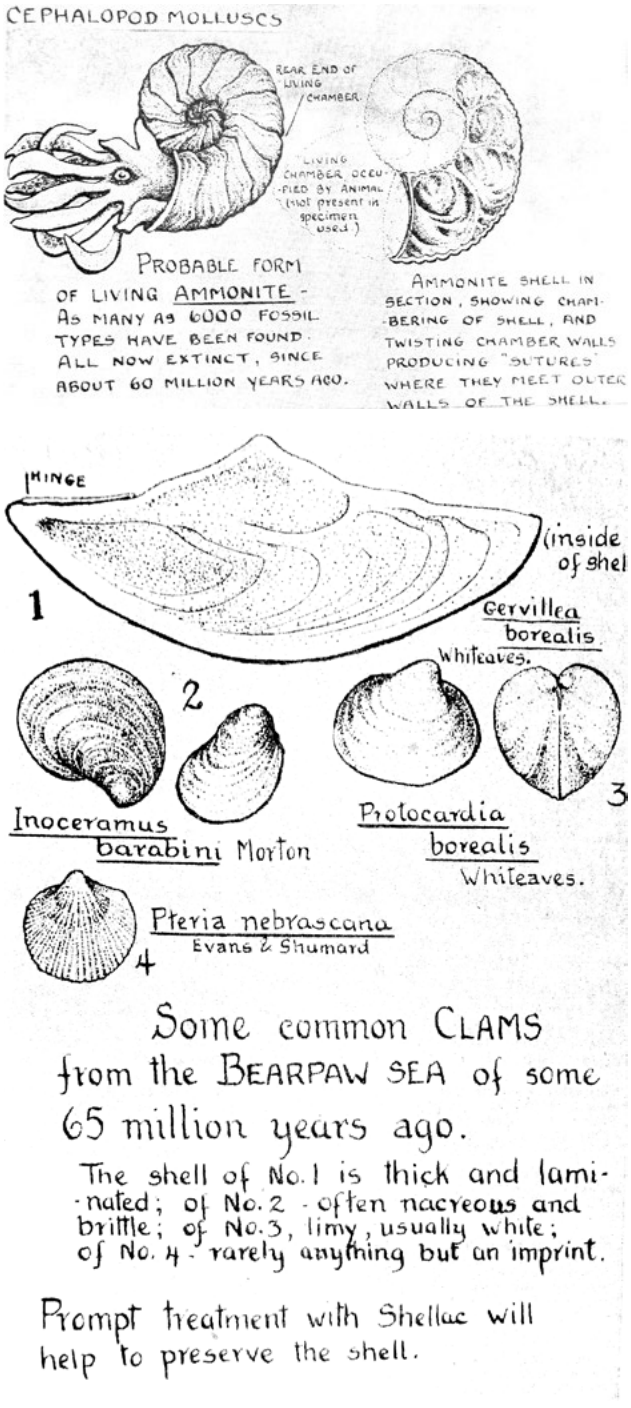


Figure 58. Two illustrations from Hope's 1961 newspaper series on geology and palaeontology of the Medicine Hat region.

of "Mountains of the Moon Coulee" is uncertain; possibly somewhere near Petrified Coulee. There is a "Valley of the Moon(s)" locality in DPP, but it is well away from the Red Deer River, a fact that adds to the mystery.

But is there really any mystery? B. Ashwick (pers. comm., 2014) pointed out that Hope was always inventing flowery place names for small patches of badlands on the South Saskatchewan River. The naming of specific, prominent or unusual outcrops in DPP has a long history. Anonymous (1920b) figures one outcrop called "The Praying Lady." This outcrop, now missing its "head" is seen today on the north end of Quarry 78, a *Chasmosaurus* partial skeleton (NHMUK R4948). There are similar descriptive names for localities in DPP such as "Valley of the Moon" and "The Citadel" (at the Quarry 143 *Centrosaurus* bonebed named "Castle Rock" in Currie, 1979 and "The Crown" by ROM 1930 field crews). Informal geographical nicknames abound: "Cathedral Ridge," "Valley of the Gold," "Petrified Teepee" and "Valley of the Castles" (Adams, 1982; Bryden, 1984). Imaginations tend to run wild in the badlands, giving birth to even more bizarre names like the iconic "The Camel" (or "Phred the Camel"; Turner, 1980), "The Submarine" and "The German Helmet." Bye (1981) and Johnson (undated 1) add places like "Pond Mesa," "Massif," "Nick Point Cave," "Conference Room," and "Big Coulee," the latter three in the eastern core area of the Park. These names do not appear among earlier fossil collectors' field notes or correspondence, suggesting they were coined later (though there is a "Knife Ridge" figured in Rogers, 1991, p. 172, near Display House #3). "Petrified Teepee" was coined in the mid-1960s (Tranter, 1992, p. 69). "Valley of the Moon," "The Citadel," "Cathedral Ridge," "Valley of the Gold," and "Valley of the Castles" were already in regular use when the author started working in the Park in 1979, some coined by Park staff in the 1970s (R. Chamney, pers. comm., 2015). The origin of some of the other geographical names remains unverified.

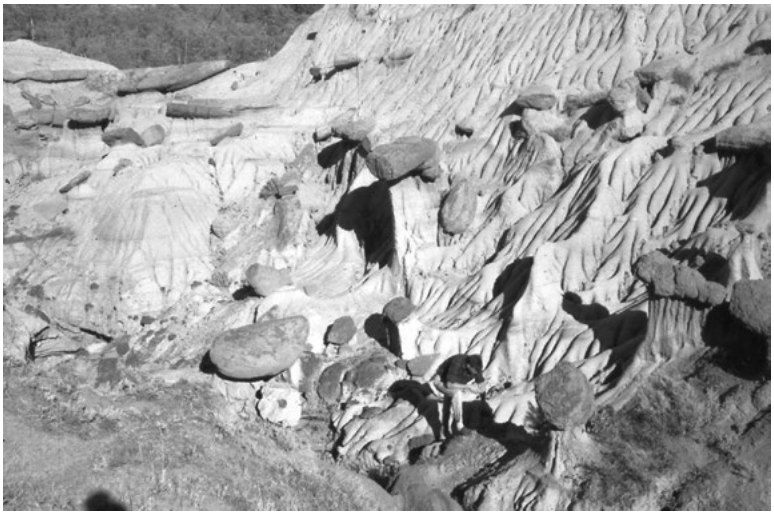


Figure 59. Gareth Jones prospecting at "Mountains of the Moon Coulee," an otherwise undisclosed locality. Hope's shadow is also visible. Image courtesy of the Johnson family.

Other fieldwork forays in 1961 took Hope to Denhart Coulee, in today's DPP; the old CPR pumphouse locality on the South Saskatchewan River; "Rose's Ranch" (location currently unknown to the author); and an unspecified site in the "Red Deer River Valley."

On the artistic front, the Ralston Art Club was enjoying great success in 1961, with a membership of sixteen—the highest number ever (Anonymous, 1961l). Hope attended the 5th Annual Community Art Class Conference in Edmonton in April. The theme of the conference was Community Art Classes (Anonymous, 1961h), something Hope was heavily involved with by this point. She also joined a painting course with Harry Wohlfarth from June 21–23 (Anonymous, 1961i). There was a show of Alberta artists' work in Ralston (Anonymous, 1961g); Hope did not display her work in this particular show, but later that year there was a two-day show of her artwork in Ralston (Anonymous, 1961f) and a group showing of the Club's art was held in Ralston just after Christmas (Anonymous, 1962p). It was also about this time that she helped develop the local school shield or crest. Others created the basic design, which was then "improved artistically by Mrs. Johnson" (Anonymous, 1962q; the author and

2015 school staff were unable to find an image of this crest).

Hope was busy with the Scouting movement in 1961. Leaders came and went—there was often a desperate call for new Cub and Scout leaders, but Hope could always be relied upon. She handed out Scouting badges at an awards dinner on June 7 (Anonymous, 1961j), where she was one of two Cub leaders (Anonymous 1961n).

There was a short-lived Ralston Science Club established during this period. It is unknown if Hope was involved, but given her interest in science it seems likely. Nothing is known of its activities.

1962

Beginning April 16, 1962, the *Medicine Hat News* ran a daily series of Hope's plant illustrations, featuring local flora. About seventy of her drawings were regularly published (Anonymous, 1962e; Johnson, 1962a-x). These small articles were technical in nature, yet readable by the general public. The whereabouts of the original drawings is unknown.

Anonymous (1962i) records that Hope had a series of her colour images of southern prairie flora on permanent exhibit at the Saskatchewan Museum of Natural History, in Regina. When she did this work is unrecorded (an undated note among her papers says "27 years ago"; the note includes data suggesting that the museum work would have been done about 1956). In her notes she records that she didn't remember how she got that job. The artwork may have been done for the museum when it newly opened in 1955 (Anonymous, 1955b; Young, 1955). The artwork is no longer exhibited but is curated in that museum's collections.

It appears that Hope only rarely painted in the field, preferring to create small watercolours or ballpoint pen or pencil sketches (with subject colours recorded as notes) on coil-bound pads, using these for reference at home. Some of these sketches are works of art themselves and some show numbered grid markings on the edges for enlargement purposes (Figure 182). She was also known to use photographs as reference material (Figure 61). An old artist friend recalled how Hope liked to sit on the ground, in a chair, or on an unpadded work bench (Figure 60) when she did her artwork, with her legs awkwardly crossed. Such a pose would seem particularly uncomfortable but evidently it suited her. The same friend noted that, while many artists add interpretation to a scene they are painting, working expressions of emotion or thought into a piece, Hope preferred to take an objective approach, being more interested in reproducing the real information that was visible in a particular scene or subject (E. Kleis, pers. comm., 2016).

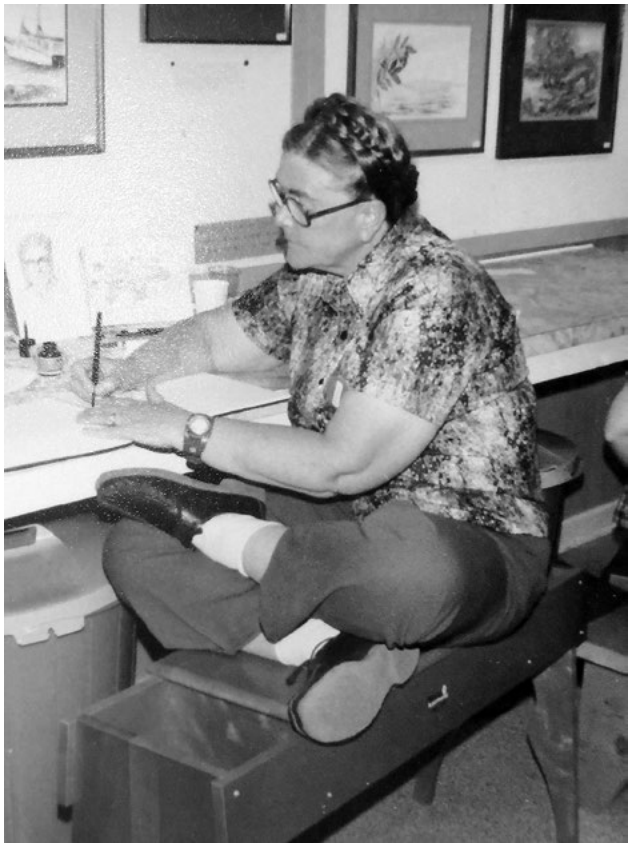


Figure 60. Hope works on an art piece at the Ralston Art Club, 1978. Image courtesy of the Johnson family.

Though the exact year is unknown, in the early

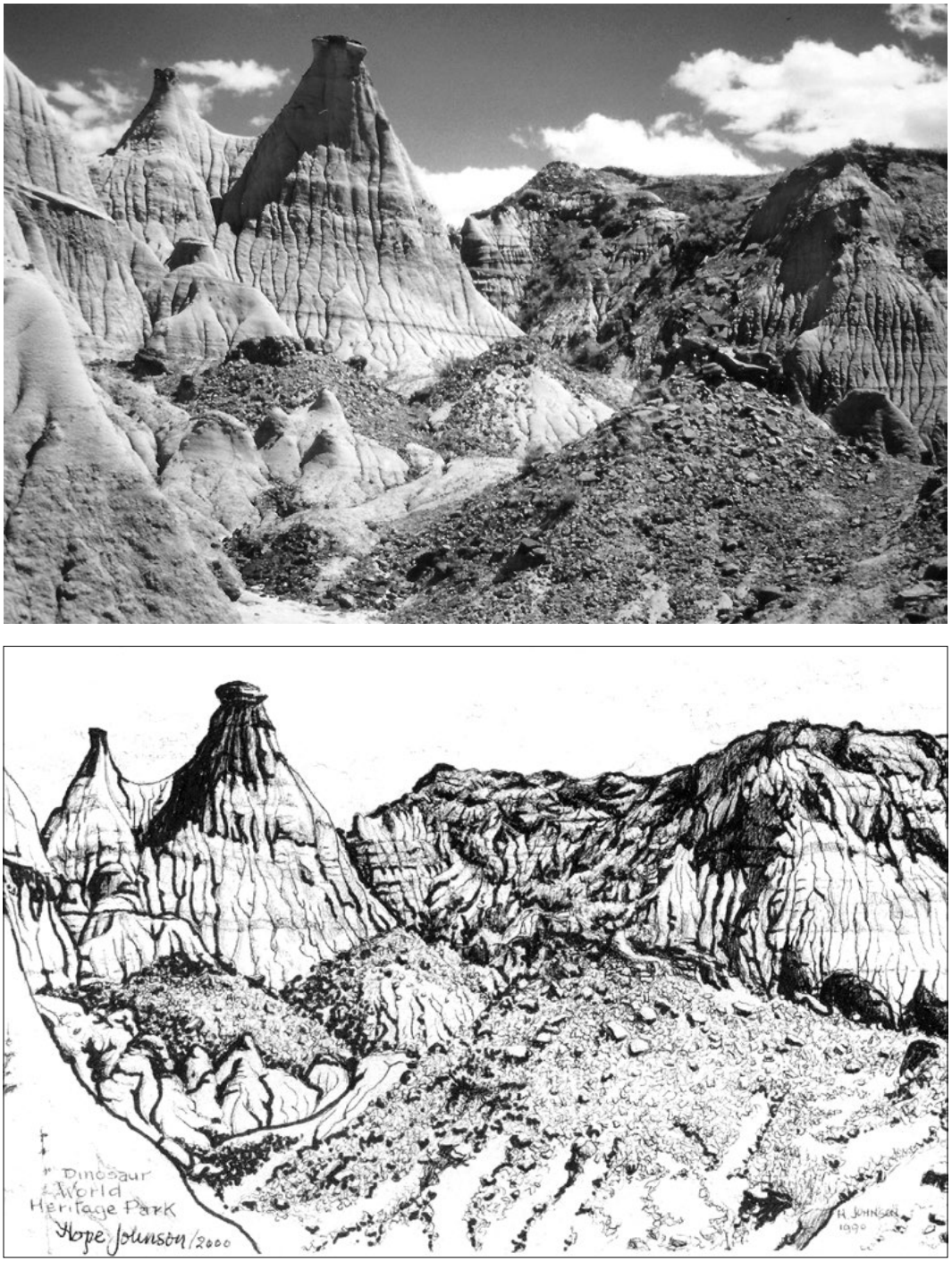


Figure 61. (Top) Reference photo of DPP badlands, undated. (Below) The same scene, drawn up and made into a greeting card, 2000. Images courtesy of the Johnson family.

1960s Hope provided much palaeontological information and drawings to a fellow amateur fossil collector and Milk River regional historian, Lawrence D. Halmrast (1925–1992), who resided in the Warner, Alberta district. He initially contacted her asking for fossil identification information which she sent in abundance. A life-long friendship ensued. When he received the Johnson and Storer (1974) fossil identification guide it became his dog-eared "fossil bible," filled with notes by him and by Hope

(Gasser, 2001, p. 38). Despite their long friendship, she is not known to have done any field collecting or fossil trading with him.

Hope delivered a lecture on an unknown topic at the Medicine Hat University Women's Club in early May, 1962 (Anonymous, 1962e, 1962f). It would be interesting to know what topics she discussed at a relatively early phase of her natural history interests. It may well have been on palaeontology, for on November 14 she gave a talk on that topic to members of the Association of Professional Engineers of Alberta and their wives (Anonymous, 19620).

During the summer, Hope's husband Olafur gave a talk at the 16th annual meeting of the Alberta Institute of Agrologists on weather forecasting for agriculture. The meeting included a pleasure trip to the Cypress Hills for the conference delegates, with Hope speaking on the geology and flora of the area (Anonymous, 1962i). She didn't record when she first discovered this rich, elevated oasis of flora, fauna, fossils and human history (Eggleston, 1951; Anonymous, 1953; Breitung, 1954; Gallup, 1965; Halladay, 1965; McCorquodale, 1965; Williams, 1965; Michael and Johnson, 1981), but it would forever be one of her favourite places to visit, explore and paint (Figures 62, 204). Scouting activities had Hope and an assistant co-leading a July outdoor camp in the Cypress Hills with some of her eighteen-member Cub pack (Anonymous, 1962r).

Figure 62. Not many photos exist of Hope painting in the field. This is the only one the author is aware of. Hope paints a watercolour high up in the Cypress Hills in 1997 (see Figure 204), courtesy of Police Point Interpretive Centre, Medicine Hat.

She led another Cub Scouts hike to the badlands on a fossil hunt when they came upon a porcupine. Rather than learn about it from a safe distance, they went quite a bit further—it was an encounter the hapless porcupine likely did not forget for quite some time. Anonymous (1962s) relates the action: "[the Cub Scouts] gave a porcupine a hard time and harvested a supply of quills as souvenirs of the encounter, fortunately none of them reached the anatomy."

Palaeontology fieldwork activities in the 1962 season included prospecting the shark and shell bed locality at the "Pat C Ranch" in March and April and seven trips to Petrified Coulee from May through September. She led two field trips for staff of the Defence Research Board at Suffield, to Petrified Coulee and DPP. In early October she explored the CPR pumphouse locality on the South Saskatchewan River.

Evelyn Kleis (1942– ; pers. comm., 2016), remembered how Hope helped her when she started out at her new teaching job in Ralston this year, noting that Hope was "generous with her time," and helped out with science topics like geology and palaeontology for the Grade 6 classes.

1963

This proved to be a pivotal year for Hope. First, on April 21, in what she called Denhart Coulee (now part of DPP), near Quarry 79 (quarry number of Sternberg, 1950) she found her first associated dinosaur specimen. This was the greater parts of the associated hind feet of an ornithomimid (her

field number 651; Figure 63). In a February 1, 1964 letter to C.M. Sternberg, Hope relates that Gord Edmund told her ". . . this is a very fine specimen and hope (suspicions follow) that you will cherish it accordingly."

While leading a Boy Scout troop on one of her many badlands field trips, Hope had found the specimen on the surface near a bonebed. The bones were somewhat broken up and loosely heaped together in what sounds like a deflation lag deposit. Six years later, Hope still considered it one of her best finds (Bush, 1969). One foot was nearly complete, the other much less so. None of the rest of the animal was found, though Hope went back to the site at least once (on May 18) and secured a few more pieces.

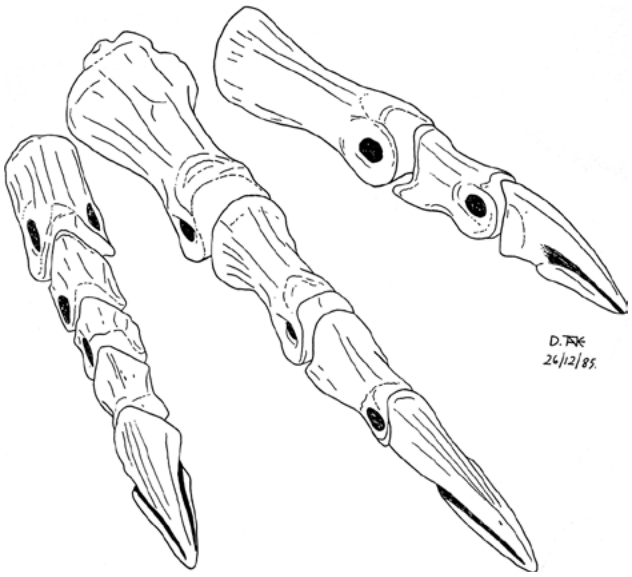


Figure 63. Sketch (by the author) of Hope's 1963 ornithomimid foot discovery, as seen on display at the RAM in 1980.

Hope spent much time studying the specimen and trying to identify it to species level. She noticed some evidence of "disease" on the unguals. Species identification was impossible given the incomplete nature and non-diagnostic components represented but she still tried, as evidenced in her letter to Sternberg. Hope states that the specimen was sent to the ROM and further prepared by technician Frank [sic] (Ralph) Hornell and returned to her. She tried mounting the right foot and worried (along with fellow collector Gareth Jones) that the specimen should go on display, but not in a display case at the Officer's Club on the Suffield base, where it "might get lost in the shuffle." It did end up there for a while (see 1965 section, Page 81), but the young fossil museum in downtown Drumheller showed interest in acquiring it, too. In the end the specimen was taken to the PMA in 1965 and was catalogued as PMA (later TMP) 1965.014.0001. The more complete foot was later put on display at the PMA (now RAM) in time for the museum's grand opening and bears the same catalogue number: RAM 65.14.1a–l, each phalanx having its own letter designation. This specimen (and/or the cast) was on public display in Edmonton for about forty years. The specimen was moulded and a cast was displayed for a time in DPP. It was deaccessioned from RTMP collections and retained by the RAM in 2001.

In another important event for 1963, Hope attended the seminal International Vertebrate Palaeontology Conference held in Edmonton and Calgary (Colbert, 1963; L.S. Russell, 1963; Devillers and Devillers, 1964; Folinsbee and Ross, 1965). She attended on Gord Edmund's suggestion (J. Danis, 1981). One of the chief objectives of the meeting was to seek outside expert opinion on whether there should be a large museum in the province highlighting Alberta's fossil heritage. Meetings were first held in Edmonton, then Calgary, with a mid-conference field trip. The group stopped at outcrops on the Red Deer River, near Scollard (Figure 64). A Late Cretaceous fossil mammal jaw was discovered at a new, mammal-rich site on that trip (Anonymous, 1963i; Robertson, 1964; Clemens, 1965). Anonymous (1963g) credits William A. Clemens with the find, but Clemens (2003) acknowledges British vertebrate palaeontologist T. Stanley Westoll (1912–1995) as the lucky discoverer. Late Cretaceous mammals were known from the province but only in limited numbers (Clemens and Russell, 1965). Thus, Hope witnessed a significant event in the province's vertebrate palaeontological history. Today, specimens of Late Cretaceous mammal teeth and jaws from Alberta exist in museum collections in the "thousands" (C. Scott, pers. comm., 2016). But at that point in time, mammal teeth were considered very rare; so much so that Fox (1976,

p. 1216) later commented: "as recently as 1963 there were no more than twenty specimens in museum collections." Knowing of their rarity and importance for research, Hope made a point of keeping a special lookout for them.

Figure 64. Hope, the only woman, standing prominently (left foreground) among palaeontology conference delegates on a field trip stop near Scollard, Alberta, August 31, 1963. Image courtesy of Hope Johnson.

The mammal jaw discovery would lead to over half a decade of work at the now famous KUA 1 (University of Kansas/University of Alberta) and KUA 2 mammal-rich microsites and other sites years later. Hope would devote much of her field time to looking for such small and rare treasures. In her February letter to C.M. Sternberg, she commented that the ROM's Gord Edmund believes ". . . we don't look for them [small mammal fossils], but we do." Up to this point in time she and her friends had found only a piece of mammal jaw with three empty tooth sockets. She quipped that she'd eventually find more small mammal specimens if she would "look hard enough and live long enough."

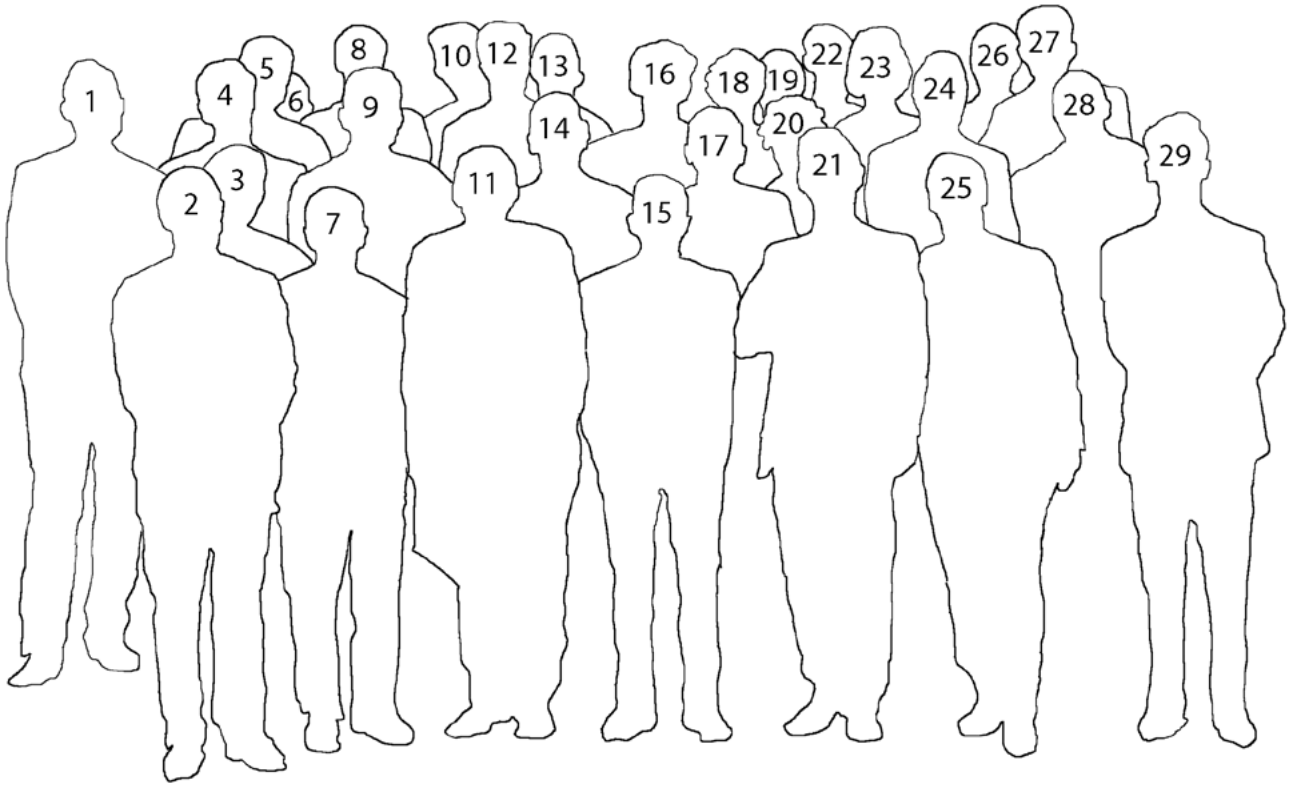
The Conference field trip group continued their itinerary with a visit to the small fossil museum in Drumheller. Then they were toured around the valley by bus (Anonymous, 1963h). A visit to DPP was arranged for August 31 (Anonymous, 1963f), then they headed on to Eastend, Saskatchewan to look at Oligocene-aged deposits and fossils there. On the way back to Calgary the group visited the badlands of the Milk River Canyon.

Hope's involvement—if any—in organizing or assisting the field trip is not recorded. Certainly by this time she would have made a valuable field guide, but she was a woman in a male-dominated profession. Hope also appears in the formal conference delegates' group photograph (Folinsbee and Ross, 1965, p. 6; Spalding, 1999, p. 118; Figure 65 here), barely visible, peeking over a man's shoulder in the back row. In the photograph there are only four women and twenty-five men. Of the women, Hope is

Figure 65. Some of the delegates of the August 29–September 3, 1963 vertebrate palaeontology conference held in Calgary and Edmonton. Image from Folinsbee and Ross (1965).

1. Robert E. Folinsbee (1917–2008)
2. Edwin H. Colbert (1905–2001)
3. Heinz T. Tobien (1911–1993)
4. Donald "Don" A. Taylor (1909–1983)
5. Rosendo Pascual (1925–2012)
6. **Mrs. Hope Johnson** (1916–2010)
7. Charles M. Sternberg (1885–1981)
8. David Bardack (1932–)
9. Charles Devillers (1914–1999)
10. Brian G. Gardiner (1934–)
11. Alfred Sherwood Romer (1894–1973)
12. Charles R. Stelck (1917–2016)
13. Walter H. Cottle (1921–2004)
14. Wann Langston Jr. (1921–2013)
15. Donald M. Ross (1914–1986)
16. A. Gordon Edmund (1924–2004)
17. Man identified only as "Leech"
18. Mrs. Mona Irene Read (1913–1965)
19. Dr. Wesley R. Read (1915–1994)
20. Mrs. Nena Westergard (1914–1981)
21. Loris Shano Russell (1904–1998)
22. Raymond O. Harrison (1925–?)
23. Mrs. Sherwood [arrangements]
24. Bruce A. McCorquodale (1924–1997)
25. Björn Kurtén (1924–1988)
26. Andy Westergard (1908–1971)
27. William A. Clemens (1932–)
28. Thomas Potter Chamney (1924–1983)
29. Charles S. "Rufus" Churcher (1928–)

Len Hills (1933–2013) and others unrecorded also attended. The Reads and Westergards were from Drumheller and affiliated with the Drumheller and District Museum Society. McCorquodale was Head Curator of Natural History and then Human History at PMA, and was later its Director. Taylor was the first Curator of Earth Sciences at the PMA. Many of the first vertebrate fossils collected for the PMA (later the Royal Tyrrell Museum) were collected by Taylor and associates (Tanke, 2008b). Hope would eventually donate no less than 2,300 vertebrate fossils to the PMA, RTMP and other Canadian museums.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the only one who could truly be considered an active amateur palaeontologist and as a woman she was clearly a minority at this conference. The other three women pictured were conference organizers or wives of other attendees.

This small palaeontological conference, the first of its kind in Alberta (or in western Canada for that matter), was attended by about thirty-five to forty delegates and included some of the leading figures and rising stars in North American vertebrate palaeontology, such as Alfred Sherwood Romer, Edwin H. Colbert, Charles M. Sternberg, William E. Swinton, Wann Langston Jr., William A. Clemens, Loris S. Russell, Dale A. Russell and Bruce McCorquodale and many others, including some of the more prominent researchers from Europe. Development of the Paleontology Department of the UA and the Paleontology Department of the PMA, both in Edmonton, were direct results of this meeting. The Paleontology Department of the PMA was transferred to Drumheller in 1982, to become the Tyrrell Museum of Palaeontology. Hope met vertebrate palaeontology legend Dr. A.S. Romer (1894–1973) at this meeting. Three years later he would publish *Vertebrate Paleontology* (Romer, 1966), a book that Hope would frequently reference over the rest of her life.

The Calgary portion of the conference culminated in a well-attended public lecture by Professor Romer. Among the attendees was an impressionable high school student named Michael C. Wilson (Figure 66). He was there to meet Loris S. Russell, with whom he had been corresponding for two years regarding some Paleocene vertebrate finds Wilson had made in southwest Calgary road cuts and residential basement excavations. Introducing Wilson to Hope Johnson after the talk, Russell spoke encouragingly about the important role amateur fossil collectors play in vertebrate palaeontology and reminisced about his own finds in the 1920s, in the Glenmore area of Calgary. Wilson went on to earn his Ph.D. (UC, 1981) and became a palaeontologist, though with an emphasis on Quaternary vertebrates, especially *Bison*. He had a long relationship with Hope over the ensuing decades (M.C. Wilson, pers. comm., 2018).

Figure 66. Michael C. Wilson in July, 1970. Image by Calgary Herald photographer Terry Cioni, courtesy of M. C. Wilson.

Hope continued to enlarge her circle of fellow amateur and professional fossil collectors. At this conference, Hope struck up a friendship with Dr. Charles Devillers[17] (Figure 110) of the University of Paris. He was there with his wife Jean; neither spoke much English. Hope spoke some French so was able to assist the couple on the tour. Years later (see Page 123) Charles and Hope went ammonite hunting in southern Alberta, then traded letters and a few fossils for a number of years. She acknowledged his help later in a scientific paper (Storer and Johnson, 1974). In 1976 Ollie and Hope visited Devillers and his wife in France. Another man attending the conference, Dr. Wesley "Wes" Robert Read (1915–1994), a medical doctor in Drumheller, was affiliated with the fossil museum there; more on him later (Page 76). Other material collected by Hope in 1963 was sent to an unidentified person in Switzerland by an "E. Naylor," possibly her friend Evelyn Naylor (1917–2013). Hope tried to drum up local interest to have Dr. Lou A. Bayrock, from the Research Council of Alberta (Edmonton) and a conference attendee, give a presentation in Ralston (Anonymous, 1963k), but it appears not to have happened.

Also during this time Hope had some direct contact in DPP with palaeontologist Charles M. Sternberg, briefly out of retirement and working over several summers on Alberta Government contracts to expose and develop four *in situ* dinosaur skeleton exhibits there (Anonymous, 1958c, 1961d, 1962a;

Nesbitt, 1963). Her fossil catalogue books indicate that Sternberg helped her with some fossil identifications when he was there. In a card written to former Alberta Palaeontological Society President Dan Quinsey, on March 20, 2006, Hope recalled these early days with Sternberg in DPP:

> "Dr. Charlie Sternberg was there, having removed the complete skeleton [this would now be TMP1980.023.0004; quarry 128, *Corythosaurus*] from the rocky wall nearby. He was stunned when an American lady visitor asked him 'Now, why would they drag a whole bunch of skeletons into a place like this?'"

At this point another important professional Earth scientist and museum worker entered Hope's life. The Alberta government hired oilman Dr. Donald "Don" A. Taylor[18] (Figures 67 and 85) to put together the palaeontology collection and exhibits for the PMA, which was slated to open in late 1967 (Spalding, 1999). RTMP fossil collection records indicate that his first catalogued fossil find was dated June 6, 1964, so he must have arrived a bit before then. Don and Hope had regular contact and his field notes and catalogued specimens reveal a wide aptitude for fossil vertebrate identification, collection, and curation; skills and knowledge he no doubt shared with Hope.

Figure 67. Dr. Donald "Don" A. Taylor. Photo courtesy of Robert Bruinsma.

Hope's twelve-part geological and palaeontological series that ran in the Medicine Hat News in late 1961 (Johnson, 1961b-m) were available in compiled tabloid form by January, 1963. Hope added some more illustrations and the final product was distributed free of charge to many local people and Medicine Hat city and district schools (Anonymous, 1963a; Johnson, 1963). The publication included a portrait of Hope (Figure 68).

She was noted as a special guest at a Girl Guide camp meeting at Elkwater in the Cypress Hills (Anonymous, 1963e). She often attended these functions to present a natural history aspect of the region, teaching about animals, geology, flora, or fossils. She continued working with her Cub "B" Troop but she is acknowledged in Anonymous (1963j) for helping with "A" Troop as well.

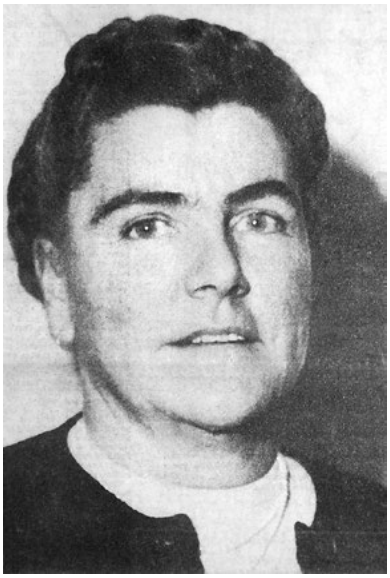


Figure 68. A newspaper portrait of Hope as she appeared in or about 1963. Image from Johnson, 1963.

A dated 35 mm slide suggests she visited Horseshoe Canyon near Drumheller on October 19. Her notes place her in Petrified Coulee at least four times this year and again at Milk River in July, but it is still unclear if this was the town site or the river. It is known that Dr. Kenneth Clementi, who helped Hope form the Ralston Art Club, moved to Milk River in 1956 to set up a medical practice there (Anonymous, 1956c, 1966b; Milk River Historical Society, *c.* 1989). Perhaps she was visiting him and fossil collecting in the same area. She collected invertebrates at "McCracken's Mine," though the date is not specified. The CPR pumphouse locality was visited sometime in September. She did some work in what is now DPP in April, August and October. A visit to Irvine, east of Medicine Hat, was conducted sometime in August. The Pat C Ranch was visited September 8, and Jenner Crossing (at that time a ferry site, now a bridge) on October 6.

Around this time, Dr. Archie M. Stalker[19] (Figure 69) began geological work on Pleistocene landforms, megablocks, gravel deposits and potential locations for water wells, modern topography resulting from glacial action, the "Foothills Erratics Train" and glacial valleys across the southern part of Alberta (*e.g.* King, 1956; Anonymous,

1957b; Stalker, 1960, 1969; 1970; 1972b, 1975, 2006; Mott and Stalker, 1972; Stalker and Churcher, 1972; Barendregt *et al.*, 1988).

It is not known when Hope and Archie first met; mid-July 1968 is the earliest the author could confirm from Stalker's detailed field notes (see 1968 entry, Page 93). They got along well and collaborated on a number of field projects with Hope assisting in any way she could. They were in professional and friendly contact for over thirty years. Stalker's work in the Medicine Hat area, which ran from 1963 to about 1994, located a number of Pleistocene multitaxic bonebed sites that produced fossils in abundance—making the Medicine Hat area one of the richest regions in Canada for Ice Age mammal remains.

Figure 69. Dr. Archibald MacSween Stalker in an undated photo. Courtesy of the Stalker family.

Stalker was assisted for much of this time, on the fossil component of the fieldwork and technical descriptions, by Dr. Rufus Churcher of the ROM. Churcher filled in for Loris S. Russell, who was too busy to assist (R. Churcher, pers. comm., 2014). Hope was involved in this fieldwork, too, but to what degree is uncertain. She occasionally helped on their fossil digs, "but not more than once a season," according to R. Churcher. She got along well with them both but was more "Archie's associate" (R. Churcher, pers. comm., 2014). Churcher crossed paths with her most summers when he was doing fieldwork in the Medicine Hat region. Bonebed excavations were only one component of the Pleistocene research being conducted, so only those seasons spent digging in bonebeds is related chronologically in the following sections.

In 1963 Hope was supported by a local newspaper editor (Anonymous, 1963b) in her personal crusade to have one of her favourite collecting areas—Petrified Coulee—protected as a park, something she had commented on previously (Bridge, 1959a; Johnson in Bridge, 1959b; Bridge, 1960; Johnson, 1961e). The idea never came to fruition. To this day the area remains distant, wild and undeveloped.

1964

Hope was briefly involved with the small fossil museum in Drumheller. Begun in the mid to late 1950s, the museum's exhibits got a major shot in the arm when the NMC set up a complete adult composite *Edmontosaurus* skeleton there in 1962 (Anonymous, 1962c, 1962h, 1962j). But smaller, high-quality fossils and other infrastructure were still required for the smaller exhibits. In a February 1, 1964 letter to C.M. Sternberg she discloses that she had agreed to provide "drawings and things" for this museum. In the same letter she asked Sternberg to send them his scientific reprints, wryly noting that the Drumheller people "are kind of short on comparative anatomy."

She donated fossils (types and quantities unknown) to the Drumheller museum as well as providing her artistic services. Her fossil notebook records preparing, mounting and sending fossils in August, 1964 for display. One was a complete hadrosaur sacrum (her catalogue number 367). She likely hand-delivered the fossil, owing to its size, fragility and weight. Hope's notes to her family say she spent about four days in Drumheller helping to make illustrations and identifying fossils for the museum with Wesley Read, a member of the museum's Board of Directors (Johnson, undated 1, p. 39).

Hope had first met Read at the Edmonton vertebrate palaeontology conference the year before (Figure 65). Upon learning that she could draw, paint, and identify fossils, Read asked her to come and do

some exhibit-related work at the museum in Drumheller. They were also interested in exhibiting her 1963 ornithomimid discovery, but that specimen ended up in Edmonton.

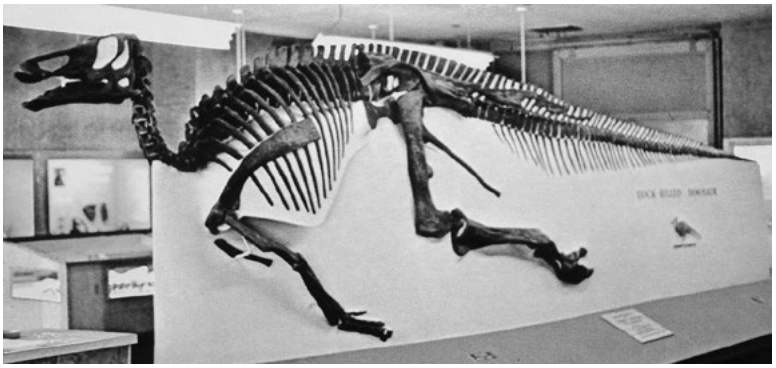


Figure 70. Composite *Edmontosaurus* skeleton at the museum in Drumheller, seen here in a 1960s or 1970s postcard view. This skeleton was installed in March 1962 and was on display there until 1997. Dismantled and stored in Drumheller for some time as a possible exhibit in the new hospital, it was returned to the CMN.

At the Drumheller museum she met a high school student and museum summer employee named Allan Jensen. Later she would have much contact with him in the Medicine Hat art community. At the museum she made coloured illustrations as backgrounds for some of the exhibit cases (A. Jensen, pers. comm., 2014) and large hand-printed labels for fossil specimens. None of these appear to have survived, at least in the public exhibit area. The Drumheller newspaper during this time frequently ran news stories on local fieldwork by professional museum teams or activities of the Drumheller museum itself, but no mention is made of Hope's work for them. While working at the museum in Drumheller she would have been able to study at length the excellent composite *Edmontosaurus* skeleton (Figure 70) installed there by Wann Langston and technician Hank Shearman two years earlier (Anonymous, 1962b, 1962d, 1962g).

In August 1964 Hope (Johnson, undated 1, p. 100) notes finding a Late Cretaceous trionychid turtle carapace (and possibly the plastron; 23 cm × 18 cm) in DPP and reporting it to Park Warden Roy Fowler (Figure 71, 1902–1975; Liddell, 1958; Anonymous, 1975b; Tanke, 2010d), another amateur fossil collector of note, who collected it. It may be the specimen figured in Vanderloh (1979, p. 198; Figure 72), as this turtle was on display in one of the small fossil sheds that Hope was heavily involved in developing and improving during the 1969–1979 time period. Another possibility is that it is PMA (now TMP) 1979.014.0851 (Figure 73), part of the Park collection which was returned to the PMA in 1979, but is now back at the Park where it serves a role in public summer lab tours. More turtle specimens are known. Walper (1980, p. 13) records another trionychid carapace and two baenid turtle carapace/plastron specimens being found in the DPP general collection. These were no doubt collected by Hope. They would require plaster jackets for safe extraction and she was the only one at the Park with the level of field experience to collect them.

Figure 71. Roy Fowler (1902–1975), DPP's first Park Warden from 1958 to 1965. Image courtesy of Donna Martin, DPP.

Hope's fossil catalogue books (Johnson, undated, 1–3) make occasional reference to her collecting outside the DPP boundary. It is clear from her comments that the Park was then much smaller than it is today and she was making a conscientious effort to respect the law and stay out of the Park. She only did so later when she had established her reputation and was employed by the Park.

She was visiting the Denhart Coulee and Jenner Crossing areas and was inspired by the badlands scenery there to paint

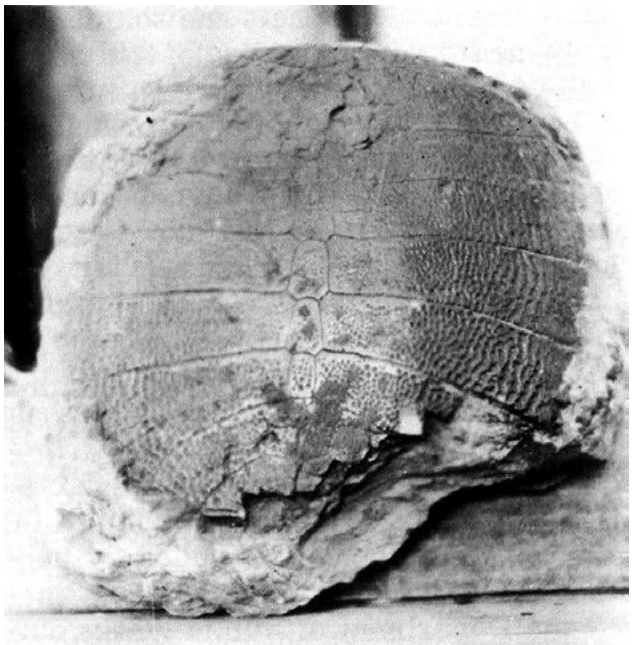


Figure 72. Trionychid turtle carapace, possibly collected by Hope. The specimen could not be located in a March 2016 search of RTMP collections. Image from Vanderloh, 1979.

Figure 73. Dorsal view of the carapace of an adult fossil trionychid turtle, *Aspideretoides foveatus* (TMP 1979.014.0851), the most common species of trionychid found in DPP. Scale bar = 10 cm. Photograph by the author.

(Figures 172, 173, 186, 201). These are among the few paintings Hope signed and dated and among her earliest known Alberta badlands renderings. One would think that the badlands would inspire many other artists to paint them, and to do so regularly, but it seems Hope ended up painting them more than anyone else. Many more DPP badlands and other southern Alberta badlands drawings and paintings would follow over the decades.

In 1963 Hope had been to Toronto and visited with fellow amateur fossil collector Hedy Hobberlin.[48] Hedy was a Toronto-based amateur fossil collector and educator of some renown (Anonymous, 1964d; Morrison, 1975; Quinter, 1975; Filion, 1981a, b; James, 1988; Irish, undated), but she is virtually unknown today. The Hobberlin collection, including Alberta specimens sent by Hope, eventually ended up in a Mississauga school (Anonymous, 2002). There, according to Hope's notes, she gave a talk to Sheridan College students about vertebrate palaeontology and visited the ROM. However, Hope must err on the year this occurred because Sheridan College did not form until 1967. In the early to mid-1960s Hope not only had a friendship, but a working relationship with Hedy, who accompanied her on a number of fossil collecting expeditions to sites that are now inside DPP. Hope's fossil catalogue books record a visit there with Hedy on June 25. Hope's fossil catalogues (Johnson, undated, 1–3) also note that she mailed Hedy Alberta fossils (vertebrate mostly) for educational purposes in types and quantities unknown, though some were apparently parts of an associated ceratopsian skull.

In a rare non-palaeontology or botany subject, Hope offered advice (Bridge, 1964) on the identity of a curious bird sighting in Medicine Hat: a pine grosbeak (*Pinicola enucleator*). Hope was a keen naturalist, but it appears she had little to do with bird and mammal natural history research in the province.

Early in September 1964 announcements were made for the soon-to-open PMA, which would include a palaeontology department (*e.g.* Anonymous, 1964c). Not long after, on October 13, Hope donated her first fossils to the museum's collections: two specimens of hadrosaur skin impressions, PMA (now TMP) 1964.013.0001 and TMP 1964.013.0002 (Figure 74), preserved in iron-rich sandstone from Denhart

Coulee, now within DPP. Though not major finds, they represent the first of many fossils she donated to the PMA/RTMP and other institutions over the years.

Hope's palaeontological fieldwork activities in 1964 included two visits to Petrified Coulee; at least eight trips to Denhart Coulee between April and November, once with a Lt. Col. Shelton of the U.S. military and another with a Defence Research Board photographer.[20] On a May 17–18 trip she may have camped in her vehicle, as she was known to sometimes overnight on a cot in her pickup truck with a box cover. On the May 18 trip she found an unusual bone; she found more pieces at the same site on June 5 and realized that they fit onto the previously collected material, resulting in a nice tyrannosaur nasal specimen (Figure 75).

She visited Irvine, Alberta briefly on July 5. On July 15 she led a field trip to Denhart Coulee for staff of the Defence Research Board at the Suffield base. On August 23 she led a small tour to DPP and viewed the "caged skeleton" on exhibit there. Forgotten by many, this was the prototype of some of the *in situ* dinosaur skeleton displays at the Park. At least one of these skeletons was not enclosed by a building as seen today, but was exposed, glued and then surrounded by a chain-link fence topped with

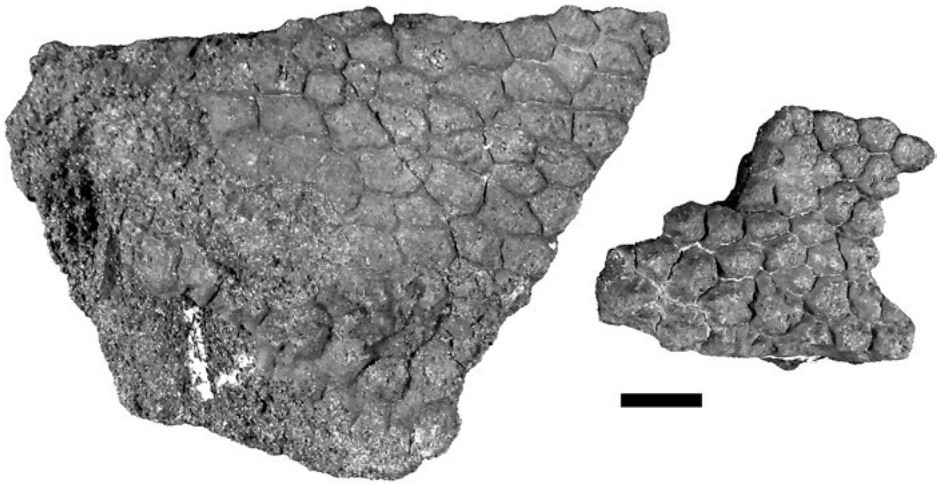


Figure 74. Hadrosaur skin impressions, TMP 1964.013.0001 (left) and TMP 1964.013.0002. Scale bar = 1 cm. Photograph by the author.

barbed wire. While relatively safe from the depredations of tourists and vandals, the skeleton(s?) was/were fully exposed to the elements (until at least August 7, 1965 according to Johnson, undated 3), so they did not fare well (L.S. Russell, 1965) and had to be properly covered later. One of the chain-link fenced exhibits appears in L.S. Russell (1990, his fig. F.2).

In July she visited the badlands along the south side of the Red Deer River, north and slightly east of Iddesleigh. The Pat C Ranch locality was revisited in September and Rose's Ranch in late October. White Rock Coulee was also visited this year.

Hope led a four-day campout and hike in DPP in August (Anonymous, 1964f). It is recorded that in October she was one of three leaders for the 1st Ralston Wolf Cub Pack, which now numbered twenty-one boys (Anonymous, 1964f). She also assisted the local Girl Guide chapter as an unspecified "tester," possibly a judge of the girls' efforts toward earning merit badges.

The Ralston Art Club displayed artwork by Hope and other club members at the Officers' Club on the Suffield base in June (Anonymous, 1964e). One of Hope's strangest paintings appeared this year (Figure 166). It is a confusing and violent scene with much symbolism and a mishmash of subjects.

In the background is an apparent nuclear blast with the classic mushroom cloud. A silver figure with outstretched arms, reminiscent of the robot "Gort" in the 1951 science-fiction movie *The Day the Earth Stood Still* rises from vapours (or blue flames?) in the foreground, along with a trident-bearing, ancient, emaciated man on an equally emaciated white horse—Father Time? A concentric circular form, perhaps an atomic structure or the sun shining through a poisoned sky, looms through an ambiguous perspective of both background and foreground. A girl in a white dress, awkwardly mounted on a galloping black horse, flies through the scene, slaughtering people with a sword while crows peck at their bodies. A dying tree sheds black leaves. All very bizarre, and one wonders if this was a Wohlfarth-inspired Ralston Art Club project themed on war and destruction. The atomic blast could easily be related to the simulated nuclear blasts going on at the Suffield base during this period, some of which Hope witnessed. Images of the fireball and mushroom cloud that formed after these blasts are truly apocalyptic. The imagery could be symbolic of the especially troubling times of the early 1960s with the Cold War, Cuban Missile Crisis, the threat of nuclear annihilation, the Kennedy assassination, social unrest and other issues during these turbulent times.

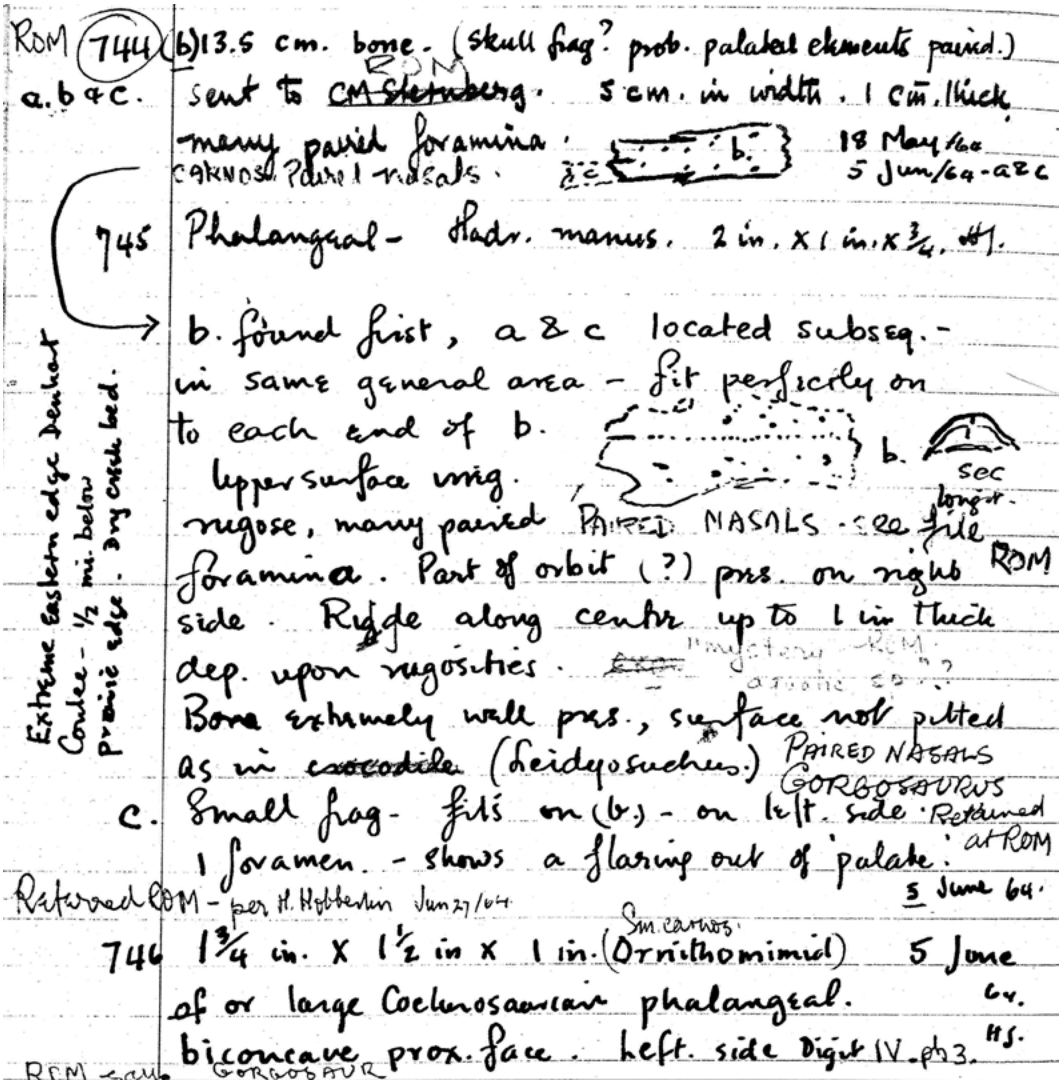


Figure 75. Detail of Hope's fossil catalogue for May/June, 1964 regarding the tyrannosaur nasals discovery. Modified from Johnson, undated 1.

1965

A little-known aspect of Hope's activities during the 1960s is that she had substantial fossil displays in the officers' mess at the then-named Defence Research Establishment on the Suffield Military base near Medicine Hat, where husband Olafur worked. Hope (Johnson, undated 1, p. 114–119) records in 1965 that there was a large number of marine invertebrates, bones and teeth on display: hadrosaur jaws, teeth, phalanges and an assembled composite pes; ceratopsian metatarsal, vertebrae, horns and teeth; ankylosaur scutes, teeth and a vertebra; large theropod vertebrae, phalanges and teeth; ornithomimid foot (later sent to the PMA; probably the previously noted TMP 1965.014.0001), other phalanges and vertebrae; crocodile jaw fragments, scutes, and teeth; champsosaur vertebrae, jaw fragments and a rib head; various pieces of turtle shell, and a variety of fish teeth and scales—about 300 specimens of all types were displayed. Bush (1969, p. 11) saw the exhibit in mid-July of 1965 and notes that these were exhibited in tall cases supplemented with Hope's specimen labels and various sketches. The cases no longer exist; when they were first set up and finally removed is unknown. They might have been taken down in 1972 when the Canadian military took over administration of the base; there were "many changes" implemented that year (P. Dudley-Reese, pers. comm., 2014).

This summer Hope began using the legal land description system of sections, townships and ranges to pinpoint fossil localities in her notebooks, but used it only sporadically, preferring hand-drawn maps and detailed directions to her sites.

Hope was called upon to give a talk on fossils and fossil collecting in southern Alberta for members of the Engineering Institute of Canada (Lethbridge branch) who attended a meeting (Anonymous, 1965a). Her presentation was entitled "The Fossil Fields of Southern Alberta." She led a Grade 7 class on a day trip to the local badlands for a day of natural history instruction and fossil collecting. Grateful thank-you comments from the students are recorded in Anonymous (1965e).

Her fieldwork activities this year included explorations at the Pat C Ranch, Denhart Coulee, DPP and surrounding area (Princess) and Petrified Coulee. She also collected some fossils on behalf of educator Hedy Hobberlin (see Page 78). Hope may have done some work in DPP in early November of this year. Anonymous (1965b) reports that she visited fellow amateur fossil collector Irene Vanderloh, who lived in Wardlow, close to the Park.

Hope had some contact with Charles M. Sternberg, who was at the Park for three months this summer, working on the last of four *in situ* dinosaur skeleton exhibits: an adult *Corythosaurus* skeleton (TMP 1980.023.0004; Anonymous, 1966a; McNeil, 1966). He again helped identify some fossils for her. A curious and incomplete bone in two parts attracted her attention this year. Its significance was not understood at the time; it and other fossils were donated to Bruce McCorquodale of the PMA and are now catalogued as TMP 1965.014.0039. Years later the bone was identified as a pterosaur ulna (Godfrey and Currie, 2005, their fig. 16.7), making this the first recovery of a pterosaur bone in Canada.

Beginning in 1965 Hope occasionally mentions finding First Nations stone artifacts. Inevitably, as one scans the ground for fossils, these archaeological objects also turn up from time to time. It is unknown what became of most of these finds or how many she found—at least one made it into an academic collection (see 1979 section, Page 132). While she is generally credited by others as having an interest in archaeology, she appears to have been a relatively insignificant participant in the field, as her name is absent among those of other Albertans acknowledged in one of Alberta's first major archaeology publications (Wormington and Forbis, 1965).

A major geological field trip was held in the Medicine Hat and Cypress Hills area this year: the 16th annual conference of the Alberta Society of Petroleum Geologists (ASPG, now CSPG; Zell, 1965). There

were 170 registrants (de Wit, 1966, p. 166). It is believed that Hope participated or helped in some way on this field trip though her activities are not specifically recorded. Her involvement is almost certain, as this was her field area and the conference leader was T. Potter Chamney, who knew Hope well (perhaps first meeting her at the 1963 conference) and he had a great respect for her palaeontological knowledge. It is also known (though again the details are lost) that Hope participated in several ASPG field trips in the 1950s and 1960s (R. Chamney, pers. comm., 2015).

On the art front, Anonymous (1965c) notes that Hope was now teaching adult beginners an oil painting class. The largely self-taught landscape and modernist artist William L. Stevenson (1905–1966) of the Art Extension Department of the UA (Edmonton) was also teaching some classes; he was one of Alberta's very first professional artists (Townshend, 2005). Hope was also vice-president of the Ralston Art Club this year (Anonymous, 1965d).

A comment in Anonymous (1965f) records that Hope was a member of the Scouting Group Committee for 1965–1966 and served as secretary.

1966

Hope reported finding a *Styracosaurus* bonebed[21] (now known to academics as Bonebed 42) in DPP, that she thought contained articulated material (Johnson, undated 2, p. 40–41). Don Taylor and technician John Poikans (1919–2007; Tanke, 2008b) from the PMA responded and worked the site September 17–21 (Johnson, undated 2). Recovered specimens are catalogued under the TMP 1966.010 series. An exasperated Taylor was evidently unimpressed with her find, writing in his field notes "it is only a bonebed!" (Taylor, 1966). He was no doubt hoping for a good skeleton to be a showpiece at the soon-to-be-opened PMA. But a good number of diagnostic skull bones were collected from this site and a few went on display in the newly opened PMA in 1967, to be exhibited there for many years (Figure 76). Several specimens were later transferred to the Oilsands Interpretive Centre in Fort McMurray, where to this day they serve a public educational purpose.

Figure 76. Dinosaur display case at the PMA, Edmonton. *Styracosaurus* material from Bonebed 42 in DPP and part of Hope's 1963 ornithomimid foot discovery are seen (bottom, left centre). Image scanned from RTMP slide PN 68.30.42. The slide bears a processing date of August, 1972. Photograph courtesy of the Royal Tyrrell Museum.

Despite Taylor's misgivings, this was indeed a significant find, as later studies would show. University of Calgary (UC) student John Visser conducted excavations and geological studies there in 1983 and 1984 for his Master's thesis on sedimentology and taphonomy (Visser, 1986), though this work was never formally published. Stratigraphically it is the lowest known *Styracosaurus* site of any type in the Park and is therefore useful for understanding ceratopsian evolution and dinosaur faunal turnover.

Skull material from the site was in-

cluded in several recent studies of *Styracosaurus* or ceratopsian bonebeds (Andersen, 2000; Holmes *et al.*, 2006; Eberth *et al.*, 2007; Ryan *et al.*, 2007; Brown, 2013) and will serve a valuable function for ongoing and future comparative ceratopsian craniology and taphonomic research. The RTMP reopened the bonebed in the 2014 field season (see online video in Anonymous, 2014i) with good results and future work by the Museum is anticipated.

On July 3, in an encounter in DPP, Hope met the recently hired Jane Colwell (later married surname Danis or Colwell-Danis; 1941–; Tanke, 2015a; 2017a, b) who had been tasked by Dr. Richard C. Fox (Scott *et al.*, 2013) to prospect for rare Late Cretaceous mammal remains and other microvertebrate fossils and sites in southern Alberta (Colwell, 1966a). Jane was uncertain about Hope at first. She was surprised by Hope's visit, as Jane's presence in the Park that summer had been kept quiet. Jane commented: "Well I just had a very interesting visitor a lady fossil hunter who knows all the wrong people. For a top secret start with just me, Toronto sure found out about it fast. Oh well, that should make it more interesting. I don't know what it means but she was nice and seemed friendly so we'll see. I guess I'd better close and drop Lance [Richard Fox] a letter and record her visit . . ." (Colwell, 1966b). Despite Jane's initial misgivings, she and Hope had a good working relationship both professionally, in DPP, and socially over many years. Jane was Canada's first formally trained female vertebrate palaeontologist, with a Master's degree from the University of California Berkeley, and globally among the first women hired into a professional vertebrate palaeontology position—indeed, Alberta led the way in hiring women into vertebrate palaeontology positions (D. Spalding, pers. comm., 2016).

Notable earlier (though not much earlier) female workers, also researching Late Cretaceous terrestrial beds when Jane started, were the Poles, Zofia Kielan-Jaworowska (1925–2015), Teresa Maryańska (1937–) and Halszka Osmólska (1930–2008); see also Turner *et al.*, 2010.

According to her notes, Hope was a member of the Society of Vertebrate Paleontology (SVP) until her death, having been nominated for membership in the summer of 1966 by the ROM's A. Gordon Edmund (Figure 54 d; Anonymous, 1966l, p. 4; also noted in Anonymous, 1966i, where she is further listed as an "amateur, interested in field paleontology and graphic restorations of fossil vertebrates" [22]). As a member of the SVP she would receive the quarterly *News Bulletin* which gave detailed reports and updates on global field, lab and research activities. Names of researchers, their fields of expertise, place of employment, and mailing addresses were also revealed, allowing Hope the chance to network with others. Some of the papers Hope penned late in life indicated that she was still a member of SVP, but with an honorary membership; this claim is unconfirmed.

Edmund was working in Hope's collecting areas during this time, but whether Hope was involved in fieldwork with him is not recorded. It is known that she visited Petrified Coulee in May, Denhart Coulee and other areas near the Park six times between April and September, White Rock Coulee sometime in May, and the Lost River badlands in September.

Also this year she did some fossil prospecting in the Killdeer badlands of southern Saskatchewan, collecting a Miocene rhinoceratid(?) phalanx from the Wood Mountain Beds. This fossil was found by the author among her collection late in 2014. In late 2015 the fossil was repatriated to the Royal Saskatchewan Museum and bears their catalogue number RSM P3208.1.

Hope penned a four-page article titled "Studying Geology and Palaeontology at Home." It would have been a useful item, but its appearance in the Ralston newspaper (Johnson, 1966), with a limited readership, severely reduced its value. She continued with her badlands paintings and fossil vertebrate drawings. According to her notes, many of the drawings that were eventually used in the Johnson and Storer (1974) fossil book were drafted during this period. It's likely that while she collected fossils with her friends and members of the public she began to realize a need for an educational publication detail-

ing what people were finding and what was important. No doubt during her badlands hikes she would be peppered with anatomical and taxonomic questions about the identity of something her companions had found. The still rapidly developing oil and gas industry and a good economy in the province meant that more and more people were moving to Alberta—commensurately there would be more people interested in fossils. She would develop a small fossil identification guide on her own in 1968 (Johnson, 1968a), but a much more ambitious publication would come six years later. In the meantime, Hope continued finding and collecting fossils, cataloguing, preparing, identifying, and drawing some of them.

Figure 77. Hope, Velma Heath and Bill Burton collecting microvertebrate fossils from an ironstone fragment-dominated site of unknown provenance, June 1968. Image courtesy of Penny Heath Torres, Nanaimo, British Columbia.

Work in the Medicine Hat region with Archie Stalker (GSC) and Rufus Churcher (ROM) continued with hundreds of bones being collected from a variety of Ice Age mammals—such as camel, prairie dog, bison, mammoth, horse and others. Many of the bones were complete long bones or jaws (L.S. Russell, 1966a). Again, Hope's involvement, if any, in this work is unrecorded.

Experimental large blasts at the Suffield base continued this year. While waiting to watch one such blast, she passed her time at the viewing area looking at rocks and plants. It was here that she first met her future son-in-law, Ed Preston, and Dr. Charles Heath (1921–1980; Anonymous, 1980) from the departments of Physiology and Pharmacology at the UA. Hope and Charles struck up a friendship that led to Charles, his daughter Deanna (from a previous marriage), his wife Velma Heath (1929–2015) and her son, Bill Burton, going on fossil collecting trips with Hope on a regular basis (Figures 77, 78).

In 1966 Hope was invited by chairman Edward McKenzie to join the Board of Trustees for the new Medicine Hat Museum and Art Gallery, then under construction with an anticipated opening on Canada's Centennial, July 1, 1967. She served with the Board for fourteen years, leaving in 1980. During her tenure, the Board decided to revitalize the Historical Society of Medicine Hat and District, which had "dwindled away almost completely." Hope and several others (Hope's notes record that two sisters, both Alberta-born teachers, Fiona and Lucy Milne, were particularly valuable) had to rebuild the Society from scratch as much of the earlier documentation was lost or never generated. Unfortunately, it appears that much of this revitalization work was not recorded or saved, either. About five years later Hope became the Society's President for two years and was also its Editor for part of the time.

In 1966 members of the Medicine Hat University Women's Club paid a courtesy visit to the Havre, Montana University Women's Club (Anonymous, 1966e). Hope tagged along as a guest speaker, but what she presented on is not recorded—fossils or botany, most likely!

Hope's art activities during this time are not well known, but she did donate a painting for a Boy Scout Association fundraiser (Anonymous, 1966h). Twenty-four boys were now in her troop (Anonymous, 1966j) and she continued on as Secretary for 1966–1967 with the Boy Scout Executive (Anonymous, 1966k). She also helped at a 4-H summer camp at Elkwater, in the Cypress Hills.

In an already busy year for Hope, the local military base was celebrating its 25th anniversary. Her role in this celebration is incompletely known. There was a concert at Easter and a later celebration on August 26. Anonymous (1966f) records that Hope was one of three members on the Easter Concert Committee. There was an exhibit of paintings by Ralston and Medicine Hat artists at the base on April 29. The base also held a contest for a Silver Anniversary logo. It appears she did not submit an individual entry, but took part in a group effort. Anonymous (1966g) writes: "credit for the anniversary crest. . . goes to a number of people. The original composition was prepared by Mrs. Hope Johnson from portions of designs submitted by Carol McCallum and Allen Davey." The crest (Figure 79) was subsequently used on base letterhead, the local newspaper masthead and commemorative souvenir items such as badges, stickers, ash trays, and coffee mugs. At a concert celebrating the anniversary, Hope and another woman served coffee during intermission (Fish, 1966).

1967

Early in the year Hope did a quick sketch for the Ralston newspaper's masthead, for which she was acknowledged (Anonymous, 1967a). She and fellow Ralston artist Mr. Amerigo (Rick) Filanti (1928–2013; Anonymous, 2013) were honoured by being selected by the Art Instructors Council of Alberta to attend a one-week seminar in Edmonton. Both became members of the Community Art Infrastructure Council. Their invitation to the seminar was significant, because only professional instructors had participated previously (Anonymous, 1967c). The seminar improved their art teaching skills.

Figure 78. There are few pictures of Hope in the field. Here she has some fun with Velma Heath (née Burton, right) and Deanna Heath, on a day trip from Ralston, Alberta in 1966 (Deanna Heath and Penny Heath Torres, pers. comm., 2017). Deanna was the daughter of Charles Heath from a previous marriage. The ladies pose with a fossil log. Image courtesy of the Johnson family.

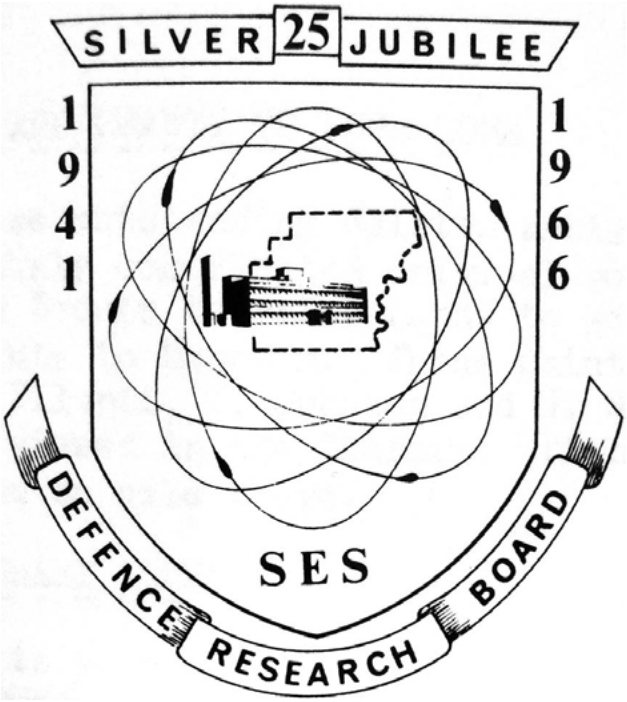


Figure 79. The Suffield Experimental Station 25th Anniversary logo which Hope finalized from other artists' ideas. Image from Anonymous, 1966g.

In early April, Hope and her daughter Barbara went to Edmonton and visited the new workshops of the PMA. She made this trip as a member of the Board of Directors of the Medicine Hat Museum. In Edmonton she examined Alberta fossils and First Nations archaeological material. She also visited Richard Fox at the UA where they discussed rare fossil vertebrates (Anonymous, 1967b). No doubt she took some interesting fossils with her to be identified. On this trip, it seems that Fox advised her to begin corresponding with Dr. Dale A. Russell (1937–), curator of vertebrate palaeontology at NMC (Figure 80), which she did for the better part of two decades, sending him many fossils for identification and even discussing items of a personal nature, such as child-rearing.

In 1967 an excited Hope noted in her fossil catalogue her discovery at DPP of a small fossil mammal tooth. She was on the lookout for these, as they were especially important for research and at the time they were particularly rare fossils in Alberta, outside of the KUA 1 and 2 localities noted earlier (Page 72). She sketched different views of the specimen (Figure 81); it was later donated to the UA. Hope also assisted the UA's Jane Colwell in looking for microsites that produced rare mammal remains. Hope reported a potential microvertebrate site and—sure enough—good mammal material was found there (Colwell, 1967, July 22 entry). She also assisted Jane in prospecting for and collecting small fossils later that summer. Colwell was no doubt happy to have Hope accompany her. It was a dry year and Jane was having a "poor summer" (Johnson, 1967b). Soaking rains were needed to wash away sediments and expose new microfossils such as mammal jaws and teeth.

Years later, Richard Fox (pers. comm., 2011) recalled his early meetings with Hope and her fossil mammal finds:

> I first met Hope Johnson in 1966, or possibly 1967; she brought a collection of microfossils with her, as she did on several subsequent visits. Among her several collections that she brought were a few Cretaceous mammal teeth each time, I believe from the railway grade on the west side of Dinosaur Park, and the Irvine locality, in a small patch of badlands at Irvine, just south of the Trans-Canada Highway. In any case, she is the only amateur besides Tim Schowalter that I know of who ever found Cretaceous mammal teeth on her own (*i.e.*, without prior instruction) in Alberta, and deserves credit for that.

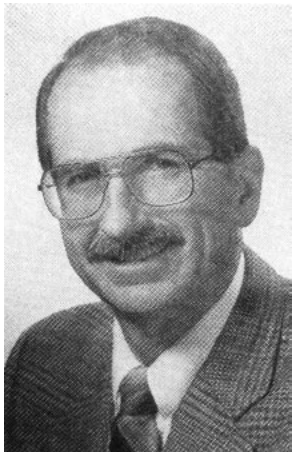


Figure 80. Dr. Dale A. Russell of the National Museum of Canada, Ottawa, in a 1989 newspaper photograph. Image from Anonymous, 1989b.

An interesting 1967 incident in DPP was related to the author by Nick Rothwell early in 2015. Nick—then a twelve-year-old boy interested in dinosaurs—and a friend were dropped off at the Park for a couple weeks of camping. Remarkably enough, they were left to fend for themselves. Their parents would check up on them every four or five days and resupply them with food. They camped, swam in Little Sandhill Creek and rode around on their bikes.

They were not totally abandoned, however, for "Hope Johnson along with a park warden and some palaeo students basically looked after us the whole time." In the course of their stay, the boys were allowed to watch ongoing digs and go exploring with the fossil experts:

> We were allowed to surface pick small bones in certain bone beds unsupervised as long as all bones were later checked (and a few were then taken away), we were taken to active digs and participated with the university students; the warden drove us around once in a while on patrol. We could sit and watch but we could NOT dig or touch anything near where they were working. We were directed to a nearby spot where a bunch of broken and eroded-out bone fragments existed. It was mainly Hope who kind of told us where to go and what we could do when we left our campsite. It was a kid's dream come true. Hope Johnson was pretty strict and tried to teach us as much as she could about palaeontology and the local flora and fauna the whole time we were there. She gave my buddy and me several samples of dinosaur bones/teeth each as gifts. Many are marked and identified in Hope's hand [Figure 82].

This story was to be repeated twice between 1967 and 1970 when the boys had parent-free holidays in the Park (N. Rothwell, pers. comm., 2015).

Hope assembled "fossil boxes" (Figure 83) for some of her fossil collecting friends during this period. Each fossil was selected from her own personal collection and carefully wrapped, with handwritten identifications of each specimen.

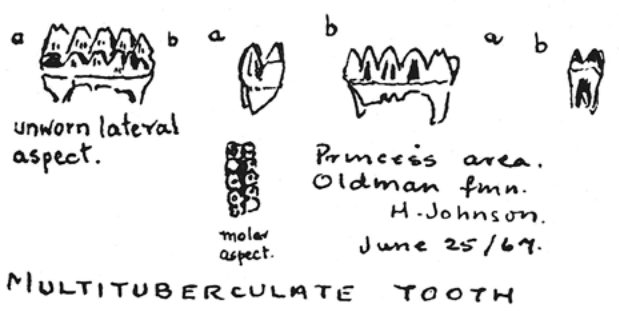


Figure 81. Hope's summer 1967 sketches (rearranged here) from her fossil catalogue, of a Late Cretaceous mammal tooth she found in DPP. While just rough sketches, they are good enough to identify it as a multituberculate lower right first molar. Her use of "a" and "b" are for orientation reference. Hope's notes state that it measured about 7 mm × 2.5 mm.

The GSC and ROM continued their work in the Medicine Hat area, finding bones of Ice Age animals in five separate bonebeds and discovering another rich site (Stalker, 1967; Churcher, 1968; Stalker and Churcher, 1968). Evidence of ground sloths was recovered (L.S. Russell, 1967; Churcher and Wilson, 1990). 350 specimens were collected from all the sites, ninety of which were from Saskatchewan. Some sources believed that Hope had worked with Churcher and Stalker this summer, but apparently she did not. Michael C. Wilson was on the field crew this year and kept good field notes. He recorded the comings and goings of people and Hope's name does not appear. Possibly she visited briefly but did not work with them this year (M. Wilson, pers. comm., 2018). She and a friend visited another ROM camp on July 23, this one on the Milk River and run by Gord Edmund (Edmund, 1967).

Her other field activities included two visits to Petrified Coulee and three to the DPP region. There she found a rare, nearly complete ceratopsian maxilla with seventeen teeth—it was sent to the UA in October, 1968 (J. Danis, 1981). In September, in White Rock Coulee, she found another major specimen—a partial *Gorgosaurus* skeleton protruding from a hoodoo. It was collected in 1968 (see Page 90). She had also visited White Rock Coulee in August and found fossil mammal material and a rare tooth-marked dinosaur bone specimen: a proximal caudal vertebra. Tooth-marked dinosaur bone is not uncommon, but the "victim" in this case was an ornithomimid, making it one of the first saurischian-on-saurischian encounters documented in the province. She later donated this one-of-a-kind specimen to the RTMP, where it was catalogued as TMP 1985.006.0158 and has been used in a number of studies (Jacobsen, 1995, 1997, 1998, 2005; Jacobsen and Currie, 2005; Jacobsen and Bromley, 2009; see also Jacobsen, 2001). Hope loaned Dale Russell a number of theropod bones (Johnson, 1967a) for a study he was doing with Edwin H. Colbert of the American Museum of Natural History (Colbert and

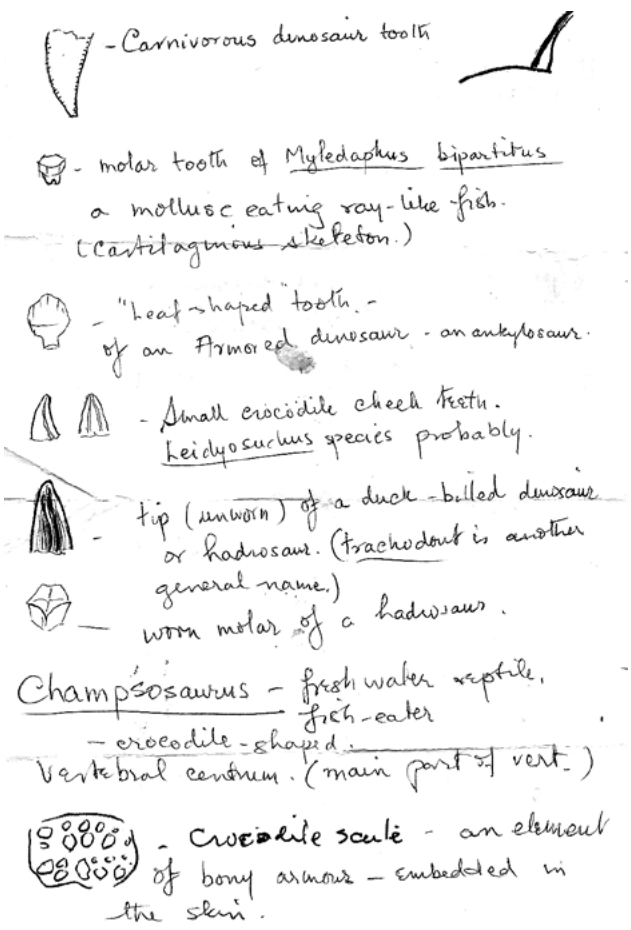


Figure 82. Not only did Hope educate and help some boys collect small fossils in DPP in the late 1960s, she sent them home with sketches and identifications of them so they would not forget their lessons. Image courtesy of Nick Rothwell, Nanaimo, British Columbia.

Russell, 1969), though she was not mentioned in the acknowledgements for that paper. Her specimens may have been used for their revision of the genus *Dromaeosaurus*. She also sent fossil fish material to Dr. David Bardack, ". . . who has apparently been doing work on identifying fish centra" (Johnson, 1967b).

In her fossil catalogue (Johnson, undated 3), Hope reports searching for fossils at a place called "Big Coulee," in July. This is not to be confused with a similarly-named locality in DPP/Denhart Coulee. This "Big Coulee" site was on the Suffield military base, near the "seismic vault." She also visited the Irvine badlands in June, the Lost River badlands in July and the Milk River area in October. On an early October visit to White Rock Coulee she collected much of the front half of an old bison skeleton. The skull was nearly perfect—missing only three incisors (Johnson, 1969b). She would eventually donate the bones to the Medicine Hat Museum (just one of a number of modern animal bone specimens she donated), but not before the skull and a few other bones of the bison and the skull of a domestic cow were put on display for comparative purposes in the Ralston Library (Anonymous, 1967d).

Buffalo skeletons, both articulated and disarticulated, and isolated bones are not uncommon in the badlands. Most appear to represent animals that had either fallen into or been washed post-mortem into deep sinkholes eroded into Late Cretaceous sediments (Tanke, personal observation). Live animals in this dire predicament die and become buried as the sinkhole gradually fills with sediment washed in from nearby outcrops; a mapped example of a filled-in sinkhole is shown in Currie *et al.*, 2016 (their fig. 1). Eventually, the walls of the sinkhole erode back, exposing its sediment fill in vertical cross-section and revealing the bones of its hapless victims (Ralrick and Tanke, 2005; see Figure 92). To the inexperienced eye, such occurrences appear to be in Cretaceous rock or occur in geographical areas that defy explanation. Hope was intrigued by one in particular (Johnson, 1969b):

> I have seen some [bison skeletal material] in the badlands which I believe must be very old, as I cannot see how a herd animal would have arrived alone on some mesa-like portion of the badlands in such an isolated area (I mean, isolated from the plain nearby, separated by precipitous coulee which must be hundreds of years old—in spite of the relatively "rapid" erosion in such places).

In the summer of 1967 Hope met Memory Myers, the Ralston librarian. Though they were casual acquaintances, Memory recalls Hope being great with children, funny in a dry sort of way, and "always good" or "always around" the library or town, ready to lend a helping hand and volunteering for

different things. If there was any type of community event, Hope could be relied on to attend and help out.

Also this year, the Medicine Hat Museum opened a new facility. Hope provided the background mural, fossils and labels for the new prehistory exhibit (Bush, 1969, p. 12). Taylor (1968) notes that one of ten display cases in the new museum was dedicated to dinosaur fossils. Hope was assisted in this work by the museum's curator, John "Jack" Barrie (Delday, 1977).

Hope's familiarity with professionals and inter-institutional politics made her think about how her relationships with various groups might be affected. She writes (Johnson, 1967b):

Figure 83. One of Hope's "fossil boxes" formerly from the collection of Velma Heath. Image courtesy of Penny Heath Torres.

> There is quite a probability that I will be collecting part time for the new Alberta Provincial Museum. I am also at the moment on the Board of Directors of the new Historical Museum. There may be a conflict of interest developing here, esp. as I have been prospecting with Jane Colwell—so I think I had better dispose of some part of my collection now to the University, since I cannot see what interest or use some of these items will be to the Prov. Museum during the foreseeable future. I feel that I can fit in the Historical Museum & the Provincial Museum without much trouble, but the University seems to be "off by itself." This is really because the museum does not foresee anything but exhibits for the public for the time being, not having the staff or accommodation for any research material.

1968

With her children now grown up and independent, Hope could dedicate more of her time to her cherished natural history-related activities. This year saw Hope's first publication on fossils: it was a booklet entitled *Dinosaurs in the Southwest Prairie Area* (Johnson, 1968a). It was a simple, eleven-page general guide to Late Cretaceous fossil vertebrates of southern Alberta, but was just a glimpse of what was to come. It was self-published by Hope in a print run of unknown number.

Palaeontology continued to interest her; during this period professional fieldwork and research in Alberta had resumed and was steadily growing. Hope was recognizing the need for a resource to identify fossil bones for the amateur-level collector and in one of her illustrations (Figure 84) from the booklet we can see the beginnings of the level of anatomical detail she was trying to convey. Perhaps there is too much information for some, but it was perfect for advanced amateurs, artists, budding museum technicians and vertebrate palaeontologists.

The recently hired David A.E. Spalding (1937–) of the PMA (Anonymous, 1968b) visited DPP on May 18 of this year and met with his co-worker Don Taylor, Hope and some of her fellow collectors. Taylor demonstrated how to remove a [presumed dinosaur] tibia from its plaster field jacket. The next

day Spalding visited the museum in Medicine Hat and saw her *Gorgosaurus* find (see below). His field notes record that Hope had made a dinosaur display case at the museum and that her small fossil guidebook (Johnson, 1968a) was available for sale there (D. Spalding, 1968, and pers. comm., 2015).

Another major discovery that Hope made was her partial tyrannosaurid skeleton located the year before in White Rock Coulee, north of Medicine Hat (Anonymous, 1968c).[23] Though incomplete it was the first, and for many years the only, associated tyrannosaurid skeleton in PMA/RTMP collections (Currie and Koppelhus, 2015). Hope revisited the site on February 10, prior to excavation and found

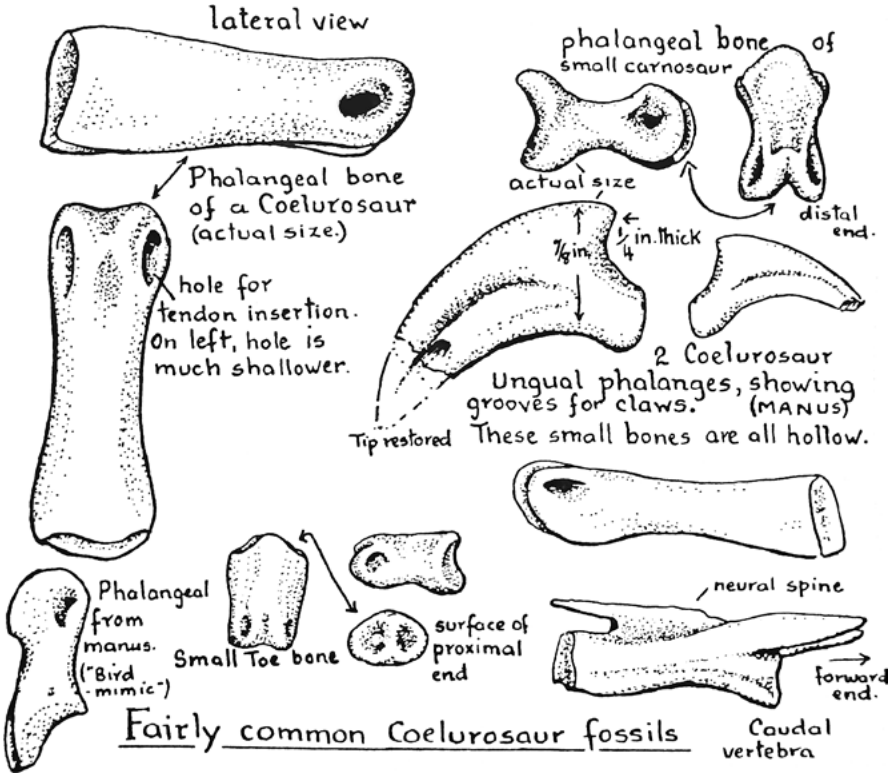


Figure 84. One of the illustrations from Johnson (1968a). While identified as small theropod phalanges, they appear to be ornithomimid, due to their relative size and fewer degrees of arc on the articular ends.

enough pieces of bone to reconstruct most of a fibula. The skeleton (TMP 1968.003.0001) was excavated in late May to early June, 1968; a quarry photograph appears in Spalding (1999, p. 128). Comprising trunk vertebrae, pelvic material, two pathologically fused caudal vertebrae and incomplete hind legs, it has played a supporting role in several tyrannosaurid studies (Britt, 1993; Mackovicky, 1995; Tanke and Currie, 2000; Currie, 2003; Erickson *et al.*, 2006). There is no written record of Hope assisting on the excavation of this specimen; but a family photograph (Figure 85) places her there and the following comments indicate that she did indeed help. She witnessed excavation activities and in an interview conducted thirteen years later (J. Danis, 1981), commented on what she felt were improper field collecting techniques:

> I was quite appalled at the state in which it arrived [presumed to mean arrived at the museum] . . . I remember being there and helping the man who was doing the collecting lift the one piece of the sandstone off of the other in which these bones were contained and I just gasped—I had never seen the inner surface of an ilium, etc.,

> when it had never been tampered with, destroyed, eroded, whatever—and the muscle scars on it were just perfect, you know, and I thought, oh, it was going to be difficult to prepare or pack or anything else. And I'm sorry to say that when I saw it when it was finally prepared this [muscle scars?] had just disappeared. It was a victim of probably indifferent handling and also great difficulties in preparation, I'm sure.

At the PMA the entire specimen was soon prepared, the pelvis and femur mounted and sent on long-term loan to the Drumheller and District Fossil Museum. There the two specimens, along with pictures of the 1968 dig, were put into a glass-fronted display case (Figure 86). They were on public display for several decades before being returned to the RTMP. In Hope's fossil catalogue notebooks for this summer she indicates that Dale Russell was openly and freely sharing his ideas on tyrannosaurid taxonomy.

On May 19 Hope led a trip to White Rock Coulee with members of the public. Jane Colwell (1968b) was not impressed: there were more than thirty people in a convoy of seven cars, some drivers getting lost *en route*, Hope losing sight of her car trail on the prairie, loose children and dogs, an abundance of rattlesnakes, people pitching tents near a rattlesnake pit, and other horrors. Hope seemed unperturbed.

On May 20 (recorded as May 18 in Colwell, 1968b) Hope led a cryptically-named "Physiology Museum Expedition" back to the *Styracosaurus* Bonebed 42 area in DPP. This was no doubt a group that included Charles Heath, a fellow collector and member of the Department of Physiology at the UA. Jane Colwell commented on this (Colwell, 1968b); she did not partake in the prospecting, but did visit Hope's group later that day to discuss the UA's 1967 helicopter lift at Dry Island Buffalo Jump Provincial Park (See Tanke and Walker, 2011) and to talk about the University's ongoing field plans.

In late May, Hope joined Gord Edmund and the ROM crew for four days as they worked at the "Hunter Quarry" in the Oligocene Cypress Hills Formation on Calf Creek, in southwestern Saskatchewan (Churcher, 1968; Edmund, 1968). Hope wrote a short paper on her experiences there (Johnson, 1969c; also noted in Anonymous, 1969f), but this was one of the rare instances in which she joined a professional crew for more than a day.

In an interview (J. Danis, 1981) Hope comments on her involvement with the museum in Medicine Hat and the PMA regarding a taxonomically diverse collection of Cenozoic mammals ("pirated out of the Calf Creek locality" according to Hope and sold to Eston, Saskatchewan station agent Frederick James, of whom the author could find little information). Hope performed this work during her tenure on the board of the Medicine Hat Museum.

Figure 85. Hope in the White Rock Coulee quarry that yielded *Gorgosaurus* TMP 1968.003.0001. The man in the hat is the PMA's Donald "Don" A. Taylor. The other man is unidentified. The image is undated, but a Spalding (1968) fieldnotes entry for May 20 records: "Take off small knob [of rock]" and that appears to have been freshly done here. Photograph courtesy of the Johnson family.

In 1968, then-UA student researcher Peter Dodson (1946–) began several field seasons of research by relocating major dinosaur quarries in DPP for his landmark paper on dinosaur quarry site sedimentology and taphonomy (Dodson, 1971). Hope had done so much fieldwork in the region by this point that a road in the area appears to have been named for her, at least informally. On July 31, Dodson (1968) notes: "Took Hope Johnson's road from Princess to south rim. . ." It is not clear from Dodson's field notes that he engaged

her to help relocate sites. Certainly in her explorations she would have seen many of the old quarries marked by a numbered metal "quarry stake" set in concrete, these placed mostly by C.M. Sternberg and his son Ray in 1936 (Sternberg, 1950; Tanke, 1994; J. Danis, 1988). In J. Danis (1981) Hope notes that she and Dodson had frequent contact and that he would visit her in her trailer, to "sit and talk away about all sorts of things" and that Dodson was a "mine of information."

In a letter, Dale Russell of the NMC in Ottawa told Hope that he and a crew would arrive in Edmonton on June 27 and visit with Richard Fox before conducting fieldwork in the Peace River district. A week's work was also planned at DPP in the latter part of July (D.A. Russell, 1968b). However, fieldwork up north was a bust, so they proceeded to DPP earlier than planned (J. Danis *et al.*, 1981). Hope joined Russell, Jane Colwell and Dr. Wilson Stewart (University of Alberta; see Page 93) in DPP in July (Colwell, 1968d), assisting Russell's crew in their quest for small theropod specimens (Johnson, 1969d). They had hoped to find a small theropod skeleton (Cooper, 1968b), but were having no luck. Some ten years earlier, Hope had catalogued some fossil discoveries made by fellow local collector Irene Vanderloh (1917–2009; Tanke, 2010a) and saw a box of pieces, including a foot, from an associated skeleton of just such an animal. She sent the grateful Ottawa field crew and the UA's Jane Colwell to meet Irene, who led them to the site. This action salvaged what appeared to be a summer of otherwise poor field results (Anonymous, 1969a–d). Russell was "ecstatic" with Irene's find and more material was found there, including brain case and pelvic material (Colwell, 1968f; J. Danis, 1981). He was quick to describe the find (D.A. Russell, 1969) and sent Hope a cast of the skull roof of this important specimen.

Figure 86. Prepared pelvis and femur of *Gorgosaurus* TMP 1968.003.0001, seen on display at the Drumheller and District Fossil Museum in the summer of 1972. Photograph courtesy of the Royal Tyrrell Museum.

Hope was such a larger-than-life character in camp that when she arrived the NMC crew would announce a "Code: Hopeless." She was in on the joke. During this time she was also called "Hopeless Johnson" by one palaeontologist who is best left unnamed, but that person came to admire and respect her and the two worked together in the field whenever they could. She visited the NMC crew in August, bringing a berry and apple pie (Colwell, 1968e). Another day she brought them fresh mushrooms she had found on the way to their camp. She cooked the mushrooms and they were eaten with some trepidation by the crew—not knowing her very well, nor in particular her expertise on mushrooms—there were some unfounded worries that they might be poisonous (G. Danis, pers. comm., 2015). The mushrooms had, however, been collected from the edge of a busy highway: the accumulated car exhaust and road grit didn't improve their palatability (Colwell, 1968f; J. Danis *et al.*, 1981).

Hope went to DPP in September and received numerous fragmentary but associated ornithomimid pedal phalanges collected by Dale Russell and crew that summer. She borrowed the lot and reassembled them at home. Parts of two feet were restored: nine phalanges from the right pes and eight from the left. In assembling the shattered feet, Hope commented (Johnson, 1968d) "it was a nice Chinese puzzle, but I was amazed how much there was." Later she commented (Johnson, 1968e) that she had "sort of" mounted the right pes and turned both over to the Park Warden, Bob Young, in early Novem-

ber. The current status of this specimen is uncertain. An articulated ornithomimid right pes from one individual and an articulated digit IV from another are recorded in Walper (1980, p. 13); perhaps the pes is the one Hope mounted. If so, it is TMP 1979.014.0701.[24] Russell later confirmed that the ornithomimid feet were collected by him and two assistants over a two to three day period near the Steveville railway grade. The rest of the specimen was buried under burlap and dirt, as the remaining bones were badly shattered (D.A. Russell, 1968c). The author is unaware of any subsequent field crews relocating or collecting this specimen.

In her fossil catalogue notes (Johnson, undated 2), Hope writes that she was scouting sites for Wilson Stewart and his students to find fossil wood. Stewart was a palaeobotanist and professor at the UA from 1966 until his retirement in 1978 and head of the Botany Department, 1966–1971. In early June she guided them to various sites for fossil wood samples, including Redcliff, White Rock Coulee and Petrified Coulee. Though she is not specifically credited for the field assistance, wood from Petrified Coulee was later described by Ramanujam (1972). Another wood specimen was described by Ramanujam and Stewart (1969). These samples were later mentioned in Koppelhus (2005). Hope also collected silicified wood samples for Stewart from the Hunter Quarry in Saskatchewan.

Sometime during the summer she visited an old coal mine near Burdett, Alberta. What—if anything—was found there is not recorded.

Stalker (GSC) and Churcher (ROM) returned to the Medicine Hat area this summer, continuing their work on Ice Age vertebrate bonebeds (Stalker, 1968; L.S. Russell, 1968, 1969). Finally, we learn something of Hope's involvement with their work. A grateful letter from Stalker to Hope in September 1968 thanks her for a technical drawing of a large cliff-face of tightly packed glacial gravels; his field notes record this drawing being done on August 15 at Mitchell Bluff (site figured in Churcher and Wilson, 1990, p. 68). The cliff face could not be photographed properly to show unusual stratigraphy so she quickly made a black and white technical drawing of the exposure and its stratigraphy for him. This event is also noted in Bush (1969). The crew had recovered evidence of early horses, camels, dwarf caribou, saber tooth cats, dire wolves, ground sloths and assorted small mammal material from various localities. Stalker attended several Medicine Hat Museum Board meetings regarding local palaeontology and his efforts to start a dig at a place known as Reservoir Gully (= Saamis Gully) in the city of Medicine Hat.

A mild winter allowed for some outdoor excursions. Hope visited White Rock Coulee on February 17, noting (Johnson, 1968b) a balmy daytime temperature of 54°F (12°C). The gravel pits around Ralston were still an attraction for their fossils, even in the dead of winter; she visited one on February 25. She also visited the Irvine badlands in mid-March.

By 1968 she was using Richard Estes' 1964 instant classic paper on Late Cretaceous microvertebrate fossils to help identify her own finds (Johnson, undated 2). Another fieldwork project Hope conducted this year was the recovery of a large bison rubbing stone from the "Little Holland" area of the Suffield Military Base. The stone, polished smooth by many centuries of itchy bison scratching against it was taken to the Medicine Hat Museum for an outdoor display. Part of this effort also involved erecting fencing around the run-down cemetery of the former community (Anonymous, 1968h).

Among other activities, she conducted a science program with the 4-H summer camp over an eight day period at Elkwater, in the Cypress Hills (Bush, 1969) and wrote a brief tourism piece on DPP (Johnson, 1968c). In November she travelled to Edmonton and visited with palaeontologist Richard Fox and palaeobotanist Wilson Stewart (Johnson, 1968e). Hope is acknowledged (Carpenter, 2005, p. 36) for speaking in 1968 to members of the Lethbridge chapter of the Alberta Archaeological Society.

Regarding her artwork, it is recorded (Sheridan, 1968) that with the Ralston Art Club she was ex-

perimenting with clay this year, with the idea that working in three dimensions would help with her painting. A photograph accompanying the Sheridan article shows how she first made drawings of a muskox and then completed the rendering in clay. The sculpture, though a bit damaged, still exists (Figure 87). The art club went on to have a section devoted to clay and the throwing of clay pots. It is not known if Hope did any other clay sculptures; this may have been a short-lived trial for her. The art club had a public showing in March at the Officers' Club at the Suffield base (Anonymous, 1968a, 1968g). More than 200 people attended the show and Hope's painting, *Forest—Cypress Hills* was named the second most popular painting on exhibit (Anonymous, 1968g). This year the art club received instructional support from German-born poet, artist and author Ursula Zandmer (1917–2004) and landscape painter Jack B. Taylor (1917–1970; see Townshend, 2005, p. 118–119). Hope was giving free art lessons to new members of the Ralston Art Club this year (Fooks, 1968).

Figure 87. Hope's 1968 abstracted rendition of a muskox in kiln-fired clay. The tips of both horns have since been broken off and lost. The sculpture is 17 cm long. Collection of Margaret Noone, Medicine Hat.

1969

This was another busy year for Hope. Early in January she was writing to UC archaeologist Dr. Richard G. Forbis (1924–1999) about the upcoming summer digs in Medicine Hat by Stalker and Churcher. She writes:

> A working group associated with the Med.[icine] Hat Historical Museum plans to operate next summer in a glacial deposit near the city, under the direction of Dr. A. Stalker and Dr. C.S. Churcher. As a member of the museum board interested in this facet of the museum's work, it is my duty to find interested persons + try to help them to interest themselves in the sort of material that may turn up in glacial deposits etc.

She goes on to ask about *Bison* axis vertebrae and how she had noticed that some have a single foramen in the lateral wall of the neural arch and others have two. This and some other anatomical differences piqued her curiosity and she sent Forbis detailed sketches (Figure 88), asking if these differences might indicate two different *Bison* species.

Early in 1969 Dale Russell sent her a plan drawing of *Saurornithoides*, a Mongolian Late Cretaceous maniraptoran, a genus (or something closely related) he thought once lived in Alberta. Hope (Johnson, 1969a) thanked him for the drawing and added that it "appeared to be one OGLY beast."

By this time, Hope was regularly providing fossil specimens to professional palaeontologists for their research; for instance, she was acknowledged by Ramanujam (1970, p. 855) in his paper. In about 1969 she was again helping UA palaeobotanist Wilson Stewart look for plant fossils in DPP.

She was also mentoring young people. Among her papers, archived at the Esplanade Arts and Heritage Centre in Medicine Hat, are some letters from then fourteen year-old Brian Keating who dreamed of building a natural history museum some day. He was born and raised in Medicine Hat until age six

when his parents moved to New York. Some eight years later his parents revisited Medicine Hat, met Hope, and told her about their son's interest in all things natural history. They asked if she could spare a few small fossils for him, but Hope did better than that—she put together a box of fossils for them to take back and later sent several other small shipments of fossils for his "museum" and a school display.

The museum Keating envisioned never materialized, but a natural history career certainly did. He went on to become the Calgary Zoological Society's Curator of Education (1981–1996) of which about 30% was vertebrate palaeontology educational programming tied in to the Zoo's new Prehistoric Park. In a 1981 letter to Hope he acknowledged her help in learning about vertebrate palaeontology, which had helped him gain his position. Around the time of his letter (*c.* 1982), either Keating was asked to give a talk to the Knights of the Roundtable Club in Calgary, or Hope was asked by them and she

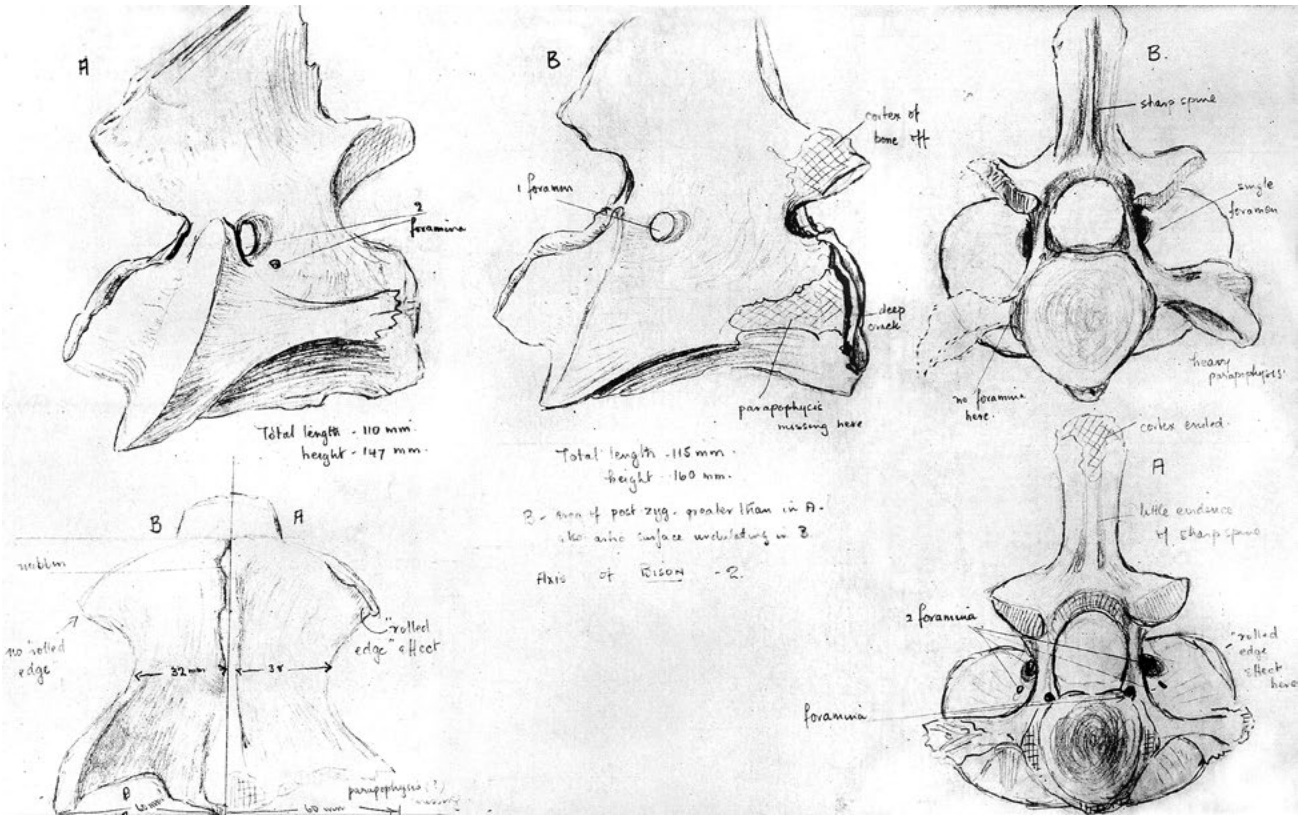


Figure 88. Sketches of Buffalo (*Bison bison*) axes by Hope with her comments on blood vessel openings. Image courtesy of Dr. Michael C. Wilson, Coquitlam, British Columbia.

recommended Brian. She introduced him as the luncheon speaker, suggesting that she was somehow affiliated with the Knights organization, too. Keating later went on to become Head of Conservation Research at the Zoo. He recently retired to focus on his professional speaking, tour guiding and media production company, "Going Wild." A long-time guest on CBC Radio's *Homestretch* program (Calgary), Keating has produced and hosted television programs and regularly appears on the Discovery Channel. He is also a public speaker and the author of a series of children's books. In addition to his many accomplishments, Keating was the *Reader's Digest* "2006 Canadian Hero of the Year." He received an honorary doctorate in 2011 from the UC (Keating, 2014).

Hope penned an unillustrated report on plants growing in DPP (Johnson, 1969e). This work is sometimes cited as an article on plants and lichens in the Park, but it was bound together with a short article

on lichens written by another person. Somehow the entire coil-bound product has become credited to Hope alone. Her contribution is a simple listing of plants and short natural histories of each. Hope could not resist adding some of her dryly humorous remarks to what otherwise would have been a rather dull technical document. For example:

> **Prairie onion**, *Allium textile:* The Indians used this as a food, and small children delight to eat it to annoy their parents.
>
> **Silverberry or wolf willow**, *Eleagnus commutata:* The dry berry has a very silvery skin, and is said to be edible, but may you never be so hungry.
>
> **Prickly pear cactus**, *Opuntia polyacantha:* In an emergency the stem lobes of the prickly pear plants can be eaten, by somehow (!) paring off the spiny skin. The insides taste rather like a tired warm cucumber.
>
> **Broomweed**, *Gutierrezia diversifolia:* Note the imposing Latin name. . . ah, well.
>
> **Colorado rubber weed**, *Hymenoyxs richardsonii*: The plant has a pungent odour and is said to be poisonous to stock, but one has the feeling that by the time one got around to eating this, one would no longer care WHAT happened, so. . .
>
> **Rabbit brush**, *Chrysothamnus nauseosus:* It is really rather too bad of the botanists to call it "nauseosus"= nauseous, the poor wretch is just doing the best it can in the appalling circumstances.
>
> **Common cattail**, *Typha latifolia* [in reference to collecting and preserving the fluffy seed stalk and perhaps spoken from bad experience] . . . do so early, or immerse the whole thing in a dilute mixture of shellac and alcohol, otherwise your cattail will come apart and make an annoying MESS! (Half an hour's work with the vacuum cleaner).
>
> **Russian thistle, also tumbleweed**, *Salsoia pestifer:* Revolting rather, but apparently romantic. Certainly a very mixed blessing.

In the spring of 1969 Hope got a paying job as the fossil custodian ("palaeontologist" in Hope's own words) at DPP.[25] She started working there in late May (J. Danis, 1981). Increased grants from the Provincial Parks section of the Department of Lands and Forests over the preceding three years had allowed all sorts of upgrades in Alberta parks, province-wide (Anonymous, 1969g). This was the first year the Park was "truly in operation" (Moser, 1974, p. 41), and saw some 8,000 visitors during the May 1 to September 1, 1969 tourist season (Moser, 1974, p. 54). According to personal notes penned to her family, Ottawa-based vertebrate palaeontologist Dale A. Russell recommended her to the Alberta Parks Department to be hired as a DPP employee, though C.M. Sternberg and former Park Warden Roy Fowler may have played parts. In a taped interview (J. Danis, 1981) Hope speaks of how she was "absolutely astounded" when DPP Warden Robert Young and southern Alberta Provincial Parks administrator John Erickson (Figure 89), from Lethbridge, showed up unannounced at her Ralston home one day and offered her a paying job as Park Palaeontologist. She was "really, really pleased" to do it, as whenever she had visited the Park she had been "rather distressed by the Park's exhibits . . ."; they struck her as quite fanciful and irrelevant on the whole—and quite messy. The pay "was quite low; it was $3.89 an hour or something like that, but it was something I was really interested in doing."

Figure 89. John Uno Erickson (1909–2007), head of southern Alberta Provincial Parks from 1965–1974. His hiring of Hope led to vastly improved palaeontology exhibits and bus tours in DPP, solidified her position as a serious vertebrate palaeontology amateur, the Johnson and Storer (1974) fossil guide and other achievements. Image from Swihart (2003).

She lived in a trailer and sometimes in the Park headquarters bunkhouse building[26] which included bunks for visiting researchers and occasional Park staff. The trailer was small but had all the amenities to get by, including a fridge and a tiny stove to heat soup. Hope would overnight there. She was given a "uniform" of sorts: a Parks shirt with requisite badges identifying her as an employee. She would wear this with jeans, the legs tucked into her high-topped, leather hobnail boots. One former Park staffer still recalls the sound of her boots (see Figure 90) clunking on the bunkhouse floor. A traditional choice, perhaps, as they were the same style of footwear favoured by early dinosaur collectors in the province. When she left the bunkhouse, other tenants would see that her hobnails had scuffed the floor, but she did not seem to notice.

Hope worked at the Park five days a week (Friday to Tuesday; Wednesdays and Thursdays off), from mid-June until mid-September. Working weekends maximized her contact with the public as an educator and let her perform other duties at the peak periods of visitation. For her job, as she described it in personal notes to her family, she ". . . re-arranged [fossil] displays, hunted fossils, made diagrams, labels, talk[ed] to visitors, etc." She usually did not collect articulated material but specimens up to the size of adult hadrosaur limb bones were collected on occasion (Figure 91). Her collecting work was occasionally thwarted by "so many tourists and so few [Park] vehicles to use." (Johnson, 1969d). The work included cleaning fossils and adding more specimens to existing exhibits. The revised "outdoor museum" exhibits were now "quite intelligible and informative" (Anonymous, 1969g). When it rained she prepared fossils that she and visiting tourists had found, cataloguing them in her instantly recognizable style of a prominent (up to penny-sized) circle of hand-applied white paint followed with a black India ink specimen number—or she penned the number directly on the specimen in black ink, sometimes with a cursive "HJ" added. Specimens in her personal collection were marked in a variety of ways (Figure 93).

Figure 90. A 1976 field photograph shows Hope with good friend and fellow fossil collector Shirley Lawson. Hope's old-style hobnailed boots are plainly visible. Image courtesy of Bill Lawson, Redcliff.

Getting to and from the Park in those days was often difficult after heavy rains, as the roads were unpaved, muddy and deeply rutted. Southern Alberta provincial park roads and some regional roads were noteworthy for their poor condition during these years (*e.g.* Anonymous, 1963d, 1968f, 1975c; see also comments in Moser, 1974, p. 56, 58). Hope liked to recall her adventures negotiating those quagmires.

Early vehicular tours for DPP visitors were casual—if not unsafe—affairs with the use of a Park pick-up truck, three people in the cab and up to ten in the box, the driver being the Park Warden (Tranter, 1992, p. 68). The Park Warden typically conducted these tours alone (Anonymous, 1969e) which distracted him from other pressing duties and palaeontology was not his strong point, so Hope assisted.

In her notes, Hope states that she also acted as a tour guide, travelling with the Park Warden and tourists in a "vehicle." More research disclosed that this was no doubt the Park's new and better-suited

Figure 91. 1969 photograph of a nearly complete adult hadrosaur tibia in DPP that Hope uncovered. One of her favourite digging tools was the bone-handled hunting knife (Bush, 1969). Image courtesy of the Johnson family.

Figure 92. Adult *Bison* skull eroding out of an infilled sinkhole in DPP; seen here in the early summer of 1976. Once a popular DPP bus tour stop, the skull has long since eroded away. Image courtesy of the Johnson family.

twelve-passenger Ford Club Wagon van, purchased in 1968 (Anonymous, 1968e; Cooper, 1968a; G. Forsen, pers. comm., 2014) and put into service June 28 or 29 (Anonymous, 1968d). With its acquisition Dinosaur Park became the first provincial park in Alberta with guided public tours (Figure 94). These loudspeaker-equipped tours lasted about one hour and travelled 4 km (2.5 miles) to the gated road and into the natural preserve area (Anonymous, 1968e; Cooper, 1968a). Tours were unscheduled but provided as demand dictated (G. Forsen, pers. comm., 2014). A small fee was charged: 50 cents for adults, 25 cents for children (Cooper, 1968a). As a guide on this van, Hope essentially became one of DPP's first naturalists and interpreters (J. Danis, 1981). A popular stop was a complete bison skull and other bones eroded out of a filled-in sinkhole deposit (Figure 92).

Park interpretive programming was not limited to vehicular tours. There were also self-guided hiking trails in the badlands, but there was a problem: a frustrated Hope wrote to Dale Russell (Johnson, 1969d) asking him to intervene on some poorly researched and written nature trail brochures published for hikers in DPP. The print run of the brochures had just run out and Hope was trying to stop plans for a simple reprint of them, pleading with Russell to write someone high up in Alberta Parks,

Figure 93. Fossil bones from Hope's personal collection showing the various ways she numbered them for cataloguing (left); even late in life and with a shaky hand she still did her best (right). These specimens are now part of the RTMP education collection. Photographs by the author.

point out all the errors and have them fixed. She was not only upset with numerous factual errors of natural history, but she was a stickler for grammar, too, pointing out for example ". . . singular subject with a plural verb, lack of parallel construction in complex sentences; omission of verbs, etc. His spelling of alluvial as 'eluvial' (twice—help!) is very sad."

It's possible that Hope's pleas had some effect and that she was tasked with rewriting this trail guide. The PMA's John E. Storer writes, in his field notes for June 22, 1971 (Storer, 1971) about the Deadlodge Canyon Nature Trail hike: "There is an excellent 8-page booklet available at the beginning [of the hike], and the job done is, in general, excellent!"

A 35 mm slide in her collection (Figure 95) shows a UA *Centrosaurus* skull quarry (UALVP 11735; Quarry 135) being worked in 1969—so she did on occasion visit professional work crews there during her paid work at DPP. In Johnson (1969d) and J. Danis (1981) Hope discusses visits to this quarry and how she helped UA technician Allan Lindoe chip out the skull and "observed his methods and that sort of thing," wanting to learn proper field technique. Regarding this skull, she also notes: "[Richard Fox] plans to return it to the Park (some pressure from the Prov. Museum but as yet they have no palaeontologist . . .)" (Johnson, 1969d).

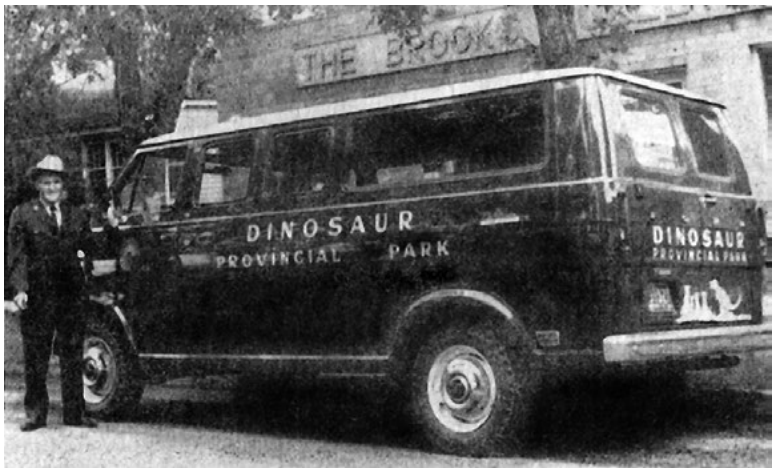


Figure 94. Mr. Robert R. "Bob" Young, Park Warden of DPP with the Park's new 1968 Ford tour bus in August, 1968 . How long this bus was in the Park is unknown, it was not there when the author arrived at the Park in 1979. Image from Cooper, 1968a.

Asked by Jane Danis (1981) for her thoughts on the fieldworkers at the PMA shortly after it opened (in 1967), Hope admitted disappointment in their field techniques:

> I have the feeling that Don Taylor and John Poikans, working with him, I'm under the impression they hadn't really any training. They were also very lacking in ordinary caution. For instance, if you wanted to loosen something up a bit, a six foot crowbar was used to knock on the rock to loosen things up a bit. This—for an amateur, who's been told that amateurs are the people who destroy fossils, etc.—was enough to stand my not inconsiderable hair on end! I was just shocked at this and couldn't believe my eyes; but what is a person to say, you know? Here are the people appointed by Her Majesty's government, provincial or not, to handle these things and this is the way it's being done. I'm sure the intentions were good, but the road to hell was being paved very thickly.

Poor field technique aside, these professional teams would have been an ideal resource for identification of problematic fossils and the latest news in the world of vertebrate palaeontology.

While at the Park, Hope went on many fossil collecting excursions with local (Steveville, Alberta) amateur fossil collector and good friend Irene Vanderloh. The two regularly hiked the Park together and collected a wide variety of Late Cretaceous vertebrate fossils in the rich badlands. Hope's most important finds during this time were a *Euoplocephalus tutus* armored dinosaur skull (TMP 1979.014.0747)[27]; severely pathological (bilaterally fractured proximally) and fused adult lambeosaurine hadrosaur ischia (TMP 1979.014.0381)[28]; and a then very rare pterosaur limb bone. In fact, this was the first recorded discovery of a pterosaur fossil in Canada (TMP 1972.001.0001; Quarry 139,[29] though it seems her 1965 pterosaur find had been overlooked; see Page 81). At the time it was so rare that it was quickly pre-

pared in Ottawa and described by top Canadian vertebrate palaeontologist Dale Russell (D.A. Russell, 1972d). The find is also noted in Danis (1973, p. 29) and Beaty (1979a, b). The specimen was further studied by Godfrey and Currie (2005) who assigned it to the giant pterosaur family Azhdarchidae, of which *Quetzalcoatlus northropi* is the best known example. Currie (1980a, p. 30) records Hope donating a rare frontal (which she nicknamed "Warty Bliggens"[30]) from a pachycephalosaur, suggestive of an undomed or flat-headed form[31], subsequently noted in some newspaper accounts as belonging to a "new type of dinosaur" and eventually named *Ornatotholus browni* (Galton and Sues, 1983).

While she was adept at finding many small and curious fossils, it seems odd that Hope never found many dinosaur skeletons. This may be related to her collecting regimen: a full review of her fossil catalogues indicates that she focused on mostly disarticulated material, probably in the Park's numerous, rich mono- and multispecific bonebeds. It does not appear that she collected material from the several dozen ceratopsian-dominated bonebeds distributed throughout the Park. In the field, Hope always kept her eyes open for quite small and especially unusual fossils, spending much time lying prone on the ground with her eyes only inches above the surface, looking for small "microvertebrate" fossils (Figure 77). The author recalls her regularly sending small boxes of mystery fossils by mail to Philip J. Currie, for us to puzzle over and identify. People frequently sent us mystery items in the mail or brought them to the museum, typically with a simple query: "What is this?" or "Is this a fossil?" But Hope's packages were always accompanied by her technical drawings and informed speculations on the fossil's anatomical identity and from what animal it may have originated. In many cases she was correct or close in her anatomical and taxonomic determinations. She continued to send Currie boxes of fossils for identification, even into her nineties. Many of these rare or unusual specimens found their way into the museum and university collections. Over the decades Hope also examined fossils other people had found and made sure they got to the correct authorities if they were deemed important enough.

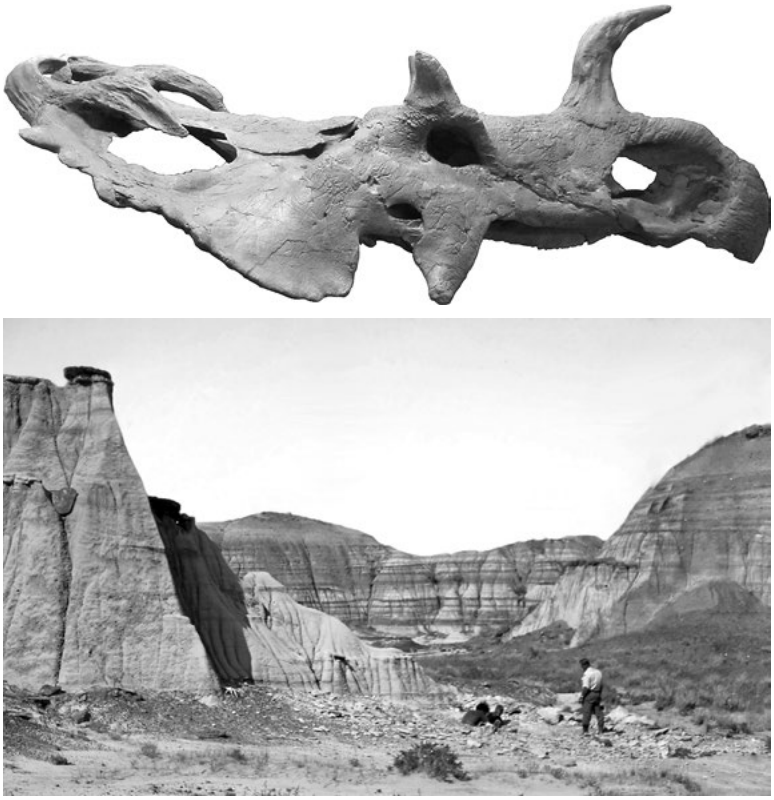


Figure 95. (Top) *Centrosaurus* skull (UALVP 11735) in right lateral view, on display at the University of Alberta, Edmonton. Photograph edited by the author, courtesy of Dr. Caleb Brown, RTMP. (Bottom) Quarry 135 being worked by the University of Alberta crew in 1969. Image courtesy of the Johnson family.

In her official role at DPP, Hope not only collected fossils, but helped in the improvement of the fossil exhibits. Certainly there were the four *in situ* dinosaur skeletons on display at this time,[32] but these each featured a single large skeleton and did not indicate the richness and diversity of the other, smaller extinct animals represented in the Park. Also, all four skeletons were kept well away from the main campground and were not immediately accessible: some only by a long walk, by car, or in one case by guided tour alone.

Hope had a small role in upgrading or supplementing these relatively new *in situ* exhibits. Certainly they would

need tidying-up and dusting from time to time. Some hadrosaur skin impressions in rock, originating from Display House #3 (old #4, TMP 1980.023.0004) has smeared brown plaster on the underside that resembles plaster work on other specimens Hope is known to have worked on. The four dinosaur exhibit houses came with their own signage and text panels, created by other artists. An image (Figure 99) shows a supplementary exhibit illustration panel leaning against the freshly stained or painted wood of a display house. The image was provided by Jane Colwell-Danis who worked in the Park from 1966 to 1968. The final of four display houses (old #4, later #3 after the demolition and removal of one of the skeletons) was new and opened during this period. Hope's illustration features *Corythosaurus* and the newest exhibit featured a *Corythosaurus* skeleton (TMP 1980.023.0004).

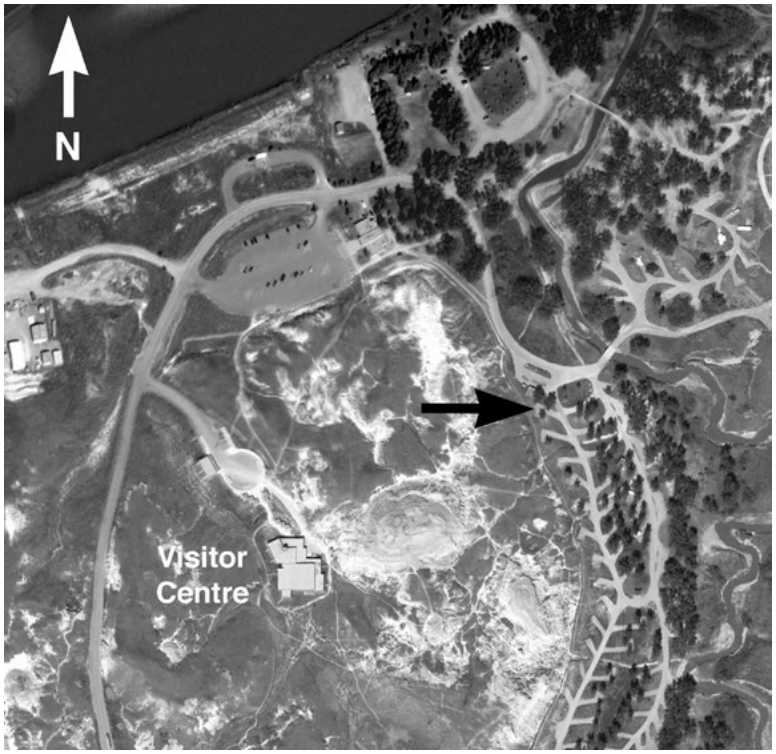


Figure 96. 2012 satellite image of DPP visitor centre and campground area, showing former location of fossil exhibit buildings (tip of black arrow). From Google Earth, reproduced under terms of use. Image © 2017 DigitalGlobe.

Supplementing these exhibits, in the main campground entrance area, was a series of three unheated, long, rectangular buildings with glass fronts that served as exhibit buildings for the tourists to see representative fossils. These exhibit buildings are discussed in more detail here, as Hope's involvement with them likely led to the Johnson and Storer (1974) fossil book and elevated her status as a serious amateur fossil hunter. A map in Maier *et al.* (1981, p. 10) showing the early 1980s configuration of Park infrastructure (Figure 96) shows a single small building labelled "displays" but there were in fact three buildings. A free pamphlet briefly describing highlights of the Park, including the contents of the exhibit buildings, was made available to visitors (Brooks Chamber of Commerce, 1965).

First Nations artifacts were also included in a small section of these exhibits (Moser, 1974, p. 48). The exhibit buildings were known to Park and PMA/RTMP staff as "the chicken coops" which they outwardly resembled. Each building was about 7.3 m (24 ft) long, 3 m (10 ft) high and about 1.5 m (5 ft) deep, with four large viewing windows. A standard-sized door allowed access for exhibit cleaning or maintenance. The buildings were lit inside for the public's ease of viewing in the evenings but the lights attracted all sorts of insects and sometimes the animals that preyed on them. The buildings (Figure 97) were loosely arranged in a semi-circle; lengths of local petrified wood logs were embedded in the soil for decoration and educational purposes nearby (Fossil, 1966). Hope's notes from 1975 (Johnson, undated 2) record that these exhibits were called "museum buildings" and each numbered 1 to 3. Exactly when the buildings and their exhibits were first installed appears lost to history, but they were certainly present by the summer of 1960, when they appeared in a newspaper photo (Anonymous, 1960i).

Student Peter Dodson considered the original Park fossil exhibits to be "not that impressive" (Dodson, 1968). But the newly-hired Hope would soon fix that. In J. Danis (1981), Hope relates how she

Figure 97. The three unheated fossil exhibit buildings near the Park entrance that Hope improved with new fossil specimens, signage and paintings. Image modified from Alberta Recreation Parks and Wildlife (Parks Division), 1979, p. 90. Image was printed backwards in that document, corrected here.

". . . did a real housecleaning; there were a lot of things in there that were irrelevant or not nearly as interesting as they should have been, things such as funny-shaped pieces of rock, lanterns hung so you could see light shining through holes in rock—rather kitschy-looking. They were not instructive at all and I felt there was a lot to be told about the Park that I could manage to get across." They required a major upgrade, which Hope had largely accomplished by about 1970 (see Page 107).

Exhibits consisted of single bones or composites of fingers, toes, and feet made by combining isolated specimens from the same families and similar age-classes. Not only did Hope collect many of the fossils used therein, she also prepared them[33] and then created the interpretive signage necessary for the exhibits (Figures 98–102). Signs consisted not only of small and large identifying text panels, but also osteological and full-body reconstructions of the animals, many rendered in watercolour and acrylic paintings. The exhibits were supplemented with prints of dinosaur life reconstructions, reprints of archival photographs and some small casts (Anonymous, 1971a) supplied by Dale Russell of Ottawa. Richard Fox of the UA would later donate a fine *Stegoceras* skull cast. Some of her painted dinosaur images were obviously copied or adapted from existing scientific papers or popular publications by palaeontology artists of the time, such as William E. Scheele (1920–1998), Zdeněk Burian (1905–1981) and possibly Charles R. Knight (1874–1953; Knight, 1942), but some were her own renditions. Of this fossil exhibit work, Hope relates:

> While I had been collecting all around the Dinosaur Provincial Park, I now had the opportunity—*indeed the duty* [author's emphasis] to find more fossils in it. I also quickly refurbished the three outdoor displays which were in a rather untidy mess. The parts supervised by Dr. Sternberg were clear and impressive, but the local assistant he had to leave in charge was "out of his depth" when it came to labeling, and unclear as to what should be on view in such a location. To give him his due, he was proud of his position. [The assistant was Roy Fowler; see Fowler, 1967, p. 326].
>
> It rained, quite hard, for the first two weeks I was there. This was great for me, since I could use the time, during the day, to clear away a lot of broken material, do a great deal of signage, and make some "cut-outs" of sturgeon and garfish, and mount the small enameled scales and vertebrae, etc. of this important genus with descendants still

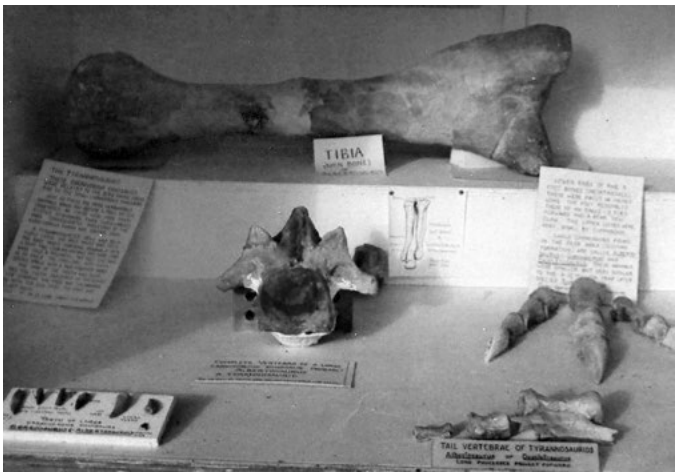


Figure 98. Some of the theropod dinosaur component of the DPP fossil exhibit buildings as seen in the summer of 1979. The largest bone is actually a hadrosaur tibia.[34] Photograph by the author.

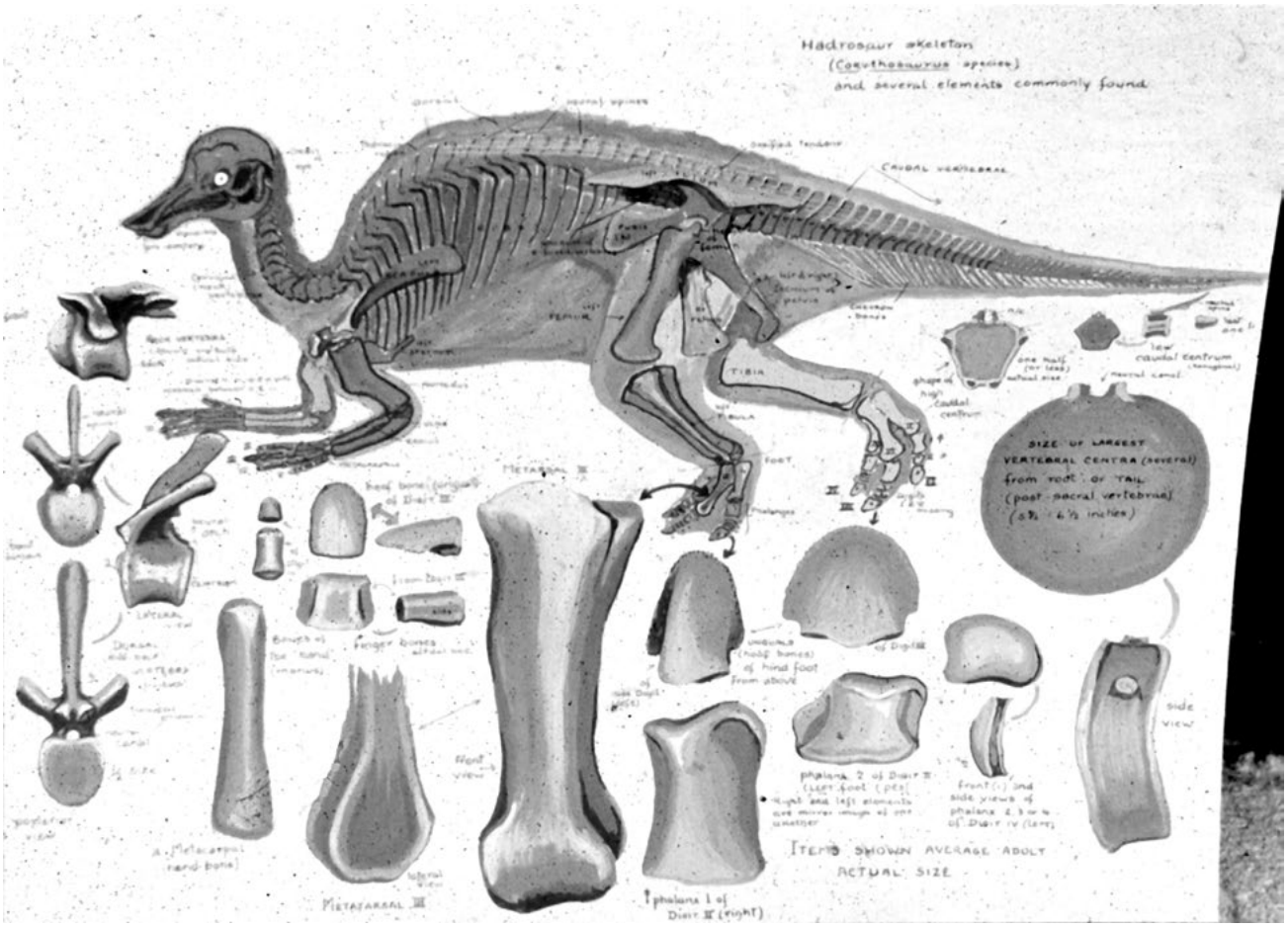


Figure 99. Detailed watercolour illustration panel of a *Corythosaurus* and hadrosaur skeletal elements rendered by Hope. The panel was probably used in DPP Display House #4, *c.* 1966. Image courtesy of Jane Colwell-Danis.

> inhabiting the rivers in this part of the World. It was reasonably cool during this period. Later, it became very hot down there in the Park (especially so in the fossil display buildings) and I can remember the sweat pouring off the end of my nose, and making damp blobs on the specimens, labels and drawings! Bats could be a problem too. They flew under the eaves at night after moths, etc. and while flapping their wings, displaced the labels. I had to fix the signs in places with white glue!

When the display cases were emptied in 1980 the fossils went to the Palaeontology Department of the PMA and Hope's exhibit graphics were no longer needed. PMA staff were shocked to learn from some DPP staff that all of her artwork and signage was slated for the garbage! The artwork (Figures 100–102, 163, 164, 168–171) was quickly saved and divided up by the PMA field staff, the pieces now treasured objects decorating the walls of some of their homes.

Some have been vocally critical of Hope's DPP dinosaur exhibit paintings as being amateurish, but they were new subject matter for her. Most early dinosaur reconstructions, even by the best artists of their day portrayed the animals as sluggish, dim-witted brutes. Hope excelled at technical pen-and-ink, bone and plant drawings and watercolours (the latter described as "delightful" by Cochran, 1978), and especially badlands landscapes. It is uncertain when she started drawing vertebrate fossils, but by 1963 she was producing publication-quality work (Figure 103). Her output of badlands paintings is also unknown, but must have been prodigious. Nearly everyone contacted during the research into this book had at least two of Hope's original paintings of prehistoric animals, badlands, or modern animals. She

usually signed her paintings with a simple "H. Johnson," but rarely with a date. The oldest signed and dated badlands painting is from 1960 (L. Johnson, pers. comm., 2010).

The Park exhibits were probably viewed by more than 100,000 visitors over the years. The author recalls that they were popular and always seemed to have one or two people viewing them whenever he passed by. The RTMP's Dr. Don Brinkman remembered how informative these exhibits were when he first saw them as a high school student (Brinkman, 2009).

Another duty she performed in the Park during this time was conducting tours for schoolchildren, which she loved doing. These were held in the old John Ware (*c.* 1845–1905; MacEwan, 1973) cabin. There Hope would talk about John Ware, First Nations peoples, plants, the history of the Park and fossils. Hope was also excited to help with the newly-developed, supervised walking tours for tourists in the restricted areas of the Park (G. Forsen, pers. comm., 2014).

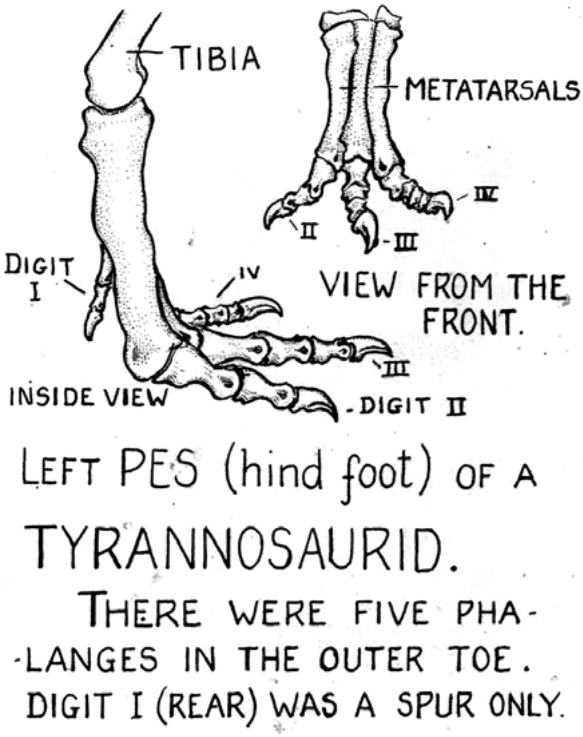


Figure 100. Tyrannosaur hind foot skeletal morphology signage used in a former DPP fossil exhibit building, *c.* 1969–1980. Collection of Clive Coy, Edmonton.

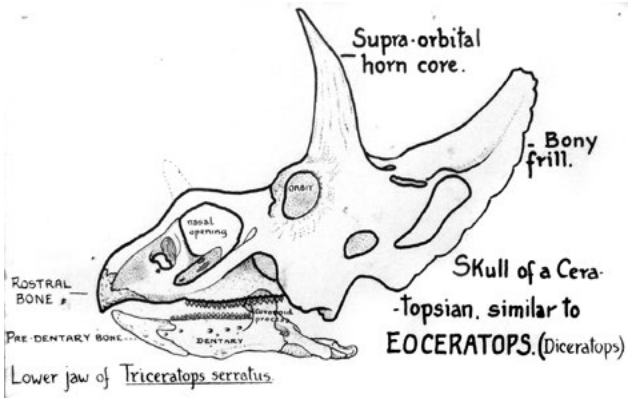


Figure 101. Ceratopsian skull illustration. Former DPP fossil exhibit building signage, *c.* 1969–1980. Author's collection.

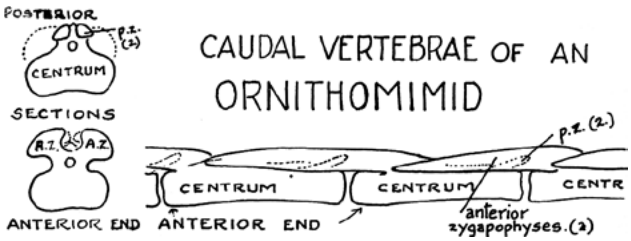


Figure 102. Ornithomimid distal caudal vertebrae illustration signage used in a former DPP fossil exhibit building, *c.* 1969–1980. The level of detail was likely too much for the average Park visitor, but for budding palaeontologists it was great stuff. Author's collection.

Hope was a great teacher to the Park Wardens. They were, of course, primarily administrators, with little knowledge about the Park's human and natural history. But Hope was able to educate them extensively on these matters: "she was very good at explaining things" (G. Forsen, pers. comm., 2014). To explain fieldwork, she would take them out into the badlands and show them how to find and identify important fossils, even how to properly uncover them and make a plaster field jacket. Some years later she marked out a badlands area that intrigued her and asked Park Warden Gerhard Forsen (1934–) to keep an eye out for fossils there. Sure enough, Hope's instinct paid off and in 1976 he found the largest-known and most complete trionychid (soft-shelled) turtle carapace (*Aspideretoides splendidus*; TMP 1976.007.0002) recovered from the Park, a cast of which later went on display in DPP. When Hope

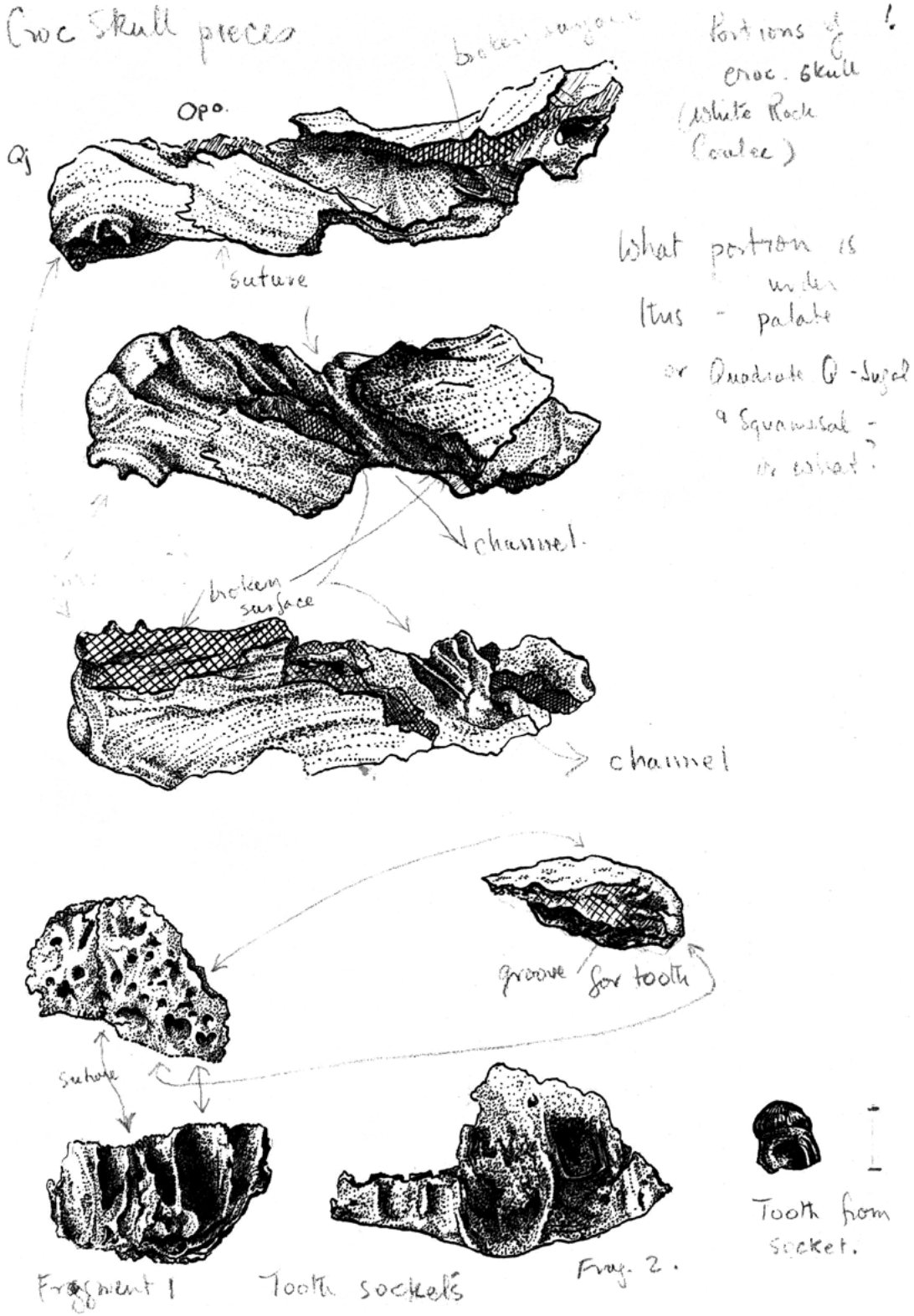


Figure 103. Hope's annotated technical pen-and-ink drawings of crocodile skull fragments, 1963 or 1964.

eventually left the Park to pursue other things, Park Warden Forsen was able to carry on with the bus tours, hikes and public fossil reports, Hope having taught him well (G. Forsen, pers. comm., 2014).

From early June to August of 1969, Drs. Archie Stalker and (much later that year) Rufus Churcher resumed their work in Pleistocene deposits of the Medicine Hat area. Stalker's field notes for May 30 record his discussions with Hope on various things, including digging at a place named Saamis Gully, formerly called Reservoir Gully. The next day Hope and four other volunteers assisted Stalker at the Saamis Gully site. Limited digging was done and about half a dozen fossil mammal specimens, mostly teeth, were found. They were fragmentary, but demonstrated that the locality was fossiliferous. A Medicine Hat Museum group comprising thirteen people returned to the site on July 5 and found more fossils and lots of chert fragments—possible artifacts.

At another site, over 6,000 m^3 of overburden was removed from one bonebed. Stalker's team found "several hundred" animal bones and apparently-worked chert stones, at the time believed to be evidence of early man in the province (Stalker, 1968), though this interpretation has since been challenged (Reeves, 1980; Wilson and Burns, 1999; M.C. Wilson, pers. comm., 2018). Another aspect of the research was to determine the veracity of the "Ice-Free Corridor" concept which postulated that early man and animals migrating from Asia over the Bering Land Bridge could pass freely along an unglaciated strip of land east of the Rocky Mountains in Alberta, through to the 49th parallel and points south (McIntosh, 1971).

A unique find this summer was a nearly complete Pleistocene frog skeleton (Nackoney, 1969), later re-identified as a toad (*Bufo* sp.). The Medicine Hat Museum staff helped on the dig this year, but it's not known if Hope took part, as she would have been tied up with her responsibilities in DPP. Nackoney (1969) identifies her as a Museum Board member with many years of palaeontological experience. Further, he states that she was the "organizer and chairman of the museum group" that helped at the dig site and goes on to say that the museum had a "palaeontology department."

Her work in DPP included some fieldwork interactions with Dale Russell and his crew. Her notes (Johnson, undated 2, p. 90) record that she showed him a string of ankylosaur caudal vertebrae but it seems that no action was taken. It appears that her job at DPP gave Hope her fill of fieldwork; only a few other sites were visited in 1969. In DPP, she found a skull dome of *Stegoceras*: "a very good one, about 140 mm long" (Johnson, 1970a). Just such a specimen was recorded in Walper (1980, p. 12), so it appears to have entered the RTMP collections. She visited the Irvine badlands in April and June. There she found a fossil mammal tooth that was sent to the UA (Johnson, undated 2).

A discovery among Hope's personal papers, found after her death, are copies of several university lecture notes, one penned by the UA's Richard Fox. Entitled "The origin of dinosaurs," it bears Fox's name and a November 28, 1969 date. Another set of notes, in the same format, but bearing no author's name is entitled "Dental formulae" and is a detailed review of placental mammal teeth, their formulae, and some discussion on morphology. Hope's handwritten notes appear on these copies. There have been suggestions that Hope took some university palaeontology courses, but the author of these notes, Dr. Fox, did not recall having her in his class. He suggested that he, or one of his students, gave copies of the notes to Hope on one of her visits to Edmonton (R. Fox, pers. comm., 2015).

The 1970s: Her career continues, development of the first fossil guidebook

1970

Watmough (1970) records that on February 9 the Nanton Rock and Mineral Society had a meeting in Claresholm, Alberta where Hope was a special guest:

> Special speaker at the meeting was Mrs. Hope Johnson, Ralston, Alberta, giving information on the fossils and dinosaurs of the badlands of Alberta, and showing some excellent slides of that area, which made a very interesting evening. During the lunch hour Mrs. Johnson examined and identified fossil specimens brought in by the members.

She also spoke—for the second time in three years—to the Lethbridge chapter of the Alberta Archaeological Society (Carpenter, 2005, p. 36).

Hope continued in her role as fossil custodian at Dinosaur Provincial Park (DPP) this summer. A welcome addition to the Park's fossil displays was a cast of the complete skull of the pachycephalosaurid *Stegoceras*. Richard Fox from the UA sent her the cast which she was to hand-paint and place on public display (Johnson, 1970c).

July 22, 1970 was likely the most important day in Hope's amateur palaeontology career. That morning, the Provincial Museum of Alberta (PMA)'s newly appointed Curator of Palaeontology, Dr. John E. Storer (1944–)[35] met Hope for the first time. Being very knowledgeable about DPP, she toured newbies Storer and his assistant, Michael C. Wilson (Figure 66), around the Park, taking them into the restricted area, to the dinosaur display houses and, coincidentally, meeting up with Richard Fox. She volunteered information on the whereabouts of microvertebrate fossil localities in the Irvine, Alberta area and commented on the northerly occurrences of rattlesnakes, noting their presence in the Steveville area (Storer and Wilson, 1970). Storer saw her technical drawings of Late Cretaceous vertebrate fossils and her arrangements of specimens in the three fossil exhibit buildings. He was impressed and recommended that her drawings should be published as a fossil identification guidebook, as nothing of this sort had been done before in Alberta. Regarding the meeting, Storer (1970) wrote:

> H.J. has been putting up displays in cases near the entrance to Dinosaur Park, emphasizing not only dinosaurs but crocs., turtles, *Scapherpeton*, fish, other small herps, and wood. It is a well rounded, informative display, and is an excellent one for its purpose.
>
> H.J. is an excellent artist, and has been putting together a set of drawings of representative elements from Upper Cretaceous faunas, *i.e.* dinosaur toes, turtle vertebrae, fish scales and spines, herp jaws and vertebrae, etc. Many

> of these are not illustrated in the literature. I spoke with her about publishing this information some day, and making it available to both professionals and amateurs in Alberta. This would, when she completes it, be an ideal Natural History publication for the museum."

Hope's notes to her family and comments in J. Danis (1981) both state that Ottawa's Dale Russell, having seen her drawings of fossils, may also have played a role in the book's earliest development. She and Storer worked together on the project for several years (Storer, 1973a-b), or in "close collusion" as recorded in Hope's personal notes. Storer wrote and edited the text and also arranged for publication, though earlier versions of the text indicated that Hope helped write some portions, too. She did additional fossil drawings as required, most at 1:1 scale and therefore useful for direct comparative purposes. Some figures needed changing or rearranging on plates to address editorial comments, some of which came from the UA's Richard Fox, who also provided photographic images of Late Cretaceous mammal teeth. Bison and horse teeth are often misidentified by amateurs as dinosaur teeth, so Hope and Storer included illustrations of those, too. Altogether, Hope rendered 687 tooth and bone drawings for the book. Twenty-one photographs of specimens curated at the UA and PMA were also included. All of this hard work resulted in the landmark softcover book, *A Guide to Alberta Vertebrate Fossils from the Age of Dinosaurs* (Johnson and Storer, 1974).[36] At a 1973 publication cost of $4,596.00, one thousand copies were printed and ready in time for the 1974 Christmas season (Anonymous, 1974i; see Figure 109 and Page 120).

The book sold quite well considering how few rockhounds and amateur and professional vertebrate palaeontologists there were in those days. It sold out and was reprinted; it remained in print for about thirteen years.

In April 1975 she relinquished all rights to her images in the book to the Alberta Government and agreed to receive no royalties from the sale of the books. There was no budget for author payment and the end result—the publication—was reward enough for all the work she had put into it, as it confirmed her reputation as a leading amateur fossil collector in the province. She did get some free copies of the book for herself and for a number of her amateur and professional colleagues across North America. Between Johnson and Storer, about forty copies were mailed out to fellow amateurs and professional vertebrate palaeontologists; forty to natural history museums and potential booksellers in the province; fourteen to amateur and scientific journals and seven to Alberta newspapers for review. The publication was reviewed by Anonymous (1974h), Churcher (1975), Kristensen (1975), A.P. Russell (1975), D.A. Russell (1975), Walker (1974) and possibly others. Each critiqued a number of relatively minor technical or anatomical errors but also agreed it was a valuable guidebook for all fossil collectors and the whole project was a definite "thumbs up" for Hope, further cementing her significance and reputation in both the amateur and professional Albertan palaeontological communities.

This excellent soft cover publication, originally sold for $5.70, became the fossil identification "bible" for several generations of Albertan amateur and professional palaeontologists alike. The author estimates he has helped train and supervise over 1700 PMA/Royal Tyrrell Museum of Palaeontology (RTMP) summer staff, field volunteers, and DPP staff from 1980 to present and the Johnson and Storer (1974) fossil guide is and was a standard training and reference manual for many of them. A photograph of a RTMP volunteer with a copy on her desk is seen in Turner (1984). Hope's books are even used in a number of national museums to help identify and curate the Alberta fossil material in their care. Used copies of Johnson and Storer (1974) are harder to come by now and fetch a good price in used book stores. Hope's personal notes to her family describe her seeing numbers of photocopies of the book in the hands of amateurs and these were cherished even when of poor quality. A late 2009 Internet search revealed one bookseller selling a good used copy for $90.00!

Hope continued to search for fossil mammals. In a typewritten document that appears to be an educational item made for Park staff (Johnson, 1970a), she writes about Late Cretaceous mammal remains in Alberta: "I have found two good long (6 cusped) molars of multituberculates, and a couple of jaw fragments. I have given these to Dick Fox because he is actively doing research on these beasts."

On August 19 she found a new and rich mammal locality near Quarry 119 at South Sandhill Coulee, now within Park boundaries. Worries about fossil poaching in the Park were raised this year and a small report on the situation was issued in 1971. According to Kondla and Crawford (1971; see 1971 section, Page 110), a large portion of the badlands was protected this summer and the public was banned from the area. They comment: "Last year, over a fear that too many of these precious fossils were being pilfered, a large part of the Park was designated restricted and closed to the public." This change no doubt reduced Hope's potential collecting area, when she was collecting as a civilian, but later she would have access to it all, having earned the trust of Parks staff and the scientific community.

Extra fieldwork that Hope conducted this year is not well known. She visited the badlands of the Jenner Ferry Crossing (now a bridge) and returned to the Park on September 22. Only a month of fieldwork in Saskatchewan and the Medicine Hat area was conducted by Stalker and Churcher in 1970. The weather was miserable, with strong winds and rain, but the crew managed to sample three Pleistocene mammal-dominated bonebeds (L.S. Russell, 1970; Stalker, 1970). Archie Stalker's note books record that Hope was in Ottawa in late November and visited him there.

As much as Hope was self-taught on many palaeontological matters, she still relied on the experts for their comments and apparently trusted them implicitly. She knew for many years that *Myledaphus* teeth were referable to rays, yet in Johnson (1970a) she states in a faunal listing with comments:

> *Myledaphus bipartitus*. A Chimaera, variously described as a "pavement-toothed shark" etc. etc., but probably nearer to the Ratfish. Only the teeth are found—that is, only these are certainly associated with the name. Chimaeras are said not to have any denticles, etc. Skeleton cartilaginous.

With these comments she states that *Myledaphus* is a chimaeroid fish, like the extant species of ratfish. Why the taxonomic change from rays? Romer's *Vertebrate Paleontology* (1966, p. 351), a book Hope owned, erroneously refers this genus to the Chimaeridae. This suggests that Hope was still dependent on experts for issues of identification and taxonomy and only later did she develop her own ideas. The genus is currently referred to the extant guitarfish family (Wilson *et al.*, 2013).

Late in 1970 Hope sent Dr. C.R. "Dick" Harington of the NMC photographs of an unusual, dark-coloured skull with "white things" found near Waterton, Alberta. It turned out to be that of an adult male domestic pig (*Sus scrofa*) that had anomalously acquired dark staining suggestive of fossilization. The "white things" were the tushes, or long, upturned tusks (Harington, 1970).

Finally, for palaeontological activities this year she traded a number of fossils to Leslie Adler (1926–), a school teacher in Fort McLeod, Alberta who would eventually use them for educational purposes. He would later become a founding and long-term Alberta Palaeontological Society member and serve as President.

Anonymous (1970b) notes that in 1970 Hope was still a member of the Medicine Hat Historical and Museum Foundation and the University Women's Club. Her activities and membership tenure with the latter are unknown. According to Hope's personal notes, 1970 was the last year of her involvement in the Boy Scouts movement. A search of the local paper (*The Ralston Co-Operator*) was made to see if she was acknowledged or thanked in any way for her fourteen years of volunteering for the Ralston Scouting movement, but nothing was found. In a letter to the editor of the *Co-Operator* (Johnson, 1970b), Hope complains about all the junk in town and the need to clean it up. She suggested that citizens and

children's groups like the Scouting movement could do this. Shortly thereafter, a cleanup did indeed take place (Anonymous, 1970d).

Concerning her art activities, Hope and two others attended the annual conference and seminars of the Alberta Community Art Clubs Association in Edmonton on April 4. At this event some of Hope's drawings were selected for exhibit at Corbett Hall and the University of Alberta (UA) campus (Fooks, 1970). Also this year she taught Composition I and Drawing I for the art club (Anonymous, 1970c).

1971

Comparatively little is known of Hope's activities this year. She did not work in DPP this summer but would the following year. The reason for her absence from the Park is unknown.

Hope continued to work with John E. Storer on their fossil identification book. She travelled to Edmonton in January to work with him on this, but approached the meeting with some trepidation, as the PMA's director had recently told senior staff that the museum was "not a research institute" (Johnson, 1971a). Storer visited her at home on June 30 (Storer, 1971). She saw him again in November (Johnson, 1972a), bringing more fossil drawings and discussing book plans. Storer had hoped for drawings of limb bones and turtle shell textures; but photographs would ultimately be used instead as her drawing technique would not be "very satisfactory."

Early in the year she received some very good news from Dale Russell at the NMC. By mail, she had been sending him unusual small bone and teeth finds for his analysis and potential study. One of these finds turned out to be quite significant. In the field it was broken into splinters but Hope had saved all she could find and sent them to Ottawa where they were reassembled. In a January 11 letter, Russell (1971a) writes: "Congratulations! You have apparently collected the first remains of a pterosaur ever to be recognized as such in Canada." Strangely, she seemed rather indifferent to this accolade, replying: "I am not really jumping up and down with glee, but will say that it is extremely fortunate that the scrap happened to have some diagnostic characteristics" (Johnson, 1971a). Palaeontologists John E. Storer (in the field) and Dale Russell (by mail; D.A. Russell, 1972c) worked with Hope to try to establish the precise spot where she found the specimen. It was actually her second pterosaur discovery but the first was not recognized for what it was until years later (see 1965 entry, Page 81).

Hope was concerned about fossil poaching for subsequent sale and rampant, excessive collecting by amateurs: something that had already been identified as a problem in the Drumheller area in the late 1930s (Anonymous, 1939c). These activities could pose a problem in a fossil-rich area like DPP. A report (Kondla and Crawford, 1971) was issued and suggested that at this point at least, such issues were minor to non-existent—even suggesting that professional palaeontologists were more of a menace! They do reveal, however, that a restricted area was established in DPP in 1970, a point briefly mentioned in Matern (1970).

Another field activity she was involved with was the discovery of a Late Cretaceous mosasaur skeleton. She thought it was the first such discovery in Alberta, but it was actually the second: the first had been collected by Wann Langston and crew at Scabby Butte in the 1950s. The fossil Hope was involved with had been found by a pair of amateur fossil collectors from Lethbridge, Gordon H. (1902–1972) and Elsie F. (1918–2004) Kitchener, whom Hope had met in DPP in 1970 (Johnson, 1971b). Elsie was a friend of Hope's and had family living near DPP (Anonymous, 2014f). The specimen consisted of an associated large dentary and vertebrae. It was found about 2.4 m above the high water level on the St. Mary River west of Welling, Alberta. Hope sent sketches to Dale Russell (Johnson, 1971c) who iden-

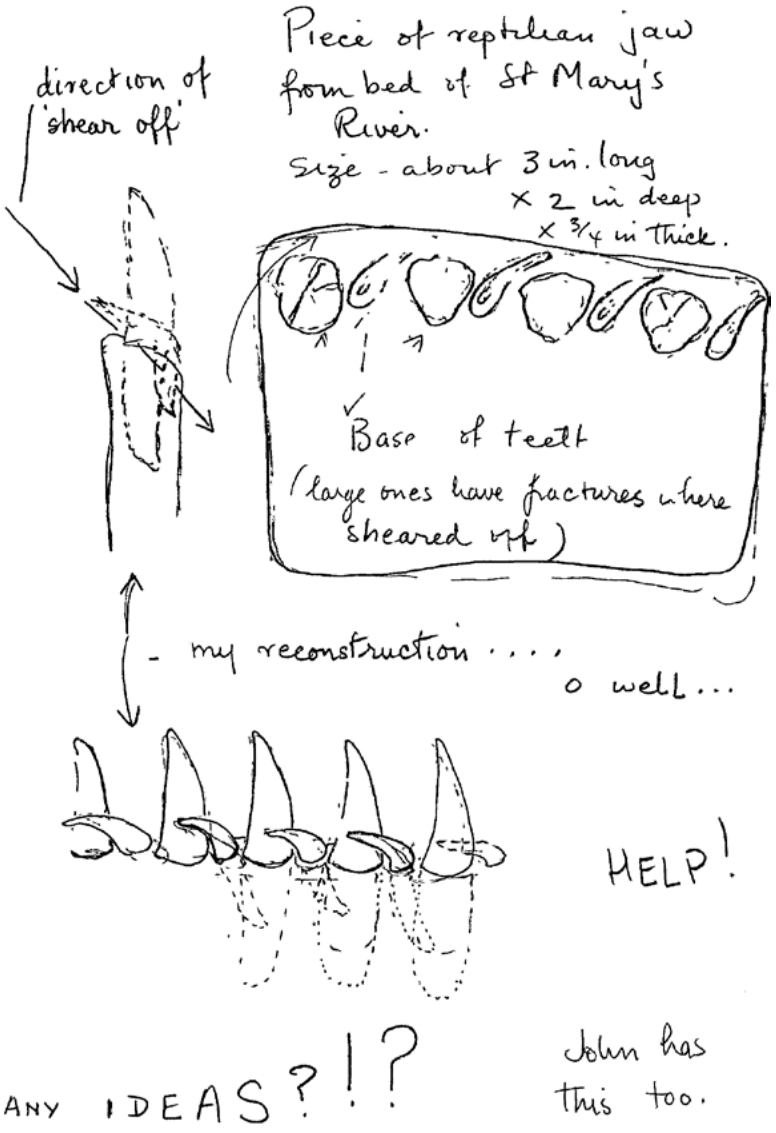


Figure 104. Hope's sketch of an unusual "reptilian jaw" fragment from the St. Mary River near Lethbridge. It has recently been identified as part of a plesiosaur skull, TMP 1971.008.0002. "John" is John E. Storer. From a letter to Dr. Dale A. Russell (Johnson, 1971c).

tified it as deriving from either *Mosasaurus conodon* or *Prognathodon* (D.A. Russell, 1971b, c). Hope first visited the locality on May 27 (Johnson, 1971c), returned to the site several times this year and collected more pieces, but it seems a full scale professional dig was never attempted. She saw to it, however, that the specimen ended up in a museum: it is currently housed in RTMP collections bearing the catalogue number TMP 1971.008.0001.

While prospecting for fossils in the autumn of 1970, at Late Cretaceous marine Bearpaw Formation outcrops on the St. Mary River west of Welling, she found on the riverbank a piece of jaw (TMP 1971.008.0002) from a large animal. Some of the teeth were eroded away, leaving only the bases of the crowns while other, much smaller teeth lying alongside are complete. This, as it turns out, was a very rare find—the first piece of plesiosaur skull recovered in the province. While some of the teeth are broken away, in a letter (Johnson, 1971c) and attached drawings (Figure 104) Hope interpreted them as possibly representing interlocking teeth. If true, as it appears to be, this fossil was derived from an articulated skull. The fossil was briefly considered by Sato (2002, p. 46) who identified it as a "small portion of a snout portion that bears 5 maxillary and at least 7 dentary teeth" of an indeterminate elasmosaurid (long-necked plesiosaur). A more recent interpretation suggests that the piece is actually from the proximal dentary near the corner of the mouth; the dentary teeth positioned there are enlarged and the occluding maxillary teeth are small (M. Mitchell, pers. comm., 2018). Sato considered the fossil reworked due to its "very smooth and worn" surface, but this is more likely a result of being exposed in the river for years, washed and gently abraded by the flowing water. This fossil, and the mosasaur associated skeleton, found around the same time and in the same general area, were precursors of spectacular discoveries of tooth-marked ammonites, fish, marine reptiles and dinosaurs made by the ammolite mining industry (particularly Korite International) in the region since 2002 (Hewitt and Westermann, 1990; Kase *et al.*, 1994, 1998; Seilacher, 1998; Tokaryk, 2001; Tsujita and Westermann, 2001; Narine, 2002; Sato, 2002; Kaufmann, 2004; Keupp, 2006; Anonymous, 2012a, b; Konishi *et al.*, 2011, 2014; Kubo *et al.*, 2012; Switek, 2012; Spivak, 2013; Tanke *et al.*, 2013; Newbrey and Konishi, 2015;

Konishi and Brinkman, 2016; Geggel, 2016; Gramling, 2016; Kilmury *et al.*, 2017; Tanke and Mitchell, in prep.); see also Sato and Tanabe, 1998.

Fieldwork by professional organizations in the Medicine Hat area was reduced. Stalker did fieldwork elsewhere; Churcher and Loris Russell did exploratory work in the Medicine Hat area but not actual fieldwork—they were busy with fact-gathering, on-site planning and writing a guidebook for a major field conference to occur in 1972 (L.S. Russell, 1971).

A visit to the Johnson home around this time is related by son-in-law Ed Preston who married the Johnsons' daughter Barbara:

> Barb and I had the pleasure of visiting the Johnsons in Ralston on a monthly basis first from Edmonton, 1968–1971 and then Calgary until 1973 when we left Alberta. I have often been in the living rooms of fine homes that were on the other hand sterile and uninteresting places to be. Not so the Johnsons'; the shelves and walls laden with paintings, photos, litre-size containers filled with Hope's polished stones, vials containing small mammal skeletons, marine fossils, teeth, buffalo bones, wonderful old books containing the works of Pepys, Coleridge, Yeats and others (even in her later years at the lodge, Hope's room was still a mini-museum). Ollie was a quiet but engaging fellow who always made sure his guest's beer mug was never empty. He made a home brew that we all appreciated, including Hope, and which I could never replicate (something to do with crock chemistry). Then there was Hope herself who loved to fill us with details of where she found this or that, the geology of the place or filling us in on some aspect of botany, history, *etc.* Her prodigious reading across a variety of fields and her memory made her something of a family search engine, long before the Internet existed. I was awed one afternoon by her detailed knowledge of centuries of British Royal history. Anyways, our weekends with the Johnsons were so much fun. Whenever we had her as a passenger in the car she was always commenting on specific plants by the roadside, or on local geology, *etc.* I remember on one occasion driving near Eastend, Saskatchewan with her when she asked us to stop the car because we could find small fossil deer bones by the roadside. Our children loved their Grandma's visits to Ottawa because of her interest in them. They loved sitting with her while she taught them how to draw or paint or would read to them, and were lucky to have such a special Grandma. We still have boxes of carefully labelled small fossils and minerals that she sent them. During some of Hope and Ollie's visits we also had the pleasure of social visits with the [Dale] Russells, [Peter] Dodsons and [Archie] Stalkers.

In August, Hope—in the role of "field captain"—led a group from the Lethbridge chapter of the Alberta Archaeological Society on a fossil-hunting day hike to the Manyberries, Alberta badlands (Carpenter, 2005, p. 35). During this period she taught a course called "Rocks and Fossils of our Area" at Medicine Hat College. The course was a hit. Anonymous (1971d) writes: "response overwhelming and the College has had to consider refusing some applications."

Hope's extra fieldwork activities this year consisted of visits to McDougall Ranch (formerly the Pat C Ranch) in April, Petrified Coulee in May, Foremost Formation exposures at Etzikom Coulee in August, marine Bearpaw Formation outcrops near McKinley Ranch, southeast of Manyberries in September, a locality a few kilometres upstream of Petrified Coulee in September, and the south side of the South Saskatchewan River, opposite the CFB Suffield pumphouse in October.

In a Ralston area gravel pit, she found an articulated tail of an ichthyodectiform fish in a Bearpaw Formation concretion (TMP 1972.012.0001). Ichthyodectiforms were primitive, elongate, predatory fishes that lived during the Mesozoic Era and had a global distribution. The best-known member of this group is *Xiphactinus*. Hope was excited to report (Johnson, 1972c) that the UA's newly-arrived fish palaeontologist Peter Forey (1945–2016) had planned to describe it. For reasons unknown, this never happened.

Hope entered some of her artwork in the September 13–25, 1971 National Exhibits of Nature Art show, sponsored by the Canadian Nature Federation. In a letter (Johnson, 1971d) she states that she submitted one badlands oil painting, two drawings and three floral watercolours. Work by Alberta

artists was eventually sent to the PMA in Edmonton and exhibited there for a time (D. Spalding, pers. comm., 2016). This show was the first formal recognition of her excellence as an artist and more professional art exhibits would follow.

The Esplanade Arts and Heritage Centre has an antique soap dish (their catalogue number 71.70) collected by Hope from McCracken's Mine this year. The museum has a small number of historical artifacts, some from Hope's own life, or curious items she found out in the field.

UA artist Harry Wohlfarth visited Petrified Coulee sometime this year and did one painting of Late Cretaceous outcrops there, presumably with Hope.

1972

In mid-April, Hope and Dale Russell exchanged several letters (Johnson, 1972b; D.A. Russell, 1972b). Russell was enquiring about local ground conditions in southeastern Alberta, emergence times of native wildflowers and logistics for aircraft availability, all for photographic purposes. Images needed to be professionally photographed for a new dinosaur book in progress (D.A. Russell, 1977). It is uncertain if this photography work happened, but Hope noted some aerial footage of badlands was taken by the Canadian Armed Forces near Munson, Alberta (Johnson, 1972c).

In May, Hope arranged for John E. Storer of the PMA to come to Medicine Hat and give a lecture on fossils to the general public (Anonymous, 1972a). Storer's (1972b) field notes do not mention this talk but he stayed at the Johnson home May 15–19 and he and Hope did some fieldwork together. On May 17 they visited, along with others, a buffalo jump excavation being worked by the University of Calgary (UC). On the 18th, Hope and Storer visited a Foremost Formation site on the South Saskatchewan River that had yielded rare *Ischyrhiza* sawfish rostral teeth that they later published on together (Storer and Johnson, 1974; see also Storer, 1972a). Storer's field notes record that "many things" were collected that day, including a rare multituberculate mammal tooth. No doubt the fossil identification book, then in progress (Johnson and Storer, 1974), was also discussed.

In 1972 the International Geological Congress was held in Montreal and as part of it Archie Stalker was to conduct a field trip in the Medicine Hat area (Stalker, 1972a). This excursion would have been of great interest to Hope as she had been involved in the fieldwork prior, but the trip was cancelled because not enough people signed up for it, though several useful guidebooks resulted (Glass, 1972; Russell and Churcher, 1972).

Figure 105. Hope with her braided hair coiled atop her head and heavy framed glasses, *c.* 1972. Image courtesy of the Johnson family.

Hope continued her summer job as fossil custodian at DPP but specific details of her activities this year are unknown. She does mention in J. Danis (1981) that she found the work physically harder for her in this, the first of

another two-year stint in the Park. It was around this time that her eyesight began to deteriorate with age, so she began wearing the heavy-rimmed and oversized glasses that (along with her braided hair) so characterized her appearance for the better part of four decades (Figures 105, 117).

Hope's work with the museum in Medicine Hat also continued. She complained in a letter to Dale Russell (Johnson, 1972c) how frustrated she was on receiving a very poorly curated amateur fossil collection, consisting mostly of Alberta and Saskatchewan material assembled by a Mr. James, a former railway conductor, that Hope had to process. Amateur fossil collectors should take note. She described the globally sourced collection as a ". . . wild, uncatalogued affair with piles of utter rubbish all mixed in with goodies." Some of the more bizarre finds consisted of "camel bones from the Oligocene of Steeveville" [sic], adding "I think you will agree with me that the 'Oligocene of Steeveville' is very elusive, to say the least." She sent the camel bones to John E. Storer at the PMA, stating that it was "VERY DANGEROUS for camels to be so early" [beds around Steveville are Late Cretaceous in age]. A black shark tooth was identified as a "fossilized bird's tongue." She continued: "The collector had a 'lamentable habit' of slapping a wad of some sort of putty on anything small enough to adhere to a wall and just slapping it on said wall." But there was good news: "It is fortunate for us that none of the collection was ever catalogued as this enables us to toss out the junk."

Her fieldwork activities this year are incompletely known. She visited the Red Rock Coulee area and Eagle Butte in June; the Bow River badlands and the Jenner Bridge area in July. Stalker and a few others did limited work in the Medicine Hat area this summer, mostly at the Galt Island Bluff locality (Stalker, 1972b; Lister and Sher, 2015). In September Hope travelled into Saskatchewan and examined the Morgan Creek badlands there. On September 17–18 she and some friends rented a holiday trailer and went fossil collecting at the Calf Creek locality, north of Killdeer, Saskatchewan. They also visited a locality near Willow Bunch (B. Lawson, pers. comm., 2015). Many of the Saskatchewan localities are Cenozoic in age. While some of her collecting friends amassed collections of disarticulated Miocene and Oligocene vertebrate material, Hope forwarded most of her finds to John E. Storer in Edmonton.

On October 7 she explored outcrops at McKinley Ranch and wrote to Dale Russell about the discovery of large elasmosaur centra there, stating "Vertebrate stuff in the Bearpaw, is really frustrating, tantalizing" (Johnson, 1972c). A centrum was moulded and cast by the PMA's John E. Storer, the original returned to the landowner; the cast is TMP 1973.006.0002.

Hope proposed teaching a course to the people in Ralston and district about the rocks and fossils of the area (Johnson, 1972d). She suggested a course of eight classes covering rock types, geological history of the area, effects of glaciation, invertebrate and vertebrate fossils, First Nations stone artifacts (briefly), and how to get more information from the experts on geological and palaeontological matters. Experience was not necessary, but ". . . it helps to have an elementary knowledge of chemistry." A minimum number of eight people were required for the course to proceed. The fee was $4.00 per person or $5.00 a couple. The course ran in 1973 and was a success, for Ralston town council minutes in Anonymous (1973c) record that "rock and fossil course by Hope is getting underway and indications are that it will be well attended." She taught a similar course in Medicine Hat (see 1973 section, Page 115); these two may be one and the same, but it could not be confirmed. The length and cost of the courses differed, suggesting that they were, in fact, separate.

John E. Storer, in his foreword to Johnson and Storer (1974) notes that Hope had some of her artwork shown in the 1972 National Exhibits of Nature Art, sponsored by the Canadian Nature Federation. This cross-Canada show stopped in Edmonton August 12 through October 15, 1972 (Beaudoin, 2012). As noted in David A.E. Spalding's foreword (herein, Page 8), Hope's two pieces were titled *Early Summer* and *Wild Dock* (Anonymous, 1972b, c).

1973

Beginning January 18, at Medicine Hat College, Hope taught an in-depth course on geology and fossils (Anonymous, 1973a). The classes ran for two hours a night, once a week, over a ten week period at a cost of $20.00. The course included the following lessons [sic]:

1. Rocks Types (Igneous, Sedimentary and Metamorphic).
2. The minerals involved in the common rocks: silica, the silicates, minerals.
3. How some rocks are found, *e.g.* metallic veins, evaporates, by decomposition of less stable compounds, *etc.* By metamorphism.
4. Our area, what types of rocks are native to the area, what types have been introduced by glaciation, streams, *etc.*
5. The geology of our area. One of the layered sedimentary rocks.
6. The fossils and characteristics of their sediments.

There is no record of course attendance or whether Hope ever taught it again.

Hope's avocational fieldwork activities were reduced this year. She was likely busy with her job at DPP and working on drawings for the Johnson and Storer (1974) book. She visited the Red Rock Coulee area in July. A trip to marine Bearpaw Formation outcrops on McKinley Ranch yielded nearly two dozen isolated mosasaur vertebrae and phalanges (Johnson, undated 2).

John E. Storer's 1973 field notes (Storer, 1973c) record three days of work with Hope in July. He spent the day with Hope on July 21 on unstated business. The next day was exciting. The pair went to DPP and met with Irene Vanderloh to see a small carnivorous dinosaur skeleton she had found. Storer recorded that "parts of at least one foot were found, and into the back [cliff] run large metapodials and long bones." The site was opened up the next year and the widely scattered partial skeleton became the type specimen (TMP 1974.010.0005) of *Saurornitholestes langstoni* (Sues, 1978; see also Tanke, 2010a). The following day they explored the badlands around the Princess Pumping Station near the Park.

Hope was involved in the discovery of one of the first good mosasaur specimens in the province, found in the Bearpaw Formation of the Manyberries area. In a letter (Johnson, 1973a) she describes a string of twenty or more vertebrae found in one place, parts of vertebrae, articular ends and centre sections of phalanges (about fourteen in number) and several larger bones, "perhaps metapodials." The vertebrae were 52–65 mm long. Two vertebrae were fused and "round-bottomed," suggesting prior injury. Hope expressed a desire to exhibit some of the vertebrae at the Medicine Hat Museum. In a subsequent letter (Johnson, 1973b), more information on the mosasaur find is given. It was found by Hope and her fellow collector friends, the Lawsons, 16 miles (26 km) east of Manyberries in what Hope described as a "dune-y landscape." More vertebrae were identified; Hope says there were sixteen to twenty in "each string," suggesting more than one string of articulated vertebrae. She makes the odd comment about the specimen: "We shall continue to search + even, alas, dig." In another letter (Johnson, 1973c) she states that all the bones were found in a six-foot (1.8 m) area. Responding to Hope's report, Russell asks for pieces of it to be sent to him for examination (D.A. Russell, 1973a). Later he determined that the material was undiagnostic but probably referable to *Plioplatecarpus* (D.A. Russell, 1973b). Curiously, it seems that this discovery was never followed up by the professional palaeontological community, despite the obvious fact that it was an associated/articulated skeleton confined to a small area. Marine reptile finds in the province were then only poorly known so it is unfortunate that it was not investigated further.

On the weekend of August 31–September 2 Hope and some friends returned to the Calf Creek locality north of Killdeer, Saskatchewan. Day trips to Petrified Coulee and the Jenner Bridge area were

conducted in October.

Hope created a pamphlet guide for a nature trail in DPP (this would have been dated 1973—none seem to have survived) and was interviewed on tape about her palaeontological experiences. The 23 minutes, 39 seconds audio tape is curated at the Provincial Archives of Alberta in Edmonton (Dryden, 1981).[37] A copy of this interview was made for the author. It appears to be the soundtrack for a half-hour Alberta history television show, possibly titled "The General Store" and hosted by Calgary CFAC television personality Jerrold Alair "Jerry" Puckett. Some years later she noted in a letter (Johnson, 1977a) that she was once the subject of a half-hour Calgary television program that she never saw; perhaps this was it. In the show, Hope talks with great detail and enthusiasm about the DPP badlands; the Bearpaw Sea; fossils of, and postulated life history of small theropods, ankylosaurs, ornithomimids, tyrannosaurs and hadrosaurs; ammonites; geology; vertebrate microfossils; volcanism and ash deposits; bentonite mud and its extremely slippery qualities when wet; depositional environments; Rocky Mountains orogeny; glacial erosion and badlands formation of hoodoos; DPP local human history (namely "Happy Jack" Jackson); and local fossil protection efforts by Park staff (Puckett, 1973). She also expressed concern over irresponsible amateur fossil collecting by those who are "souvenir-mad," some using large hadrosaur caudal vertebrae as doorstops or others who were "whacking at pieces of fossil wood" for lapidary purposes (*e.g.* Anonymous, 1947b), but she remarked that the fossil wood is actually useless for such purposes. Finally, she comments on the numerous and mostly garish concrete dinosaur models then displayed at the Prehistoric Park at the Calgary Zoo (Figure 106), stating that some of them are "not very faithful representations" and "some people are rather cross about them," but adds that a benefit of them is that it gets people thinking and talking about dinosaurs.

Figure 106. Vintage postcard of "Dinny" the *Brontosaurus* (*Apatosaurus*). This giant concrete model, built in the 1930s, is the only survivor of the Calgary Zoo's original Prehistoric Park. Image courtesy of Peel's Prairie Provinces, a digital initiative of the University of Alberta Libraries. **http://peel.library.ualberta.ca**

Hope worked in DPP again this summer, her last season as fossil custodian there. Why the job ended is unknown; none of her family could recall. But she would continue to return as a private citizen and to do volunteer work as needed, or if asked (G. Forsen, pers. comm., 2014). In J. Danis (1981) Hope relates how, during this time, the Park experienced repeated disappointment with promises of "all sorts of wonderful or vast" improvements, surveys to be done, and none of it happening. She says that in 1969 there were sixteen DPP staff of all types (assistant wardens, students, naturalists)—even including a cook—but not long after, staffing was reduced by two thirds, making it difficult for the remaining overworked staff to do their jobs. It also created security issues, due to reduced patrols, and exposed the fossils to the activities of poachers. Morale was bad. It is possible that Hope's job at the Park was a victim of cutbacks, but she does not specifically state that in her interviews or papers.

In July she met with John E. Storer and showed him a number of sites and part of a hadrosaur skeleton that turned out to be a 2 m (six foot) length of tail that was not collected. However, one prospect was good: it was an immature tyrannosaur with an eroded pelvis, both hind limbs, and the base of the tail (TMP 1973.030.0001), found earlier by her friend Irene Vanderloh. Later that summer, while a PMA crew was working on the find, she visited the site and took some pictures. Hope's shadow appears in

RTMP photographs of the quarry, the distinctive braided hair coiled atop her head (Figure 107). The location of this quarry is presently unknown. The specimen is figured in Eberth and Currie (2005, p. 462; their fig. 24.2D), but is misidentified as a tyrannosaur forelimb.

Hope collected a 56 cm-diameter *Placenticeras* ammonite using her son's horse to haul it out of the badlands, as Alberta's early dinosaur collectors did long ago. It marks one of the rare uses of horses in a badlands setting to recover fossils post-1954, when the ROM last used horses in DPP, until more recently when the RTMP employed horses to haul bulk microvertebrate matrix bags for wet screening (Fotheringham *et al.*, 2014). And it appears to be the first and only documented use of horses in a field fossil recovery context by an amateur collector in Alberta. The fossil was needed for a flat-case display of ammonites Hope was making for the Medicine Hat Museum. This was one of several she was developing for the Museum; another was to feature local Late Cretaceous vertebrate fossils and included small bones, teeth, and scutes. A third displayed Oligocene mammals from near Eastend, Saskatchewan—she credited the PMA's John E. Storer for help developing that exhibit. Yet another case was to show Pleistocene-aged mammal teeth, including "horse, camels, beavers, tigers, and caribou" (Anonymous, 1973b) which were being collected at the time from Mitchell Bluff, north of Medicine Hat. Some of the items displayed in this exhibit were casts provided by the ROM. For this Pleistocene exhibit, Hope made a copy of a painting (Figure 179) of mammoths and sabre-toothed cats by Zdeněk Burian.

Figure 107. TMP 1973.030.0001, subadult tyrannosaur pelvis and hind limbs in left latero-ventral view in DPP prior to plastering by the PMA crew. Image courtesy of Hope Johnson, whose shadow appears in the photograph.

Hope acknowledges receipt of a cast of a 1:1 scale *Dromiceiomimus* skull reconstruction which was then exhibited in the Medicine Hat Museum. This replica was made in part for Dale Russell's research on ornithomimid dinosaurs (D.A. Russell, 1972a). A cast also found its way to the fossil exhibit buildings at DPP, but it is uncertain if Russell sent Hope two casts, or if the one used in Medicine Hat was later repurposed for DPP.

It appears that Hope was attempting to mould and cast replicas of fossils around this time. In the last boxes of her fossil collection taken to the RTMP in 2014 was an open, plastic bag of "Casty Craft Insta Mold." The bag bore a 1972 copyright date and that bag was stored within a plastic grocery bag. The grocery bag bore a "6-73" number, presumably meaning June 1973. A good-quality painted cast of a small ammonite, several centimetres across and a larger one were found in her collection. The smaller example bore the number "73.98.24" written on paper and glued onto the back, but no fossil moulds were found, nor, despite appearances, was it a RTMP catalogue number. It does not appear that Hope pursued large-scale moulding and casting of fossils, at least none that the author can confirm. The ornithomimid foot she found in 1963 was moulded and cast by persons unknown; perhaps Hope did this. Two painted plaster casts of upper molars (Figure 108), attributed by Hope to the early three-toed horse, *Merychippus* from the Miocene of the Calf Creek Quarry, north-northwest of Eastend, Saskatchewan, were also found in the last remnants of her fossil collection.

Her professional working relationship with Archie Stalker continued. His notes record that on July 17 they were mostly digging at Galt Island (near Medicine Hat) and that he led Hope and a small group to the dig site that evening, though not much was found. On August 3, he and Hope investigated a bone site she had found.

Figure 108. Two painted casts of *Merychippus* upper molar teeth found in Hope's collection, seen here in occlusal view. Scale bar = 1 cm.

Hope is credited with discovering a rare plant, the tiny cryptanthe (*Cryptantha minima*) on July 22, near Suffield (Bradley, 2004, pp. 8, 28). This was the first report of its occurrence in Alberta since 1894.

Little is known of Hope's art activities in 1973. A label on the back of a *c.* 1975 oil painting of a great horned owl has a note in her hand indicating that she exhibited in the *Nature Art Canada '73* show. Details of her contribution to this show are unknown.

Surprisingly, some people considered Hope to be a woman of just average intellect, but the breadth of her knowledge and reasoning is amply revealed in an undated letter to John E. Storer, where she addresses the proper use of English and linguistic history regarding a simple caption for one of the drawings in the Johnson and Storer (1974) book:

> Re "An Herbivorous Dinosaur." I really don't like this. I'm aware that many Albertans, and rural people from some parts of England, and some Americans (how many??), say "erb." We have an American Dictionary which gives both pronunciations, but our English dictionary gives only HERB. I'm sure that most educated English people are really worried about dropped h's (aitches), since this is a sign of humble origins, and exceedingly prevalent among uneducated people there. Canada follows the British usages about 85% at least on spelling, etc. I know it is derived from the French herbe and that they write "sur l'herbe" etc indicating not even a French aspirated h (as in le hibou = the owl). Probably 'erb went out in England with the Hanovernian [sic] Kings, who being Germans, didn't lack aitches. (No—I don't say 'Honour') . . . (How about 'A plant-eating dinosaur'?).

1974

On the May 17–19 long weekend the 13th Regular Federation of Alberta Naturalists Board of Directors held their Spring Field Meeting at Elkwater in Cypress Hills Provincial Park. A brief report on the meeting says it was well attended (Anonymous, 1974d); Hope gave a lecture at this event (D. Spalding, pers. comm., 2018), though the subject of her presentation is unknown.

It is recorded (Anonymous, 1974f) that in the late summer of 1974 Hope gave a guided tour in DPP to some seventy-two elementary students from Lethbridge. There she talked about fossils and the reconstructed cabin of the African-American cowboy, John Ware. The cabin had been moved to the Park in 1960 and became a tourist attraction (Anonymous, 1960f, 1960g, 1960h).

In an internal report, Moser (1974) credits Hope for "assistance and guidance and reviewing some of the material" for a report on future interpretive development in DPP. Some of Hope's writing—just over four pages—destined for a "pamphlet presentation" were used in the report (Moser, 1974, p. 30). Moser's notes intimate that Hope had, on her own, developed a pamphlet on the dinosaurs and some other fossils of the Park and the report voices a wish that these could be developed into a temporary brochure for tourists until something more formal was published. Unfortunately the idea "did not come to pass" (Moser, 1974, p. 58). However, it seems that Hope's efforts may have been developed later, in an abridged

form, for sale in the Park (see 1977 section, Page 128).

One person interviewed for this book recalled Hope being a frequent visitor to DPP during this period. She was given access to Park infrastructure and was allowed free rein to go hiking and collecting anywhere in the Park (C. Virovec, pers. comm., 2014). The Park Warden at this time was Gerhard Forsen. His daughter Carolyn remembers going out exploring for fossils with Hope. She recalled finding a tiny *Myledaphus* tooth and how an excited Hope took the time to carefully explain what she had found, how the teeth were arranged and the animal's life history (C. Virovec, pers. comm., 2014).

Ron Chamney, who worked in the Park from 1974 to 1978, remembered her as an unofficial consultant at the Park on many things palaeontological. While the PMA in Edmonton at this point had a field program, it was small and only visited the Park on occasion, to collect mostly small specimens. There were still serious differences of opinion between some Park staff and the PMA palaeontologists, regarding the best ways to protect the fossils (D. Spalding, pers. comm., 2016; see also comments in Kondla and Crawford, 1971). The more frequent presence of PMA field crews would have been welcomed by other Park staff, who personally craved palaeontological knowledge, but Edmonton was a long drive away and the Park/palaeontologist relationship was still uneasy. Hope was much closer and she made more frequent appearances at the Park, which was always appreciated. If she went out into the badlands she was quite insistent that a Park staff member come along with her. "The hikes in the hills with Hope were always fun and informative" (R. Chamney, pers. comm., 2015). In the Park she was doing fossil preparation, too. In an early September letter to friends and fellow collectors Charles and Velma Heath, she notes that she was using white glue in the preparation of larger dinosaur bones and mentions doing plaster [reconstruction] work as well.

Curiously, there is no record of Hope ever joining a professional palaeontological field crew in Alberta (*contra* Sanderson, 1999) to assist on excavating major dinosaur specimens or assisting with extended fieldwork. Some early 1980s RTMP field notes record her occasionally joining the crew for a day in DPP, looking at microfossil sites, conducting general prospecting with RTMP staff or visiting research scientists working with RTMP (she did some work with Dr. Peter Dodson who was part of the RTMP crew in 1982). But it appears she never joined on as a full crew member for an extended period of time. As mentioned earlier (see 1968 section, Page 91), she did join Dr. Gordon Edmund and his crew in Saskatchewan on a Royal Ontario Museum (ROM) dig, collecting Oligocene-aged mammals, where she described her experience (Johnson, 1969c), but this seems to be her only extended work with professionals. In Alberta she would assist palaeontological crews from time to time on short stints—pointing out sites of interest to crews from the NMC, PMA, ROM, RTMP, and UA; visiting with palaeontology workers on a regular basis at her home; or showing and donating her finds to them.

Rather than become involved on professional digs in a big way, it seems Hope preferred to stay more on the sidelines, working alone, or more with fellow amateur fossil collectors[38] and family when she conducted her fieldwork. It is possible that this was on account of the then-minor role women played in palaeontology: it was very much a male-dominated field. While there were rare, prominent exceptions, it was not until the 1980s that women played a more noticeable, steadily growing and integral role, though parity has still not been reached, even today. She also occasionally led field trips for military staff members and families of her husband's Defence Research Board colleagues. As noted, Hope shared her passion for fieldwork with her children, though none developed an interest in the field professionally.

Churcher (1973) acknowledged Hope anonymously and later by name (Churcher, 1974) for her role in reporting a new Quaternary mammal site found by some of her friends in the Medicine Hat area. This important work was later written up in Barendregt *et al.* (1988; see also Lister and Sher, 2015).

Subsequent work was done at a variety of sites in the area by Rufus Churcher and Archie Stalker, but it is unclear to what extent, if at all, Hope was involved in these digs (*e.g.* Nackoney, 1969; and see 1969 entry, herein; Francis, 1971; Bilodeau, 1979; Stalker, 1984). Overall, it seems that Hope had little to do with Quaternary vertebrate palaeontology on her own time. The glenoid region of a mammoth scapula (TMP 1990.005.0084), recovered from an Empress, Alberta gravel pit is credited to her.

A fact often overlooked is that Hope co-authored a scientific paper in a peer-reviewed journal. Storer and Johnson teamed up again in 1973–1974, this time describing and illustrating (by Hope, of course) some rare *Ischyrhiza mira mira* (sawfish) tooth material that she and others had collected in various localities in the Late Cretaceous brackish-water deposits of the Foremost Formation of southern Alberta (Storer, 1972a; Storer and Johnson, 1974). It remains the definitive paper on Late Cretaceous sawfish from Alberta.

In early September Hope gave a geology and vertebrate palaeontology talk and slide show to members of the Xi Gamma Chapter of Beta Sigma Phi. The presentation was on the geology of Dinosaur Park, the Drumheller valley and other areas in the southern part of the province. She also discussed the erosive effects of glaciers on the region. She used some of her own illustrations of dinosaurs as well as actual fossils in the presentation (Anonymous, 1974g).

She attended the annual meeting of the Society of Vertebrate Paleontology in Flagstaff, Arizona, which included a field trip to early Mesozoic sites in northern Arizona in and near the Navajo Reservation. They looked at exposures of the Triassic Moenkopi, Chinle and Kayenta formations, the latter to see *in situ* coelurosaur tracks, and the Petrified Forest member of the Chinle. At a lunch stop on Late Cretaceous Mancos Shale exposures, a fish microsite (including pycnodont remains) was discovered by Albertan Michael C. Wilson. There, Hope discussed with Wilson possible hybodont shark teeth she had found in southern Alberta and had sent to John E. Storer. She later visited the Los Angeles County Museum to see their spectacular fossil exhibits.

During this busy year Hope had some of her art pieces featured in the 1974 show "Canadian Nature Art '74." This was an annual nation-wide travelling exhibit put on by the Canadian Nature Federation and the National Museum of Natural Sciences (now the CMN). Three of Hope's pieces were included (out of 39 total pieces): one of the DPP badlands (in oil), one of Bearpaw Formation invertebrate fossils (ink and wash) and a pen-and-ink rendition of the rock clematis plant, *Clematis columbiana* (Anonymous, 1974m). The fourteen-city tour ended in Ottawa on March 8, 1976 (Walker, 1976). Hope's art was featured in at least three of these annual shows (Delday, 1977). The 1974 show travelled to the PMA and was on display there July 3–24, 1974.

There was an art show and sale by Ralston Art Club members on December 1 (Anonymous, 1974l). The Club's activities were reported less frequently in the local newspaper during the early to mid-1970s, so Hope's involvement is not well known. She created a new masthead for the Ralston newspaper—a Buffalo Bean plant drawing (Anonymous, 1974k) and wrote and illustrated an article on the same plant in the first issue featuring the new masthead (Johnson, 1974).

Her role in the painting of background sets for the Ralston Elementary School production of *The Mikado* is acknowledged this year in Gilbert (1974). Photos of the students in Japanese costume with partially completed background painting by the uncredited Hope appear in Anonymous (1974j). Her fine set work for the play was even acknowledged twelve years later (Mumford, 1986).

Her biggest event of the year was no doubt the release of the Johnson and Storer book (published in 1974, though occasionally misdated—even in Hope's notes—as 1973), *A Guide to Alberta Vertebrate Fossils from the Age of Dinosaurs* (see Page 108). A newspaper advertisement (Figure 109 a) suggested that it would make a nice Christmas gift. She was very proud of this book (Figure 109 b). But in a pub-

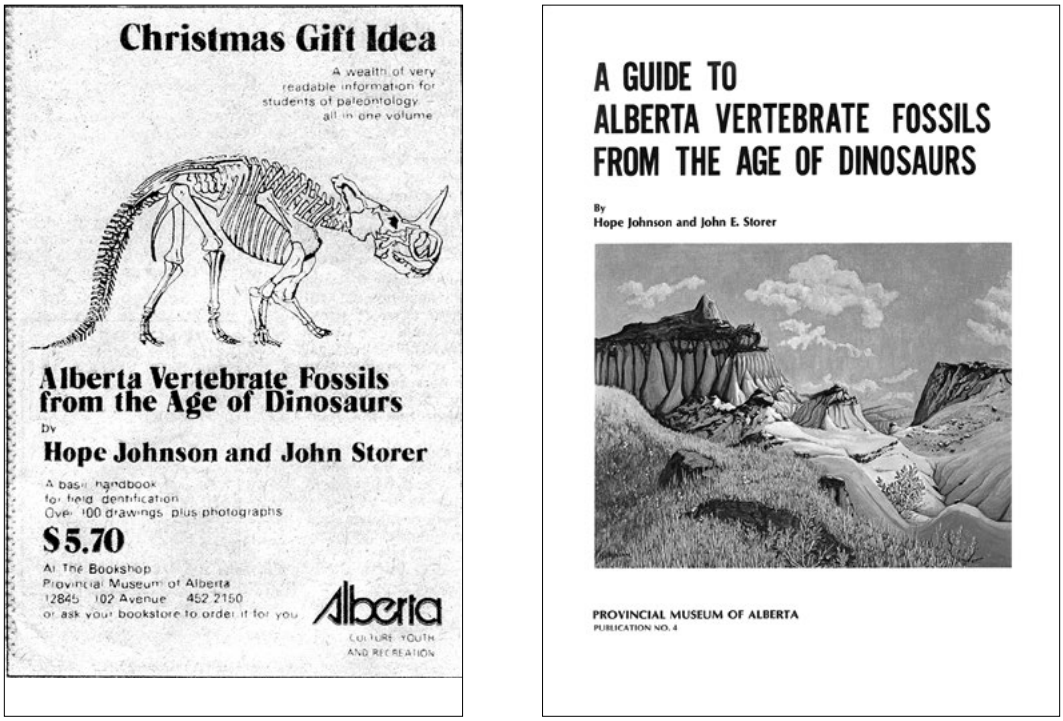


Figure 109. Newspaper advertisement (left) for the 1974 Johnson and Storer fossil identification book (right). Figure 109 (left) from Anonymous (1974i).

lication with so many drawings, taxonomic names, anatomical identifications and ongoing research, a few factual errors inevitably crept in. Sending a complimentary copy of the book to some amateur collector friends in 1974 she wrote in a cover letter: "I have amended it in a couple of places, since, in spite of our efforts to get everything right, there are a few errors we overlooked. Ho hum." When the author asked her to autograph his own copy long ago, she did just that—and then proceeded to open the book up to a page near the middle and penned a notation next to a figure of a vertebra that had been misidentified. Questioned on this, she went into great detail to explain why the identity had been changed, describing the morphological features of the specimen that she now better understood. She definitely wanted readers to be well informed.

Her fossil catalogue book does not list much fieldwork outside of DPP this year. She visited a location called Bare Creek, 40 km south of Elkwater, Alberta and collected some marine invertebrates there (Johnson, undated 2). Four dried plant specimens curated in the UC herbarium (see Page 188) were collected and donated by Hope this year; two were collected at the McCracken's Mine locality in May and June, two from DPP in August.

1975

Early this year something very peculiar happened in DPP. The Park Warden and his daughter were at the viewpoint overlooking the badlands on the night of February 8 when they claimed to have seen a lighted UFO, hovering about 2 km east of the Park headquarters, near the archaeological "Dreambed" site. The light illuminated the entire ground; no other details were given. It was not the first observation of such an object, but it was the most famous. The incident inspired Hope to write a short piece for the Ralston newspaper. In a letter to Ottawa's Dale Russell (Johnson, 1975b), she joked: "I am hoping they

come back in the summer + I shall ask them—whoever is aboard to help me find small carniv[orous] dinos remains . . . I am quite curious about them but I cannot believe they are harmful (unless swallowed!) and anyway they are quite fun." Russell had a discreet sideline interest in cryptozoology and UFOs. Is it just coincidence, then, that his "Dinosauroid" (Russell and Séguin 1982; Naish, 2008, 2012) made its appearance not long after?

Hope was rarely seriously ill, but she was admitted to hospital on April 16. She had been suffering from recurring abdominal cramps and pain for some time. She was released after a six-day stay and a gall bladder operation, in which an astonishing total of 165 stones with an average diameter of 4 mm each were removed (Johnson, 1975c). She had healed well enough by late May to enjoy a four-week holiday to Europe to visit her daughters and their families, as well as some museums (Johnson, 1975c).

She reported the Earth science highlights of her museum visits in northwestern Europe to Dale Russell (Johnson, 1975c). She wanted to visit the museum in Maastrict, Netherlands, but it was closed for renovations. In Brussels she saw the famous *Iguanodon* dinosaur herd and mosasaur material, and in London the Natural History Museum fossil vertebrate and geological exhibits.

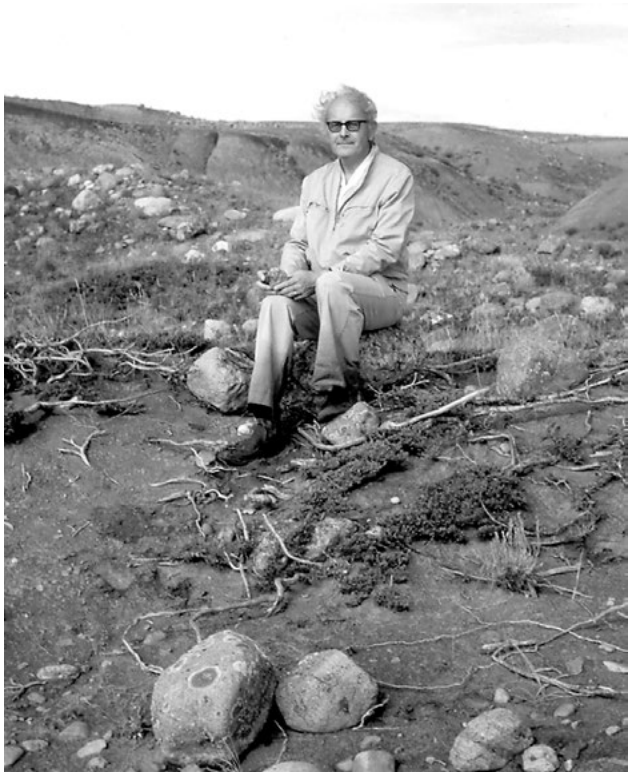


Figure 110. Dr. Charles Devillers, seated in the marine Bearpaw Formation outcrop area near Manyberries, Alberta in August, 1975. Image courtesy of Hope Johnson.

She reveals in a letter (Johnson, 1975a) that early this year she was an active member of an *ad hoc* committee to develop a park encompassing the badlands along the South Saskatchewan River, from just upstream of Redcliff to ten miles north of Medicine Hat, all within Improvement District No. 22. The park was to be administered by the provincial government and the City of Medicine Hat. The public were invited to make recommendations on developments they would like to see in such a park. Suggestions had already been submitted when Hope penned her letter. How long this committee lasted is unknown and the park concept never came to fruition. It appears to have been yet another, larger effort to preserve and protect parts of this region, a wish Hope had voiced previously (Johnson in Bridge, 1959a; Bridge, 1960; Johnson, 1961e).

Hope continued contributing to scientific publications by co-writing a paper with Beryl Hallworth (*c.* 1911–2000), Assistant Curator of the herbarium at the UC, on a rare and endangered plant (small-flowered sand verbena) in the southern prairies region (Johnson and Hallworth, 1975). She then donated the technical illustrations from that paper and some other floral drawings to the UC herbarium. A little-known fact is that Hope also collected and pressed an unknown number of plant specimens for the herbariums at the UC and the PMA. She also ran another series of plant drawings in the local Ralston paper (Johnson, 1975d-i). The series was later amalgamated and republished in edited form in the same paper in the early 1980s. She coloured these drawings with watercolours and put them on display in the local library. Two sets of these wonderful sketches exist: one is curated at the Graham Community Library in Ralston (see Figure 177); her daughter Margaret has a set in Medicine Hat. The

Ralston Art Club participated in a juried show and sale in Medicine Hat in June (Anonymous, 1975a).

Hope seems to have done little fieldwork this year. She was in the badlands on the periphery of DPP in June and other times with Park staff (R. Chamney, pers. comm., 2015) and along the Bow River near Hays, Alberta. She visited Petrified Coulee in August.

An overseas visitor this year was Charles Devillers, whom she first met at the palaeontology conference in Alberta in 1963 (see Page 74). He was teaching a summer course at the UA with Dr. John R. Nursall, so Hope invited Devillers to come down for a six-day visit. They did fieldwork in the marine Bearpaw Formation south of the Manyberries area in August (Figure 110). He had an interest in ammonites, specifically studying the living chamber region (B. Lawson, pers. comm., 2015). An unidentified professional colleague of Devillers who lived in Dijon, France was also interested in Alberta *Placenticeras* ammonites. The pair were working on a study of evolutionary trends in ammonites and Hope said the "*Placenticeras* ammonites seem sufficiently plentiful and varied to be quite a help [for his research]" (Johnson, 1975j). She further noted that Devillers was "great company & we really enjoyed his visit."

In a letter to Dale Russell (Johnson, 1975a), Hope mentions scattered mosasaur material found in the riverbed gravels of the St. Mary River near Lethbridge. Another interesting find was the distal half of a subadult tyrannosaur femur found in the marine Bearpaw Formation about thirteen miles (21 km) east of Manyberries. The specimen had been identified by Loris S. Russell, but its current status is unknown. People are sometimes confused about how a terrestrial dinosaur finds its way onto the sea bottom. Dinosaur carcasses found in Alberta were typically buried in freshwater river deposits; but sometimes one was carried far out to sea by water and wind currents. A bloated carcass would bob in the seawater, shedding skeletal parts as it decayed and was scavenged, eventually sinking to the bottom as trapped gases escaped (*e.g.* Horner, 1979; Pasch and May, 1998, 2001; Lucas *et al.*, 2006; Brown *et al.*, 2017; Kilmury *et al.*, 2017).

By this point in her career, Hope could recognize pathological dinosaur bones, finding a large tyrannosaur pedal phalanx (digit III, phalanx 2 or 3) with an expanded proximal end and its distal end "stunted." Sometimes other unusual bones intrigued her and she would sketch them in her fossil catalogue books. Some of these were never identified in her time, but are now recognized as badly deformed hadrosaur distal caudal vertebrae, likely trampled and crushed by other hadrosaurs and often with irregular healing (Tanke and Rothschild, 2014). Such affected specimens with their unusual bone growths and distortions can exhibit a bewildering array of malformed shapes and cortical textures related to post-trauma fracture callus formation or osteomyelitic conditions (*e.g.* Rothschild and Martin, 1993; Rothschild *et al.*, 1997; Rothschild and Tanke, 1992, 2007; Tanke and Rothschild, 2010, 2014), sometimes confusing even the experienced fossil collector.

Figure 111. Dr. J.-P. Christopher Jackson in 2016. Photo by the author.

Another important person would enter Hope's life this year. Dr. J.-P. Christopher Jackson (1946– ; Figure 111), an avocational palaeontologist and new assistant professor of the Music Department at the University of Lethbridge, announced in a letter to Calgary lapidarists and fossil collectors his imminent arrival to Alberta and his desire to meet other fossil collectors (Jackson, 1975). Hope's son Lawrence was taking some music appreciation courses at the University and on learning of Jackson's interest in fossils, introduced him to Hope. What ensued was a friendship and dozens of fossil collecting forays from

1976 until 1982 or 1983 (Jackson, pers. comm., 2006). Jackson was impressed with her dedication, art and scientific knowledge of plants and fossils. It was his initiative that would result in Hope being honoured by academia (see Page 136).

Hope worked on new, local vertebrate palaeontology-themed exhibits at the Medicine Hat Museum. One concerned the evolution of the horse. Images of a skull of *Equus lambei* were sent to her by NMC palaeo-mammalogist Dick Harington that spring (Harington, 1975) for use in the display. In a later letter to Dale Russell (Johnson, 1975c), she noted that the ROM and Rufus Churcher had been very helpful in the completion of that exhibit.

1976

Little is known of Hope's activities early this year. In a May 19 letter to Dale Russell (Johnson, 1976a), she disclosed that on that very day she had received a letter from Edmonton-based "Television North," asking to film her in DPP in August (see following page).

Figure 112. Dr. Philip J. Currie, age 27, in the late summer of 1976. Author's collection.

On July 5, Dr. Philip J. Currie (Figure 112) began his vertebrate palaeontology career in Alberta (Carroll, 2001), replacing John E. Storer, who moved to the Saskatchewan Museum of Natural History in Regina. Most Albertan Earth science professionals contacted for this book noted that they had heard of Hope within a few weeks of starting their careers. On July 9, 1976, only four days into his own career, the newcomer met Hope while he was being toured around palaeontological sites and introduced to useful contacts by David A.E. Spalding (Spalding, pers. comm., 2016). A few weeks later, Currie was dropping off a palaeontological permit application at her home (Currie, 1976). He would have professional contact with her for the next 34 years.

In a letter (Johnson, 1977a), Hope related her first impressions of Currie and how she thought he would get along with Alberta Parks staff, whose policy in those days was to discourage collecting in DPP by professional palaeontologists and to let the exposed fossils erode where they lay. She wrote: "I have met him, he was at McGill. Seems very nice, but I doubt he will give the Park's people any trouble. He seems suitably passive. (No doubt carefully selected as a 'non-rocker of boats')." Any reader who has knowledge of Currie's non-passive work ethic and his resultant and ongoing massive contribution to vertebrate palaeontology can see with the advantage of hindsight that Hope seriously erred in her first impressions of him. She was to become "a great admirer of his work" and followed it closely, showing her son Lawrence copies of Currie's new papers whenever he visited her in later years (L. Johnson, pers. comm., 2018).

Permits were now required to excavate fossils in the province, as dictated by the Historical Resources Act. This legislation meant that amateur collectors could no longer excavate their finds unless they were greatly experienced and working directly with a researcher. Hope received a permit and commented (Johnson, 1977a) that it rather vaguely covered "the area south & east of Medicine Hat." She also expressed disappointment that professional Earth scientists very rarely did fieldwork in her rich collecting areas and stated that southeastern Alberta was called "next year country," signifying that field crews would explore there "next year," but never did. Later she was acknowledged for her active role in finding fossils, reporting them to authorities and being granted a permit to excavate at specific sites (Currie, 1984).

On June 9 she met with Alberta filmmaker Warren Graves (1933–2008; Whittaker, 2010) in DPP. This was likely the "Television North" production noted earlier. Graves' crew were actually doing a piece on an inter-school drama competition held in Drumheller and wanted to get some footage on Drumheller dinosaurs. They later decided, according to Hope, to "do it right" and film in DPP. Apparently the plan was to run a video camera along a 244 m-long cable so they could follow and interview Hope as she walked through the badlands exploring for, finding, and discussing fossils. This was all initial planning and Hope expressed doubt that the filming would take place that year. It is unknown whether the filming ever actually occurred.

In 1976 BBC Horizon filmed and produced a television documentary, *The Hot-Blooded Dinosaurs*. Spalding (1976a) visited Hope at her home on July 12 and noted that she and Dale Russell were doing a film for the BBC. Hope had expressed some anxiety about being filmed, stating in a letter to Russell, "I don't suffer from nervousness over TV, or talking to people, generally. Maybe I ought to but I don't! I am probably ignorant about many things, but I realize there are indeed degrees of ignorance." (Johnson, 1976c). D.A. Russell (1976) replied that Robin Bates, from BBC London, had commented "she was a natural" and added that "she should breeze through the Edmonton TV thing," in reference to the "Television North" production.

BBC's *The Hot-Blooded Dinosaurs* first aired in Britain on November 1, 1976 and then under PBS's NOVA banner in North America, on January 12, 1977. Hope was interviewed and filmed in DPP. She appears in the NOVA version near the beginning of the one-hour program and in a one minute-forty seconds segment she shows the viewer a hadrosaur skeleton with well-preserved, three-dimensional skin impressions and discusses dinosaur colouration and taphonomy.[39, 40] Many dinosaur documentaries have been made on or include aspects of vertebrate palaeontology in Alberta but this was, as near as can be determined, one of the first in the province[41] and, in fact, one of the first anywhere in the world featuring a woman. A Park employee and helper recalled that the film crew were puzzled about the appearance of a certain dinosaur, so Hope found a nearby flat, silty pediment area at the base of an outcrop and accurately "drew" the animal at 1:1 scale by scraping into the sediments with a pointed tool (R. Chamney, pers. comm., 2015). The eroded sediments were used much as a teacher would use a blackboard, a technique the author independently "invented" and uses on occasion.

Dale Russell (1976) comments on a curious maniraptoran frontal that Hope had sent him. He thought it might be something new, writing:

> HOWEVER, the frontal is most interesting. If I were incautious, I would slap a new species, genus, and family on it with no further ado. That's what it represents. A broad-skull theropod with an incipient forehead, cerebral hemispheres halfway between those in *Dromiceiomimus* and *Dromaeosaurus* in size, and a very heavy "bridge" across the region where the olfactory tract should be. Another one of those brainy theropods—there is a similar right frontal in our collections which is about twice as large, although the cerebrum is about one-third larger lineally which is what one would expect. Probably a different species. I would love to stick a binomial on this, although I suppose I shouldn't because it is so incomplete. But it is characteristic. If it can be separated from *Oviraptor*, the temptation would be great.

Found by Hope in DPP, the current status of this interesting find is unknown. Hope sent it to Russell with some hesitation, as the Park was becoming more restrictive about the collection of fossils by professionals. But at the time there was no infrastructure for the Park itself to safely curate such material, so she felt it was better off in a museum.

Now aged 60, perhaps she was beginning to feel her age in the badlands. In a letter to Dale Russell (Johnson, 1976b) she complained of painful arthritis or gout symptoms at the base of her right big toe. In an interview this year (Smith, 1976) Hope commented:

> Digging up dinosaur bones is a terrible business for an old lady. People have no idea what it entails. There you are, hacking away by the hour and you mustn't break anything. It all has to be carefully plastered. It's one thing to discover something, great fun and all, but one always hopes to find a small, rare, neat dinosaur that will go into a packet, not weighing over 50 pounds, including the plaster!

Despite these comments, she would continue to hike the badlands for decades to come. Around this time (1975 to 1984) Hope was using an antique car for fieldwork (Figure 113). Some people interviewed for this book thought that Hope did not have a driver's license and could only do fieldwork when she

Figure 113. Hope takes a break near the badlands with the Johnson family's 1956 Chevrolet. The photograph is undated, but the license plate indicates the picture was taken no earlier than 1975. Being thrifty, the Johnsons used this car for some 20 years. Spear grass (*Stipa comata*) awns are stuck in the car's grille. If matted together like this near anything hot underneath a car they can create an extreme fire hazard. Image courtesy of the Johnson Family.

could entice others to join her. This is not true: she did in fact have a driver's license, but did not get it until the late 1950s (B. Preston, pers. comm., 2014), though she did not like driving and would often tag along with others on fossil collecting field trips.

Barrie (1977) records that in the summer of 1976 Hope, John Barrie (Director of the Medicine Hat Museum) and several others collected some fossil mammoth material in the Medicine Hat area on behalf of colleague Archie Stalker. They collected a large thoracic vertebra and a section of skull "comprising the sheath of a tusk plus some 14 inches of the tusk itself." Of this event, Stalker recorded in his field notes:

> Sept. 13, 1976—To museum and saw Mr. Barrie and phoned Hope Johnson. Made arrangements to meet at Hargrave Mitchell's farm, to show them the elephant tusk and vertebrae. I have not the time to take them out and pack them, and so was hoping they would collect them for the museum. They were quite impressed, and will collect the tusk and sheath. I took out the vertebrae in good condition and gave it to them.

The mammoth bones were displayed for a time at the museum (Stalker saw them there on June 12, 1977) along with other mammoth material from Babine Lake, British Columbia.

Hope's other fieldwork activities this year are not well known. In May she led a tour into the restricted area of DPP (Johnson, 1976a). As in the preceding several years it seems that her fieldwork was reduced, not only in sites visited, but in the number of days she spent outdoors; this year much of her time was taken up with packing for the move to Redcliff (Johnson, 1976b) and a trip to Europe. Even while on holidays she could not help but look down and find a fossil: part of a Jurassic(?) sea urchin preserved in chert, from gravel at Châtres Cathedral, in France (Johnson, 1977a). Her fossil catalogue book (Johnson, undated 2) records only that she visited White Rock Coulee once this year and a fossil site at Irvine in May (Johnson, 1976b).

She again had her art featured at a major exhibit. Anonymous (1976a) notes that Hope was: "one of nine Albertans among thirty-seven artists successful in competing for inclusion in *Canadian Nature Art '76*." This show was held at the PMA in Edmonton in June and July. She was interested in borrowing birds (presumably taxidermy specimens) as models for painting (Spalding, 1976a, July 12 entry). It is not known if such loans ever took place. She probably exhibited her art at a small show in the Medicine Hat Art Gallery this year, though details are lacking. One of two labels on the back of a *c*. 1975 45 cm × 61 cm oil painting of a great horned owl, written in her hand, indicates that there was a show called *Celebration '76*. Details of her contribution to this exhibit are unknown, but this owl painting was one. In Brooks, this autumn, Hope began teaching oil painting and drawing for artists with little experience (Anonymous, 1976b; Boquist, 1976).

This year on July 2 the Johnsons moved to 436 7 Street (Figure 114) in Redcliff, just west of Medicine Hat—a "ridiculous upheaval!" according to Hope (Johnson, 1977c). It was the year of Olafur's retirement. A new lawn and garden had to be put in, trees, shrubs, and flowers planted. It was quite trying for the Johnsons, now in their sixties. The gardening was hard work with poor, "indifferent soil" and hot, dry winds. The lawn was planted from seed and Hope (Johnson, 1977c) described it as "more trouble than a new baby!" with errant dogs digging it up and children running across it. At least the vegetable garden was a success. Hope would live here for about twenty-six years.

Figure 114. The Johnsons' home in Redcliff, seen here in October, 2015. Photograph by the author.

Before the Johnsons left Ralston, a local reporter interviewed her at length about her life (Smith, 1976). Details from that interview confirmed many of the stories related by Hope herself and others and provided a few clues herein. One would have expected a send-off for the Johnson family's departure from Ralston, but the local newspaper took a summer break soon after the interview so we don't know if such an event took place.

1977

Early this year (or late in 1976) Hope was invited by David Spalding of the PMA to hold a solo show of her paintings. She commented (Johnson, 1977a) that she'd have to ". . . scratch together about 30 paintings or drawings—not too easy." The show ran from November 1978 to January 1979 (see Page 130). Hope continued teaching classes for amateur oil painters in Brooks, Alberta, in the spring (Anonymous, 1977b) but it appears that this was the last time she held classes there.

Hope and another woman taught art classes this spring and summer. Students attended field trips to the Elkwater area in the Cypress Hills for wildflower painting workshops (Anonymous, 1977c).

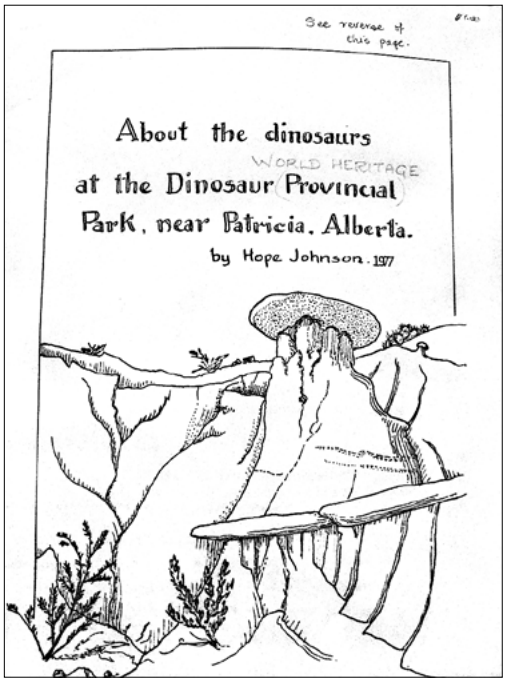


Figure 115. Cover of a small 1977 dinosaur pamphlet curated at the Esplanade Arts and Heritage Centre Archives in Medicine Hat.

She put together a small dinosaur and fossil guide for the younger crowd (Johnson, 1977b). The illustrated guide was simply photocopied and stapled in one corner. It sold for $1.00. Few copies of it exist. Notes penned by Hope on one copy (Figure 115) at the archives in Medicine Hat state that the pamphlet was written in 1977 for children of Medicine Hat School Districts No. 4 and No. 76. If so, the presence of a sale price is interesting. Perhaps some were sold in DPP, too.

Hope is seen in a newspaper photo examining a bison skull found exposed near a bridge in Redcliff (Anonymous, 1977d). She reported local bison skull discoveries in the area to the UC's Michael C. Wilson (Johnson, 1977c).

Another important and continuing role Hope played during this time was as a Board member for the Medicine Hat Museum. They worked hard for five years to develop and have the federal and provincially-funded "National Exhibition Centre" added to the museum grounds to house local and travelling exhibits (Anonymous, 1977a). Many of Hope's art pieces would be exhibited there at various shows. She was also involved in the development of six fossil display cases at the museum, and though the themes for all of these are known, the timing of their public debut is forgotten. One exhibit involved the evolution of horses (see 1979 entry, Page 132), another the evolutionary relationships of unspecified 30 million-year-old animals. Locally collected Ice Age mammal fossils filled a third case; mammoths were the main topic of a fourth. Hope arranged the fossils and artwork for the displays (Figure 119). The Johnson family personal archive includes old pictures of Hope's family history and penned documents. Among these is a picture of Bearpaw Formation fossils from southern Alberta (Figure 116); this may have been the theme for another exhibit case.

Hope did more fieldwork in 1977 than in the preceding years, visiting Petrified Coulee three times. An unusual specimen she found there turned out to be a partial trionychid turtle skull and skull fragments (Gardner, 1992), which she later donated to the RTMP (TMP 1977.030.0001). She visited DPP (R. Chamney, pers. comm., 2015) only a few times this year; the specific dates are unknown. Whenever she announced a visit the Park staff would "roll out the red carpet" (R. Chamney, pers. comm., 2015). She also made trips to White Rock Coulee and badlands on the Bow River.

She visited a new locality at Murray Lake, near the town of Seven Persons, Alberta, but did not find much there. In early to mid-August she explored a site near Irvine and recovered sixteen reworked

and presumably unassociated Late Cretaceous mosasaur vertebrae from a gravel wash. The fossils were found in Pleistocene "Saskatchewan gravel stones." Visits to the Lost River badlands and nearby Canal Creek were made in August. There one of her companions found an 11.5 cm-long, rooted tyrannosaur tooth; Hope commented, "one of the largest I've ever seen and in good condition." At McCracken's Mine, southwest of Ralston, she found the remains of a ceratopsian skeleton in the Foremost Formation. Hope (Johnson, undated 2, p. 99) states that it was destroyed by erosion and could only be recovered by "removal and casting," presumed to mean removal of the eroded bones and salvaging data by painting latex rubber into the bone moulds, letting it cure fully, then peeling out the latex bone casts. The author does not believe this find was ever pursued by professionals. Hope collected a few scraps from the specimen (cervical neural spines and a partial phalanx).

Later in the year she explored outcrops near Chinook, Montana, in the company of fellow collectors Leroy and Rena Anderson, with whom she traded fossils. She collected a partial bison skull in southern Alberta that went to the "Officer's Club," presumably at Suffield, revealing that despite Olafur's retirement, she was still supplying them with conversation pieces.

Figure 116. A variety of Bearpaw Formation marine fossils arranged in an unidentified museum display, possibly one Hope set up for the Medicine Hat Museum and Art Gallery. Image courtesy of the Johnson family.

Hope attended the Society of Vertebrate Paleontology meetings in Los Angeles on November 10–12. At this event she made contact with vertebrate palaeontologist William J. Morris (1923–2000) from the Natural History Museum of Los Angeles County.

1978

Early in the summer Hope met an entourage, including high-ranking Lieutenant General Sir Peter Hudson (1923–2000), the Deputy Commander-in-Chief of the United Kingdom Land Forces. She gave them a nature hike and badlands tour for which they expressed appreciation.

Archie Stalker's field notes for June 9 record that Hope joined him and Dr. René W. Barendregt at the H. Frisch site, northeast of Irvine, where Hope collected a few bones. The trio continued to Golden Valley Bluff where Hope made an outcrop sketch for them.

Hope and some family members visited Philip J. Currie and crew at two active fossil digs at Sandy Point, Alberta, in early August. Currie visited with Hope on August 9 to discuss an upcoming art show at the PMA (see below). At this point, she was still visiting DPP on a regular basis (R. Chamney, pers. comm., 2015).

She participated in the fifth AMQUA (American Quaternary Association) conference and field trip (August 31–September 3) and attended this and/or another conference with field trips (the 1985 CANQUA (Canadian Quaternary Association) courtesy of Archie Stalker, and paid for entirely by grateful conference organizers. She did not make a formal presentation at the AMQUA meeting but drew a field trip guidebook figure and was listed as one of eight co-leaders for the pre-conference field trip, examining Quaternary-aged deposits and landforms between Medicine Hat and Edmonton. Stalker's

field notes for August 31 state that at the AMQUA meeting, "René (Barendregt) and Hope's slides went over very well," indicating that she contributed images for a presentation at the conference or gave an after-supper slideshow. Another conference she attended was the Society of Vertebrate Paleontology (SVP) meetings, held in Toronto this year. She never presented at the SVP meetings, but educated herself by attending the talks and networking with attendees, both new and old.

From November 1978 to late January 1979, a gallery exhibition of Hope's artwork, entitled "Images from Alberta's Southeast" was featured at the PMA in Edmonton. This was Hope's first formal art showing, outside of various locations in southern Alberta, and her first solo exhibition. It was yet another important recognition of her work. The significance of the exhibition is impressive, considering that during this period the only other art shows held at the PMA featured the likes of Robert Bateman, Charles Knight and Carl Rungius (D. Spalding, pers. comm., 2016).

Figure 117. Hope Johnson, pencil self-portrait, 1978. Image courtesy of the Redcliff Museum, Redcliff.

Philip J. Currie delivered the art pieces from her home in Redcliff to the gallery on October 6, on his way home from fieldwork (Currie, 1978); David Spalding hand-delivered them back to her in March, 1979 (D. Spalding, pers. comm., 2018). The gallery featured paintings of badlands, flowers, portraits of two First Nations men, some figures from the Johnson and Storer (1974) fossil identification guide (Fischer, 1978) and even a self-portrait (Figure 117). Cochran (1978) features a picture of a prickly pear cactus drawn by Hope and shown in the exhibit and notes that her rock outcrops are painted in bold, broad forms, while plants and other natural features such as flowers, grasses and rocks are more highly detailed, usually placed in the bottom left quarter region of the painting.

The show was facilitated by David A.E. Spalding, then Head Curator of Natural History for the PMA. Of twenty-five pieces exhibited, twelve were of badlands scenery (Hrapko, 1978). Some of her fossil finds were also incorporated into the show. Currie (1978) notes picking up four or five fossils from her collection that needed further preparation before they went on display. At the same time he received Hope's donation of some fossils for the PMA vertebrate palaeontology collections. She was always good this way and encouraged it in others—donating fossils if a researcher felt they should be curated into their museum's collections.

Hope took part in an "Arts in Action" show held by members of the Hat Art Club (Anonymous, 1978b) from July 7 to August 6, at the National Exhibition Centre in Medicine Hat. It featured artwork by Hope and three other local artists in an exhibit cleverly titled "Four drawn from the Hat" (Anonymous, 1978c; Bilodeau, 1978; Cochran, 1978; Fischer, 1978). She was also co-leader of a group of amateur naturalists on a trip to study and paint wildflowers and landscapes (Anonymous, 1978a) and taught pen-and-ink drawing in the fall (Anonymous, 1978d).

DPP planners held a variety of meetings and think-tanks in the 1970s, planning for ways to improve and expand the Park (Alberta Recreation Parks and Wildlife, 1978; Anonymous, 1979f). It is known that Hope assisted with these efforts around 1973, but documents stating her involvement could not be located. Big changes were about to happen. One decision made in the 1978 government study was to dismantle the three fossil exhibit buildings that Hope had worked on in the late 1960s and early 1970s. The buildings were to be replaced with something more permanent, but that idea never came to fruition. However, later plans (Alberta Recreation and Parks, 1983 among others) resulted in the Field Station that exists at the Park today. The fossils and exhibit signs in the small exhibit buildings were removed in 1980, displays temporarily upgraded, and the buildings themselves removed a few years later. It is unfortunate that they were eliminated, but other Park infrastructure changes, such as a revised main entrance, necessitated their removal. The fossils contained in the buildings, as well as all the other Park fossil collections which at that time were stored or exhibited at seven different locations in the Park, went to the PMA in Edmonton, later to be transferred to the RTMP in Drumheller. The Park was also under consideration for a UNESCO World Heritage Site designation at this time (Anonymous, 1979e; 1982b). Hope would have been an invaluable resource during these later phases, but it does not appear that her vast experience was put to use in promoting the prestigious designation, other than citations of some of her earlier research relevant to the Park.

Figure 118. Sixty-two-year-old Hope with butte evening primrose (*Oenothera caespitosa*) in the Cypress Hills. Image courtesy of the Johnson family.

In the 1978 field season she visited Petrified Coulee in May and found a juvenile hadrosaur metatarsal, 5.5 cm long. In early June she hiked the Cypress Hills with Ollie and members of the Historical Society of Medicine Hat and District. They stopped near Irvine and found some butte evening primrose (*Oenothera caespitosa*) blossoms. Someone photographed them with Hope (Figure 118). Two dried plant specimens collected by her this year, at Petrified Coulee and Redcliff, were donated to the UC herbarium (see Page 188).

July was a busy month for fieldwork, with exploration near DPP on July 5 and a number of visits to the Bow River badlands and Petrified Coulee. She visited Philip J. Currie and his PMA crew at Sandy Point on August 1 and was given a tour of two active quarries (Currie, 1978). Archie Stalker's field notes from August 9 show that Hope and Tom A. Willock (of the Medicine Hat Museum) helped complete the collection of an elephant (mammoth) tusk; this may be ROM 44912. She also visited White Rock Coulee in September. Dinosaur Park Formation outcrops near the Mattson and McKinley Ranches were examined in early November. The Rose's Ranch locality was visited sometime this year, too.

1979

In the 1965 section (Page 81) it was noted that Hope occasionally found First Nations stone artifacts and other natural history items of interest. A recent search by archaeologist Bob Dawe (pers. comm., 2016) disclosed the following:

> . . . she did donate a single item, on January 12, 1979, rather unspectacular but a donation nonetheless. A split chert pebble core from the Suffield area (Royal Alberta Museum accession no. H70-020). To her credit, though, the artifact she donated would not be recognized as something significant by the vast majority of people, particularly at the time she collected it, when there was only a handful of people in the province that could identify it as an artifact. So clearly, she had a keen eye.

In April and May of 1979 Hope developed and assembled an exhibit (Figure 119) for the Medicine Hat Museum on the evolution of fossil horses, using casts and original specimens on loan from the ROM. She drafted life reconstructions of the extinct species and made specimen labels. She received praise for the small but very informative exhibit in a June 25 letter from Dick Harington of the NMC in Ottawa.

Figure 119. Image from a 35 mm slide showing the "Evolution of the Horse" exhibit that opened at the Medicine Hat Museum and Art Gallery early in 1979. Image courtesy of the Johnson family.

Late in 1979 the city of Medicine Hat appointed Hope and two others to three-year Advisory Board terms for the Museum and Exhibit Centre (Anonymous, 1979d). Her work in this capacity is not known; the author could find no documentation describing her role. This year she drafted a detailed map showing part of the city and the location of the museum, for a tourist brochure.

Her palaeontological fieldwork activities for 1979 are not well known, but she visited White Rock Coulee in June and found rare fossil mammal material there. She also discovered an unidentified dinosaur skeleton, consisting of nine eroded neural spines. She felt that more was there, but that either the head or tail was missing. The site is not known to have been investigated. In July Hope led a small group to visit a dig site (Frisch Site) being worked by Archie Stalker and others, then gave the participants a

tour of the museum in Medicine Hat. She visited Petrified Coulee on July 28 and joined the PMA crew on August 22, prospecting for microvertebrate fossils with Jane Danis (*née* Colwell) (Currie, 1979: Figure 120). Three days later she joined the PMA crew at Sandy Point, northeast of Medicine Hat, for their end-of-the-field-season party.

Large-scale excavations for Pleistocene vertebrate material at the Maser-Frisch site near Irvine were conducted by Stalker, Churcher and others this summer and in 1984; see Barendregt *et al*., 1988. Discoveries of possible imperial mammoth and large felid material in the area no doubt inspired a painting Hope completed this year (Figure 179).

Philip J. Currie paid her a visit at home on September 18—yet another of a long series of visits by palaeontologists to her homes in Ralston and Redcliff to see her fossil collection and to acquire specimens for their museums. He was on his way to examine dinosaur footprints exposed on the South Saskatchewan River near the Redcliff Golf Course. It is unclear from his notes if Hope accompanied him.

Figure 120. Hope and Jane Danis in 1979, working in the restricted area of DPP. Images courtesy of Bill Lawson, Redcliff.

A particularly rare find that Hope made was accessioned into PMA collections this year: a pair of fused dentaries (TMP 1979.008.0622) of a caenagnathid dinosaur she found in DPP, the smallest ever found in Alberta (Currie *et al*., 1993, p. 2258). The specimen is so strange in appearance that Hope tentatively identified them as turtle jaws. In a letter to the author (Johnson, 1996) she recalled that the specimen "was completely baffling to me at the time, but I saw that it was part of a central structure." They are numbered as "526" in her hand, but no such entry exists in a chronological listing in her fossil catalogue books. There is evidence therein to suggest she found them in the early 1960s. They are figured in Alberta Palaeontological Society (2009, p. 123). Since their original description they have reappeared in the scientific literature (Lamanna *et al*., 2014). Currie (1980b, p. 18) notes that Hope and fellow collector Shirley Lawson (1944–2012; Figure 90) donated a number of significant finds this year, including a "distinctive" ungual from an otherwise unidentified small theropod.

The DPP fossil collection that she had largely assembled on her own in the late 1960s and early 1970s was taken to Edmonton late in the summer of 1979. The collection had been stored "in seven build-

ings" within DPP (Walper, 1980). Philip J. Currie's field notes (Currie, 1979) for September 19 relate the activity: ". . . these specimens had accumulated over the years in the garage and behind the displays in the Park office." The fossils were processed (minor preparation, repair, identification and cataloguing) in early 1980 and eventually became part of the TMP 1979.014 series. A number were selected to create a smaller and more manageable comparative reference collection for seasonal Parks and RTMP staff volunteers, visiting palaeontologists and students. They were repatriated to DPP under the TMP 1980.008 series.

Slowing down? 1980 to 2000

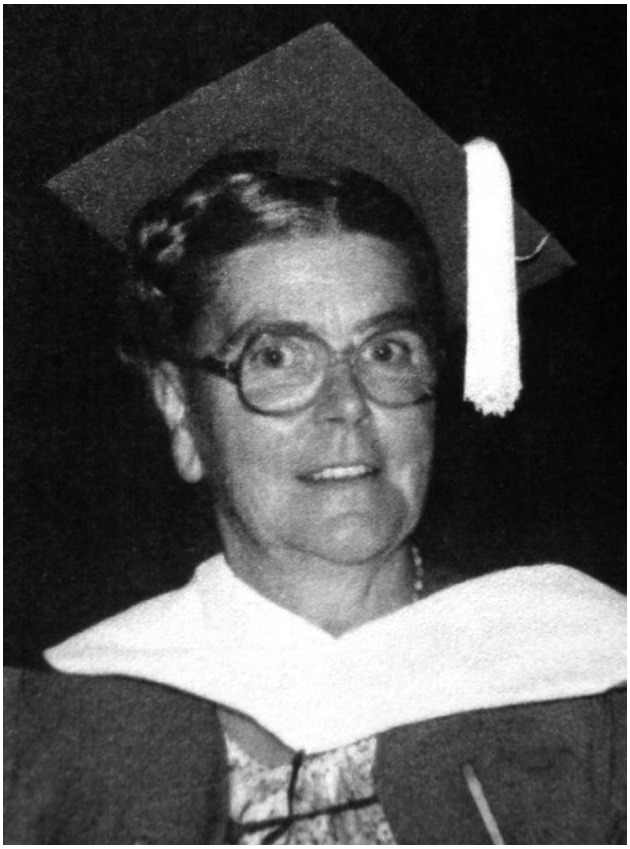


Figure 121. Hope in her University graduation gown. Image courtesy of Tom A. Willock.

This period marks a quieter time in Hope's life, palaeontologically speaking, and a time of well-deserved and important recognition by her peers. Certainly she continued to do fieldwork and donated fossils to various institutions. But in 1980 she was sixty-four years old and becoming less active in collecting fossils, though she still actively hiked in the badlands. Many of her field companions were aging, too, and becoming less interested in travelling over the rough terrain. Some had passed away. But Hope stayed active in other ways.

1980

Hope invited people to bring her photographs they had taken of local wildflowers, for identification. She wrote several sections for an Alberta nature guide, covering her favourite hiking and fossil exploring grounds: the Red Rock Coulee and Irvine areas (Anonymous, 20140; Johnson, 1980a-b). She attended the UNESCO World Heritage site designation celebrations at Dinosaur Provincial Park (DPP) on June 19, 1980.

The numerous exhibits Hope developed in the three unheated "museum buildings" at DPP were removed this summer and exhibits therein were "modified" (according to the author's notes) but exactly what changes were made inside the buildings is long forgotten. Some of the fossils were replaced by high-quality casts to keep the originals safe from vandalism, theft and the vagaries of climate. This removal, upgrading with casts, and other salvage work on the endangered Park fossil exhibits by the Royal Tyrrell Museum of Palaeontology (RTMP) led to erroneous claims that the Museum was pillaging the Park of its bones (Turner, 1984).

Hope's field activities this year are not well known. Two trips are confirmed. In the Lost River badlands a piece of dinosaur egg shell was collected (Johnson, undated 3), one of the first recognized in the province. She examined outcrops along the Canada/United States border near Rudyard, Montana, late in August and met up with Jack Horner (1946–) and his good friend Robert "Bob" Makela (1940–1987).

Dr. Peter Dodson sampled Dinosaur Park Formation fossil vertebrate microsites in DPP between 1980 and 1983. In 1984 microsites from the same formation further south in Alberta were examined and work upriver in lower Scollard Formation beds began. Hope assisted with some of this work. Specific details of her involvement are lacking, but she is known to have worked with Dodson in 1982. Her field assistance in showing him some sites is acknowledged in a report to the National Geographic Society (Dodson, 1985, p. 19). Dodson's project did not continue beyond 1984.

1981

This was a year of triumph and tragedy for Hope. Triumph, in that she was recognized for all her hard work and volunteerism in science and other fields and was awarded a well-earned honorary doctorate degree (LL.D.) from the University of Lethbridge, on May 16 (Anonymous, 1981c–f; Figure 121). Her friend, J.-P. Christopher Jackson, from the University of Lethbridge conceived of the idea that Hope should be formally recognized by the academic community and awarded an honorary doctorate from his University. Jackson gathered an extensive amount of supporting evidence including letters of support from University faculty, administrators, community members and Earth science professionals such as Philip J. Currie. In a congratulatory letter, Dale Russell commented that: "You have served your province well, and I wish there were more of you."

Before she received this honour, some introductory remarks were made by Dr. Jackson, Associate Professor in the Department of Music. There are some minor historical errors (not corrected here, so they differ from the larger text), but the introduction is given below, as spoken:

> Madam Chancellor,
>
> When I came to Alberta five years ago, my avocational interests in palaeontology and botany soon led me to an awareness of the accomplishments of Alberta naturalist and paleontologist Hope Johnson. It was Hope's son Lawrence Johnson (himself a gold medalist from this University), who provided me with this opportunity to meet this most extraordinary woman.
>
> Hope Johnson truly exemplifies the all too seldom found "expert amateur." She is eminently an artist scientist who practices many skills, most of which involve aesthetic as well as scientific understanding. As such, Hope Johnson is a superb example of the following excerpt from the University of Lethbridge statement of philosophy:
>
> ". . . the University regards learning as an end in itself, not merely as a means to material end. Its primary aims are to foster the spirit of free inquiry and the critical interpretation of ideas."
>
> Hope Johnson has consistently demonstrated her love of knowledge and has maintained a quality of work which has earned her international respect from professional and academic communities and the public at large.
>
> Hope Johnson began collecting fossils in the Ralston area in 1949 and since that time has collected throughout southern Alberta, frequently in direct collaboration with field teams from the National Museum of Canada, the Royal Ontario Museum, the Geological Survey of Canada, the Alberta Provincial Museum, the universities of Alberta, Toronto, Pennsylvania, and Paris. She is responsible for many unique discoveries for Canada.
>
> In the 1960s she co-authored a guidebook of Cretaceous vertebrate fossils published by the Provincial Museum of Alberta. In 1964 she was elected to membership to the Society of Vertebrate Paleontology.
>
> Mrs. Johnson has served as Curator and Park paleontologist at Dinosaur Provincial Park. Her work in this context has resulted in the Park being named a World Heritage Park by UNESCO. In addition, she has been closely connected with the Medicine Hat Museum for several years and contributed significantly to its development.
>
> Dale Russell, eminent palaeontologist and Chief of the Palaeobiology Division of the National Museums of Canada, expressed Hope Johnson's achievements best when he wrote:

"Through the example of her enthusiasm, and her organizational and artistic talents, Mrs. Johnson has enabled the people of the prairies to better appreciate both the reality and the richness of an organic world which was once typical of western Canada. She has amply earned our respect and admiration."

Madam Chancellor, in the name of the Senate of the University, I ask you to confer on Barbara Hope Johnson, the degree of Doctor of Laws, *Honoris Causa*, in recognition of a most distinguished naturalist, artist and palaeontologist."

This was a well-earned and huge recognition for Hope. But tragedy soon followed, with Olafur beginning to forget things and then being diagnosed with Alzheimer's disease. An increasing proportion of her time was spent caring for him over the next six years, although in a candid interview later in life she noted he was not as badly affected as some of the worst cases. She spent this time reading many medical documents trying to understand the disease and how to better care for Ollie, whose memory loss Hope notes was "profound," and ". . . made life difficult for us both." Toward the end he could not dress himself or even remember common words (Campbell, 1996). She was supported by the local chapter of the Alzheimer's Society during this difficult time and served as its President, on the urging of others, from 1981 until 1992 (though she is recorded as President as late as 1995; see Campbell, 1995), and was a member of the Medicine Hat Alzheimer's Society, formed in late 1985 (Stubbs, 1986). In 1991 Hope delivered at least one public presentation on Alzheimer's disease to Brooks area residents (Buday, 1991). A news story (Stubbs, 1986) was published on Hope's and Fran Hart's work supporting others married to people with Alzheimer's and its debilitating effects on all involved. Her public talks on the disease were done in conjunction with the Southeastern Health Unit. She would volunteer with the Alzheimer's Society for about eleven years.

On the heels of her honorary degree she joined the PMA field crew in DPP in June, prospecting in the core area of the preserve and revisited one of her old hunting grounds, the *Styracosaurus* Bonebed 42 (G. Danis, 1981; Strong, 1981).

In the late summer of 1981 Hope and Ollie visited family in Manitoba and Ottawa. While in Ottawa she also visited Charles M. Sternberg, partly on behalf of the PMA's Jane Danis who was compiling historical information on the early dinosaur collectors in Alberta. Jane had questions for Sternberg. His health was failing rapidly and it seems quite likely that Hope was the last person with a vertebrate palaeontology background to see him alive. The visit was a sad one and Hope recorded it in detail in a letter to J. Danis (Johnson, 1981l), shared here [author's correction and comments in square brackets]:

While in Ottawa, my daughter Barbara and I went to see dear old Dr. C.M. Sternberg, who was a patient at the Glebe Centre, just opposite Lansdowne Park. Barbara is a nurse and through a friend she had heard he was in good health, but when I telephoned and enquired about him, they told me he was failing. This was indeed the case . . . on the 24th August (I think it was the date—it may have [been] a day later), we got to the Glebe Centre . . . We found the old man sitting in a wheelchair in the hallway near his room, with a small carton of chocolate milk. He was dozing, and it was with difficulty we roused him. He had become extremely deaf—he was a little slow of hearing years ago, but no more than I am, I think—and I had to shout right into his ear. At that, he was very slow to respond. One eye—his left—remained closed the whole time.

At first he couldn't make out anything I said, but after awhile he became more alert. Perhaps he was sleeping when we arrived. It seems possible. He couldn't remember me, although I am certain that had he been able to see, he would have. Asking him questions was difficult indeed, and seemed an imposition. However Barbara said we should persist—she deals a great deal with the elderly and impaired (in one way & another!)—and I got a firm answer on whether he knew Roy Chapman Andrews—"Yes. (pause) I knew him very well." Then I asked him if Chapman ever came to Alberta. After a long pause he said "I can't remember." I also drew a blank on "Montana" Johnson [cook and field assistant to Barnum Brown in Alberta, 1913–1915]—but he said he remembered the name, but could remember nothing about him. He was obviously discouraged with himself, because at one point, when I enquired about something else, he said very firmly—"No. I don't know, and if I ever did,

> I've forgotten!" He didn't come out with anything we don't already know. I felt that we were disturbing him, but then, he took my hand, and that of Barbara (he had never met her, nor knew of her existence), squeezed them, and said, "I want to thank you two girls (sic) for coming to see me." Poor old dear, we both felt so sorry for him. He was quite recognizable you know—I knew him at once when we caught sight of him at the end of the hall, but he was a shadow of his former self.

A few weeks later Hope was at the University of Lethbridge and added:

> On Saturday morning, Dr. Bob Hall, our host, told me that Dr. C.M. had died. We had been talking about him at supper the night before, and I told them how much he had failed in the past 7 or 8 months, since when Barbara's friend had been working at Glebe Centre, he had been well and alert. The attendants at the Centre told us the same story. Consequently, on hearing of his death, I told Liz and Bob Hall I was really quite relieved to hear he had gone—his life must have been just frustration . . . I am glad we went. I think the old man was lonely. He grasped well that I had come from Alberta etc., but his certain memories seemed all to pertain to the distant past. At one point, he said "I haven't been very well since I was a young man." By my reckoning, this was probably all of five years ago!—maybe even less.

A newspaper article records that in the fall of 1981 Hope agreed to be a Team Captain for the United Way, overseeing forty door-to-door canvassers (Harbron, 1981). The author of the article further noted that Hope "has been a community activist in the Medicine Hat area for many years."

Hope and Ollie are recognized in a good number of newspaper articles for making the most generous of donations to the Salvation Army's Santa Claus Fund for needy families during the holidays, even though both are remembered as being thrifty. In his eulogy, son Lawrence recalled Hope donating money to many charities—so many, in fact, that she began to lose track of which she had donated to. She made a particularly large financial donation late in her life (see the 2010 section, Page 162). Originals, and more frequently, reproduction prints of Hope's artwork were often donated for charitable auctions by local art shops.

In 1981 came the release of *Down the Years at Elkwater* with author Hope Michael (Michael and Johnson, 1981). It is usually stated that Hope Johnson "helped" Michael, but examination of the final publication reveals that Hope Johnson actually did most of the work. She wrote eleven chapters of the fifteen in the book (Johnson, 1981a-k), drafted two detailed maps and contributed several pages of wildflower illustrations. It is recorded that she helped to rewrite some of Michael's contributions for the book, too. In 1988 she received word that the book had been selected, among about 130 other titles, by Washington University in St. Louis, as one to be studied by the Northern Plains Culture History Project with regard to the settling of the Canadian and American West (Montana, Alberta, and Saskatchewan; see Bennett and Kohl, 1995).

This summer a dedication cairn with a plaque was unveiled, commemorating a historical river crossing site in Medicine Hat. This work was done by the Historical Society of Medicine Hat and District, with which Hope was involved. Hope helped search for suitable glacial erratics for the cairn's construction. She designed, drew and lettered the cover art for *River Crossings 1883–1980*, a small document related to the history of bridges and early modes of transportation across the South Saskatchewan River in Medicine Hat. Though she is not credited for this work, she penned her name on the cover art, "B E H Johnson" (Milne, 1981).

Hope joined the Hat Art Club in the summer of 1981 and stated that the club "didn't seem as serious-minded as the Ralston one." Through the Hat Art Club she won first prize at an art show for a watercolour bought by the Medicine Hat News. The piece was a badlands scene on the Krause Ranch (where vertebrate palaeontologist David Krause grew up) near Hilda, Alberta. Hope further added in her notes this year that she found watercolours easier to use, "because my back is arthritic."

Figure 122. *In situ* extraformational clasts seen at Petrified Coulee. Such rock deposits are dominated by cherts and quartzites. Sometimes the clasts contain Palaeozoic fossils such as solitary and colonial corals. No glacially-derived granites, gneisses, limestones, or schists here! Photograph by the author.

A visit to Petrified Coulee was made in July. In addition to fossils, she was intrigued by *in situ* extraformational clasts—rounded river rocks from older deposits washed far downstream during the time of the dinosaurs and entrained in the Late Cretaceous sands. She noted stringer deposits of quartz pebbles, greenish, black [chert] and white especially, and correctly notes ". . . these are a component of the Cret[aceous] deposits there." The author saw such deposits in abundance at the base of the Dinosaur Park Formation exposures during a fall, 2014 visit to Petrified Coulee (Figure 122). Hope led a Medicine Hat Museum group field trip to Petrified Coulee again at the end of October.

Hope also made a valuable observation at Petrified Coulee—the presence of hatchling-sized hadrosaur material. She wrote the author (Johnson, 1996) about how a fellow collector had found a 40 mm-long hadrosaur humerus and she had seen two hatchling hadrosaur dentaries (their completeness unstated), both specimens about 35 mm long, one with nine tooth grooves preserved and the other with eleven. She concluded, "Their nesting area must have been very close by," a conclusion also reached by the author regarding hatchling-sized material in DPP (Tanke and Brett-Surman, 2001). In researching this book the author visited Petrified Coulee for the first time and within minutes found the end of a tibia of a very small hadrosaur, suggesting this may prove to be an important locality for baby hadrosaur material.

1982

Anonymous (1982a) records that Hope and others conducted up to a dozen nature and human history hikes across southern Alberta and Saskatchewan for anyone interested in joining, but examination of her fossil catalogue (Johnson, undated 3) demonstrates a further decline in badlands hiking and fossil collecting from this year onward. She offers some experience-based safety tips on hiking in the Medicine Hat area in Appelbe (1982). A brochure shows that Ollie and Hope were judges at the 7th Annual Regional Science Fair at Medicine Hat College. Their participation in other local science fairs is unknown.

About this time Hope was a volunteer naturalist at Police Point Interpretive Centre in Medicine Hat. She did this on Sundays, though the duration of her volunteer work there is apparently unrecorded.

Hope was able to use her influence and contacts at the RTMP to get the son of her collecting friends, Bill and Shirley Lawson, a brief summer job. Al Lawson was hired for one month and first worked at the *Centrosaurus* bonebed (Quarry 143) in DPP and was then transferred to the crew working in the Crowsnest Pass area collecting a scattered *Tyrannosaurus* skeleton (TMP 1981.006.0001; "Black

Figure 123. A 35 mm slide image of the Quarry 143 *Centrosaurus* bonebed, here seen cleaned off for VIP tours in mid-June, 1982. Image courtesy of the Johnson family.

Beauty") from a large quarry in hard, white, sandstone. The hot and arduous work gave him "many lasting and rewarding memories" (A. Lawson, pers. comm., 2015).

She visited Writing-On-Stone Provincial Park in August. Hope visited this place, with its unusual, eroded Late Cretaceous outcrops and First Nations petroglyphs (Anonymous, 1954) from time to time, for artistic purposes. A single visit to the Irvine badlands in August is recorded, though during this time she would also occasionally join the PMA/RTMP crews working in DPP. It is known that she did a little work with Dr. Peter Dodson there in August; she visited the Quarry 143 *Centrosaurus* bonebed, snapping a picture of the bones all carefully exposed and glued for an upcoming VIP tour (Figure 123). Though Hope enjoyed exploring multi-generic bonebeds, she recognized the scientific importance of ceratopsian mass mortality bonebeds, having been involved in the discovery of a *Styracosaurus* bonebed in 1966 (see Page 82).

Now in her mid-sixties, she was still able to hike the badlands along with the museum crews without trouble. RTMP technical staff recalled being assigned to her as assistants on occasion and in a way were her "caregivers," given her age, legendary status, and concerns that she could suffer a mishap; but those worries were wholly unfounded: she got along quite well, in fact, often leading the way on hikes.

1983

In January Hope received a glowing letter of thanks from the Medicine Hat Historical and Museum Foundation as she had made dinosaur puppets for children's school programmes held there. This is but one example of Hope going above and beyond when it came to children's education. The long-awaited cultural centre opened in town (Lynch, 1983), a project Hope helped develop.

In what turned out to be a busy year, she teamed up with seven other local women artists to create a mural, *Odyssey*, a gift to the people of Medicine Hat on the occasion of the city's centennial (Anonymous, 1983c). The mural was described as a "kaleidoscopic vision of mankind." The current location of this mural is a little-used hallway in the lower level of the Medicine Hat Public Library. The mural measures 22 ft, 10 in (6.95 m) by 4 ft (1.2 m) and is painted on two 4 ft × 8 ft (1.2 m × 2.4 m) masonite boards and other irregularly-shaped "floating" boards which form the ends of the mural. Several peculiar things are noted. Despite Hope being an artistic contributor, her input on it seems reduced—her name appears on artwork in only a few places, whereas the other artists signed their names multiple times. The mural includes a palaeontological theme, including some dinosaurs; yet curiously Hope did not paint all of that section and what she did paint was of a surprisingly primitive style, with respect to anatomy and use of paint, reminiscent of her work in the 1950s. Examples include Devonian fishes *Pteraspis* and *Bothriolepis* (Figure 195), Egyptology (Figure 194) and samples of script in global languages.

In the summer, Hope contributed some western-themed paintings for a show entitled Western Interpretations, held at a ranch near Medicine Hat (Appelbe, 1983b). In November her art work was displayed at the Police Point Interpretive Centre (Anonymous, 1983l) and early December saw Hope giving a lecture on Christmas card art (Anonymous, 1983k) and classes on Christmas card design (Anonymous, 1983j).

On March 3 of this year she was presented with the "1982 Civic Recognition Award" and honoured at the award banquet for her long-term volunteerism (about forty years), historical and naturalist work in the city of Medicine Hat and region (Caldwell, 1985b). Hope's picture and a story appeared in a 1983 book of thirty-five honorary doctorate recipients awarded over the previous sixteen years by the University of Lethbridge (Anonymous, 1983b). In May she received a thank-you letter from Margaret Wooding School in Redcliff, where she made a presentation to students as part of Career Week.

RTMP crews were doing summer fieldwork in southeastern Alberta, looking for any specimens that might be suitable for the galleries of the new RTMP, slated to open in a couple of years. Hope suggested an area worthy of museum investigation and several mosasaur specimens and a substantial portion of a hadrosaur in marine sediments were soon recovered there (Anonymous, 1983d–i; Currie, 1983; Harbron, 1983a–c). One, a *Mosasaurus conodon* (TMP 1983.064.0001) in a hard concretion was prepared in relief, but it lacked the front part of the body and was not used for exhibit purposes. A second specimen, a *Plioplatecarpus* partial skull (TMP 1983.064.0002) was also collected and prepared for research purposes only (Tanke and Mitchell, in prep).

She made a trip to Irvine at the end of August and explored the Eagle Butte area in September (Johnson, undated 3).

Hope, commenting on geography of the Cypress Hills, is called the "district geologist" in Strickland (1983, p. 22). Hope was not only useful in recommending field areas she had seen many times before, but was also good as a liaison person, as she knew most of the landowners quite well and had established a good rapport with many.

As if palaeontology, artwork and other activities were not enough to keep her busy, Hope was also active in a number of other natural history research fields, especially rare plant conservation (Alberta Sustainable Resource Development, 2004). She donated botanical specimens (types and numbers unknown) to several provincial academic institutions. In 1983 her drawing of the Butte Evening Primrose was featured in the local newspaper (Appelbe, 1983a) with an announcement of the forthcoming release of her book on prairie plants. This was the plant identification guide, *Prairie Plants of Southeast Alberta* (Johnson, 1983), fully illustrated by her. It was no small undertaking, with 125 pages of text and 356 plant drawings, of which 123 were primary drawings and 233 secondary drawings or sketches. Hope wrote the text, commenting on the plants' habitat, appearance, potential poisonous properties, historical comments on the naming of the species, difficulties in rendering drawings, *etc.* While the text is hers, the script typeface, sometimes credited to her, is not. She had some serious issues with the typeface regarding legibility and its lack of certain punctuation marks. The book went on sale for $9.95 in early 1984. It must have sold well, as Caldwell (1985b) records that 100 copies of the book were originally printed by the publisher, but Hope had another 500 produced by a private printer in May, 1984. Another printing of unknown quantity was undertaken in December, 2003. It is not known who paid for these reprints or if Hope received any royalties or payments from them. Some people are lucky enough to own special personalized editions where Hope thoughtfully hand coloured some of the illustrations with watercolours. Several such copies are known to exist.

1984

In mid-April Hope donated artwork for a silent auction, proceeds of which went to the Medicine Hat Museum and Art Gallery building fund (Anonymous, 1984a).

On May 25, Hope and Olafur attended a convocation reception at the University of Lethbridge for her friend and professional colleague, Archie Stalker, who received an honorary doctorate.

In June she visited Rufus Churcher and crew at the Pleistocene "Heather Valley Site" near Irvine, Alberta. Barendregt *et al.* (1988) described the site; the authors thanked Hope for her "interest and encouragement" during the project. Large-scale excavations for vertebrate material were conducted there in 1979 and 1984. Churcher (1984) notes various visitors, but Hope is not listed among them. She certainly visited the site, as she snapped a picture of the field crew.

During this time she was corresponding with Dr. Karl M. Waage (1915–1999) from Yale University about ammonites and squids. In an April 12 letter to Hope, Waage compliments her: "I would like to see the interest and obvious talents you possess in all my graduate students."

She visited the Irvine badlands in August and again in September, at which time she was rewarded with a dinosaur skin impression fossil. Red Rock Coulee, one of her favourite localities, was visited at least once, in October.

On Hope's recommendation the RTMP visited Late Cretaceous marine Bearpaw Formation outcrops on the McKinley Ranch area, east of Manyberries, Alberta, where they found and recovered a partial mosasaur skeleton (TMP 1984.102.0076).

1985

A 1985 art show titled "Prairie Plants of Southeast Alberta" exhibited thirty of Hope's pen-and-ink drawings; the show ran from August 10 to September 8 in Medicine Hat (Anonymous, 1985a; Caldwell, 1985b). These were no doubt from her 1983 prairie plant book (Johnson, 1983). At least one image was made into a postcard, distributed by the Medicine Hat Museum and Art Gallery. Her show ran concurrently with a Manitoba Museum of Man and Nature exhibit, "Collecting our Natural Heritage."

Hope helped form the Grasslands Naturalists group this year (Lindores, 1997). The stated objectives of the group, which focuses on the prairie environs of southeastern Alberta, include the following:

- To encourage study and protection of the natural world.
- To provide educational opportunities such as lectures, field trips and publications.
- To encourage the preservation of natural habitat and aesthetic natural features.
- To assist in the collection and provision of records of species for statistical and educational purposes.
- To act as stewards of the environment.

In co-founding this group it is possible that with her advancing age, she realized that her days of active hiking in the badlands were nearing an end so she chose a field activity that had always interested her, but required less "up and down" exploration.

Also in 1985, the Alberta Foundation for the Arts purchased one of Hope's watercolours for its collection of Alberta artists' works (Alberta Foundation for the Arts, 2014; Anonymous, 1985d). She was Vice-President of the Hat Art Club in 1985 (Caldwell, 1985a). Hope was also featured in an exhibit entitled "One Hundred Years of Alberta Women," chronicling 200 influential Alberta women. The exhibit ran from December 14 to January 6 at the Medicine Hat Museum Art Gallery (Caldwell, 1985c).

This year she began a fourteen-year run of teaching oil painting for seniors (aged fifty-plus), two courses of ten sessions each. Classes were "very popular" and consisted of ten to twelve students. She was paid for this work which ended in the fall of 1999. Classes were held in the Veiner Centre, Strathcona Centre and the Pavilion at Strathcona Park, Medicine Hat.

In 1985 Hope donated money to the University of Lethbridge's "Leave a Legacy" project. She and other honorary doctorate recipients were given the opportunity to be the first to donate toward a $2,000 investment certificate to be buried in a time capsule. When opened in 2067 the investment was expected to be worth some $2 million, based on the 1985 interest rates (Anonymous, 1985b, c).

From August 18 to 28 Hope participated in the inaugural CANQUA (Canadian Quaternary Association) conference in Lethbridge and took part in several field excursions to Medicine Hat, Lethbridge, Waterton Lakes townsite and various localities in and near these places. The conference was entitled *The Paleoenvironmental Reconstruction of the Late Wisconsin Deglaciation and the Holocene*. The field trips (one before and one after the actual technical meeting) examined evidence of Ice Age geology, geography and fossil sites across southern Alberta. Hope was a pre-conference field trip co-leader, but her ability to participate was uncertain, so her name is not included in the final conference abstract volume. The field trip included visits to sections at Taber, Medicine Hat (Barendregt, 1985) and Red Rock Coulee. She helped co-lead the field trip, "but not much to the fore" (R. Churcher, pers. comm., 2014). Three other scientists dealt with the technical aspects; Hope talked about what specimens came from some sites, and she would have been the local expert on matters of natural and human history.

Her only known field activity this year, besides the CANQUA event, was an August visit to Red Rock Coulee.

The RTMP in Drumheller opened its doors on September 25, 1985 and Hope would be involved with that institution too, but in a more limited way, to be discussed later.

Figure 124. Public education display case of local and mostly vertebrate Late Cretaceous fossils at Police Point Interpretive Centre assembled by Hope, seen in the fall of 2014. It is usually protected by a sheet of clear Plexiglas, removed here for the photograph. Photograph by the author.

1986

In April 1986 Hope helped develop a small fossil collection exhibit at Police Point Interpretive Centre in Medicine Hat. This consists of a 67.3 cm × 104 cm sliding drawer filled with specimens and handmade labels, all carefully glued down so as not to become dislodged or mixed up (Figure 124).

In 1986 one of her abstract paintings, *Cloud Shadows over the Badlands,* was shown at a western-themed art show in Medicine Hat (Purdey, 1986).

The Alberta Palaeontological Society, based in Calgary, was formed this year and Hope, who joined in 1987, would play a prominent role with that organization in later years.

Before summer holidays, she spoke to four classes of Redcliff junior high school students about dinosaurs and fossils from the area.

Limited correspondence between Hope and Yale University's invertebrate palaeontologist Karl M. Waage during this period indicates Hope's renewed scientific interest in the Late Cretaceous Bearpaw Formation invertebrates of southeastern Alberta. Waage indicated that he was interested in trading Alberta fossils for some American specimens but it is unknown if this ever took place. Her only documented fieldwork this year took her to Red Rock Coulee, in June. She could rarely leave home now, as Ollie's health continued to deteriorate and he required her care. Her name appears in a guest register of July 23 at the Tyrrell Museum.

1987

Hope visited the new Tyrrell Museum of Palaeontology in March and attended the opening of the Museum's Field Station in DPP on May 15 (Anonymous, 1987a, b; Blakey, 1987a-b; Kooy, 1987; Plumtree, 1987a). She remarked to the author how much the Park had changed for the better, with the addition of museum-quality exhibits, since the days she had been involved with the little exhibit buildings nearly twenty years earlier.

Anonymous (1987e) shows Hope, then President of the local chapter of the Alzheimer's Society, presenting the Executive Director of the Medicine Hat chapter of the Victorian Order of Nurses with a cheque for $2000. This donation was for daycare for Alzheimer's patients and respite for their caregivers. It is believed that the donation was made on behalf of the Society.

Hope's husband Olafur, ill with Alzheimer's, passed away on July 27 at the age of 78 (Anonymous, 1987d). They had forty-two years together, though he had been ill the last six years; the final four months of his life were spent in hospital. People described Hope as a strong woman, but the two had been very close and his death had a deep impact on her. On March 22, she wrote a touching poem to him, about four months before his passing, which shows the deep feelings and pain she was then going through:

On My Alzheimer Patient
(My husband Olafur Johnson—1909–1987)

Who sits there in that shell,
Half familiar, yet other?
One whom I knew, and well.
Now unresponsive, strange,
At times even threatening.
A fleeting look crosses that face,
A gesture, half-familiar.
Reminds me of a happier time
Forever gone.
Friend, lost in time,
All memory razed,
It's best you do not know
Because your tired heart might break.

Another project Hope was involved with was Elderhostel, through the Community Education Division of Medicine Hat College. Elderhostel is the largest non-profit educational travel organization for adults over 55 years old. Hope taught courses entitled "Geological Heritage of Southern Alberta" and "Cypress Hills" and received high praise from the mature students who participated. She was involved with the Elderhostel tours for many years, but their duration is not entirely known. A fun story about Hope and the tours was related by Dennis Baresco of Medicine Hat, who writes:

> From about 1994 to 2004 I was Chief Interpreter at the Medicine Hat Interpretive Program, working out of the Police Point Park Nature Centre. I was also one of the guides on the annual local Elderhostel tour. One of the spots we went to was Red Rock Coulee. Since Hope was possibly the world's best expert on that site (she'd written a great little document explaining its origins), and since I knew Hope fairly well, I'd asked her to accompany my Elderhostel group on one of the trips. So when we got there, and just before we all got off the bus, Hope apologized and mentioned that because her arthritis had been acting up, she would probably not be able to lead them around that terrain as well as she liked. That seemed to be fine with the group, since, after all, she was the oldest of all of them. So we all got off the bus and Hope seemed to gain new life, as she went up and down and around, back and forth, and stooping to point out nature nuggets on the ground. By the time we got back to the bus, about two hours later, most of the seniors were exhausted—except Hope, who was beaming. One of the seniors said to me: "If this is what she's like with arthritis, what's she like when she's feeling good?

She submitted some of her artwork for a competition at the Medicine Hat Hospital. Eventually the hospital bought some of these pieces and they are displayed there today.

Archie Stalker's field notes for November 21 note that he had a "talk with Hope Johnson and she was happy and described a new cave they had found, etc. Phoned back in evening and had a long talk." Details of this cave are presently unknown.

Christmas this year was her first without Ollie in forty-two years so she travelled to Ottawa to spend the holidays with her daughter and family.

1988

Hope was acknowledged in a paper dealing with rare and endangered flora and fauna in southern Alberta (Wallis and Wershler, 1988). She did drawings for a book on Alberta berries that came out this year (Droppo and Johnson, 1988). In a family copy of this book, Hope's dry humour is again revealed in comments she added about the actual edibility of some of these berries, no doubt based on personal experience:

> Page 56. Fairy bells, *Disporum*. The book suggests the berries can be eaten raw, to which Hope adds: "How hard up can you get?"
>
> Page 84. *Lonicera caerulea*. The book again states that it is edible; Hope disagrees and remarks: "Take no chances."
>
> Page 126. *Ribes glandulosum,* skunk currant: Hope suggests the fruit should be "left to the skunks."
>
> Page 168. *Shepherdia canadensis,* Canada buffalo berry or soapberry: "Bears love them. Let them have them."
>
> Page 192. *Symphoricarpus occidentalis,* buckbrush. In the book, the fruit is identified as "famine food." Hope agrees, and adds: "Indeed. Eat your shoes first."
>
> Page 196. *Trillium ovatum*: On their edibility, Hope considers them "useless."
>
> Page 198. A recipe for blueberries and cranberries suggests cutting the amount of sugar in half or substituting one teaspoon of salt to each half cup of sugar, to which Hope adds: "How is your blood pressure?"

She records in her notes that she began painting watercolour landscapes this year, though she had much prior experience with the medium, doing high-quality and life-like renditions of flowers and local plants.

She received a letter of thanks from the Superintendent of Schools for Medicine Hat School District No. 76 for her help in the Early Childhood Services Program. Hope gave talks to children on dinosaurs, fossils, rock and plants.

Hope's fossil catalogue (Johnson, undated 1) records that she helped facilitate the donation of the Heath family collection to the RTMP. It likely occurred at the end of March (Johnson, undated 3). The Heaths were fellow amateur fossil collectors who accompanied Hope on many fossil collecting trips (see Page 84). Thirty fossils from the Heath collection were catalogued into the TMP 1988.060 series.

Dated photographs show that she visited Petrified Coulee in May. In July and August she was joined by some friends, exploring badlands on the periphery of DPP in the Princess area. By about this time she was using a cane regularly and needed help getting down into the badlands and out again at the end of the day. She would also occasionally have a hard time getting into a sitting position on the ground. With the tip of her cane she would point out what looked to her fellow collectors like an obscure piece of junk bone and ask them to pick it up for her examination. To their surprise it would often turn out to be something of interest. One fellow collector from this time called her "The Vacuum" as she could spot fossils in unlikely places while walking toward the richer, established collecting grounds.

Figure 125. Hope with a *Triceratops horridus* skeletal facsimile at the Natural History Museum in London, England, early October, 1988. It is a *papier mâché* reconstruction; see Miller, 2015, for an interesting discussion on this mount. Image courtesy of the Johnson family.

An overseas trip to Broadstairs, in Kent, England, allowed her, on October 3, to explore the marine Jurassic Purbeck Beds where she collected a small ammonite. This is one of the few fossils she is known to have personally collected outside of North America. On October 8 she visited the Natural History Museum in London and posed with the *Triceratops* skeleton there (Figure 125).

Hope attended the Society of Vertebrate Paleontology meeting at the RTMP in Drumheller, October 13–15. Later that month she wrote a letter (Johnson, 1988) to Dr. Michael C. Wilson indicating confusion and frustration with the new provincial legislation regarding registration ("disposition") of private fossil collections (fossils collected before July 5, 1978 versus final disposition of fossils collected after that date), transfer of specimens upon the death of the collector, and especially the prohibition against sending fragmentary fossils out of the province for educational purposes, while others were selling similar fossils in the province. In 1988 Wilson (then at the University of Lethbridge) was chair of the Alberta Palaeontological Advisory Committee (APAC), who as a group had signing authority over fossil excavation permits in the province and advised Alberta Culture concerning palaeontological policies. A serious issue being addressed was outright vandalism (Wilson, 1986) of Alberta fossils and their illegal sale. Another issue was the disposition of private fossil collections by a deadline of Decem-

ber 31, 1993 (Braunberger, 1993). Hope wrote some impassioned letters to Wilson, both as a friend and professional colleague. She wrote as an advocate of amateur collectors province-wide and was worried that new regulations regarding disposition of private collections could result in confiscation (*e.g.* Anonymous, 1988; Bolvikoski, 1988a). She felt that legal efforts should be directed toward individuals who were digging up Alberta fossils illegally and selling them.

1989

The Medicine Hat Rock and Lapidary Club was formed this year. Some people thought that Hope was a member, but a brief interview with a long-standing and founding member confirms that she was not a part of their club, though she did do some of her own lapidary work (Smith, 1976). She did not lead any field trips for them either, even though they visited some of her favourite haunts such as Red Rock Coulee (Figures 180, 205).

The final entry in her third and last fossil catalogue book (Johnson, undated 3) records her co-leading an Alberta Palaeontological Society field trip (Figure 126) to her old stomping grounds at Petrified Coulee and Manyberries on August 19–20, where it was noted "she had a sharp eye for finding small material" (Morgan, 1989). At this point she was 73 years old. The last entry in her catalogue book occurring this year suggests that she stopped collecting fossils. If true, it would mark the end of thirty-seven years of a rewarding hobby. She had not entirely given up palaeontology and did venture out into the field on fewer occasions; but details of most of these later trips are unknown. From this year onward she produced more and increasingly technical line drawings and paintings in all media.

Figure 126. Hope on a field trip with members of the Alberta Palaeontological Society, August, 1989. The photographer is unknown; image courtesy of the Johnson family.

A show at the Medicine Hat Art Gallery featuring her watercolour paintings and those of Deanna Lemna opened on September 23, according to Archie Stalker's field notes. The show was titled "The Prairie in Relief" and ran until mid October (Anonymous, 1989c). Hope received a Heritage Merit Award for her longtime historical and education work in the Medicine Hat area in the early summer of 1989 (Anonymous, 1989a). The same publication states that the Johnson and Storer (1974) fossil identification book is "now published through a Bozeman, Mont. museum." A look into this revealed that Jack Horner of the Museum of the Rockies had approached his museum to republish the book as part of their occasional paper series, but the idea was vetoed by the museum's Director (J. Horner, pers. comm., 2015). Anonymous (1989a) also states that her artwork had been exhibited in Bonn, Germany, a claim that the author could not substantiate. In a *Medicine Hat News* supplement published in May (Heuft, 1989), Hope comments on local flora and promotes her 1983 book on southeastern Alberta plants, noting that two bookstores in Medicine Hat were carrying it.

1990

Hope and others taught pen-and-ink art classes through the Hat Art Club, most of the month of March (Anonymous, 1990a). She again received a letter of thanks from the Superintendent of Schools for Medicine Hat School District No. 76 regarding her participation in May with the Early Childhood Services Program. Around this time she read through and improved a draft of a popular natural history book (later published as Willock, 1990) by doing a general review and making comments regarding the geology.

In 1990 she was involved to an unknown degree with local television. Anonymous (1990b) notes a half-hour television program entitled *Primetime: The Red Rock Coulee visit with Dr. Hope Johnson*. The program ran for several weeks. Other notes suggest that Hope was a regular on the local CHAT television show, *Our Town* hosted by Lorne Havard.

Her confirmed field activities include at least three visits to Red Rock Coulee in June, July and August. It was a favourite painting locality for her; there is no record that fossils were collected on these visits. Hope was in the Irvine badlands, east of Bull Trail, in July. She also visited the East Coulee area.

1991

Not much of Hope's activities were recorded this year, save a visit to Petrified Coulee in July. She wrote an article on the geology of Red Rock Coulee which was published this year and reprinted in 1992 (Johnson, 1992b). By this time she needed a walking stick to safely traverse the badlands. Now in her mid-seventies, many would have given up on fieldwork years earlier, but not Hope—she would continue hiking the hills for another decade, albeit with increasing difficulties with balance and pain.

Found among her personal papers after her death was an issue of the Icelandic weekly newspaper, *Lögberg Heimskringla* from November of this year. She had a paid subscription, indicating that she was still following Olafur's Icelandic heritage, news of his homeland and of other Icelandic Canadians.

About this time she visited the Royal Tyrrell Museum and asked for the author at the front desk. I came from the preparation lab and greeted her in the lobby. She extended her right hand and I did the same, giving her a friendly, firm handshake. She immediately jerked her hand away and doubled over with pain. I was mortified. Not only had I hurt Hope, but an artist's hand as well! Recovering after a few moments she stood up and gave me a stern lecture on my ignorance that many senior citizens have arthritic hands. It was the only time I saw her angry. I was humbled: it was a hard lesson for me and one never forgotten.

1992

In 1992 Hope's excellent ink montages of Bearpaw Formation invertebrates were cover art for two issues of the Alberta Palaeontological Society *Bulletin:* Volume 7, number 1 (March 1992; Johnson, 1992a) and Volume 7, number 3 (September 1992; Allen, 1992; Figure 127). The author recalls the latter being reproduced on 8.5 in × 11 in (21.6 cm × 28 cm) light card, presumably for resale and he has seen some colour versions that appear to be hand-tinted. Hope sometimes had her dinosaur and badlands scenery drawings printed up into small greeting cards with no text inside.

This year her landscape paintings, done in watercolours, pen-and-ink and oils, were a featured ex-

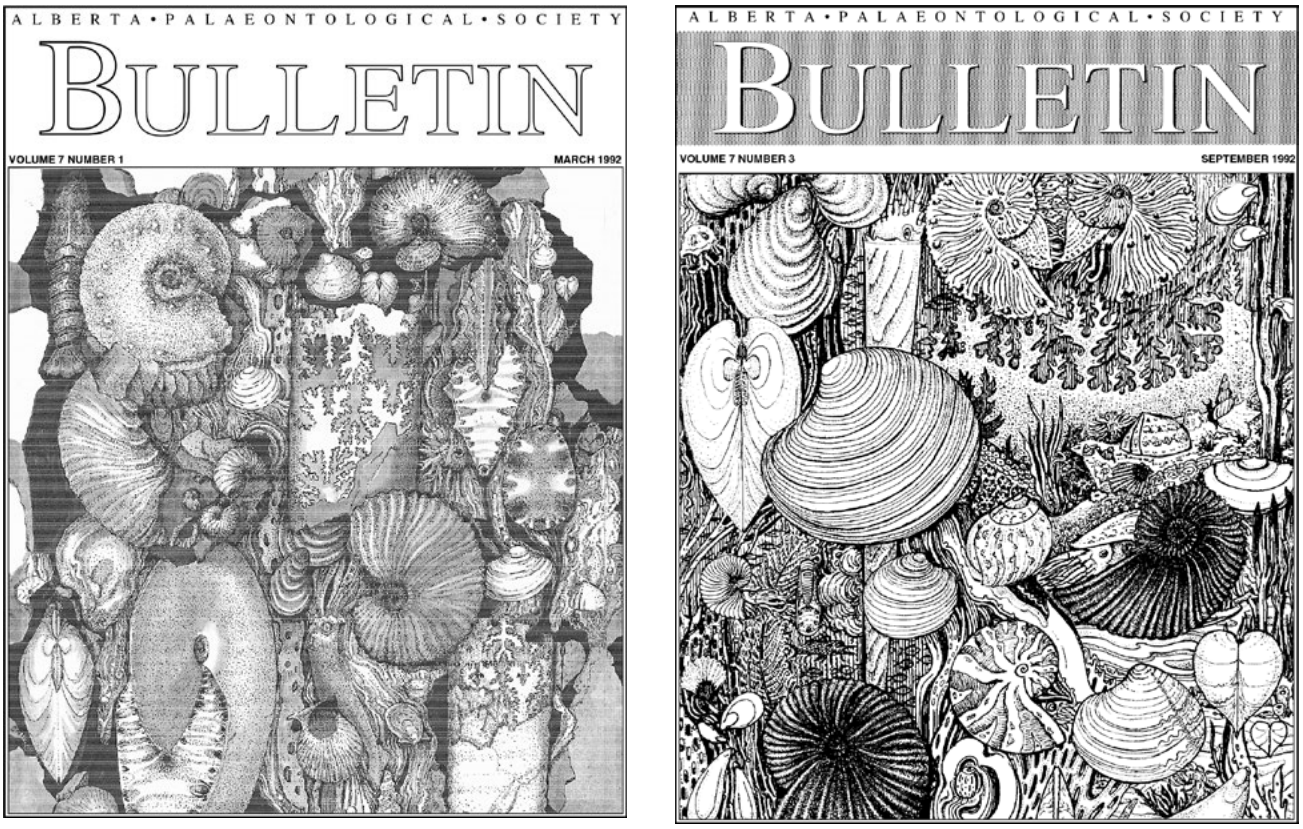


Figure 127. Montages of Late Cretaceous Bearpaw Formation invertebrates done by Hope. Left: in ink and dilutions of India ink wash; *c.* 1982. From Johnson, 1992a. Right: in pen-and-ink; drawn in 1986. From Allen, 1992.

hibit at Alberta Agriculture's Agri "Culture" Gallery in Medicine Hat from April 2–30 (Anonymous, 1992a). A 1991 22 in × 28 in (56 cm × 71 cm) oil painting of eroded sandstone scenery at Writing-On-Stone Provincial Park received an honourable mention at this show—the original is now part of the University of Lethbridge art collection. Also this year, she taught oil painting to seniors (Anonymous, 1992c).

Hope resigned her presidency of the Alzheimer's Society this year, after about eleven years of volunteer work, jokingly stating that it was "too much paperwork."

She did fieldwork at Red Rock Coulee in July and August and visited Petrified Coulee in October.

1993

The author worked with Hope in the spring of this year, exchanging correspondence in an effort to determine the exact location of her Quarry 139 (TMP 1972.001.0001), a pterosaur bone discovery in DPP (see Page 99). Hope, now seventy-seven, was still hiking the hills in the mid to late 1990s ". . . wearing out younger seniors . . . on group hikes." A small fossil with detailed collection data, retained by her family, shows that she visited a "University of Alberta" site in the Irvine badlands on June 23 and collected a possible fossil bird bone there.

In the summer, she responded to a taxonomic blooper deliberately planted in the previous issue of the Alberta Palaeontological Society *Bulletin* by the editor simply to see if any readers were paying attention. Hope certainly was! (Johnson, 1993a). She pointed out another media gaffe wherein dinosaur

fossils were called "artifacts" (Johnson, 1993b), pointing out that artifacts are objects made by man. Also this year, she appeared in a Medicine Hat local television production, *Art in Profile—Hope Johnson* (Anonymous, 1993).

She gave a talk on rocks to young children at a daycare this year (Figure 128). She was very kind to children and they were attracted to her motherly demeanor and her long, braided hair, which continued to fascinate. A number of interviewees remarked how Hope's braids made such an impression on them. One woman described how, when she was a girl, she had encountered Hope undoing her braid and combing out her hair, which extended down to near her knees. The girl, not recognizing Hope, was terrified by this apparition and fled. Hope instantly comprehended the child's alarm and ran after her, shouting "It's me! Mrs. Johnson!"

Figure 128. Hope and her good friend Isobel "Teddy" Vano (1924–2015), a fellow fossil enthusiast, showing rocks and fossils to children of the Herald School Daycare, Medicine Hat, in 1993. A subsequent thank-you note addressed Hope as "Dear Rock Lady." Image courtesy of the Johnson family.

1994

This year Hope continued teaching seniors' painting classes at the Veiner and Strathcona Centres in Medicine Hat (Segal, 1994). She participated in the 1994 Ranchlands and Badlands Art Show and Sale held on March 17. Thirty art pieces were entered in the show. Hope entered two and won first prize for a watercolour, *Crowded Hoodoos*. At the show she was recognized by a local radio station for her lifelong work in the community and was awarded the 1994 New Country 1270 CHAT Radio Collection Award. A fine portrait photograph was taken of her about this time (Figure 129). She also participated in a juried art show entitled "A Quality of Wildness" and a silent auction put on by the Grasslands Naturalists group late in the year (Anonymous, 1994b).

She visited Eastend, Saskatchewan in August. Hope's work in Saskatchewan is not well known but from interview comments in J. Danis (1981) we know that she was as familiar with fossils in that province as she was in Alberta. Archie Stalker's field notes record that he visited Hope in late September and that she "looks very well, and is keeping busy."

Figure 129. A beautiful portrait of Hope taken in Medicine Hat about 1994. Here she is 77 or 78 years old. Image from Dynes Gallery and Framing (1994).

Her membership with the Alberta Palaeontological Society lapsed late this year. She was still active with the organization, but from afar. Hope wryly admitted in an interview this year (Segal, 1994) that she had one reservation about fossil hunting: "My one real objection to palaeontology is that it's a heavy hobby."

1995

Hope visited the RTMP in June for a day of intensive library research for dinosaur bone drawings. This mission was related to a revised fossil identification book she was working on (Johnson, 1996).

In the early summer she was conducting wildflower identification seminars at the Police Point Interpretive Centre (Anonymous, 1995). She was always encouraging interest in others in the identification of plants and making people aware of their important role in natural ecosystems.

In the mid to late 1990s Hope embraced digital technology and used the Internet to educate readers in palaeontology. Friend and fellow amateur fossil collector Shane Leuck of Medicine Hat created several palaeontology web pages (now closed). His site, *Fossils in the Gateway Zone* and *Red Rock Natural Area* featured Hope's thoughts and writings on southern Alberta Late Cretaceous geology and palaeontology, supplemented with some of her artwork. She even commented once in the online Dinosaur Mailing List, a public dinosaur forum, responding to a query about the Judith River Formation in Canada and the United States but that was her only participation.

She visited the Royal Saskatchewan Museum's *Tyrannosaurus* "Scotty" quarry near Eastend, Saskatchewan in September.

1996

In her eightieth year, Hope was realistic about her own mortality. Invited by the author to participate in an event to be held four and a half years hence, she accepted—with the qualification, ". . . assuming I am 'spared' . . . at 80 years old, one never knows and is acutely aware of it" (Johnson, 1996). For her birthday, some fellow amateur fossil collectors gave her a hardbound copy of the large technical book, *The Dinosauria* (Weishampel *et al.*, 1990). It was not a book to be simply put on the shelf: she pencilled notes within and these give us an indication of what dinosaurs and aspects thereof she was most interested in at this point in her life. Most, naturally, were Alberta-related and dominated by theropods; but some common Alberta dinosaur families are not labelled at all in the text. She emended some figures and notes:

- Tyrannosaur and *Allosaurus* crania, cervical vertebrae and manii.
- Ornithomimid crania and postcrania.
- Oviraptorid crania.
- Troodontid crania and pedes.
- Dromaeosaur pedes.
- Pachycephalosaur crania.
- Ceratopsian crania and postcrania.

Additionally, she made brief comments on the discovery of new *Spinosaurus* material (which she called "astonishing"), the rediscovery of the AMNH 1910 *Albertosaurus* bonebed at Dry Island Buffalo Jump Provincial Park, Alberta and the discovery of a complete ornithomimid skeleton with skull in DPP (in 1995).

She was made a Life Member of the Alberta Palaeontological Society in November 1996 (Osztian and Allen, 1997; Figure 130). She had been a member or had long-term involvement with the Society since 1987, soon after its inception.

A small Alzheimer's care facility opened in a private residence near Medicine Hat this year; Hope offered her support and some advice based on her personal experience (Ehrkamp, 1996).

1997

At a Grasslands Naturalists meeting on March 25 a standing-room-only crowd watched and celebrated as Hope received the prestigious Loran L. Goulden award for her "many years of service and long list of accomplishments in a variety of natural history fields" (Anonymous, 1997b). She continued teaching art classes to seniors over a three month period this year (Anonymous, 1997a).

Hope attended a field trip to Red Rock Coulee, where she "briefly introduced the site" for the Alberta Palaeontological Society (Marsovsky, 1997). She wrote and illustrated a brief article on unusual tyrannosaur unguals she had found, to alert other amateur collectors to be on the lookout for these rare items (Johnson, 1997). This was one of the very few articles Hope wrote about her field experiences and finds, which is unfortunate given her vast experience.

Palaeontologists from the PMA spent three weeks digging up Pleistocene mammal material in Medicine Hat this summer (Berezowsky, 1997a, 1997b). The newspaper article acknowledges help from local citizens, but Hope is not believed to have participated; given her age and health issues it seems unlikely.

Her long-time friend and field colleague Archie Stalker writes in his notes that he had a "nice talk" with Hope on November 14. This appears to be the last time they had contact over some thirty years.

1998

Late in the year, Marsovsky (1998) reported that Hope was "currently working on her next book, which will continue on the same theme as her first palaeontological book." This was another attempt at a revised version of Johnson and Storer (1974). Her knowledge of anatomy and the science had so improved over the years that the original book, while still useful, could have used a thorough revision. The 1974 book was long out of print and there had been an increase in the ranks of amateur fossil collectors in the province. People got by with photocopies of the book to aid in their own fossil collecting hobby. Science had made numerous new discoveries since 1974 and Hope had drawn addi-

tional fossils. A revamped and enlarged book was needed. Previous attempts at a renewed book had failed, so Hope was trying to go it alone—which, given her age, would have been difficult for her. But help was coming.

She received her Life Member recognition plaque from the Alberta Palaeontological Society on August 22, 1998 (Marsovsky, 1998) in recognition of her lifetime contribution to Alberta palaeontology (Figure 130).

Sometime this year a journalist visiting her home reported, ". . . now we're in the basement where every available surface is covered with fossils. I pick up one or another because of an interesting shape or surface texture. Each fossil is numbered and recorded." (Forbes, 1998).

None of Hope's other activities for this year are recorded.

1999–2000

Around May, 1999 her artwork was featured in Lethbridge in a show titled "From Fossil to Fine Art" (Anonymous, 1999a; Forbes, 1998). The show also travelled to the Drumheller Public Library (August and September, 1998; Anonymous, 1998), Medicine Hat College and the Graham Community Library (March, 1999) in Ralston. Fifteen pieces were exhibited, ranging from pen-and-ink illustrations of fossils to oils and watercolours of rocks, badlands and wildflowers.

Figure 130. Hope presented with her Life Membership recognition plaque by Vaclav Marsovsky, then President of the Alberta Palaeontological Society. Image courtesy of Vaclav Marsovsky.

In a tongue-in-cheek letter to the editor of the local newspaper (Johnson, 1999) Hope, always the perfectionist, chastised the paper for its habit of neglecting to check the copy before a print run and pointed out several gaffes where sentences were incompletely printed.

Later in 1999 Hope packed up the last of her personal fossil collection. The specimens were wrapped in newspapers and stored at her daughter Margaret's house or at granddaughter Rachel Noone's home, both in Medicine Hat. From there the author received them in early October 2014 and November 2015 and delivered them to the teaching collection of the Education Department at the RTMP. Several years earlier the same had been done with all the figured specimens in the Johnson and Storer book (1974), save for an unusual pathological ceratopsian phalanx Hope had collected from White Rock Coulee in 1968 (their fig. 122; Alberta Palaeontological Society, 2009, their fig. 126), now catalogued as TMP 2014.666.0005.

The last year Hope hiked the badlands in her relentless search for rocks, fossils and plants was 1999, or perhaps 2000. Any future trips were of short duration and distance as it had become too painful for her to continue. Thus ended over fifty years of active, dedicated field exploration.

In 2000 she could no longer take care of her house alone, so in her mid-eighties Hope left her Redcliff home and moved into the Valleyview Retirement Community in Medicine Hat, settling down into Room 109 of the Assisted Living wing. She enjoyed it there, with a small single bedroom and a side studio where she could continue to pursue her research, writing, socializing, reading, drawing, and painting. She described her room to the author (Johnson, 2007) as ". . . crowded with books, several

on palaeontology, but also on history, simple geology; reference books; some botany; Jared Diamond's *Guns, Germs, & Steel* and *Collapse*; your [author Tanke's] book with Carpenter [this is the 2001 *Mesozoic Vertebrate Life*, honouring Philip J. Currie on the occasion of his 25th anniversary as a vertebrate palaeontologist in Alberta; *e.g.* Currie and Brett-Surman, 2001]; an old but useful text by Alfred S. Romer, plus several more." People who visited her at the Valleyview residence found it an interesting and eclectic place with books, papers, drawings, fossils, rocks and paintings—"a museum itself." When she became too frail to go out and teach art, she had five or six students come to her for a semester, once a week for a few hours, for what would be her final lesson—thereby ending nearly fifty years of teaching drawing, painting and watercolour to many people, much of it without financial compensation. This year she donated to the University of Lethbridge about five of her flat art pieces, such as the watercolour *Burmis Tree* (Figure 188), other unspecified paintings and one pen-and-ink plant drawing (Anonymous, 2014p).

Despite a period of relative quiescence in the last decade of her life, Hope would surprise many and gain prominence one final time.

2001 to 2010: Publication of the new fossil guidebook and the closing chapter of a life well lived

At some point, Hope began taking advantage of advanced photocopying technology and made laser prints of her black and white and colour artwork and selling these. In the summer of 2001, on the heels of a surprise party for Philip J. Currie's 25th anniversary as a vertebrate palaeontologist in Alberta, Hope made an appearance at a Patricia, Alberta bed and breakfast with samples of her work for viewing and sale (Figure 131).

For the next segment of Hope's life story, we first need to backtrack to 1986. That year the Alberta Palaeontological Society (APS) was formed; Hope was a member from 1987 and quite likely its most prominent amateur member. The Society comprises Albertan amateur fossil collectors, university students and professional Earth scientists.

In the 1980s and 1990s—though the date it began is now forgotten—efforts were made with American vertebrate palaeontologist Dr. Peter Dodson to update and republish the Johnson and Storer (1974) book (Anonymous, 2009). Notes penned by Hope about 1983 suggest she had already drafted forty-eight pages of new drawings (more than seventy-five by November 1996; Johnson, 1996) and Dodson was rewriting parts of the text. The project dragged on for quite some time. By 2001 she was also writing chapter texts. Philip J. Currie did some maniraptoran fossil identifications based on her drawings alone: they were anatomically and technically good enough for him to identify the specimens to species and body position. Much work was done over several decades for a revised book, but for reasons unknown it never met fruition, perhaps due to numerous delays brought on by Dodson's professional commitments (L. Johnson, written comm., 2010). As already noted (Page 147), in 1989 Jack Horner had tried to get support from his mu-

Figure 131. Hope displaying and selling laser prints of her art work to her palaeontological friends and colleagues, July 8, 2001. The author is on the right wearing the white T-shirt. Photographer unknown.

seum to reprint the Johnson and Storer (1974) book, but this did not happen. So two efforts to revamp and republish the book had failed. Among Hope's notes it appears she may have tried to get the revised book done on her own; there are a number of handwritten book chapters known. That effort did not reach a successful conclusion, either. But all was not lost.

By 2004, board members of the APS had proposed to revise, expand and republish the Johnson and Storer (1974) fossil guide as a twentieth anniversary project, 2006 being the Society's anniversary year. This project was first announced by APS President Dan Quinsey (2004).

By January 2005 the APS was in discussion with the Royal Alberta Museum (RAM) to secure permission to reprint parts of the Johnson and Storer (1974) book and a "first outline" of the book was already in existence by April, 2005 (H. Allen, pers. comm., 2014). The proposed 2006 publication date was not met, but the revised book finally went on sale in 2009.

In one of her last, if not the final, formal art shows, in early 2001 (February 3 to March 18) the Medicine Hat Museum and Art Gallery ran a show of Hope's work entitled "Hope Johnson: A Retrospective" (Petrie, 2001; Page, 2001). Two newspaper articles, each with a colour picture of Hope and her art resulted (Anonymous, 2001a, b), the former showing her holding a ceratopsian horn core. A detailed note penned by Hope on the back of an exhibit photograph states that there were about fifty paintings and pen-and-ink drawings in the show, as well as some "fossil drawings, framed with actual small fossils" (Figure 132). In a letter (Johnson, 2003) Hope disclosed that at the show she ". . . was able to sell a great many pieces, thus reducing what I had to accommodate [in her new retirement apartment]." A brochure, printed for the show, included an interview with Hope; some of the dialogue, regarding Hope as an artist, is excerpted here:

Figure 132. *Marine Life with Fossils* from Hope's 2001 art show, "Hope Johnson: A Retrospective." Here Hope not only draws the Late Cretaceous invertebrate fauna intricately, but uses actual fossils from the region in a shadow box to supplement the art piece. Image courtesy of the Esplanade Permanent Collection of Art.

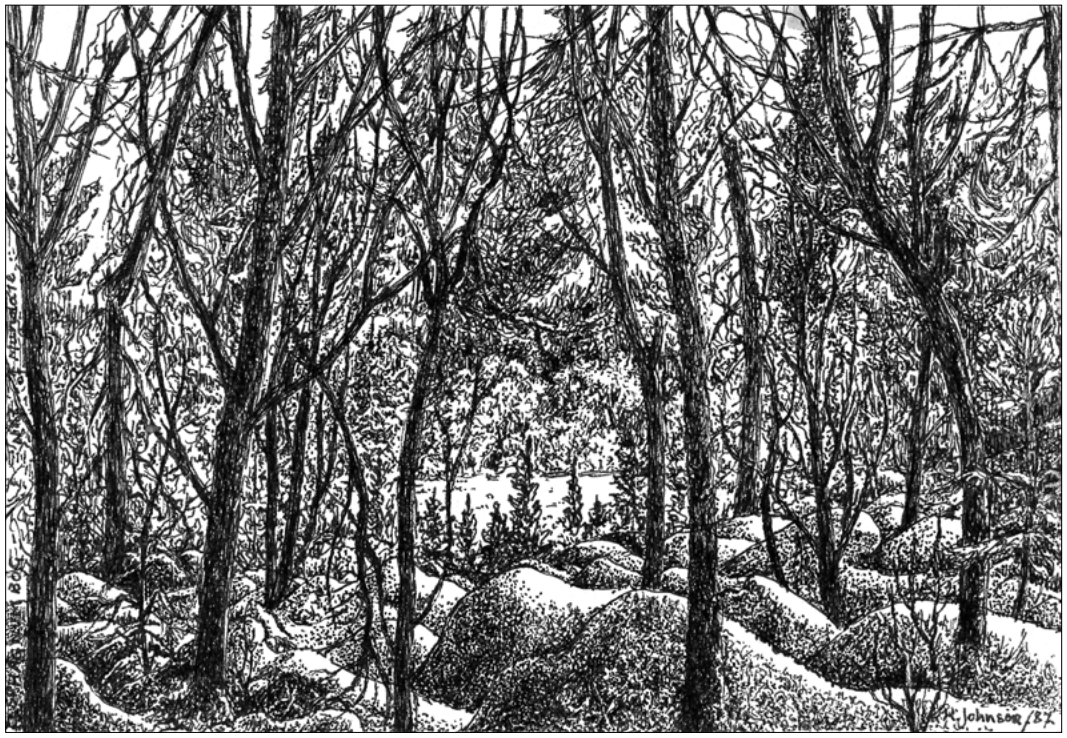


Figure 133. Small 1987 image entitled *Sphagnum Bog*. On the back, Hope identified this as a preliminary sketch based on memory. The field locality is 140 km northeast of Edmonton, Alberta. The original of this felt pen image measures 8.5 X 6 inches. Collection of Shane Leuck, Medicine Hat.

Interviewer: Looking back over the last thirty years of your work, there seems to be clusters of subjects—botanical studies, fossil studies, many paintings of coulees, *etc.*—and in all of them there is close attention to detail. Is this perhaps related to your scientific practice of observation?

Hope: Yes, for instance I've seen paintings of wildflowers, but when you look at the wildflower, say anemones or something, you might be able to see one anemone but the rest of it is one subdued mass of colour—it's pretty, but what do you learn about anemones? Nothing, I think. So I shy away from that approach. I want the thing to be recognizable. That little painting over there (*Quaking Bog with Moss Mounds*, 1998) that was done from memory [see Figure 133]. A young chap came here with Professor René Barendregt, from northeast of Edmonton, and he wanted to know where the place was—he said "there's a place just like that on our property." And so I felt that even though I had done the painting maybe four or five years after being there, I still had this image . . .

Interviewer: What drives your work then, is . . .

Hope: To let other people see what I see. A shared visual experience.

Interviewer: What leads you to choose a particular subject?

Hope: Because I like the place. In fact, I don't do things that are particularly ugly or dismal. Although I think there's sense in doing that sometimes. If you think of Picasso's *Guernica*, I think that has political and artistic value. I've never been assaulted by that kind of vision, but I think it's worthwhile.

Interviewer: But you're not just after the exactitude of a photograph.

Hope: Well, I do photographs and sketches, but I don't copy photographs of places I don't know anything about.

Blocksage (2002) reports that in September, 2001, Hope gave a talk—on belemnites, of all things—to the Southeast Alberta Archaeological Society in Medicine Hat.

In 2002 she sent staff of the Royal Tyrrell Museum of Palaeontology (RTMP) a custom-made Christmas card (Figure 134). It featured an ink wash rendition of rock outcrops at Writing-on-Stone Provincial Park. On the back of the card was a detailed description of the scene. Inside she reminded the field staff about how the bentonite slopes in the badlands become slippery when wet: "Try to keep off

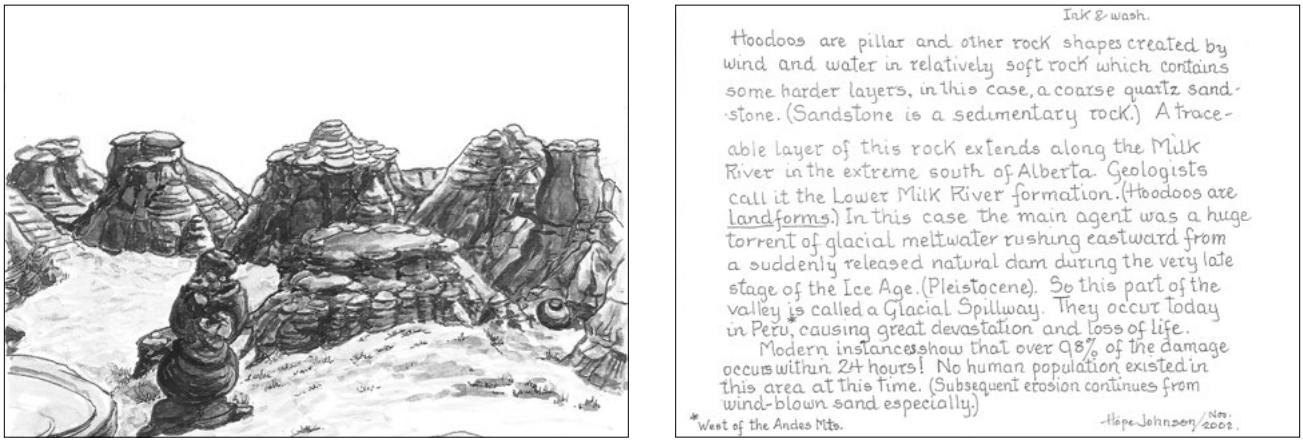


Figure 134. Front and back of Hope's 2002 Christmas card. Her penmanship is exquisite for someone eighty-six years old!

bentonitic mud—real or metaphorical!"

About this time she was also reviewing a canoeing and natural history book on the South Saskatchewan River (Dickinson and Baresco, 2003). She assisted in the research on an endangered grasslands plant and a report was issued (Bradley, 2004), wherein Hope is acknowledged.

She attended a Canadian Nature Federation field trip in 2003 (Figure 135).

In 2003, at the Valleyview residence, she was asked to paint a colourful mural in the recreation room. She was on the project right away. Within two hours she had drafted preliminary sketches and a list of the required paints. This was no small undertaking for Hope, who was now eighty-seven. The mural measured 5.2 m long by 1.9 m tall. Low scaffolding had to be built so she could reach the upper parts of the mural. With much vigour she a painted the *Valleyview Trees of Life* mural over July and August. She dedicated much of her time to this project, sometimes putting in long hours each day. It was dedicated to her friends and other residents of the facility (Anonymous, 2004; Figure 137). When a new resident moved in, they were photographed, their picture placed on the foliage of one of two trees, representing the assisted living and continuing care wings of the facility. When a resident passed on, their picture was removed and added to a new page in a memorial binder, along with a short biography, newspaper obituary, and funeral leaflet.

In 2003 fossils were still an important part of Hope's life: in a letter (Johnson, 2003) she stated, "I am still dealing with fossil finds in [the Medicine Hat] area."

She told the author that she enjoyed her walks around the Valleyview property but found it harder to deal with any inclines, especially downhill ones. She kept active, attending exercise classes three mornings a week. It was important to her to keep active both mentally and physically, despite the progression

Figure 135. Hope and Dr. René Barendregt (University of Lethbridge) at the 2003 Canadian Nature Federation field trip in Medicine Hat. Image courtesy of Dr. Barendregt.

Figure 136. Hope (with her hair uncharacteristically down) and Frederick "Fred" Zybryski at a Halloween party in 2005. Image courtesy of the Johnson family.

of physical ailments that come with advancing age.

While she liked living at the Valleyview and had more time for drawing and painting, she found it was not easy to make small talk with other residents. Hope was well educated and had many interesting experiences during her life. Many of the other residents were people who were not as worldly as Hope, so they often had little in common with her. What she really craved was intellectual conversation and she found it with Frederick "Fred" Zybryski, a resident at the Valleyview, a retired electrician and hotelier. His favourite reading subject was science (Stubbs, 1984), so the two got along famously (Figure 136) and became close friends. Hope was the more mobile of the two so she visited him regularly. He passed away in 2007, at the age of 97 (Anonymous, 2007). She spent much of this period of her life working on her many art projects.

2005

In April 2005 Hope flew with one of her daughters to Switzerland to visit her son Lawrence, who was working there (Johnson, 2005b). This was her last overseas trip. In September the Royal Tyrrell Museum hosted the Dinosaur Provincial Park conference (Anonymous, 2005). Almost 200 delegates attended, two of whom were Hope and her friend and fellow amateur fossil collector, Sharon Stiem, of Medicine Hat (Stubbs, 1993). Given her esteemed status, Hope was a special guest and was approached by many friends and colleagues from the palaeontology community. She noticed this, too. In notes to her family she writes, "I was a celebrity—I [was] also astonished. Photographed with admirers or alone! And had to autograph original numbers [of Johnson and Storer, 1974]."

It seems that the last forays into her beloved badlands took place this year. In a 2006 letter to a local professional colleague she wrote: "I think last year—2005—was the last time I [have] been a few yards into the badlands."

She attended an art awards ceremony in Medicine Hat this year. Details of the event are not known, but she exhibited at least one piece of her flat art.

Also this year she made comments in a Letters to the Editor column of the Medicine Hat newspaper on the situation of children, schooling and the importance of a stable home environment (Johnson, 2005a).

Now 89, Hope was assisted by a close friend to

Figure 137. Hope with her "Tree of Life" mural at the Valleyview Retirement Community. Cameo photos of new residents are added as they arrive. Image courtesy of Sheila Schwab, Medicine Hat.

deal with all the paperwork she had accumulated over a very busy life (Johnson, 2005b). Her papers were distributed among family, friends and the archives at the Esplanade Arts and Heritage Centre in Medicine Hat.

2006

Hope donated part of her vertebrate fossil collection to the teaching collection of the Alberta Palaeontological Society (Fortier, 2006); at least four specimens were later included in short articles by Allen (2015, 2017a, 2017b, 2018). Hope was visited in June by senior members of the APS to discuss the revision of the fossil identification book and to pick up new drawings Hope had completed for that project (Figure 138).

By 2006 her body was failing her more and more. Her mind was still sharp and she had good vision with her progressive lens glasses. Her hearing she described as very poor: even with two hearing aids, she could "barely sing a note" and her legs were very painful from "osteoarthritis." But she could still draw and spent considerable time carefully rendering new fossil drawings for the much-anticipated fossil book. She wrote to the author about this time, declaring that "getting old is not for sissies." She maintained an interest in reading German; her name appears in shaky handwriting from late in life on the inside cover of an old German language book on ballad poetry (Bucheim, 1893).

Figure 138. Hope with APS members Wayne Braunberger (left) and Dan Quinsey on June 1, 2006, as they picked up some artwork for the new version of the fossil identification book. Image courtesy of Dan Quinsey.

2007–2008

Hope was quite active during this time, making new drawings of fossils for the upcoming book. It was important to her that the new fossil identification guide should be useful to collectors at home and in the field, too. The eventual coil-bound format of the book not only saved money on bindery costs but was practical in the field, as the relevant page could be examined with the rest of the book folded behind (H. Johnson, pers. comm. to S. Leuck, *c.* 2007; S. Leuck, pers. comm., 2014). The 1974 book lacked this useful feature, making it awkward to use in the field, especially on windy days.

Also during this period, she made medium-sized prints of some of her plant drawings, hand tinting them with watercolours and making them available for sale.

Her health continued to decline. In a 2007 letter to a professional colleague she revealed that both her knees had given out—at times she could barely walk and was "no good on uneven surfaces." Despite her two hearing aids she described herself as "very deaf," which she found "annoying and isolating." She could not hear music well, writing to a friend in 2007: "I have lost the upper third and lower third of the gamut." She signed the letter"Your ancient friend." In a 2007 letter to the author, she said that she "misses the badlands greatly."

2009

On March 28, 2009, not long after Hope's 93rd birthday, the revised and greatly expanded fossil book project, now renamed the *Guide to Common Vertebrate Fossils from the Cretaceous of Alberta* (Alberta Palaeontological Society, 2009), featuring hundreds of Hope's fossil drawings, was officially launched at the Esplanade Arts and Heritage Centre in Medicine Hat (McCuaig, 2009; Figure 139).[42] This greatly expanded sequel to the 1974 book is to the credit of various members of the Alberta Palaeontological Society; to Hope, who rendered many new illustrations late in life; and to others who put in many volunteer hours to see the project completed (Marsovsky, 2009). About fifty people attended the book launch[43] (and birthday party), most from the local Medicine Hat arts community in which Hope was active for many years. Hope was even up to giving a short speech.[44] The author was very much relieved that Hope had lived to see her upgraded book published; it was worrisome that she may not live to see that happy day. She did, and was obviously pleased with the result; after the formalities, she approached the author, holding the new book in one hand, giving him a sparkly-eyed wink and his arm a warm squeeze, exclaiming "We did it!"

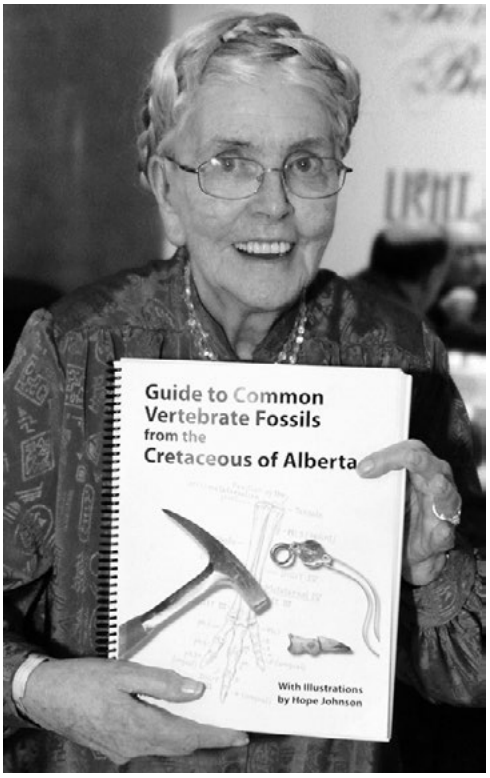


Figure 139. Hope on book launch day with her revised fossil identification manual. Image from McCuaig (2009).

Advances in printing technology mean that Hope's guidebook did not have a single, limited print run, eventually to be sold out. More can be readily printed and bound on demand, keeping inventory—and costs—low. Four hundred copies of the book were made initially and sold very quickly. New books are printed off in small batches and it continues to sell well; 1083 had been sold by the end of 2018 and the book is still in demand (M. Marsovsky, pers. comm., 2019). Hope was shown the final version one week before the official book launch, thereby reducing the risk she'd be overwhelmed with the excitement. She was "ecstatic" to see the completed final version (D. Quinsey, pers. comm., 2011). Some of those present at the book launch had the mistaken impression that she was bewildered and, given her frailty, even supposed that she was not totally aware of what was going on. But she was very much aware of the proceedings.

Figure 140. Hope with her final fossil exhibit, and the revised fossil book. Images courtesy of Sheila Schwab, Medicine Hat.

The new book contains 993 of her fossil drawings; some are small, simple line drawings and others are rendered in great detail. Most of the original Johnson and Storer (1974) drawings were reused, but there are many new ones and one can easily see how Hope's drawing

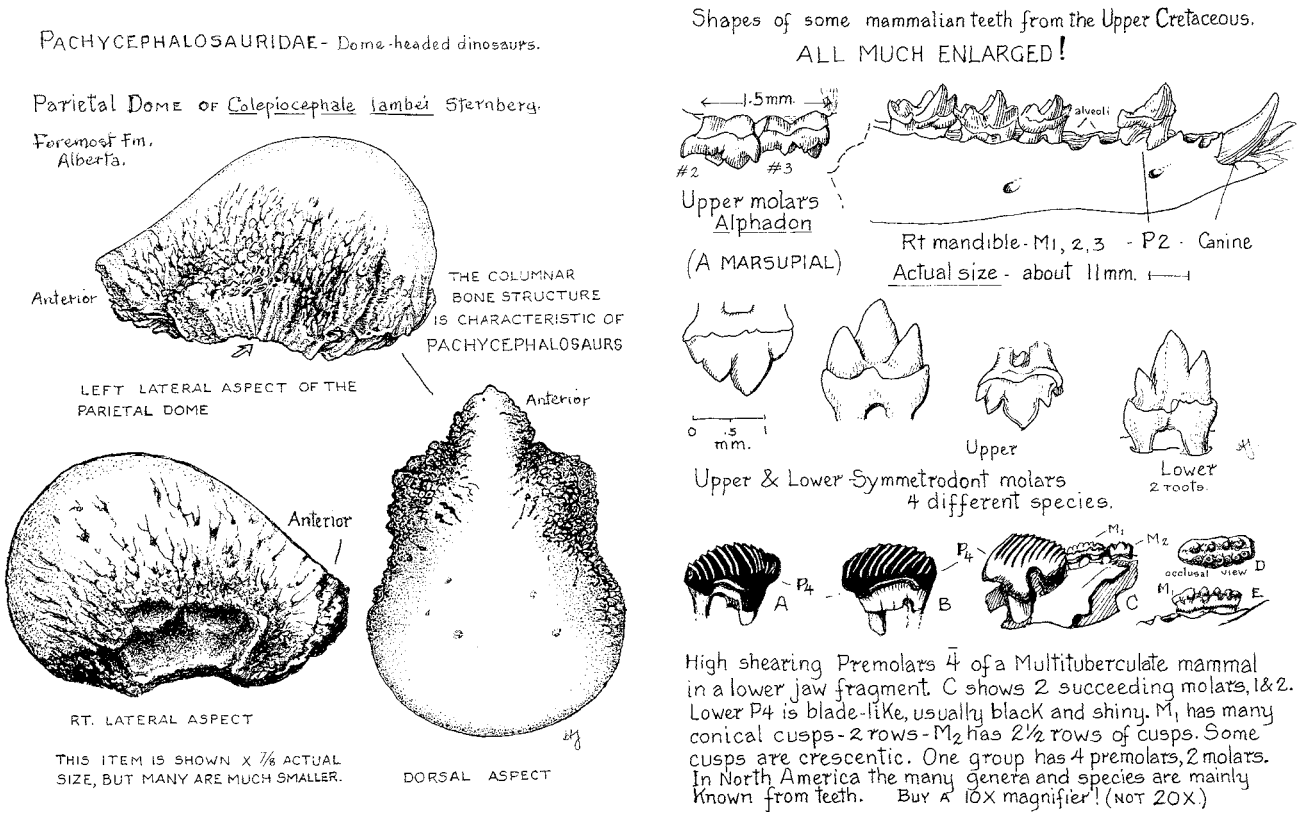


Figure 141. Some of Hope Johnson's drawings from the 2009 fossil guidebook.

style as a technical illustrator had evolved and improved with time (Figure 141). Hope penned some figures for the book when she was in her late eighties and early nineties and while some of the last were drawn with a slightly shaky hand, visible on some of the labelling, the drawings themselves are perfect. During the new drawings phase of the book, Hope had to undergo eye surgery to correct a glaucoma-like condition. Some illustrations could not be completed by Hope in time, so a few were done by APS Past-President Vaclav Marsovsky (D. Quinsey, pers. comm., 2011).

One might think that was the end of her story. But no, Hope had one more fossil display, this one at the Valleyview Retirement Community, where she lived the last decade of her life. On March 25, 2009 she set up a small table of various Alberta vertebrate fossils and one of her badland paintings and proudly showed off the new fossil identification book (Figure 140).

2010

A catastrophic earthquake levelled much of Port-au-Prince, Haiti on January 12, 2010. In her own last charitable act, Hope donated a large sum of money toward the earthquake relief efforts there. Others commented on the large amount of the donation, compared to her many earlier charitable acts but she said bluntly, ". . . not compared to what they need."

Her long hair, a trademark for much of her life, became too much for Hope and the ladies at the Valleyview to care for, so it was cut much shorter not long before she died. The result was quite surprising and took years off of her appearance (Figure 142). One of her braids was saved and framed in a shadow box along with several pictures of her. It has been kept by one of her daughters.

Figure 142. One of the last photographs of Hope, taken about four months before her passing (on or about April 12, 2010), showing her shortened hair. Image courtesy of the Johnson family.

Hope kept active in her reading, studies, and artwork until a few months before her death. Near the end she was dehydrated, had flu-like symptoms and had trouble swallowing and talking, possibly due to a thyroid tumor. She suffered some seizures and a long-standing cranial meningioma was also diagnosed. She left the Valleyview Retirement Community facility for hospital-based care, but never returned. She passed away peacefully in hospital in Medicine Hat on August 24, 2010 (Anonymous, 2010a-c). Sadly, it was son Lawrence's birthday. A non-denominational memorial service was held in Medicine Hat on September 4. About 100 friends, family, and members of the art and amateur and professional palaeontology communities attended (Quinsey, 2010).

As we silently drove back to Drumheller, lost in our own thoughts, I could not help but think it was a fitting day to say our final goodbyes: sunny and hot with a high of about 30°C, followed by an isolated late afternoon thundershower, barrelling in from the west to cool things off. Her ideal day, Hope once told the author, was to be hiking the hills of DPP or some other far-off place in southern Alberta, looking for that next fossil treasure, colourful stone, or rare plant.

At the Alberta Palaeontological Society's open house, held on September 17, Hope was remembered by Society members and visitors and a minute of silence was observed in her honour.

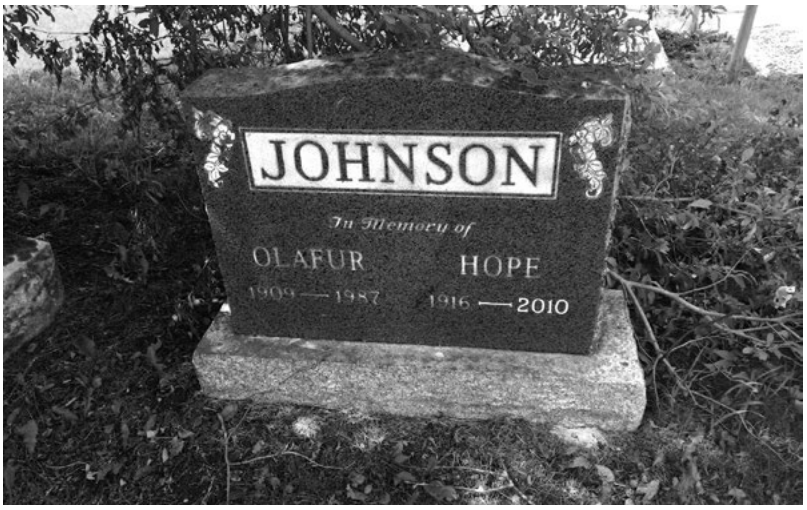


Figure 143. Olafur and Hope Johnson's headstone at the cemetery in Arnes, Manitoba in July 2017. Photograph courtesy of Lawrence Johnson.

Hope's cremated remains were interred in the family plot at the Arnes Cemetery (CanadaGenWeb's Cemetery Project, 2014a) in Arnes, Manitoba, on the west shore of Lake Winnipeg, joining Olafur and other family members (Johnsons and Thorvaldsons). Olafur and Hope's headstone (Figure 143) can be viewed at CanadaGenWeb's Cemetery Project (2014b).

In a final charitable act, after Hope's passing, the Johnson family allowed a showing of twenty pieces of her artwork at the Cultural Centre in Medicine Hat from October 28 to November 5, 2011, culminating in a closing reception and silent auction. All proceeds generated from the sale of her art went to support the Hat Art Club in Medicine Hat (Watson, 2015). Some of her last pieces, unfinished or nearly finished but unsigned artwork, were donated to the Hat Art Club in mid-November, 2015.

The following quotation was cited by Smith (1976) in an interview with Hope; he saw this inscription on a wall at the Ralston Art Club and thought Hope had posted it.

The wonder of the World, the beauty and the power,
The shapes of things, their beauty, their colour,
Lights and shades,
These I saw.
Look Ye also while Life lasts

— Old tombstone in Cumberland, England

Figure 144. Hope canoeing in southeastern Alberta badlands, August, 1966. Image courtesy of Penny Heath Torres, Nanaimo, British Columbia.

Overview

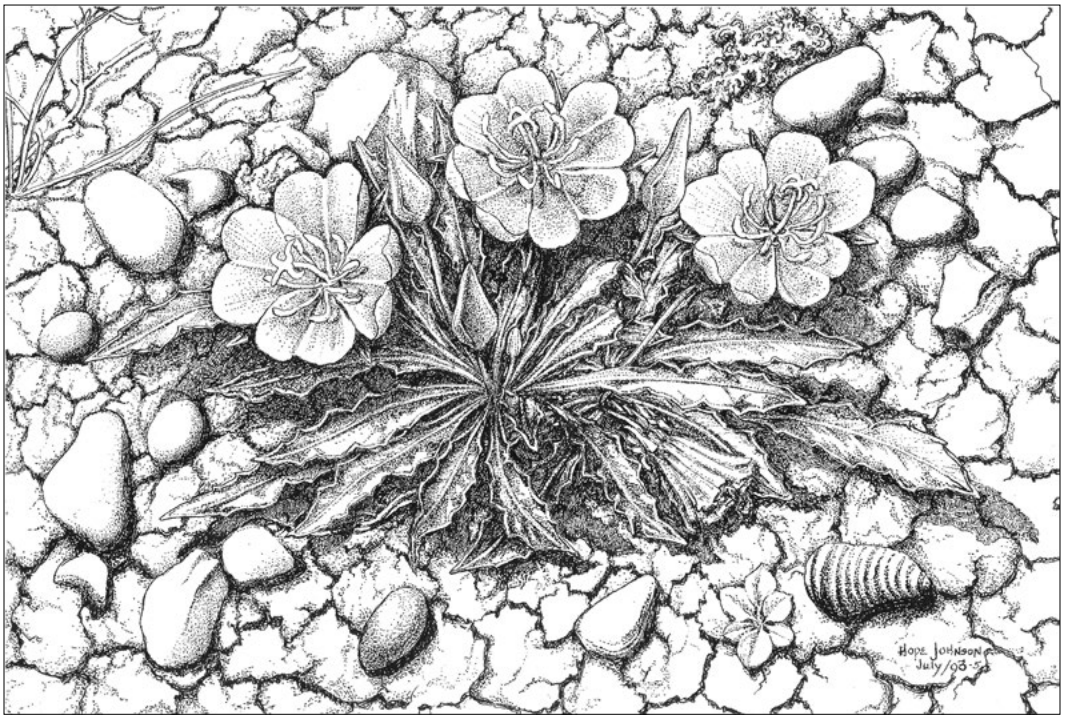


Figure 145. July 1993 technical pen-and-ink drawing of Butte Evening Primrose, *Oenothera caespitosa*. Image courtesy of Shane Leuck, Medicine Hat.

While Hope did work in Dinosaur Provincial Park (DPP), most of her prospecting took place in the Manyberries, White Rock, Petrified, and Red Rock coulees and the Irvine, Alberta areas where she prospected for Late Cretaceous (Campanian) fossils in the marine Bearpaw and terrestrial Foremost and Dinosaur Park Formations. Though not well known to history buffs, the Irvine area was one of the first areas in the province to be prospected for Late Cretaceous vertebrate fossils. Explorations took place there as early as 1883 (Anonymous, 1883; Panton, 1884, 1885, 1888). Hope made trips to Saskatchewan (for Late Cretaceous, Oligocene and Miocene-aged vertebrates) and Montana (Late Cretaceous vertebrates), but details of most of these activities are lost to history.

By about 1989 Hope had 2,366 fossils listed in her catalogue books (Johnson, undated 1–3). Her last year of fieldwork was 2005 (S. Leuck, pers. comm., 2010). How many fossils Hope collected for professional institutions is uncertain, but it is a large number (see Table 2, Page 189). RTMP catalogue records credit her with collecting and donating about 1,180 catalogued fossils of all types to which her name is attached as collector or co-collector. The first for that institution[45] was collected in mid October, 1964. Many more could likely be added to this number from the general DPP collection (TMP

1979.014 series; count = 947) which the PMA Palaeontology Department (later RTMP) acquired in the fall of 1979 (Danis and Walper, 1980). It is believed that many of the fossils in that series were collected by Hope during her years of employment and additional volunteer work at the Park. Many were of the common types of fossils found in the Park (*e.g.* hadrosaur pedal phalanges), but they provided a robust and statistically significant sample size required for some studies—for example, the frequency and occurrence of osteochondrosis lesions affecting the toes of hadrosaurs (Rothschild and Tanke, 2007).

Other professional institutions were queried about Hope Johnson-collected fossils in their care. Some odd discrepancies turned up. The University of Alberta (UA) found nothing on their database (C. Coy and N. Howard, pers. comm., 2010), yet Hope's fossil catalogue books indicate that she sent or relinquished at least twenty-four fossils to that institution over the years. Similarly, the Canadian Museum of Nature (CMN) found only one item—yet she worked with Dale Russell of that museum in DPP in August 1969 and ". . . spent several days with him, hunting for the very rare remains of small carnivorous dinosaurs, with some small success." (Bush, 1969). Her fossil catalogue books suggest that about twenty-one fossils went to the CMN, some used by Dale Russell for research toward a redescription of the maniraptoran theropod *Dromaeosaurus* (Colbert and Russell, 1969), though there is no record of them in the museum's current catalogue. Her books also indicate that about 110 fossils were sent to the ROM; sixty-six are presently catalogued there, suggesting that some were returned to Hope or deaccessioned for educational purposes.

Seventy-six microvertebrate fossils donated by Hope are accessioned in the Alberta Palaeontological Society (Calgary) collection (APS.2006 series; H. Allen, pers. comm.). Hope did some collecting in Saskatchewan, but no fossils collected by her found their way to the Royal Saskatchewan Museum (Tokaryk, pers. comm., 2010). The Esplanade Arts and Heritage Centre in Medicine Hat online collections database records sixty-eight fossils collected by Hope in their collections. Early in 1961 Hope (Johnson, undated 1, p. 63) records sending unspecified "Stenonychosaurus" (now *Troodon*) material to University of Paris Professor Charles Devillers. He is identified later as receiving other minor vertebrate and invertebrate material (count = 8) until about 1977. Some of her collection went to granddaughter Rachel Noone and fellow amateur fossil collectors Shane and Vicki Leuck, all in Medicine Hat. The Leucks used the collection to teach classrooms of children about vertebrate palaeontology (Page iv).

Regarding her two-dimensional art, the volume she produced is unknown, but must number in the many hundreds, or even thousands, if all her line and detailed technical drawings of fossil specimens and plants are taken into account. Much of her artwork was produced during the long Alberta winters. It is said that she did not like drawing or painting anything with straight lines—such as houses, and especially picket fences, which she is known to have deliberately excluded in one art piece (Figure 191; see also Figure 193).

Her artwork is mostly renditions of badland scenery, fossils, rocks, plants and flowers (Figure 145). Badland paintings in landscape format were the most popular subjects that people purchased (A. Jensen, pers. comm., 2015). She was "not a fast painter" and worked on "two or three pieces at a time, but would focus more on one if she was intrigued by the work" (A. Jensen, pers. comm., 2015). She told Medicine Hat artist Allan Jensen that the "temperature of the day influenced the colours she used in her paintings." Based on what the author has seen or heard, paintings of animals of any type are relatively rare. The most common (and only four at that) were owls (Figures 146, 189 a, b).

About five very colourful seahorse paintings are known (Figure 196), as is an etching of pronghorns (*Antilocapra americana*), dated 1964 (Figure 147). She rarely painted or drew portraits, though she did some of First Nations men (see Figure 178). Anonymous (1978b) shows her with a self-portrait in the background (see Figure 117). While she greatly enjoyed socializing with others, she did not consider

them as potential art subjects. In that regard, Hope stated that "most people are not very interesting" (E. Kleis, pers. comm., 2016). When the Ralston Art Club class drew a live human subject, Hope rarely appeared. Some unfinished landscape works were inspired by a 1974 trip to Arizona and an unfinished Machu Pichu painting exists at the [Medicine] Hat Art Club. A number of people believed she never painted mountain scenes, but the author has seen about half a dozen examples (Figures 165, 188, 190).

In her final years Hope remained active socially and scientifically though she grew progressively more frail and deaf, requiring two hearing aids. She could no longer appreciate music, which was a great loss for her. Her hips and knees gave her problems; a wheelchair was used to get her around on occasion. But her mind was clear and sharp and her hands remained steady enough to do detailed artwork. At the age of ninety she spotted a picture of a cat in a newspaper. She liked it and quickly did an accurate and steady rendering of it using pencil crayons (Figure 162).

Figure 146. Ballpoint pen sketch of an unidentified owl, undated.

She moved from her Redcliff home on January 18, 2000, to the Valleyview Care Centre in Medicine Hat where, at the age of eighty-four, to her amusement, she was considered too active by the other residents! (H. Johnson, pers. comm., 2008.) Being a stickler for spelling, she chafed at errors on the daily menu board at the Valleyview dining room. So she volunteered to write the menu each day, which entailed stretching to reach the top of the board. While engaged at this one day, a vertebral disc ruptured and caused her much trouble (along with arthritic knees) for the rest of her life.

Hope was duly recognized by society for her long dedication and scientific contributions to vertebrate palaeontology and natural history. As noted in an earlier chapter (Page 136), she was presented with an honorary Doctorate of Laws (LL.D.) from the University of Lethbridge (not the UA, as reported in Paulter, 2005) on May 16, 1981, during spring convocation (Anonymous, 1981a–d; Baillie, 1981;

Watmough, 1981; Anonymous, 1983a; Anonymous, 2014q), for her palaeontological work and public education on prairie plants. She is seen holding her degree along with other honorary recipients (Anonymous, 1981f). A celebration of this milestone was held by her friends on May 31 (Anonymous, 1981e). Though she was honoured and very proud to receive this recognition, having been unable to pursue a University education, in notes to her family Hope seems disappointed that she wasn't recognized for her prolific artwork. She writes: "They never mentioned my paintings, although I had about five one-artist shows over a period of twelve years"(see Hrapko, 1978). In a 2006 note to her family Hope made a revealing comment about her degree: "I was honoured by the U. of Lethbridge (1981) and received an honorary degree of Doctor of Laws. Unfortunately, this tends to awe people so I don't talk about it. Oddly, it was for public education, esp[ecially] in palaeontology, a very unpopular science in Medicine Hat, and even, I hear from the RTMP staff, in Drumheller!" She continued on this topic in other notes: "I find [the] main effect [of the honorary doctorate] is to separate one from those who do not have a Doctorate. This is awkward and unhelpful."

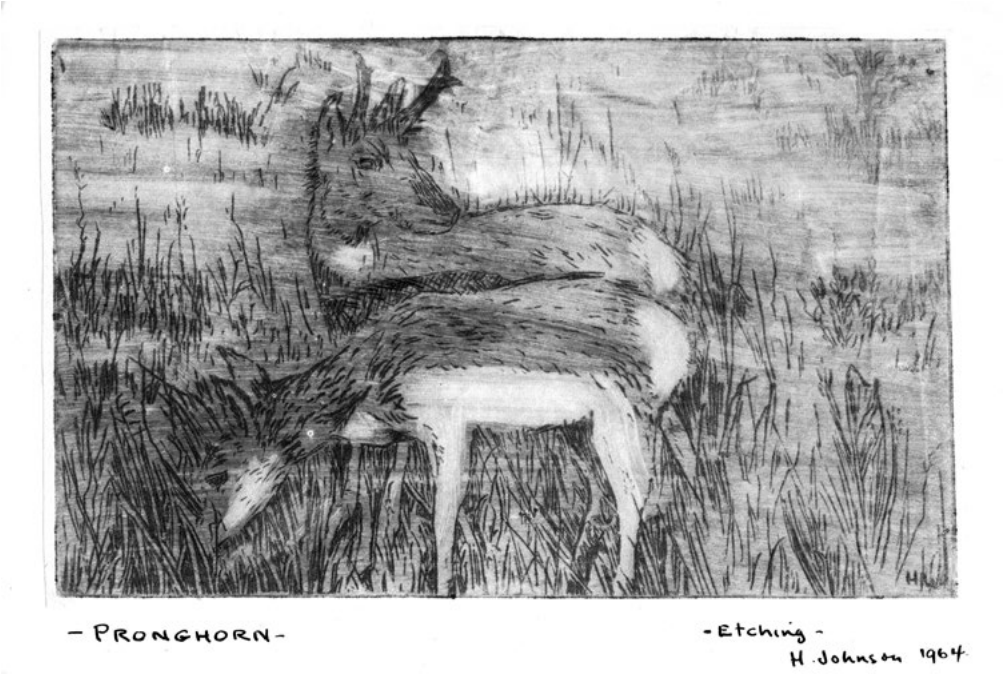


Figure 147. *Pronghorn*, etching on paper, 12.5 × 20 cm. Etchings were an uncommon medium for Hope. Image courtesy of Tom and Susan Willock, Banff.

Another unfortunate aspect of her honorary LL.D. was that people mistakenly assumed she was a doctor of medicine (Segal, 1994) and asked her for medical advice. It didn't help that during this time she was President of the Medicine Hat chapter of the Alzheimer's Society and a public speaker on the subject, which only reinforced the idea in the minds of some people that she was a medical doctor.

Hope was certainly one of a kind. Although there were many amateur fossil collectors in Alberta prior to Hope and others collecting during her active period, she always stood apart from the rest. She set a new standard of amateur excellence, responsibly surface collecting and respecting the law regarding excavation of *in situ* fossils without a permit. She carefully identified, catalogued and drew her fossil finds. She also worked closely with academics, attended professional conferences, gave talks to the public, co-wrote or assisted with scientific publications and was always willing to share her vast

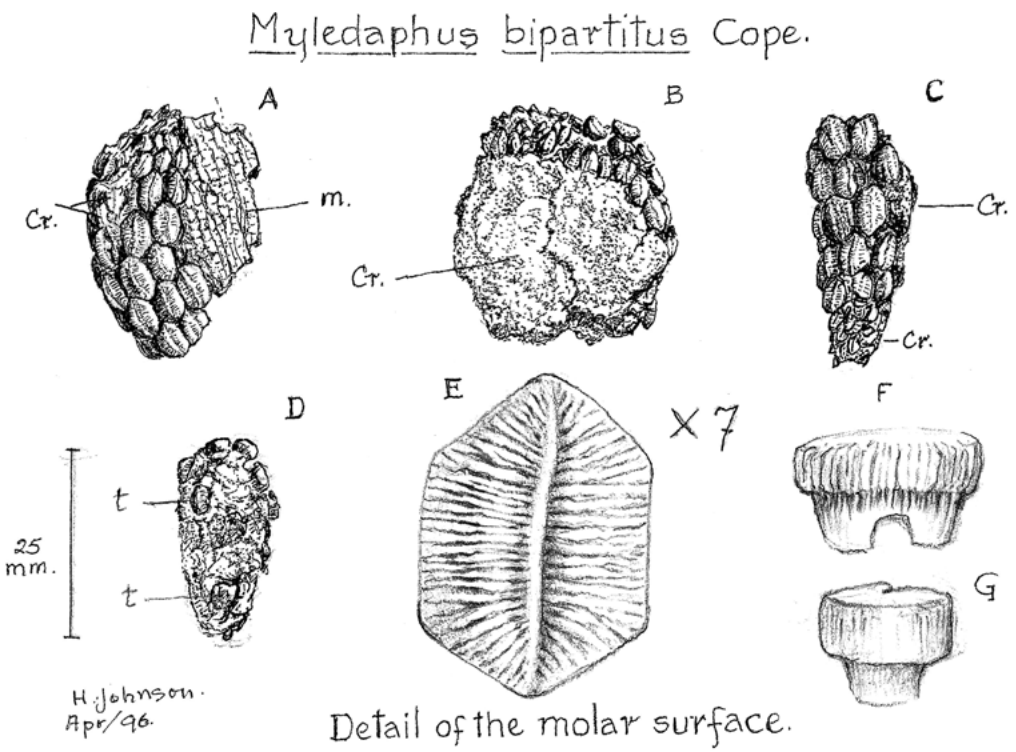
knowledge, experience, passion and enthusiasm—and all in a simple, matter-of-fact, professional manner. For these reasons the author considers Hope an extraordinary amateur palaeontologist, though she certainly excelled in other fields, too. It will be a long time before another of her calibre comes along.

I first met Hope at the end of the 1979 PMA field season party. She was asked to tell all she knew about dinosaurs—boy, did I get an earful of information! Hope was a teacher to many of us but was always eager to learn from other amateurs and especially from the professional community, with which she had contact for about fifty-five years. A specific incident comes to mind: long ago she brought me a curious, thickened, triangular-shaped and symmetrical piece of dinosaur bone that greatly puzzled her. She thought it was the midline section of an uncommon hadrosaur predentary.[46] It was indeed hadrosaur, from the midline, and was a skull bone—not a predentary, but a not often seen or understood supraoccipital. This bone forms the roof of the foramen magnum—the opening at the back of the skull where the spinal cord joins the vertebral column.

At this point, I had been in the field about seven years and here was the legendary Hope Johnson, with forty years experience. How would she respond to my correction? We stood face to face, those deep-blue eyes locked with mine as I explained it was not part of a predentary, then explained what it really was and why. When discussing fossils, she was always a very good student, taking the lesson in before asking questions. The age difference between Hope and her many colleagues did not bother her at all, especially if she stood to learn something or could share it with someone else. And she did indeed share this particular lesson. Years later I overheard a young RTMP volunteer in Dinosaur Park discussing with others a most puzzling bone someone had found that day. As I approached and saw that the mystery piece was a hadrosaur supraoccipital, the volunteer said to the others, "It's a hadrosaur supraoccipital—Hope told me." The volunteer then pointed out its salient features, describing it almost word-for-word as I had taught Hope in that briefest of lessons years earlier. I could not help but walk away with a big grin.

In other cases, she was not afraid to defend her position on a bone's identification. Regarding an opisthocoelus vertebra of larger size that I considered a hadrosaur cervical, she differed, writing to me (Johnson, 1996): "I'm having a hard time with your suggestion . . ." She went on to point out anatomical features she thought were important to identify the fossil. The matter remained unresolved and we agreed to disagree.

Hope got her start during a comparatively quiet phase of vertebrate palaeontology in Alberta. Things have certainly changed for the better. Through her life she experienced the explosive growth of palaeontology in the province. Over six decades she also witnessed the remarkable and long overdue rise of women in many levels of vertebrate palaeontology. She may have known Alberta's first geologist and *ad hoc* provincial palaeontologist, John A. Allan (1884–1955). After many years of fruitless efforts by others, she witnessed the Steveville badlands finally become a provincial park, in 1955. She attended the seminal first vertebrate palaeontology conference in Alberta in 1963 (see Page 71) and stemming from that, witnessed the development of the vertebrate palaeontology program at the UA in Edmonton (1965 to present), followed quickly by the Provincial Museum and Archives of Alberta (PMA; opened December 6, 1967; now the Royal Alberta Museum [RAM]) and its then largely Late Cretaceous-focussed palaeontology department (1967 to 1982). She helped and worked with those departments and others. She was involved in a number of oil patch field trips in the 1950s and 1960s (R. Chamney, pers. comm., 2015). She saw the rise of dinosaur research, the "dinosaur renaissance" from 1975 to the present and was there when DPP became a UNESCO World Heritage Site in 1980. She saw the Palaeontology Department of the Provincial Museum of Alberta (PMA) transferred to Drumheller in the fall of 1982, to become the Tyrrell Museum of Palaeontology (opened September 25, 1985) and the launch of its



A. Cr. stands for crystalline material coating part of 3 teeth. Either calcite or silica.
m. for matrix. Very fine striations running from the upper end, as viewed, down. Mainly parallel, but a bit of crowding is present. (about level with the —m.) This area may be the remains of cartilage. It has many crosswise fractures on each strip.

B. Note many tiny teeth at the top left. The teeth enlarge downward and to the right. Cr. This is an area covered with many tiny 'crystals' - but they may be very tiny denticles. Some have translucent tips.

C. This aspect shows the item in its narrowest view. The largest teeth shown in A. are now at the top. A-Cr. here is at the bottom in C.

D. Opposite side. Blocky fractures. 2 sections of (t) larger teeth. The lower one shows a split surface.

H. Johnson.

Figure 148. April, 1996 technical drawings (rearranged from the original) of Late Cretaceous guitarfish skate, *Myledaphus bipartitus* teeth and jaw pieces with teeth. Image courtesy of Shane Leuck, Medicine Hat.

satellite facility, the Field Station at DPP (opened May 15, 1987).

She also saw the rise in the 1970s and 1980s of the recovery and understanding of older Ice Age mammals in southeastern Alberta (Anonymous, 1979a–c) and, though little-known to many in the scientific community, she contributed to that work, too, participating in several symposia related to that research.

She attended the Dinosaur Provincial Park symposium at the RTMP in 2005. She died just one month before the Royal Tyrrell Museum celebrated its 25th anniversary, where she would have been an honoured guest. When Hope began her hobby, there were no vertebrate palaeontology students in Alberta. When she died, there were about two dozen M.Sc., Ph.D. and post-doctoral vertebrate palaeontology students active in Alberta or leaving that year to pursue higher degrees outside of the province. Through her collected specimens and especially her fossil guidebooks, many of these students, those prior—and those to come—owe a debt of gratitude to Hope as those guidebooks supplemented their own formal education. Hope may not have realized it, but a good number of leading vertebrate palaeontologists today learned their basic dinosaur anatomy in the laboratory and field with Hope's books in hand. Her memory will live on through her artwork for many years to come. The author is grateful to have known Hope and to have learned much about basic Late Cretaceous vertebrate anatomy through her artwork and one-on-one discussions.

Hope Johnson illustrated Alberta berries for another major field guide (Droppo and Johnson, 1988). She also assisted in some other native small-animal research projects, namely conservation of eastern short-horned lizards, *Phrynosoma brevirostris* (Powell and Russell, 1991) and the plains spadefoot toad, *Spea bombifrons* (Lauzon, 1999).

She was interested in researching and writing about local human history, co-writing (with Hope Michael) *Down the Years at Elkwater* in 1981. For all these activities she was again recognized, receiving the Loran L. Goulden Award in 1997 from the Federation of Alberta Naturalists (Anonymous, 1997b; Lindores, 1997) and in 1982 the city of Medicine Hat gave her a civic recognition award for her cultural work. In 1981 she was recognized by the Alberta Historic Resources Foundation as one of fifty women who made an outstanding contribution to the province's society. She was again recognized as one of 200 remarkable Alberta women (Sanderson, 1999). As if all this were not enough, she found time to be a leader in the Scouting movement and was involved in the arts and theatre as an actor and scene painter and enjoyed gardening. Exhibits of her paintings and drawings in the Medicine Hat area were frequent through the 1980s and 1990s and she gave art instruction at the Veiner Centre in Medicine Hat.

Hope liked to write and there are a number of her unpublished short manuscripts curated in the Esplanade Arts and Heritage Centre Archives in Medicine Hat. One, untitled and thirteen pages long, was about dinosaurs and aimed at children. It was penned in June, 1986; perhaps she had another book project in mind. Another, written when she lived in Redcliff (1976 to 2000) was twenty-six pages long, with pictures, entitled *Red Rock Coulee—a preserved natural area*. A real surprise was another, undated, three-page manuscript on *Bessarabean Germans in the Medicine Hat area*. The author did a double-take when he saw that and even questioned Archives staff as to its possible misfiling, but it had Hope's name on it. In discussing this further, it was thought by an old friend, who recalled her interest in the plight and immigrations of these people, to be from the late 1950s. Other manuscripts include six pages on *The Irvine area* (*c.* 1977); eight pages on *The Pleistocene or Ice Age* (undated); and *Richardson's Ground Squirrel Prairie Gopher, Spermophilis richardsonii*, undated and two pages long.

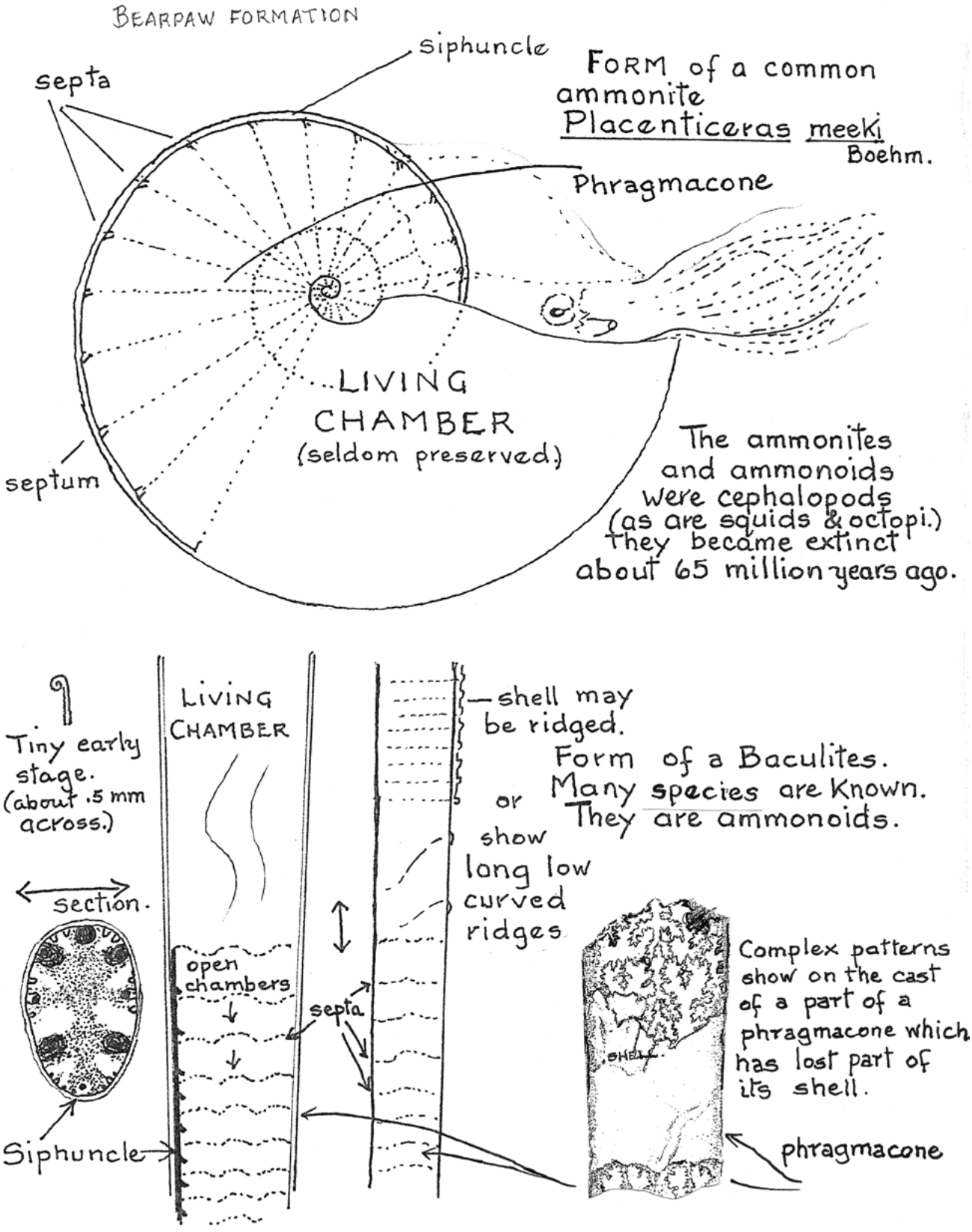


Figure 149. Hope had a penchant for Late Cretaceous ammonites from Alberta, so a number of her art projects featuring them are included here. This is an undated technical piece, which appears made for a book, but was never used. Collection of Shane Leuck, Medicine Hat.

Her character and personal anecdotes

Many of us know Hope through her art and palaeontology-themed technical drawings, but there was much more to the woman, and some aspects are quite surprising. Following is a hodgepodge of "Hope Johnson, the woman" trivia, in no particular order.

Hope did not seem to have been involved in any community sports, though she was an active volunteer at Sports Day at school. Her children enjoyed horseback riding when they were young (Olafur and Hope owned a horse for a while); Hope loved horses, but apparently did not ride them herself (F. Bickerton, pers. comm., 2014).

She did not always endear herself to those she met. Some in palaeontology found her a bit coarse, which is understandable given her gender and the time period in which she was active. Vertebrate palaeontology (and all Earth sciences for that matter) in the 1940s to 1980s—and right back to its earliest roots in Alberta—was very much a male-dominated arena and if a woman wanted to be noticed and taken seriously by her male colleagues, she had to stand out and make herself heard and accepted.

In the 1930s, Hope's long-time friend Irene Vanderloh wanted to become a palaeontologist and was told by Levi Sternberg that palaeontology was no place for a woman (Tanke, 2010a). Hope was "no shrinking violet" according to long-term friend and colleague Jane Colwell (Danis) who worked with her in the field in DPP, mostly in the mid to late 1960s. Hope was a big and outgoing woman and unafraid to speak her mind, especially against political authority within the Alberta Government, when she thought they were wrong regarding palaeontological legal matters. She was a little ahead of her time but recognized the eventual strong role women would play in the Earth sciences and particularly vertebrate palaeontology. She was thus a good friend and strong supporter of Jane Colwell when Jane worked in the Park, from 1966 to 1968[47] and of other female amateur collectors such as Irene Vanderloh and Medicine Hat's Shirley Lawson (1944–2012), Sharon Stiem (Stubbs, 1993), and Vicki Leuck (1961–2014).

Another family Hope collected fossils with were the local artist Luke Lindoe (1913–2000; Anonymous, 2018a) and his son, Allan Lindoe (1944–) of Medicine Hat (A. Jensen, pers. comm., 2015), though their names do not appear in her detailed fossil catalogue books. Luke, as well as being a well-known artist, made some significant palaeontological finds in the area. His son Allan went on to have a long career under Richard Fox at the UA as an excellent and highly regarded fossil preparator.

Regarding languages, she could speak some French and some German phrases (she taught German grammar and conversation to military members in the early 1940s). She tried Welsh, but ". . . it was a disaster." (B. Ashwick, pers. comm., 2014).

Some comments various people made about Hope: "She was SMART!"; "unique"; "always eager to learn"; "very well read"; "wonderful"; "I loved her to bits"; "could carry a conversation on pretty much anything"; "would tell it like it is"; "a strong woman"; an "Earth Mother"; "had a photographic [or wonderful] memory"; "I was in awe of her"; "a walking encyclopedia" (five people independently made this comment); "a real powerhouse"; "was an amazing woman, I am still in awe of her"; "animated"; "very curious mind"; "very kind and helped me through a bad time"; "I liked how much she was special with children and spoke to them whenever they had a question"; "loved the mystery of the [badlands] landscape"; "went above and beyond in many things she did"; "when she spoke, everyone listened"; "a community supporter"; "a bulldozer"; "down to earth"; "no nonsense"; "efficient"; "She very much knew her stuff"; "moved beautifully through the badlands"; "multidimensional"; "had a grace about her"; "supportive of children's education"; "consumed by what she did"; "just liked to have fun learning about most anything"; "knew her manners"; "had a big heart"; "classy"; "her smile lit up a room"; "She

was a lady"; "spoke with precision and conviction"; "gave all of herself"; "She is that rare person who takes the time and trouble to explain things to which our eyes are not always open" (Smith, 1976, p. 3); and ". . . felt lucky to have known her. Through her we were able to see many interesting fossil localities and met museum scientists and technicians—people we otherwise would never have met" (B. Lawson, pers. comm., 2015). Several within the Alberta professional vertebrate palaeontology community also independently described Hope to the author as a "tough old bird."

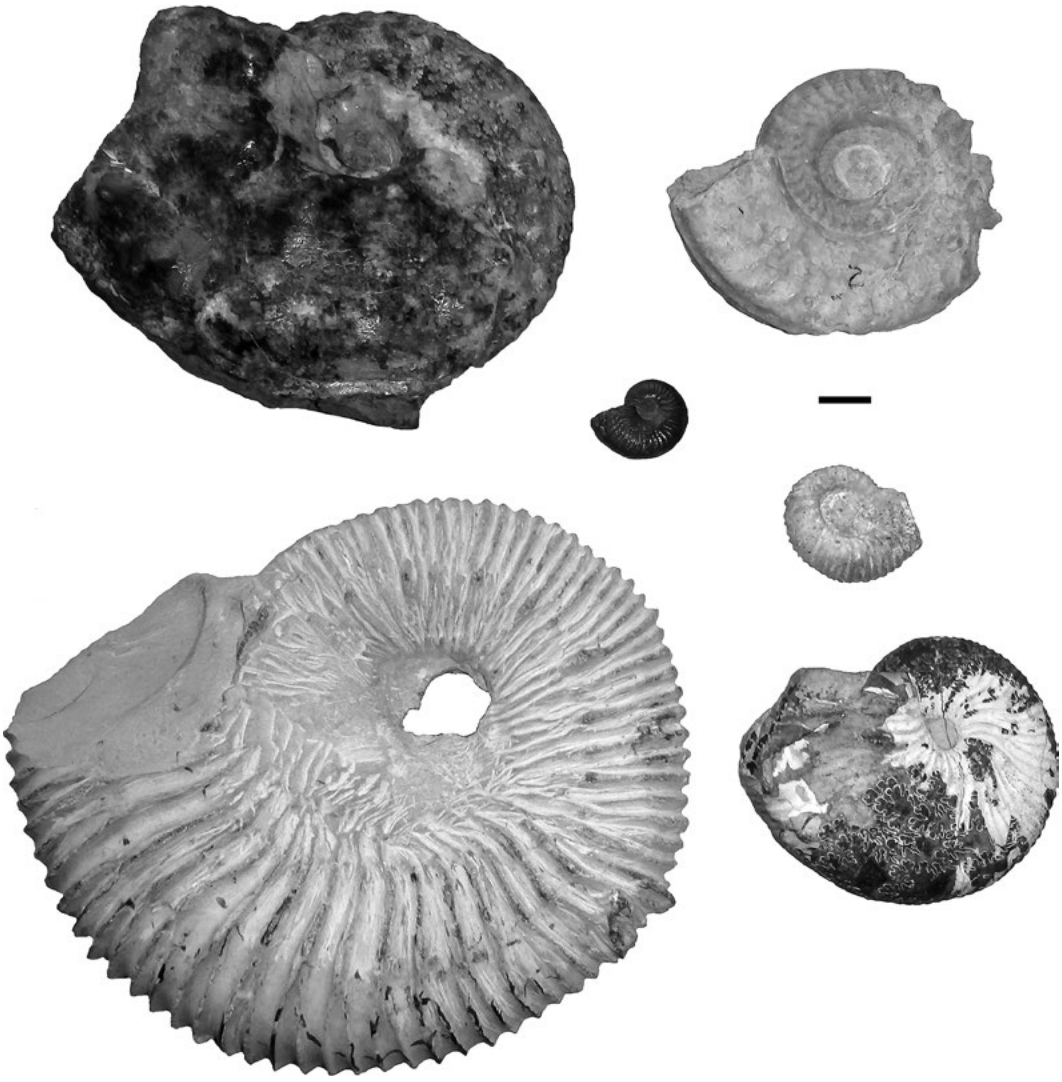


Figure 150. An assortment of ammonite fossils found in Hope's collection after her passing. Most have catalogue numbers, but their record book is missing. Upper left: *Placenticeras intercalare*; bottom right: *Jeletzkytes* cf. *crassus*; others are unidentified. Scale bar = 1 cm. Photograph by the author.

On religion, she was brought up in a religious family (her mother especially so), but as an adult was not religious herself (nor was Ollie) and had many questions and doubts about it. According to son Lawrence (pers. comm., 2018), "She respected the spiritual aspects, but was disdainful of dogma and upset at times about the behaviour of some organized religions." However, she was very knowledgeable about scriptures and could quote them.

As to Hope's memory, Lawrence relates (pers. comm., 2018), "I was often astounded at her ability to remember all the Latin names of prairie plants and birds and dinosaurs (and scripture and poems and . . .). She told me she could often see in her mind the position on the page where something was written in a kind of photographic fashion."

Her children described her as supportive, giving them space to resolve their own problems, but let them know she was there for them if need be. Lawrence recalls (pers. comm., 2018): "When I was a rebellious high school student in the mid 1960s (not rebelling against my mother—we got along famously), I grew long hair and was kicked out of school for this. My mother called the school and they said I wasn't conforming to the dress code. She asked, 'How are his grades?' The principal replied, 'They're fine, it's just that we don't allow long hair.' My mother repeated: 'How are his grades?' This went on until they backed down."

Others recalled Hope at work in the field. People were surprised by how quick and nimble this larger-sized woman was when scrambling across treacherous badlands slopes, covered with loose, roundish ironstone fragments and cactus, an agility honed over years of prospecting. Others recalled her field footwear: high-topped leather boots complete with hobnails (boots of the style worn by the early collectors in the Park); and a large hunting knife on her belt (Bush, 1969). The high-topped boots were not for show, but to guard against rattlesnakes, for which Hope had a high respect. Whenever walking through brush near the badlands, she would toss pebbles ahead of her to alert the snakes to her presence (Bush, 1969, p. 11). She was known to pop small fossils in her mouth for safe-keeping while scrambling up hills using all four hands and feet (Forbes, 1998). One recalled her curious habit of picking up large but unwanted fossil specimens or pieces thereof and placing them atop outcrops. The purpose of this was not clear, but perhaps it was used as a waypoint, as Hope noted several times in her life (Johnson, in Bridge, 1959b; Johnson, 2009) that it was easy to get lost in the badlands. The author has seen a number of such placed fossils in DPP—perhaps Hope put some of them there.

In the parched badlands on really hot days, she was not unduly affected by the heat, commenting to hot and exhausted fellow collectors, "it's nothing." The few field photographs of her never show her wearing a hat, so she certainly was unaffected by the sun's intense rays.

While interested in all fossils, she was particularly fond of ammonites (S. Leuck, pers. comm., 2014). A small number of them were found in her fossil collection after her passing (Figure 150).

In the late 1970s and early 1980s it was customary for the PMA staff to hold a big end-of-the-field-season party. People from far and wide were invited and they tended to dress up. Hope and Olafur attended all of these events. Hope would be dressed up, but just in fancier field gear and her broken-in field boots. While she was there just for the party, you knew that if she was invited to take part in a field project, she was ready to go right there and then. She was known to carry a small pair of pliers to pull out the inevitable cactus spines acquired while hiking the badlands.

Several recalled her piercing, bright blue eyes. Like many women, Hope liked to receive flowers, but not the usual florist bouquets—she preferred wildflowers (D. Quinsey, pers. comm., 2010). That said, she was partial to roses, and Ollie bought her some every year on her birthday. Perfume and makeup were rarely part of her personal grooming. She wore jewellery but it was usually simple in style and not worn to be showy.

Hope was curious about pretty much everything, except ". . . cars and the stock market." (quoted in Segal, 1994).

Some commented that her house was a mess: she was averse to dusting and vacuuming—and she knew it too, hanging signs on the walls, with mottoes like "Dull women have the cleanest houses" or "Come in and converse, don't complain about the mess, sometimes it's even worse." Friends who knew

Figure 151. Hope and a fellow CWAC at the Suffield base, in 1945, navigate a large mud puddle by swapping footwear; Hope, in a show of strength, then carries the other woman across. Images courtesy of the Johnson family.

her from 1962 to 1975 commented on ". . . arriving at the Johnson home in Ralston one day and being quite shocked by the condition of the dining room. Every inch of space on the dining room table and buffet was covered with piles of books and papers. I don't know where the family ate." (D. and P. Currie, pers. comm., 2016). Recent research (McKay, 2014; Vohs *et al.*, 2014) has suggested that untidy people are often the most creative, which Hope certainly was. Regardless of her housekeeping skills, she was very well organized—but most of that organizing was done in her head. Fortunately for us in later years she wrote much of it down.

Regarding food, Hope was perhaps not the best of cooks, but the children were always well fed. She made a fine minced beef hash and her prodigious baking was superb, with notable hot cross buns and cinnamon twists; her homemade bread was described as "wonderful." Meal preparation was often accompanied by her singing an old English folk song. She was a big time tea drinker, always at noon and four o'clock, but would have the odd beer with husband Ollie on weekends as they discussed politics, which interested both, being Liberal Party supporters at the provincial and federal levels. At the Suffield base bar on Friday nights, she would down tea or the odd Scotch. Despite growing up in a time when nearly everyone smoked, she never got hooked on it. In the early 1960s she once tried making a liquor from the berries of pincushion cactus (*Escobaria vivipara*), but it was a disaster. An unfortunate taste tester (B. Ashwick, pers. comm., 2014) relates:

> Hope had us all picking the berries, and using a very Medicine Hat system of measure, collected four large ice cream pails full. The berries were washed, mashed and allowed to ferment until just about Christmas. The "mash" was screened and the very slimy fermented juice tasted. Disgusting. The alcohol level was measured and was about 2%. Gareth [Jones] supplied half a bottle of Benedictine and Brandy liqueur, which fortified the mix. It tasted like brandy-flavoured snot. The viscosity index was off the scale—it formed "strings" from the glass to the lips. An empty B&B bottle was stripped of its label and Hope made a new label with a burnt-edge antique

decal look. The label read, "Made by the mystified natives of the South Saskatchewan River." It was an awful experiment—not to be repeated. As Gareth stated: "To be held in reserve for less than discriminating guests."

This experimental disaster aside, Olafur and Hope were known to always bring bottles of good wine to social events.

Hope was robust in build, fit and very strong. Men on the Suffield base (or sometimes their wives) would challenge her to an arm wrestle and she beat them most of the time, even men larger than her! The wife of one friend was herself a strong woman and at the Officer's Club pointed out a big wood and leather easy chair and told Hope she could lift that—and did, after some huffing and puffing. Hope did, too—but with only one arm! Another example of her strength is comically illustrated in Figure 151. An old friend recalled a community picnic where Hope joined a tug-of-war team: she was too light on her feet for pulling but, given her size, "made a good anchor."

Hope had deep respect for First Nations peoples. Among her archived papers in Medicine Hat and in the care of family members are postcards and colour prints of western North American First Nations people (some saved from calendars) and an article on Blackfoot First Nations people (Schaeffer, 1958). She is known to have painted a few portraits of them, too (Figure 178). It is not known if she had direct contact with First Nations in southern Alberta; her surviving papers offer no evidence that she did.

A number of people said she was very good with kids; the kids "just loved her" and she loved them back. She was popular with her Cub Scouts and children's school groups at DPP. In later years she would visit school classes of young children and talk to them about rocks, fossils, dinosaurs, plants and nature (Figure 128). She is known to have given one-on-one instruction with children and their fossil finds as late as about 2002, providing identifications and sketches of the living animals their fossils represented. Some interviewees remembered as children being quite fascinated with her long silver hair, braided and coiled atop her head—a Hope trademark since the early 1950s. She was taught this "milkmaid braid" style by friend and fellow Suffield CWAC, Flo Bickerton (F. Bickerton, pers. comm., 2014; see also Weiss, 2012).

Hope was gentle, motherly and soft-spoken with children; she could speak at a child's level and keep their attention. She was known to work extra hard with kids, especially those who expressed an interest in art or nature—even more so for rocks and fossils. "Anything she could do for children, she did," said one interviewee, adding that often when she walked through Ralston to the store or library, a line of children was seen trailing behind her, asking all sorts of questions about pretty much everything. Others recalled that "anyone who found anything of a palaeontological or botanical nature would take it to Hope for identification. This included children and adults; she was always very patient and took time to explain the find." (D. and P. Currie, pers. comm., 2016). When Ollie and Hope moved away from Ralston an interviewer (Smith, 1976) lamented the loss of contact she had with local children: ". . . [moving] far enough away that small boys will no longer be able to knock at her door at any time with interesting bits picked off the prairie: 'Please, Mrs. Johnson, is this a dinosaur bone?'"

An old friend recalled how Hope, without any formal medical training, correctly diagnosed bipolar disorder in a young girl, based on the unusual colours the girl used in her paintings. Hope advised the parents of this and recommended the girl see a doctor. The parents did just that and they were able to get the girl treatment for a condition no one knew she had (P. Dudley-Reese, pers. comm., 2014).

Hope loved cats and dogs and owned a Maine Coon cat when she lived alone at home later in life. When she went into the Valleyview Retirement Community she was able to take the cat with her. She found cats "good to look at and for watching their behaviour" (E. Kleis, pers. comm., 2016). "She was quite concerned about the welfare of animals," according to her daughter Margaret (M. Noone, pers. comm., 2014). Another person commented that she had a habit of picking up stray animals.

Early in the 1960s she gave her son Lawrence a copy of *The Lord of the Rings*—which he loved—and encouraged his interest by learning Elvish from Tolkien's appendix, then writing down names, genealogies and poems in the florid High Elvish text with great calligraphic skill. A drawing may originate from this time (Figure 152).

Hope travelled overseas to Europe at least three times, visiting friends and family in England, Wales (visiting with Gareth Jones several times after Ollie's passing), The Netherlands, France and Switzerland. A real botanist's treat for her was visiting Kew Gardens in England, in the fall of 1988. She painted some watercolours of old buildings when in Wales the same year (Figure 190).

A real surprise, discovered during research for this book, was that she really liked Halloween and dressed up as a witch each year at the Valleyview residence, even undoing her braids and letting her hip-length grey hair flow (Figure 136). Some years she added "bloody" ketchup streaks down her face. Before a party she would disappear most of the afternoon. Then, as festivities commenced, she would appear in costume, making a grand entrance. She painted an unusual watercolour of witches among the skeletal remains of dead cottonwood trees (Figure 192), complete with an inventive, eleven-page story describing the scene in detail, with names and activities of the witches and their magical attributes.

Hope had serious concerns about the illegal collection of fossils, especially those procured for sale. Over half a century ago she publicly spoke out against such activities, which she felt was going on with DPP fossils (Johnson, in Bridge, 1959b). Furthermore, Hope and John E. Storer spoke out against the selling of fossils (Johnson and Storer, 1974, pp. 2–3). She revealed her anger and disgust to the author when the owner of a fossil shop in Drumheller was caught and convicted for illegally excavating vertebrate fossils in DPP that were destined for sale (Anonymous, 1987f, g; Mofina, 1987; Plumtree, 1987b; Watts and Smith, 1987; Spalding, 1999, pp. 271–272).

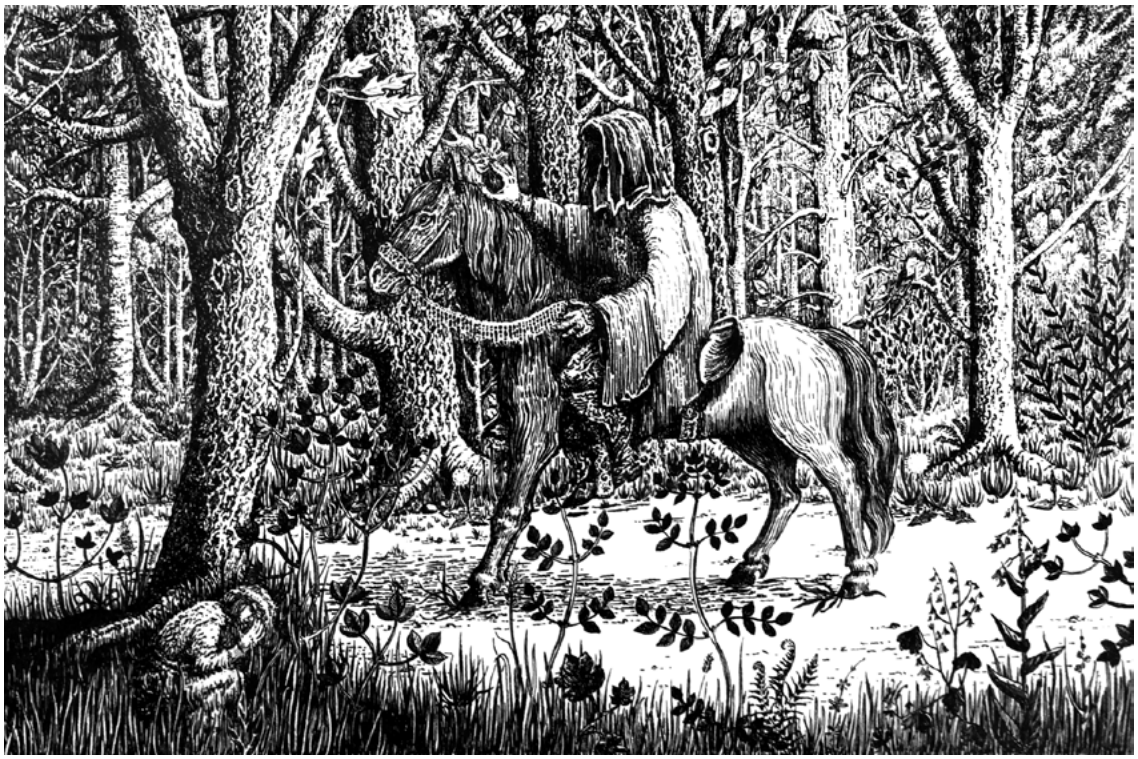


Figure 152. Tolkien-influenced artwork by Hope, *Frodo and the Black Rider*, undated. Frodo can be seen hiding at the base of the tree at left. Image courtesy of the Johnson family.

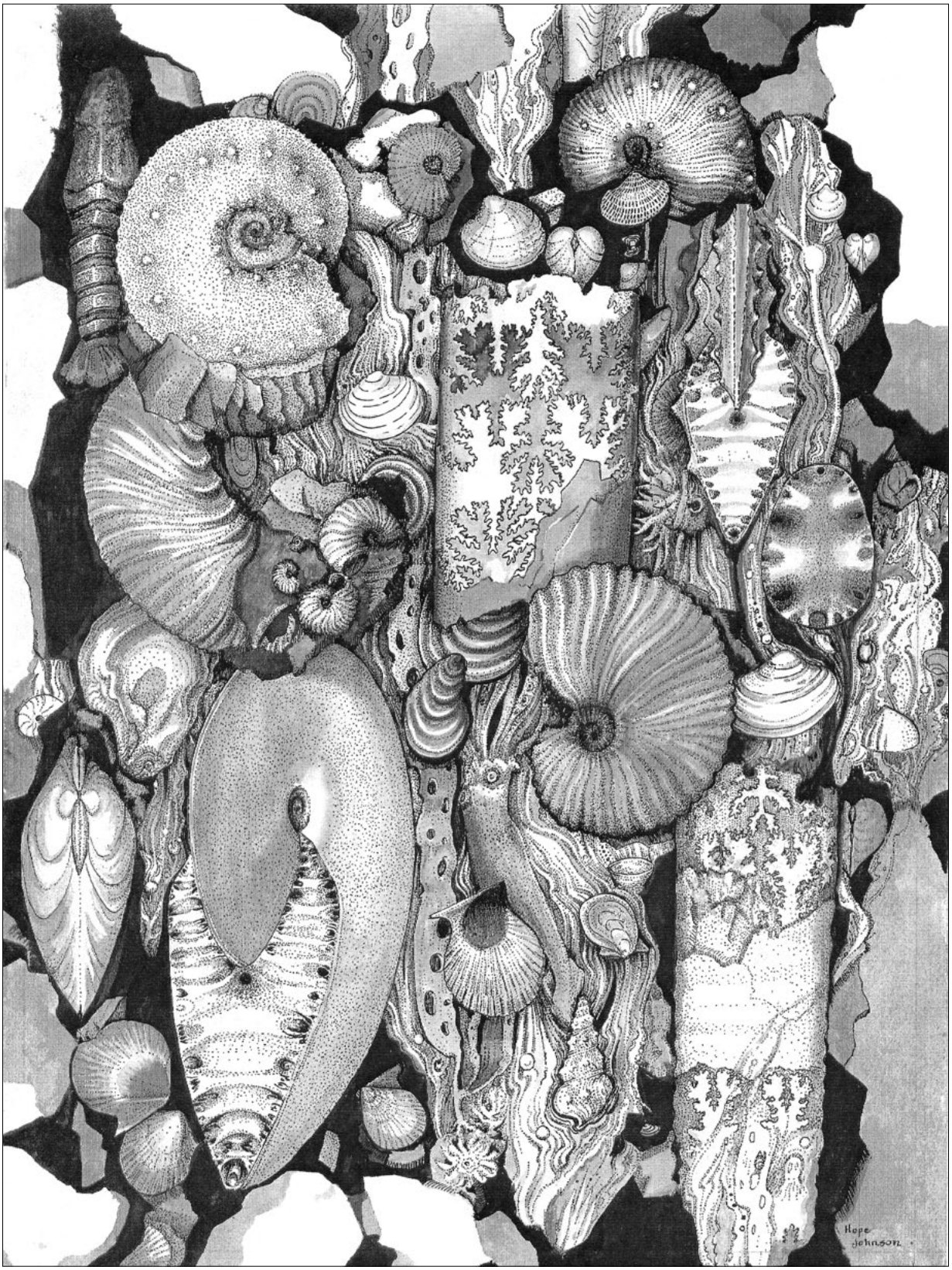


Figure 153. Montage of Late Cretaceous Bearpaw Formation invertebrates. Laser print of a pen-and-ink and ink wash. Collection of Allan C.J. Jensen, Medicine Hat, who said Hope was particularly pleased with how it turned out; it was one of her favourite pieces. The fossils illustrated here are identified in Johnson, 1992a.

She collected fossils on her own time, at her own expense and was always willing to part with important finds if academic institutions wanted them; she imparted that responsibility to other amateur collectors. Hope encouraged the pleasure and thrill of amateur fossil collecting, but emphasized that the scientific importance of such finds should not be overlooked and that important discoveries should be shared with museums and scientists (S. Leuck and B. Lawson, pers. comm., 2015). The palaeontological community recognized that she was an amateur collector, but she was accepted into their ranks as reliable and trustworthy and there was none of the suspicion, mistrust, and conflict that is unfortunately a recurring problem between amateurs and commercial fossil dealers and the academic community (*e.g.* Poulton, 1986; Wilson, 1986; Braunberger, 1987; Rivers, 1987; Bolvikoski, 1988a, b; Anonymous, 1987c, 1988; Curtis, 1988a, b; Miller, 1988; Wilson, 1988; Jones, 1989; McCartney, 1989; Struzik, 1990; Engberg, 1998; Dyke, 2015). Well into her senior years, she told the author that she found all this "troubling" and noted that like herself, some members of the general public were good at reporting their finds to the proper authorities (Maier *et al.*, 1981, p. 18), yet many amateur fossil collectors rarely do, thereby creating division.

Hope mailed fossils to various museums and there is no record among her papers of even being reimbursed for postage. Hope traded fossils with other amateurs—a common activity—and she did trade a few fossils to the University of Paris. Importantly, Hope's fossil catalogue books (Johnson, undated, 1–3) record the dispersal of some of her fossils. Some were sent to major museums and institutions of higher learning; others went to small, local museums, local elementary schools, or—importantly—to private individuals directly associated with children's education, such as Hedy Hobberlin.[48] It was her wish that the remainder of her fossil collection should go to the RTMP after she passed on. This was fulfilled and her fossils are now teaching the next generations of amateur fossil hunters (and hopefully some professionals, too!) through the RTMP's Education Department as hands-on interpretive specimens. This programme and Hope's fossils enrich the museum education experience of some 5,600 students annually (E. Wiebe, pers. comm., 2015). Earle Wiebe, the RTMP's Head of Education commented to the author, in 2005:

> . . . we have about fifty fossils catalogued in our education teaching collection. These fossils are used as "hands-on" specimens in our school and public programs. We have used about 350 uncatalogued Hope Johnson fossils in education for our simulated dig programs to give students an opportunity to experience what it is like to be a palaeontologist. The majority of her donation, however, is used in our simulated indoor digging programs. We have used (and continue to use) hundreds of her fossils as specimens that are buried in our matrix and students have the opportunity to uncover them and learn about palaeontology. Hope's fossil donation is very much appreciated, and has provided us the opportunity to create wonderful learning experiences for our visitors.

While Hope encouraged other amateurs with their fossil collecting hobby, she did have concerns about rampant, unethical collecting more than fifty years ago. Incidents were recorded prior to this (Anonymous, 1937d), but gained momentum around this time and continue up to the present (see comments by Chamney and Sternberg in Langston, 1965, pp. 13, 14). There were worries about over-collecting of fossils in Alberta, with people making off with petrified logs and dinosaur bones to end up as personal garden ornaments, rockeries (Anonymous, 1930a, 1934, 1936a, 1940c, 1955c, 1956b, 1962k, 2014j) or stone fences; permineralized dinosaur bones and wood being cut up and polished for jewellery (though Hope polished stones, according to Smith, 1976) or fossils ending up as trinket souvenir items (Anonymous, 1959a, 1962l, 1962m, 1962n, 1963c, 1964b, 1966c, 1975c; Green, 1969; Matern, 1970; Osgood, 1965; Scott, 1965; Simaluk, 1974; Jamsa, 1976). The practice extends outside of Alberta, too (*e.g.* Anonymous, 1957d; Wood, 1957; Anonymous, 1966c, 1974e, 1976c, 1994a, 1997c, 2000a, b; Frederick,

1973; Hillinger, 1990; Thompson and Taylor, 1993; Rushlo, 1999a, b; Wright, 2000; Vosepka, 2001). Regarding Drumheller, Anonymous (1957c) reported that "Visitors have taken away countless souvenirs such as pieces of petrified [dinosaur] bone and trees. The practice is still allowed, but no digging is permitted, nor are articulated skeletons allowed to be touched."

She was particularly concerned about other Albertan amateur palaeontologists and rockhounds irresponsibly collecting fossils in great abundance with no real purpose and especially doing so with little to no documentation, despite her efforts to encourage the latter in others (*e.g.* Johnson, 1961j; Johnson and Storer, 1974). As an example, she recorded in her fossil catalogue (Johnson, undated 2, p. 22) in the fall of 1965, her discovery of a large dinosaur limb bone in today's DPP:

"1 metre long. 32 cm across artic. surf. one end. Both joints there . . ." A footnote follows: "Subseq. visit noted limb bone removed. Hadros. tibia I think. (No great loss!)"

But this incident and others did bother her. About that time, in an unpublished manuscript (Johnson, *c.* 1965) curated in the Esplanade Arts and Heritage Centre Archives in Medicine Hat, she writes about this issue:

> A plea to the amateur fossil collector.
> The writer has referred to [collecting] activities, but make sure that these aren't mere depredations. In many places vast quantities of shells make it apparent that the collection of some of these is [a] perfectly legitimate thing —however—when it comes to bones of ancient animals, much more care and study should be taken . . . Take thought that you do not damage or remove fossils for which you have no real use or storage space. If you want only conversation pieces, don't take animal bones. Any large bone collection—a skeleton, a skull or sizable portion of either should be reported . . . If you decide to be a collector, make yourself a catalogue . . . As an amateur, you can contribute to the science of Palaeontology, because the area is so large and erosion rates very great, meaning a constant supply of fresh material. Just as rapid burial by sands and muds of Late Cretaceous days provided ideal conditions for the preservation of fossils, so rapid erosion of these rocks is now uncovering this record of Alberta's past.

In J. Danis (1981), Hope relates how the Park Warden befriended a nineteen-year-old man he met in the Park who was interested in fossils, so he had Hope talk to him. It was soon revealed that the young man had found a theropod dinosaur tooth in the Park and sold it for $20.00, thinking it a good price. She reminded him of the law and the Warden gave the young man the "bum's rush" out of the Park.

Hope had a quirky sense of humour or way of seeing things that is revealed in some of her correspondence and fossil catalogue comments:

- Referring to the identity of a problematic dinosaur centrum in July 1958 (Johnson, undated 1, p. 22) she comments: ". . . probably immature caudal—but not ceratopsian nor hadrosaur nor nodosaur nor hypsilophodont you guess now—well hell."
- Regarding an internal semi-translucent agate [her description] cast of a small theropod limb bone she says (Johnson, undated 1, p. 59), "just a bit, like a candle without a wick."
- Appearing on the top of page 100 of her fossil catalogue, book #1: "It ain't what ya don't know that hurts ya. It's what ya do know that ain't so."

Evidently there are now no palaeontology museum exhibits in Medicine Hat. Fossil exhibits that Hope built or filled with fossils in previous years have been taken down as the museum moved to new quarters and the exhibits were not replaced. She was not happy about this. In notes to her family, penned in 2006, Hope commented on this situation and on the future of palaeontology exhibits and education in the Medicine Hat area. Sadly, these comments are not very encouraging. She writes:

> The new Esplanade's museum section does "not envision any displays of dinosaurs" and other creatures supposedly drowned in "Noah's Floodde." Maybe the College here will venture to have a discreetly placed case of these inconvenient creatures. However, I was invited to speak to schoolchildren about them (Grade II). Hopefully they will forget about such foolish fancies by the time they begin high school.

Though long recognized for her artistic abilities, one interviewee commented that she was in awe as she watched Hope from start to finish draw a complex flower so expertly, on the first attempt: "It was perfect, and she never even had to use an eraser!"

On a more serious note, this quotation by Dr. Vicktor Frankl (1905–1997), a Jewish WWII concentration camp survivor, was found jotted down in one of her artist coil-bound sketch pads:

> We who lived in concentration camps remember the men who walked through the huts comforting others, giving away their last piece of bread. They may have been few in number, but they offer sufficient proof that everything can be taken from a man but one thing: the last of human freedoms—to choose one's attitude in any given set of circumstances, to chose one's own way.

Memories of Hope Johnson from a neighbour boy
By Jim Beare, Logan Lake, British Columbia

For a long while I have recognized the immense positive impact and how incredibly fortunate I was to have known and interacted with Hope Johnson during my life. Born in 1962, I spent the first fourteen years of my life living in Ralston on the Defence Research Establishment Suffield at 16 Kingston Avenue, three doors down the street from the Johnsons' 13 Kingston home (Figure 44). Our families socialized on a regular basis.

Perhaps my earliest memory of Hope is being invited into her living room where she painted a dragon for me. I clearly recall being so amazed to see her make this dragon appear out of nowhere before my eyes on a blank, white canvas (Figure 154). I wanted to take it home with me after she finished, but Hope explained that I would have to wait until the oil paint had dried. I also recall that it seemed like an eternity before I was finally allowed to take it home. This was in the summer of 1967 and I was not yet five years old, but I still remember this contact. She captured my imagination instantly.

A Cat Named "Poo"

Hope loved animals, and animals loved Hope. Probably about the same time she painted the dragon, I was introduced to her many cats. She took in strays and gave them a home. When she introduced one cat to me named "Poo" I asked her why the cat would be named after poo(p). She informed me that it was not Poo as in poop, but Pooh as in Winnie-the-Pooh.

Knock Before Entering

Hope let me know that I was welcome to come over to their house any time I had a question about something or wanted to look at her collections. So one day I walked down the back alley to the Johnson house and entered through the back door, startling the family at their lunch table. She made it very clear to me that it was important to knock before you enter someone else's house!

A Rattlesnake

Again, at about the same time, Hope started taking me down to the South Saskatchewan River valley to look for fossils. While in the coulees with her one day I lifted up a slab of sandstone with a small

space under it that sheltered a rattlesnake from the heat of the day. I was likely more startled than the snake when it coiled and rattled because I dropped the rock and ran back to the safety of Hope's big green 1950s Chevrolet (Figure 113). That was the end of fossil collecting on that particular day. A lesson well learned.

Dinosaur Park

At Ralston School, the staff would often load us kids up into these big old green army buses to go on field trips. One destination that was always a favourite was Dinosaur Park. Hope conducted the tours, and there she would introduce us to the amazing, fossilized prehistoric world that was on display in the seemingly endless maze of canyons and hoodoos. I was always impressed that she was able to find the bus at the end of the day.

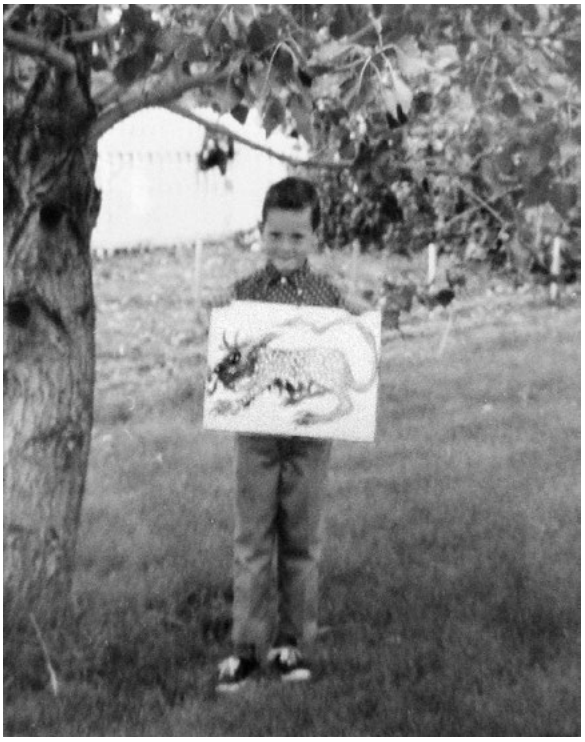


Figure 154. Jim Beare in 1967 with Hope's dragon painting. Photo courtesy of Jim Beare.

Round Rocks

About one mile south of Ralston was an intermittently active gravel pit with a crusher. During the times the pit was inactive, some of us kids would explore the diggings where we could find a variety of fossilized materials. Petrified wood was a treasured find as were the nearly perfect spherical concretions—or "round rocks" as we called them. From time to time, when we split open the round rocks we'd find fossil clam shells preserved in the middle. Hope was always available and genuinely happy to help identify our finds and to explain the geological history and origin of the materials at the site.

Library Mural

On the walls of the Graham Library in Ralston, Hope had painted a large mural (Figure 167) that brought the prehistoric world of the region to life. It was a place where we could visualize how the fossils that we found in the river valley, gravel pit and Dinosaur Park might have looked in life. It also

allowed us to imagine the ancient environment they lived in. At the time, we could name all the different life forms depicted on the mural and had found for ourselves fossilized evidence of many of them.

"Lady"

Sometime in the mid-1970s, an older black and white dog named Lady, belonging to the McGarrys who lived on Larkhill Lane, adopted Hope. For a while Lady would spend the day with Hope and then go home to the McGarrys at night. As time went on, it was agreed between Hope and the McGarrys that Lady should stay with Hope. Lady was so dedicated to Hope that she followed Hope wherever she went. When Lady became so old and crippled that she could not keep up with Hope's pace, it was a common sight to see Hope walk by our house on her way to the Recreation Centre, followed a few minutes later by Lady, waddling down the alley with her tail spinning around in circles. When Hope returned home from the Rec' Centre, there would come Lady again, waddling about five minutes behind Hope, her tail still spinning in circles. Hope loved animals, and animals loved Hope.

City Hall

Excavation spoil from the construction of a new City Hall in Medicine Hat was dumped on the prairie bench above Police Point Park. As had become second nature for me, I scrambled through the newly exposed material and discovered a large cobble with an unusual fossil of what appeared to be a cluster of tube worms or a coral of some sort. I packed the find home and called Hope to let her know I had found something unusual and asked if she would be interested in having a look at it. Upon her arrival at the house I was out in the backyard scrubbing the muck off the cobble with a bristle floor broom and garden hose. Hope took one look at the operation and immediately admonished me for my sloppy handling of the fossil. It was the only time in all of the years I knew Hope that she was even the slightest bit sharp with me. Apparently I had discovered something that needed to be handled with more care. It was the only specimen I had found in all those years of fossil hunting that Hope felt was of significance to palaeontology and so I immediately relinquished it to Hope's care.

Red Deer Drive

In her later years, Hope resided in a seniors' residence along Red Deer Drive near the Medicine Hat Regional Hospital. I contacted her during a visit to Medicine Hat and asked if it would be OK to stop by and say hello and have a short visit. As we sat and visited, I commented how much I enjoyed our outings when I was young, to hunt for fossils along the river valley. I suggested that I could somehow show my appreciation by taking her out to the river valley for a hike. She looked at me and let me know in the way only Hope could that she was far too old to be rambling around in the coulees.

Summer Lake

I was up the Bull River drainage in British Columbia, truck-camping at Summer Lake when Hope passed away and her funeral was held. I received the news when I came down out of the mountains and phoned my mother to see how she was doing and let her know I had returned to civilization. I'm torn to this day that I could not attend Hope's funeral and pay my deepest respects in person for someone who had given me so much in life. How eternally grateful I am for her unwavering patience, commitment to accurate knowledge, deep respect for the natural world, and for selflessly sharing all of that with me.

When I look at the artwork that Hope created during her life, it stands out to me that it consistently represents both an interpretation of the reality of our part of the Earth as it may have existed in prehistoric times, and an accurate representation of the present. To the best of my knowledge, the only

art she produced that may have deviated from this style was the dragon that she painted for me so many years ago. At no time during the years of interacting with her did I witness anything other than a strict adherence to the hard reality of the situation at hand; there was no fantasy. I can still picture her skillfully chipping away with her rock hammer to release a piece of the past and then carefully examining the liberated material with her hand lens before saying anything about it. I have come to believe that the dragon was intended to capture the imagination of someone she had selected for mentorship. While she may have expressed the past and present in art, it seems she invested in the future by openly sharing her vast knowledge of the natural world with the youth of the day who showed an interest in such matters.

Hope's lifetime work and closing comments

A synopsis of Hope's busy life and various recognitions is presented below. It is a long list but likely incomplete. It is based on Hope's notes, military records, letters to others, acknowledgments in scientific papers, personal communications, photographs and newspaper articles.

Volunteer work

Duration of some of these activities is incompletely known. All occurred in and around Medicine Hat, unless otherwise noted:

- Boy Scouts—assistant and "Akela" (leader) for seven or eight years, Victoria; 1934–1940/41.
- Taught German classes for two years under the auspices of Victoria Canadian Legion Services according to her military records. It is uncertain if this was volunteer or paid work.
- Four years in Ralston (Alberta) Drama Club (1950s).
- Panel member of the Ralston Home and School Association (1958–?).
- Boy Scouts—assistant and leader; Ralston (*c.* 1956–1970/72).
- Tester (of girls being awarded badges) for Girl Guides (1970s–1980?).
- Volunteer work with 4-H club (dates incompletely known, but summers of 1966, 1968 confirmed).
- Medicine Hat Museum board member (1966–1980).
- University Women's Club (1962?–1966?–1970–?).
- Appointed by City of Medicine Hat with two others to three-year Advisory Board terms for the Museum and Exhibit Centre Board (1979–*c.* 1982; Anonymous, 1979d).
- Alzheimer's Association, Medicine Hat Chapter (1981–1992, ?1995; served as President).
- United Way (1981).
- Police Point Interpretive Centre (?–1982–?).
- Medicine Hat Alzheimer's Society (1985–?).
- Medicine Hat Historical Society—reorganizer with help (1966), President, Editor (1966–*c.* 1980).

Military career

Canadian Women's Army Corps (CWAC; October 1, 1941–July 15, 1946); left military with the rank of Second Lieutenant. She served at seven bases in Victoria (Esquimalt); Sainte-Anne-de-Bellevue, Quebec; Kitchener, Ottawa, Peterborough and Brockville, Ontario and Suffield, Alberta. At most of the Quebec and Ontario bases she taught first aid, typing, stenography, and served in more senior administrative and management roles. She is known to have been second in command and supervisor to all the CWACs (in numbers unknown) on the Suffield base. Military records state that she was employed

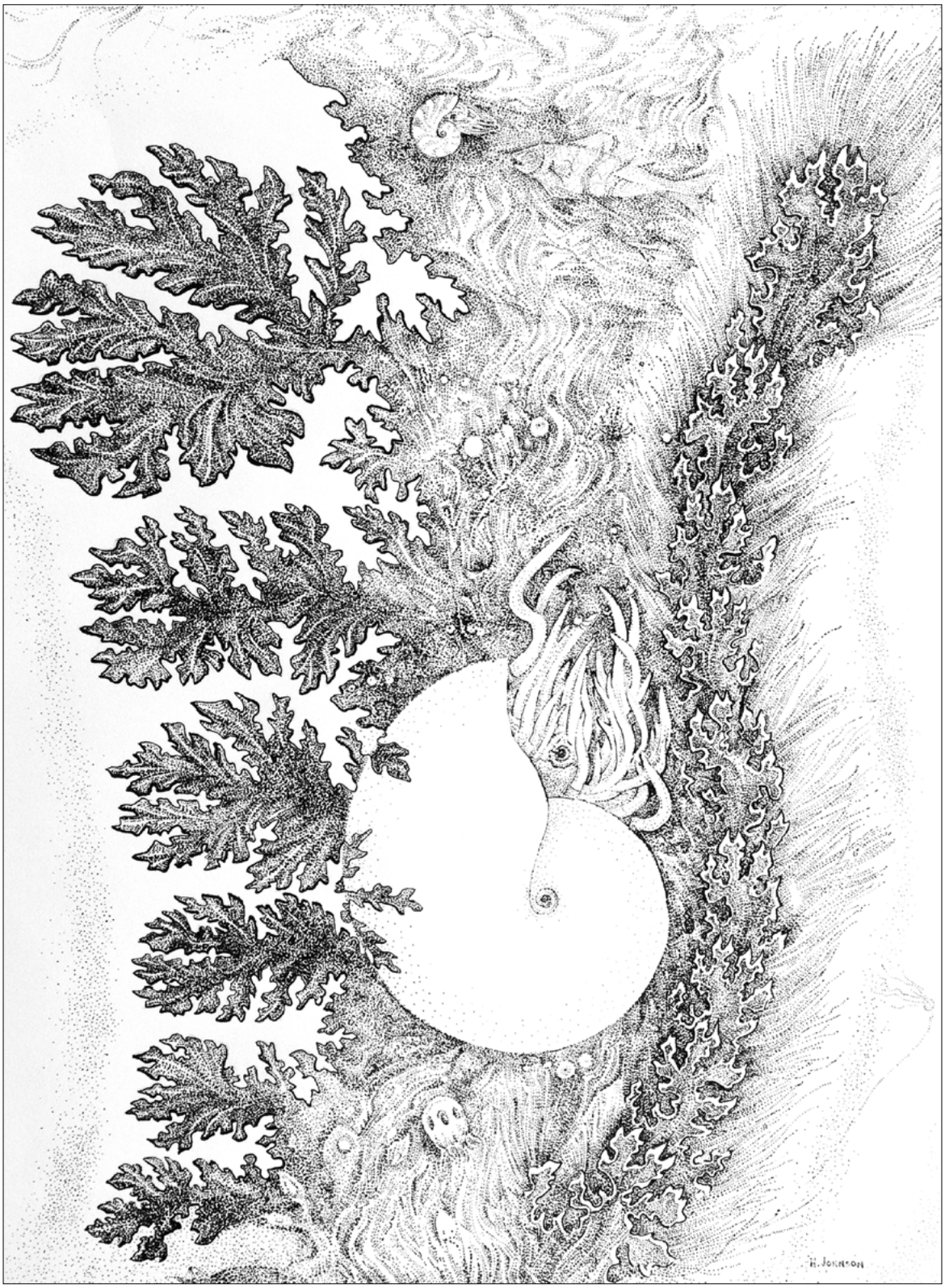


Figure 155. On the back of this undated pen-and-ink piece Hope writes: Title: "Variations on the theme of an ammonite by a primate" Suture of *Placenticeras meeki* Boehm." Note how the ammonite sutures are used on the right as a seaweed design. The back of the framed picture bears a July 1978 travelling exhibit museum sticker, so this work must predate that. The intricate piece measures 11.75 in × 9 in. Collection of Allan C.J. Jensen, Medicine Hat.

as an instructor for seventeen months and then as a Platoon Officer for three years. She served a total of fifty-four months of military service. She also served two years in the British Columbia Women's Service Corps, though not an organized military organization *per se*, attaining the rank of Sargeant. This group met and served on a part-time basis and later became the CWAC.

Formal art exhibits including museums, and less formal art shows and sales

- Armed Forces Art Exhibition, Ottawa, Ontario, 1944.
- Several 1950s art shows in Ralston, Alberta; no other details are available.
- Unidentified art show in British Columbia, 1958.
- Saskatchewan Museum of Natural History (botanical paintings), Regina, *c*. 1959–?
- Two-day art show of her works in Ralston, 1961 and the work of Hope and others in the same town later that year.
- Ralston Art Club art show at Officers Club at the Suffield base, 1964.
- Ralston Art Club art show at Officers Club at the Suffield base, 1968.
- Hope's art selected for viewing at Corbett Hall, UA, in conjunction with the Alberta Community Art Clubs Association Conference, 1970.
- 1971 National Exhibits of Nature Art, Ottawa. Alberta art shown in Edmonton.
- 1972 National Exhibits of Nature Art, Edmonton.
- Canada Nature Art 1973 (no details known).
- Canadian Nature Art 1974, Ottawa, then travelling across Canada.
- Ralston Art Club show, 1975.
- Canadian Nature Art 1976, Ottawa, then travelling across Canada.
- Celebration '76 (no details known; possibly a gallery show in Medicine Hat).
- "Art in Action" show, Medicine Hat, 1978.
- "Four drawn from the Hat," Medicine Hat, 1978.
- "Images from Alberta's Southeast," Edmonton, 1978.
- Art show at Police Point Interpretive Centre, Medicine Hat, 1983.
- "Prairie Plants of Southeast Alberta," Medicine Hat, 1985.
- Western-themed art show near Medicine Hat, 1986.
- Art competition/show at the Medicine Hat Hospital, 1987.
- "The Prairie in Relief" (with artwork by Deanna Lemna), Medicine Hat, 1989.
- Alberta Agriculture's Agri "Culture" Gallery, Medicine Hat, 1992.
- Ranchlands and Badlands Art Show and Sale, Medicine Hat, 1994.
- "A Quality of Wildness," Medicine Hat, 1994.
- "From Fossil to Fine Art," Lethbridge, Ralston and Drumheller, 1998–1999.
- Show of her watercolour paintings, Medicine Hat, 1999.
- "Hope Johnson: A Retrospective," Medicine Hat, 2001.
- "Show and Tell" event, Valleyview Retirement Community, Medicine Hat, 2001.
- Art Awards Ceremony, Medicine Hat, 2006.
- Art show and silent auction benefitting the Hat Art Club, Medicine Hat, 2011 (Watson, 2015).

Hope's artwork was also reportedly shown in Bonn, Germany, 1989 or earlier (Anonymous, 1989a). The author could not confirm this claim.

Art teaching activities

- Taught directly or helped teach art at Ralston Art Club (*c*. 1954–*c*. 1975 and 1981?).

- Member of the Community Art Infrastructure Council (1967–?).
- Medicine Hat Art Club (1981–?).
- Taught seniors at Veiner Centre, Strathcona Centre and the Pavilion at Strathcona Park in Medicine Hat (1985–1999).
- In her last classes, students came to her at the Valleyview Retirement Community, Medicine Hat, in 2000. Many of these classes were taught free of charge.

Art murals completed

- Vertebrate palaeontology/nature/human history mural, Ralston library, 1959–present.
- "Odyssey" mural co-painted with seven other women as a gift to the people of Medicine Hat on the occasion of the city's centennial (Anonymous, 1983c).
- "Valleyview Trees of Life" mural, Valleyview Retirement Community, Medicine Hat, 2003.

Original art curated at various institutions (visit websites to see examples of her work)

- Alberta Foundation for the Arts.
- University of Lethbridge: **http://artcollection.uleth.ca/collections** [search Hope Johnson].
- Esplanade Arts and Heritage Centre: **http://archives.esplanade.ca/art/** [search Hope Johnson].

Figure 156. Sketch of a pine cone Hope collected in the Netherlands, originally in blue ballpoint pen; undated.

Groups and clubs that Hope co-founded

- Ralston Art Club (*c.* 1954–1976 or 1981?).
- Grasslands Naturalists (1985–present).

Conservation research on extant fauna (directly involved or assisted)

- Eastern short-horned lizards, *Phrynosoma brevirostris* and northern scorpion, *Paruroctonus boreus* specimens sent to Zoology Department, UA, 1961.
- Eastern short-horned lizards, *Phrynosoma brevirostris* (Powell and Russell, 1991).
- Plains spadefoot toad, *Spea bombifrons* (Lauzon, 1999).

Botanical activities

- Discovered the second occurrence in the province of the rare tiny cryptanthe, *Cryptantha minima* (Bradley, 2004); the plant had not been reported in almost 100 years.
- Donated plant technical illustrations from Johnson and Hallworth (1975) and other floral drawings to the herbarium at the University of Calgary (UC).
- Collected, pressed, and donated plant specimens for the herbarium at the UC (**https://library.ucalgary.ca/digital?group_id=14797** [search "herbarium Hope Johnson"]) and the Royal Alberta Museum.
- Illustrated and co-authored major field guide to Alberta berries (Droppo and Johnson, 1988).

Botanical publications (non-newspaper)

- Johnson, 1969e (plants in DPP).
- Johnson and Hallworth, 1975 (small-flowered sand verbena in Alberta; peer-reviewed).
- Johnson, 1983 (prairie plants of southeast Alberta)
- Droppo and Johnson, 1988 (field guide to Alberta berries).

Duration of her palaeontological career

Hope began fossil collecting in *c.* 1948; her first catalogued fossil was dated 1952. Her last notation of being in the badlands (albeit briefly) was in 2005, totalling fifty-seven years of fieldwork in the province.

Number of fossils collected/donated to academic institutions

Calculating this number is problematic. Her fossil catalogue books (Johnson, undated 1–3) list some 2,366 fossils. Hope donated many fossils outright to museums and sent others to experts at those institutions for identification; some of these were retained and some were returned in numbers unknown.

Table 2 (below) is a summary of Hope Johnson-collected fossils presently curated in various institutions or societies. Figures shown are specimens presently recorded. Hope's notes record that she donated specimens in numbers that do not coincide with the tabulated figures. The reasons for the discrepancies are unknown. The TMP (originally PMA) 1979.014 series includes what became known as the "Dinosaur Park Collection." It is believed that most of those specimens were collected by Hope during her employment there in the late 1960s to early 1970s. Some of the fossils later received individual catalogue numbers but others were bulk-numbered as they were stored in vials or boxes, so the actual number of specimens Hope-collected in the TMP 1979.014 series could be higher.

Institution	Number of specimens curated
Royal Alberta Museum	29
PMA/Royal Tyrrell Museum (RTMP)	1,180
TMP 1979.014 series	947
Canadian Museum of Nature	3
University of Alberta	0
Royal Ontario Museum	66
Alberta Palaeontological Society	76
Hedy Hobberlin education collection	dozens?
Drumheller Fossil Museum	0
University of Paris, France	8
Esplanade Arts and Heritage Centre	68
Redcliff Museum	27
Police Point Interpretive Centre	70
Subtotal	**> 2,474**
Posthumous donations to RTMP	**448**
TOTAL	**> 2,922**

Table 2. Vertebrate fossils collected and donated by Hope Johnson to museum and university collections and private or public educational facilities. Counts are minimums, as some donated fossils were uncatalogued in teaching collections and some catalogue numbers comprise multiple items; for example, several teeth or fragments were assigned the same catalogue number. [49]

Included in this tabulation are the specimens that appeared in Johnson and Storer (1974). These were kept separately in Hope's personal collection and after her death they went to the RTMP's Education Department. Fifty of those were catalogued, mostly for study and comparative purposes, and 350 uncatalogued fossils are used for children's programming there. In late 2014 and late 2015 the author received about four dozen more fossils from her estate which also went to the RTMP Education Department. If all this is correct, then over 3,000 fossils of all types were collected by Hope in Alberta and found their way into educational centres, local museums, and institutions of higher learning: an impressive number from this dedicated and responsible amateur collector.

Institutions and educational groups benefitting from fossil specimens she donated

Alberta Palaeontological Society; Canadian Museum of Nature; Hedy Hobberlin teaching collection; Medicine Hat Historical Museum; Police Point Interpretive Centre; Redcliff Museum; Royal Ontario Museum; Provincial Museum of Alberta; Royal Tyrrell Museum of Palaeontology (research and educational collections); University of Alberta; University of Paris (Dept. of Comparative Anatomy), France.

Fossil exhibits she improved, developed or constructed

- 1964—assisted with creation of exhibit case background paintings, fossil donation(s?), fossil identification and signage for same at the small fossil museum in Drumheller.
- Dinosaur exhibit at Medicine Hat Museum (reportedly seen in 1968; see Page 90).
- 1969–1973—improvement of three "fossil museum" buildings (not the present-day *in situ* dinosaur exhibits) in DPP through the addition of new specimens, educational exhibit signage (paintings of life reconstructions of dinosaurs, technical illustrations of fossils and labelling of same) and repair and maintenance of these exhibits.
- 1965?–*c.* 1972?—exhibit cases of some 350 vertebrate (mostly) and invertebrate fossils on display at Officers' Club on the military base in Suffield, Alberta.
- Mid- to late 1970s to early 1980s?—produced six fossil exhibit cases at the Medicine Hat Museum (later Medicine Hat Museum and Art Gallery). Their themes were on dinosaurs, 30-million-year-old mammals, mammoths, evolution of the horse, regional Bearpaw Formation marine invertebrates and local Pleistocene-aged vertebrates.
- 1986 to present: display drawer of invertebrate and vertebrate fossils at Police Point Interpretive Centre, Medicine Hat.
- ? to present—provided invertebrate and vertebrate fossils for a small fossil display at Redcliff Museum, Redcliff, Alberta.

Palaeontological teaching and interpretation

- Gave formal fossil interpretation to the public in DPP beginning in 1969 (and 1970, 1972, 1973), including one-on-one, to families, to larger groups and on DPP bus tours. During this time she also assisted in DPP with the tour bus (15 passenger van)—the first tour vehicle (beginning in 1968) in any Alberta Provincial Park. The Park Ranger drove the van and acted as the guide, but he had no real palaeontology/field experience, so Hope did the interpretive work. As such, Hope was likely among the very first vehicle-based interpreters in the Alberta Provincial Parks system and the first true palaeontology interpreter in DPP or the Alberta Provincial Parks system. She also shared her vast experience with other amateurs and some professionals (on conference field trips) in southern Alberta as she assisted them in the hunt for fossils.
- Independent teaching *via* one-on-one contact, illustrated lectures, her publications, and mentor-

ing children who exhibited a special interest in Earth science.
- Lectures to the public, professional organizations, clubs, children, *etc.* These were slide shows supplemented with actual specimens and Hope's technical drawings. The earliest confirmed slide show occurred in May, 1962 (Anonymous, 1962e, 1962f). There is no evidence that she made any formal presentations at professional palaeontological symposia, but may have contributed images and/or made an informal "after dinner" presentation at one field trip.

Employment in a palaeontological capacity
- Seasonal work for the Alberta Provincial Government (Alberta Parks) at DPP, summers of 1969, 1970, 1972, and 1973.

Degreed Earth (and other) scientists/academics she is known to have worked with in the field and/or for research or consulting purposes

Hope had regular contact with a number of scientists of various academic backgrounds, with whom she collaborated for geology training, fossil identification, networking, *etc*. In palaeontology, they include: Donald Baird, David Bardack, Donald B. Brinkman, T. Potter Chamney, Rufus Churcher, Jane Colwell (Danis), Philip J. Currie, Charles Devillers, Peter Dodson, Gordon Edmund, Richard C. Fox, C.R. "Dick" Harington, Len Hills, Jack Horner, Gareth H.S. Jones, Roy R.H. Lemon, Bruce A. McCorquodale, Chris McGowan, William J. Morris, Chakravartula G.K. Ramanujam, Reddy P. Ramchandra, Dale A. Russell, Loris S. Russell, Bob Schaeffer, David A.E. Spalding, Archie M. Stalker, Charles R. Stelck, Charles M. Sternberg, Wilson N. Stewart, John E. Storer, Donald A. Taylor, Karl M. Waage, Percival S. Warren, and Michael C. Wilson. In addition to the author, she also had contact with a number of CMN, MOR, ROM, RTMP and UA senior and junior technical staff over the years. Though they must have known of each other, the author could find no evidence of Hope having any interactions with Ottawa's Wann Langston Jr. (1921–2013).

She also had contact with biologists of extant fish species to help with the identity of fossils: Edwin J. Crossman (1929–2003, regarding sturgeon) and Tom A. Willock (1943–, regarding bowfin), as well as plant biologists. Among botanists, her professional contacts are less well known, but include Cheryl Bradley, Beryl Hallworth, Julie Hrapko, and C.G.K. Ramanujam.

Academics from the Physiology Department at the UA (Drs. Charles Heath, Ed Preston, and others) doing research at the Suffield base became fellow fossil collectors and one (Preston) married into the Johnson family. Fellow fossil collector William Lawson (1937–) worked in the Chemistry Department on base for thirty-eight years. She also had contact with Dr. Keith Hage of the Geography Department at the UA who was an occasional fellow fossil collector.

Professional conferences and seminars attended
- Several Alberta (now Canadian) Society of Petroleum Geologists field conferences in the 1950s and 1960s (R. Chamney, pers. comm., 2015).
- Fifth Annual Community Art Class Conference, Edmonton (1961).
- Vertebrate Palaeontology Conference, Edmonton and Calgary (1963).
- Art Instructors Council of Alberta Seminar (1 week), Edmonton (1967).
- Alberta Community Art Clubs Association Conference, Edmonton (1970).
- Federation of Alberta Naturalists, Elkwater, Cypress Hills, Alberta (1974).
- Society of Vertebrate Palaeontology: Flagstaff, Arizona (1974), Los Angeles (1977), Toronto (1978); Drumheller (1988).

- American Quaternary Association, Lethbridge (1978).
- Canadian Quaternary Association, Lethbridge (1985).
- Dinosaur Provincial Park Conference, Drumheller, (2005).

Society Memberships

- Society of Vertebrate Palaeontology (1966–?).
- Alberta Palaeontological Society (1987–1994, Life Member 1996–2010).

Major specimens found

This is a difficult list to compile. Hope found many specimens as an amateur collector, but many were common, run-of-the-mill-type fossils such as teeth, centra, scutes, plates and phalanges, which on their own have limited scientific value, but combined together in some studies become more valuable. Hope found a number of mammal-producing microsites that were visited and sampled by professionals later. The importance of some of her finds may yet be realized.

- Conus or guard of coleoid squid, *Actinosepia canadensis*; now in RTMP collections.
- Rare Late Cretaceous tree bark specimen, *Cupressinocortex albertaense* gen. et sp. nov., Redcliff, 1960 (Ramanujam, 1970) and possibly another bark specimen (Ramanujam and Stewart, 1969).
- Parts of two associated ornithomimid pedes, DPP, 1963.
- Trionychid turtle carapace (and possibly the plastron; possibly TMP 1979.014.0851), DPP, 1964.
- Partial *Gorgosaurus* skeleton, White Rock Coulee, 1967 (collected 1968).
- Tooth-marked ornithomimid caudal vertebra, White Rock Coulee, 1967 (Jacobsen, 1995, 1997, 1998, 2005; Jacobsen and Currie, 2005; Jacobsen and Bromley, 2009; see also Jacobsen, 2001).
- Pair of rare fused dentaries of a caenagnathid dinosaur (Currie *et al*., 1993, p. 2258).
- First discoveries of Late Cretaceous pterosaur bones in Canada: 1. TMP 1965.014.0039, though not at the time recognized as such (Godfrey and Currie, 2005). 2. TMP 1972.001.0001, found in 1970; date from Hope's writings in Moser (1974, p. 33); (D.A. Russell, 1972d; Godfrey and Currie, 2005).
- Plesiosaur jaw fragment, TMP 1971.008.0002 (Sato, 2002).
- Uncollected, eroded ceratopsian skeleton, Foremost Formation, southwest of Ralston, 1977.
- Uncollected, unidentified dinosaur skeleton, White Rock Coulee, 1979.

Extinct species named after Hope

- *Procleptes hopejohnsonae* McKellar and Engel, 2014 (Late Cretaceous Cuckoo Wasp in amber; Figure 157).

Palaeontology publications (non-newspaper)

- Johnson, 1968a (dinosaurs in the southwest prairie area).
- Johnson and Storer, 1974 (vertebrate fossils from the age of dinosaurs).
- Storer and Johnson, 1974 (sawfish teeth from southern Alberta; peer-reviewed).
- Johnson, 1980a, b (palaeontology and natural history of Red Rock Coulee and Irvine field areas).
- Alberta Palaeontological Society, 2009 (Alberta Cretaceous vertebrate fossils).

Vertebrate specimens collected by Hope that found their way into scientific publications or theses

This list was compiled by RTMP collections staff, who are thanked for their effort. RTMP specimens prior to 1982 are formerly PMA specimens. Nine *Ischyrhiza* (sawfish) teeth were included in a study (Storer and Johnson, 1974) but not specifically identified, figured or described so are not included here:

1	TMP 1965.014.0039	Pterosaur ulna	Godfrey and Currie, 2005
2	TMP 1972.001.0001	Pterosaur phalanx	Godfrey and Currie, 2005
3	TMP 1972.014.0001	*Meniscoessus* (mammal) molar	Fox, 1976
4	TMP 1977.030.0001	Trionychid turtle skull	Gardner, 1992
5	TMP 1979.008.0622	Caenagnathid dinosaur jaws	Currie *et al.*, 1993; Lamanna *et al.*, 2014
6	TMP 1979.008.0631	*Parasaniwa* (lizard) dentary	Gao and Fox, 1996
7	TMP 1979.008.0635	*Troodon* tooth	Currie, 1987
8	TMP 1979.014.0074	*Euoplocephalus* skull	Arbour and Currie, 2013
9	TMP 1979.015.0015	Theropod phalanx	Currie, 1992
10	TMP 1980.013.0033	Tyrannosaurid tooth	Jacobsen, 1995
11	TMP 1980.016.0929	Fish (sturgeon) parietal, frontal	Grande and Bemis, 1991
12	TMP 1980.016.0930	Ceratopsid vertebra, caudal	Jacobsen, 1995
13	TMP 1980.016.0938	Tyrannosaurid premaxilla	Jacobsen, 1995
14	TMP 1980.016.1010	Ceratopsid vertebra, caudal	Jacobsen, 1995
15	TMP 1980.016.1030	Ceratopsid vertebra, caudal	Jacobsen, 1995
16	TMP 1980.016.1644	Ornithomimid ungual, manual	Longrich, 2008
17	TMP 1980.027.0025	Hadrosaurid phalanx, manual	Jacobsen, 1995
18	TMP 1981.016.0126	Tyrannosaurid tooth	Jacobsen, 1995
19	TMP 1981.024.0003	Tyrannosaurid vertebra, caudal	Jacobsen, 1995; Jacobsen, 1998
20	TMP 1981.038.0004	*Parasaniwa* (lizard) tooth, jaw	Gao and Fox, 1996
21	TMP 1985.006.0049	Tyrannosaurid tooth	Jacobsen, 1995
22	TMP 1985.006.0068	*Albertosaurus* tooth	Jacobsen, 1995
23	TMP 1985.006.0069	*Albertosaurus* tooth	Jacobsen, 1995
24	TMP 1985.006.0070	*Albertosaurus* tooth	Jacobsen, 1995
25	TMP 1985.006.0084	*Plesiobaena* (turtle) dentary	Brinkman, 2003
26	TMP 1985.006.0130	*Albertosaurus* tooth	Jacobsen, 1995
27	TMP 1985.006.0134	Tyrannosaurid tooth	Molnar and Carpenter, 1989
28	TMP 1985.006.0158	Ornithomimid vertebra, caudal	Jacobsen, 1995; Jacobsen and Bromley, 2009
29	TMP 1985.006.0176	Tyrannosaurid tooth	Jacobsen, 1995
30	TMP 1985.006.0178	Tyrannosaurid tooth	Jacobsen, 1995
31	TMP 1990.005.0079	*Aspideretoides* (turtle) plastron	Gardner, 1992; Gardner *et al.*, 1995
32	TMP 1999.062.0001	*Stegoceras* skull, incomplete	Sullivan, 2003
33	Various TMP numbers	Hadrosaur pedal phalanges	Rothschild and Tanke, 2007

Figure 157. *Procleptes hopejohnsonae* (type specimen) in amber and interpretive drawing. Length of body is 1.14 mm. Images courtesy of Dr. Ryan C. McKellar, Royal Saskatchewan Museum, Regina.

Human history preservation

- Worked with Medicine Hat Historical Society, 1966–1980.
- Worked with Historical Society in Medicine Hat which placed a dedication cairn with plaque at

a historical river crossing site there. She helped search for suitable glacial erratics for the cairn's construction and also drafted the cover art for a related document, 1980.
- Co-wrote, greatly expanded and edited a local history book, *Down the Years at Elkwater* (Michael and Johnson, 1981).

Mentoring and outreach activities
- Gave talks to kindergarten to grade 12 schoolchildren, alone or with others, speaking about science, careers, geology, palaeontology, natural history and art. This was done on a volunteer basis, with Hope going to schools, or when she was paid at DPP.

Television (all local CHAT-TV station/documentary appearances except where noted)
- 1961, regarding discovery of horned lizards in area (Bridge, 1961b).
- About 1973 on half-hour television program *The General Store* with host Gerry (or Gary) Puckett (Dryden, 1981; Puckett, 1973).
- 1976 BBC Horizons television documentary, *The Hot-Blooded Dinosaurs*.
- 1990 half-hour television program, *Primetime: The Red Rock Coulee visit with Dr. Hope Johnson* (Anonymous, 1990b).
- 1993 program *Art in Profile—Hope Johnson* (Anonymous, 1993).
- Hope was a regular subject on the local show, *Our Town* hosted by Lorne Havard; but specific dates and topics discussed are presently unknown.

Recognition and awards received
- Honorary Doctorate of Laws, University of Lethbridge, 1981.
- Recognized by the Alberta Historic Resources Foundation as one of fifty women who made an outstanding contribution to the province's society.
- 1982 Civic Recognition Award, Medicine Hat.
- Included in travelling exhibit, *One Hundred Years of Alberta Women*, chronicling 200 influential Alberta women.
- Heritage Merit Award, Medicine Hat, 1989.
- New Country 1270 CHAT Radio Collection Award, Medicine Hat, 1994.
- Life Member of the Alberta Palaeontological Society, Calgary, 1996.
- 1997 Loran L. Goulden Award from the Federation of Alberta Naturalists.
- Recognized as one of 200 remarkable Alberta women (Sanderson, 1999).
- Portrait and short biography in RTMP's temporary *Women in Palaeontology* exhibit, May 2011 to May 2012.
- Featured (June, 2013) in *Herstory: The Canadian Women's Calendar*.

Afterword

Even after Hope's death, her artwork and dedication to Earth science in particular still educates and inspires us. She and her friend and fellow amateur collector Irene Vanderloh were featured with about eighteen other women in the Royal Tyrrell Museum of Palaeontology (RTMP) photo gallery *Women in Palaeontology* exhibit, which ran from May 2011 to May 2012. A line drawing of small-flowered sand verbena graces the cover of a federal government technical paper on the endangered species (Environment Canada, 2012). Figure 78 in this book was also used in the 2013 *Herstory: The Canadian Women's Calendar*. The author was contacted in early December 2011 to explore the possibility of Hope being used in the inspirational women's calendar. He provided more biographical information on Hope, and asked that her picture be used for one of the summer months in the calendar, reflecting Hope's love for summer and the palaeontological and natural history work she did outdoors at that time of year. The request was granted and Hope's image appears on that calendar for the month of June. A benefit auction for the conservation of ring-necked pheasants featured a framed Hope Johnson signed watercolour (Chinook Pheasants Forever, 2013; Dahlman, 2013). The Police Point Interpretive Park in Medicine Hat has included her art (Figure 179) and specimen labels from a small exhibit on woolly mammoths held there in February, 2014. The character "Great Aunt Mary" in a children's book (Foster, 2014), released in the late summer of 2014 was loosely based on Hope (D. Foster, pers. comm., 2015).

As the writing of this book neared completion, the author was both surprised and delighted to learn that Hope had a tiny, Late Cretaceous wasp preserved in amber from southern Alberta named after her: *Procleptes hopejohnsonae* (McKellar, 2014; McKellar and Engel, 2014; Figure 157), related to an extant family commonly called cuckoo wasps.

The author is hoping to eventually develop a Hope Johnson page on Wikipedia and he nicknamed a Foremost Formation ceratopsian skull (TMP 2010.088.0001) "Hope." This fossil was discovered by the author shortly after her death. To ensure that her legacy is not forgotten, the author has also given a number of talks at the Royal Tyrrell Museum (Tanke, 2015c), DPP, Alberta Palaeontological Society and elsewhere, on Hope's life and many achievements.

With the 2009 revision and republishing of the expanded fossil guidebook (Alberta Palaeontological Society, 2009), Hope's drawings and educational legacy will live on for many decades to come. Given the amount of time and work expended to draw all these fossils, it seems unlikely the book will be replaced by another that is illustrated by different artists for a very long time, if ever.

She was a responsible amateur collector, respecting the laws governing the excavation of fossils and always reporting significant finds. Amateur fossil collectors and volunteers have been, are, and will always be an important aspect of professional palaeontology when they report significant finds. After all, as history has repeatedly shown, their experienced eyes often lead to major new discoveries. The scientific community can only hope that the amateurs Hope Johnson inspired will respect her legacy and follow suit, practicing their hobby responsibly, cataloguing their finds, reporting important discov-

eries to the proper authorities and thereby making valuable contributions to the field of palaeontology.

Regretfully, the author never had the opportunity to really work in the field with her. She is truly one of the most extraordinary people he has ever met and considers her a provincial treasure. Because of such dedication and her continued recognition, no doubt Hope's legacies within the arts, conservation, and palaeontological communities shall live on for a long time to come.

There is still more to know about Hope and a number of questions about her life remain unanswered. Information was still coming up long after the manuscript was first submitted for editing, so more information is still out there. Though this book is believed to be a relatively full accounting of her life, the author will continue to seek out new information on a casual basis and welcomes anything readers can add to her story. Information on her life that is not known, or not well known, includes:

1. The identity of her biological parents.
2. Her employment activities in the last half of the 1930s.
3. Pretty much any information regarding her art from 1920 to 1960.
4. Her role and work within the British Columbia Women's Service Corps (pre-WWII to early WWII) and her involvement in the early development of the CWAC in WWII.
5. Her specific work duties as a CWAC at the military base at Suffield, Alberta.
6. Her early contact and subsequent correspondence with Earth scientists at the UA. Early correspondence with the ROM's Gord Edmund would also prove enlightening. Correspondence with other palaeontologists may be archived at their respective institutions.
7. Information on her activities with the Drumheller and District Fossil Museum in 1964.
8. Information and imagery regarding her large fossil exhibits at the Suffield base in the 1960s.
9. Her activities within the Grasslands Naturalists organization.
10. Her contributions to Alberta archaeology and historical archaeology are only poorly known.
11. She appeared once in the local *Redcliff Review* newspaper (Anonymous, 1977d). Many of the issues are missing or have yet to be digitized. These may contain articles regarding her activities.
12. Finally, her significance and impact as an artist outside her home area. Certainly she was well known and respected in the Medicine Hat area, but she is rarely mentioned outside of that corner of Alberta.

Hope is survived by three of her four children: Barbara (Ed) Preston, Margaret (John) Noone, and Lawrence (Brenda) Johnson; as well as eight grandchildren and ten great grandchildren. Eldest son Robert passed away in October, 2018 (Anonymous, 2018b).

Most papers and documents related to Hope's life and achievements are curated at the Esplanade Arts and Heritage Centre Archives in Medicine Hat, where she and others began donating material related to her life's work beginning in late 1968. The author has, and will continue to donate his accumulated research material on Hope to that institution. Additional material can be found at the Graham Community Library in Ralston; the Redcliff Museum in Redcliff; the Police Point Interpretive Centre; and the Valleyview Retirement Community, Medicine Hat.

A few of her friends also have art and other important documents related to her life and it is hoped that these, too, will someday find their way into a museum archive.

Notes

1. (Page 18) Online records (**http://www.freebmd.org.uk/cgi/search.pl**) indicate that Herbert was born in September 1864; the Johnson family thought 1863.

2. (Page 21) Agoraphobia is an anxiety disorder defined by a fear of open spaces, crowds, or uncontrolled social conditions. Apparently Rachel's father's condition was so bad it eventually ruined his promising career as he was unable to argue his legal cases in open courtrooms, barring him from becoming a fully-fledged lawyer. Hope writes that, according to Rachel, he thought he was cursed. Several baby boys fathered by him died at birth. He began drinking and lost job opportunities at work. He later became a wine merchant, according to Herbert and Rachel's 1912 marriage certificate.

3. (Page 26) Information about this school proved elusive for quite some time. An article in a recently digitized newspaper article, posted online (Anonymous, 1936c), provided many clues. Classes were taught in the Larges' rented home. Children ages 4–12 were accepted. Kindergarten and primary activities included reading, writing, arithmetic, English (with special attention to literature and speech), history, geography, nature study, art (drawing and brush work), class singing, folk dancing, and elementary French and Latin.

4. (Page 30) The castle exists as a tourist attraction, drawing roughly 150,000 visitors a year; see **http://en.wikipedia.org/wiki/Craigdarroch_Castle**

5. (Page 35) The "Guild of Friendship" played a role in Hope's life as a young adult, so a bit more about it is related here. The Guild of Friendship was created by Hope's mother Rachel in late January, 1939. It was a small service group or club/society with a strong religious bent. Regular meetings (up to twice a month) were held at the Larges' home and outdoor events were also conducted. Some larger presentations by invited speakers were held in local halls. One series of talks was given under the general title "Democracy and the Kingdom of Heaven" (Anonymous, 1941f). Social teas and lectures by clergy, military men and female members of the guild were regular features. The guild also held events such as thrift sales and dances and invited musicians to raise money for the church and to support social issues—in particular, soldier benevolent societies and for victims of German aerial bombing in WWII England. For example, Anonymous (1941b) notes that the guild collected money to go toward the making of "bombed-out kits" and materials to make "comfort for overseas use." Many of the ladies also sewed items for war victims in England. Palmistry and teacup reading seems to have been a regular feature at the social teas. How many members the guild had is unknown and it apparently ended in late August, 1943 (Anonymous, 1943e): ". . . it was decided to

suspend all activities for the time being." Financial difficulties are a likely reason for the cessation: Hope's mother bought the Guild's piano "thus augmenting the guild's fund."

6. (Page 36) The Canadian Women's Army Corps (CWAC) was formed on August 13, 1941. Almost 22,000 Canadian women served in its ranks. They were assigned clerical and other duties, most serving in Canada. Their crucial work here allowed more men to go overseas and fight (for a personal account of life as a CWAC, see Bowman, 1994). In May 1945 CWAC ranks accounted for 2.8% of the Canadian Army. Twenty-five CWACs died during the war of accident, disease or injury during active service. None died in military skirmishes though several were injured slightly in a V-2 rocket attack. See Bruce, 1985; Conrod, 1983; and Gossage, 1991 for more information.

7. (Page 46) The author could find no evidence of an RAF mine-sweeping unit at the Debert, Nova Scotia base. Number 31 RAF OTU (Operational Training Unit) began at Debert in May, 1941 as part of No. 3 Training Command for general reconnaissance crews. They patrolled in Lockheed Hudsons and Avro Ansons for German U-boats and floating mines; several U-boats were attacked by planes carrying depth charges from this base (Hadley, 1985, pp. 45, 111; Madigan, 2017). From May 1944 until it was disbanded, it flew the de Havilland Mosquito. Olafur's duties within this unit are unknown, but given his next posting to the Suffield, Alberta base, perhaps they were meteorology-related.

8. (Page 52) In her archived notes, Hope makes the following comment about Ollie: "As a married woman, I must give credit to my long-suffering husband who has always been most supportive."

9. (Page 52) Dr. Percival Sidney Warren (1890–1970) was an eminent researcher of western Canadian stratigraphy. He was affiliated with the Department of Geology at the University of Alberta (UA, Edmonton) for fifty years and was the head of that department starting in 1949. Broad-minded, he felt no branch of geology less important than others and as a result was well-versed in a breadth of topics from Precambrian to Pleistocene and petrology to palaeontology. His style of teaching was welcomed by new students. His research focused on invertebrates, especially ammonites and brachiopods. For more information see Stelck (1970). It may be that a copy of the Warren letter(s) Hope so treasured is filed in the UA archives.

10. (Page 53) Dr. John Andrew Allan began work at the UA (Edmonton) in 1912 until his retirement in 1949. He published the first coal and geological maps of Alberta. He field surveyed much of Alberta and British Columbia and founded today's Canadian Society of Petroleum Geologists (CSPG). He died from a heart attack in 1955. For more on Allan see Anonymous (1999b) and other online sources.

11. (Page 56) Dr. Gareth Hubert Stanley Jones (1924–1997) was an interesting character and had a huge impact on Hope's early Earth sciences education, so he is examined in more detail here.

He was born in Wales to Basil and Katie Jones. His father was a civil engineer in charge of roads, bridges and tunnels in North Wales and was highly respected. Gareth himself was a brilliant but quiet man who acquired military training later in WWII and possibly saw some active war service. He was a Captain in the REME (Royal Electrical and Mechanical Engineers) in the British Army and had experience with explosives (Price, 2000).

By his own account he came to Canada in early 1957. He was awarded his Ph.D. on November 9, 1963 at the fall convocation of the University of Alberta (Calgary campus) under geophysicist and geologist John Tuzo Wilson (1908–1993; West *et al.*, 2014), for his thesis (Jones, 1963) entitled *Strong motion seismic effects of the Suffield explosions*. His paper *Geophysical research at Suffield* earned the "Best Paper Award" by the Canadian Society of Exploration Geophysicists in 1965 (Canadian Society of Exploration Geophysicists, 2016). In Canada he was freed of his obligations to the REME and was a civilian worker, though he worked directly with British, American and Canadian military staff and had full access to the officer clubs.

He was involved in the detonation of a number of large TNT bombs, up to 500 t, at the military base in Suffield, Alberta in the early to mid-1960s. These included various Cold War studies (Hopkins, 1959, 1961; Moss, 1961; Anonymous, 1964a, 1974c; Price, 2000—Price's chapter 5 is based largely on Jones' work), including (for other researchers) blast effects on various biological organisms (including, unfortunately, humans) and military infrastructure; but his specific interests were cratering effects (Anonymous, 1966d; Diehl and Jones, 1967a-b; Jones, 1970, 1976, 1977; Price, 2000) and battlefield trench design (T. Willock, pers. comm., 2016). He was intrigued by the small post-blast uplift produced in the centre of the Suffield craters and correctly likened them to moon craters with a central uplift, such as the crater Copernicus. NASA became interested in his pioneering research and later sent teams of astronauts to view similar explosions and study cratering research, by then also being conducted by others. The astronauts were sent to Suffield to look at these craters and to learn their morphology and development (Anonymous, 1970a; Kelley, 1971). Some of Hope's long-term friends believe that, during these NASA visits, she wined and dined with NASA's elite—some of whom would later fly to and walk on the moon. Jones also had ties to fellow crater scientist Eugene Shoemaker, namesake of the Shoemaker-Levy comet that broke apart and crashed into Jupiter in 1994.

He also had an interest in archaeology and is acknowledged in one of the first major published works on this field in Alberta, for his assistance with work done by the Glenbow Museum between 1957 and 1964 (Wormington and Forbis, 1965, p. viii).

Jones never married; ". . . all his energies were focused on ferreting-out universal geophysical truths." (B. Ashwick, pers. comm., 2014). He left the Medicine Hat area around 1967. Though he was a prolific letter writer, any correspondence between Jones and Hope has unfortunately not survived. Jones later worked for Emergency Preparedness Canada, but what he did there is unknown as it was top secret. However, he disliked the job, as it gave him no opportunity to be a scientist and do research: he was "bored and disgusted" (B. Ashwick, pers. comm., 2014).

He retired to his native Wales on January 29, 1985 and later bought a small house on the flanks of the Cypress Hills, south of Medicine Hat. He lived there part of each year, relaxing, writing up his research (Hope was apparently acknowledged in some), hunting fossils (sometimes with Hope) and big game. Jones was a chronic smoker, going through three or more packs of cigarettes a day. He was diagnosed with lung cancer in Medicine Hat in early October, 1996 and returned to Wales where he died several months later. Posthumously, his paper (with Yitian Xiao) *Predicting dolomite geobodies: Exploitable insights from reactive transport models* won the best geological paper award for 2006 (Canadian Society of Exploration Geophysicists, 2016). See Ashwick (2000) and Price (2000) for more details.

Knowing the end was near, one of his last letters to longtime friend Brian Ashwick, in Medicine Hat, ends with the A.E. Housman (1859–1936) poem, *Now hollow fires burn out to black*:

Now hollow fires burn out to black,
And lights are guttering low:
Square your shoulders, lift your pack,
And leave your friends and go.

Oh never fear, man, nought's to dread,
Look not to left nor right:
In all the endless road you tread
There's nothing but the night.

12. (Page 57) Among her notes, several times Hope expresses her displeasure at the name for this fossil locality, "Petrified Coulee." After all, a coulee itself cannot be petrified. Its name and the presence of fossils there goes back to at least 1902 (Kadatz, 2012) and 1923 (Morrow, 1923). It was known by Earth scientists by that name as early as 1930 (Williams and Dyer, 1930). L.S. Russell (1940, p. 72) was aware of the badlands there and thought them worthy of investigation by professionals, but the author is unaware of any such explorations there.

13. (Page 59) It is uncertain how aware Hope was of the technical literature. By the mid 1960s she had many of C.M. Sternberg's papers and was requesting more titles from him. She also had ROM publications on Late Cretaceous turtles and various dinosaurs from Alberta, and the 1907 Hatcher, Marsh and Lull Ceratopsia monograph. She ordered the Skinner and Kaisen *Fossil Bison of Alaska* monograph in early 1969 (Johnson, 1969b). Found among her personal papers after her death were original offprints of:

Sternberg, C.M. 1921. A supplementary study of *Panoplosaurus mirus*. Proceedings and Transactions of the Royal Society of Canada, 3rd series, vol. 15, pp. 93–102.

Sternberg, C.M. 1932. Two new theropod dinosaurs from the Belly River formation of Alberta. Canadian Field-Naturalist, vol. 46, no. 5, pp. 99–105 (sent to Hope with the author's compliments).

Sternberg, C.M. 1945. Pachycephalosauridae proposed for dome-headed dinosaurs, *Stegoceras lambei*, n. sp., described. Journal of Paleontology, vol. 19, no. 5, pp. 534–538.

And photocopies of:

Lambe, L.M. 1919. Description of a new genus and species (*Panoplosaurus mirus*) of an armoured dinosaur from the Belly River beds of Alberta. Proceedings and Transactions of the Royal Society of Canada, 3rd series, vol. 4, pp. 39–50.

Russell, L.S. 1948. The dentary of *Troödon*, a genus of theropod dinosaurs. Journal of Paleontology, vol. 22, no. 5, pp. 625–629.

When she acquired these titles is unknown. The photostats suggest some antiquity. Notes penned in the theropod papers reveal Hope's special interest in *Troodon*.

14. (Page 60) T. Potter Chamney (1924–1983) was a field assistant to C.M. Sternberg in 1947 in the Trenville Park, Alberta area. Chamney made several significant *Leptoceratops* skeleton finds that summer. He later went on to work in the oil sector where he spent most of his career. His son Ron knew Hope, describing her as one of his "favourite characters" in DPP. Ron also had the hot and dirty job of cleaning out the dead flies and dust in the three DPP fossil display buildings that Hope upgraded. Hope was particular about the presentation and cleanliness of these public exhibits (R. Chamney, pers. comm., 2015).

15. (Page 60) This is Dr. R.R.H. (Roy Richard Henry) Lemon (1927–2001), formerly of the Division of Invertebrate Paleontology at the ROM. He started at the ROM in this capacity in the 1957–1958 fiscal year (Heinrich, 1958), specializing in Neogene and recent molluscs, particularly those from Central and South America. He worked with the ROM's Ralph R. Hornell and Gordon Edmund at the Pleistocene Talara tar pits in Peru in 1958. He did some fieldwork in Saskatchewan, Alberta and Montana for the NMC in the summer of 1962, looking at Cretaceous and Cenozoic fossils. He left the ROM as an Associate Curator in June, 1967 and did exploration geology work in South America. From there he finished his career as Professor and Chair of the Geology Department at Florida Atlantic University in Boca Raton (Anonymous, 1982c).

16. (Page 62) The author recalls only two cases where it was allowed: in August, 1975 (Storer, 1975) when a ceratopsian sacrum and articulated ilia (TMP 1975.012.0001) were recovered; and a RTMP site investigation after some dinosaur bones were found by soldiers in 1992 (Anonymous, 1992b).

17. (Page 74) Dr. Charles Devillers (1914–1999; Figures 65, 110) was a palaeontologist, comparative anatomist and theorist at the University of Paris, France. He studied mosasaurs, fossil horses, therapsids, early synapsids and ammonites, and was a researcher and teacher of evolutionary theory. For more information on him see de Ricqlès (2000).

18. (Page 75) Donald "Don" Alfred Taylor (1909–1983; Stelck and Burwash, 1983) earned his M.Sc. thesis, entitled *Detailed stratigraphy of the Edmonton area* from the UA in 1934; he was an Honours graduate in geology. Around that time he secured support from the University's President Robert C. Wallace and the Geology Department's John A. Allan for his application to work on palaeontology and geology projects under a Carnegie scholarship at the ROM (McTavish, 2013). He also found the time to visit some American museums, such as the Chicago Field Museum, and did some post-graduate work at the University of Minnesota. Soon he was in charge of the UA's new geology and palaeontology museum, which opened in 1935.

His Ph.D. plans were interrupted by WWII. By 1941 he was involved in several decades of geological and oil exploration activities in Venezuela and Peru, working for Socony Vacuum Oil (Mobil). He returned to Alberta in 1960 and set up the revised Geology and Palaeontology Museum in the basement of the Agriculture Building on the UA campus. He was Curator of Geology at the newly-opened Provincial Museum and Archives of Alberta from 1966 to 1974. Hope Johnson had regular contact with him during this period when he collected numerous, mostly small, Late Cretaceous vertebrate specimens, mostly in southern Alberta, with his technician assistant John Poikans (Tanke, 2008b) and other museum colleagues. This collection formed the nucleus of today's fossil vertebrate collection at the Royal Tyrrell Museum. From 1975 onward Taylor was a Research Assistant in the UA's geology department. After retirement he taught courses on rocks and minerals to seniors. A stroke incapacitated him for a year and another took his life.

19. (Page 75) Dr. Archibald "Archie" MacSween Stalker (1924–2010), was a geologist with the Geological Survey of Canada (GSC) from 1945 to 1987. Ottawa-based, he worked mainly on a variety of Pleistocene Earth science-related issues in southern Alberta including glacial landforms, buried river valleys, possible occurrences of earliest man in Alberta, stratigraphy, fossil vertebrates and giant glacial erratics, all studied over forty-two field seasons. He started CANQUA (Canadian Quaternary Association). For more information, see Anonymous (2014m).

20. (Page 79) It is not known if these photographs were later used in a military magazine article or perhaps for an exhibit she had on the base. Hope makes no other mention of it in her many notes.

21. (Page 82) There is some confusion over who found this bonebed. Colwell (1967, August 27 entry) says Hope's friend and fellow collector, Charles Heath from the UA found it, yet Hope's own fossil catalogue says she found it and that she reported it. In reality it was undoubtedly first seen by palaeontological field crews who worked these badlands in the early twentieth century, but it is not believed that those crews collected anything from the bonebed.

22. (Page 83) Hope's notes to her family state that she joined SVP in 1963 but she is in error. Each SVP news bulletin listed new members and none of the 1963 issues record Hope joining. She did join in 1966 as indicated in Anonymous (1966l). In notes to her family, Hope also records that she attended several SVP meetings, but provides few details. She apparently attended the 1977 SVP meeting in Los Angeles, where she hand-carried some small specimens to be identified (Johnson, undated 2, p. 118). She and Ollie visited with two of his sisters who lived in the area (Johnson, 1977c). Brief notes and interpretations thereof indicate that she also attended meetings at Flagstaff, Arizona (1974, with husband Ollie); Toronto (1978); and Drumheller (1988).

23. (Page 90) There are a good number of 35 mm photographic slides of this excavation in RTMP library photographic archives. See catalogue numbers TMP PN 68.2.1–7; 68.31.1–18, 21–23, and 25–32.

24. (Page 93) Hope's handwriting and rough preparation work appears on two other articulated pedal digits of ornithomimids. TMP 1979.014.0783 is a complete left digit IV and TMP 1979.014.0864 is three phalanges of a single left digit III. These are unusual finds. After thirty-seven field seasons of work in DPP, the author has never seen isolated and articulated single digits of ornithomimids; they are either found as common isolated phalanges or as articulated/associated feet occurring with a fuller skeleton. Possibly TMP 1979.014.0783 and TMP 1979.014.0863 are from the same individual or even the opposite foot of TMP 1979.014.0701. All these specimens were from the old DPP collection which regrettably did not always include locality or other critical information. RTMP catalogue records, based on the poor DPP records, do not shed any light on the possible reassociations noted above. However, all of the phalanges catalogued under three separate TMP numbers, are the same general colour, the correct size for a right and left pair and bear Hope's handwriting and her reconstruction work. Therefore TMP 1979.014.0701, TMP 1979.014.0783 and TMP 1979.014.0864 are tentatively referred to a single individual.

25. (Page 96) Her employment marked the second time a woman was hired into a paying vertebrate palaeontology position in Alberta, the first being Jane Colwell in late 1965. Some accounts state that Hope was hired in 1978 to 1981, but these are in error, as the palaeontology department of the PMA had assumed these responsibilities by that time. It may be that the erroneous dates refer to the transfer of the large vertebrate fossil collection from DPP to the PMA (Danis and Walper, 1980).

26. (Page 97) Rapid southward-directed erosion and continual collapse of the clay and silt cutbanks on the south side of the Red Deer River threatened this perfectly good headquarters building. For safety reasons it had to close and was torn down around 1983. The 1979 to 1982 PMA crews used it for showering, laundry and occasional cooking, but especially for fossil cataloguing and prepara-

tion on rainy days and evenings. The DPP and PMA staff held some loud and raucous parties in the building.

27. (Page 99) It is the most severe dinosaur postcranial osteopathy with advanced healing the author is aware of. It is described and figured in Tanke and Rothschild, 2014, p. 561; their figure 33.34.

28. (Page 99) Colwell (1968a) mentions seeing this specimen on display at the Park on June 17, 1968. Hope, writing about ankylosaurs (Johnson, 1970b) said: "FAR OUT BEASTS [her emphasis] and completely improbable. Saw a skull in the R.O.M. made me think of a stone pig."

29. (Page 99) The pterosaur bone site (Quarry 139, DPP) was permanently marked with a metal quarry stake by the author, but is not the actual discovery site. Later in her life, Hope was physically unable to show RTMP staff the exact site, so she described it and drew a detailed map of the area. It is felt that the quarry stake marker is stratigraphically correct and spatially accurate to within tens of metres of the true site. A cast of the specimen is at the CMN in Ottawa and bears their catalogue number CMNFV 29415. The original specimen is curated at the RTMP. Hope (in Moser, 1974, p. 33) indicates that this specimen was collected in 1970.

30. (Page 100) "Warty Bliggens" is a toad, a character in a poem written by poet, humourist, author, newspaper columnist and playwright, Don Marquis (1878–1937).

31. (Page 100) At the time of her discovery, it was not well known that some dinosaurs underwent remarkable cranial ontogenetic changes as they were reaching or reached adulthood and in some cases continued as the animal neared the end of its life. "Flat-headed" pachycephalosaurs without a dome are now understood to be juvenile forms that had not yet developed their fully domed skull. Therefore, the small and flat-headed "Ornatotholus" is actually an immature stage of the larger and fully-domed *Stegoceras*, making the "Ornatotholus" name a junior synonym.

32. (Page 100) These four display houses were:

1. Quarry 125 (TMP 1980.023.0001; originally numbered as Display House #1); headless hadrosaur. Torn down c. 1992.
2. Quarry 126 (TMP 1980.023.0002; originally numbered as Display House #2); headless hadrosaur.
3. Quarry 127 (TMP 1980.023.0003; originally numbered as Display House #3); ceratopsian scattered skeleton.
4. Quarry 128 (TMP 1980.023.0004; originally numbered as Display House #4; now called Display House 3); *Corythosaurus*, much of skeleton and skull. In the mid to late 1960s this display house was also known as the "South Dinosaur" (Johnson, undated 2).

33. (Page 102) Hope's fossil preparation skills were not very good by today's exacting standards, but not bad for the times, and considering her lack of professional training and supervision in this area. Several specimens have parts reattached backward or rotated relative to their correct anatomical position. Technical papers and books on fossil preparation existed early on (*e.g.* Camp and Hanna, 1937; Kummel and Raup, 1965), but Hope may not have been aware of, or had access to these. Fortunately for her, many Late Cretaceous fossils found in Alberta are well preserved and found in relatively unconsolidated sediments, so rigorous preparation work is often not required for smaller specimens.

In DPP there was no dedicated fossil preparation area, so Hope did all her fossil cleaning and

restoration in the Park's maintenance shop/garage (G. Forsen, pers. comm., 2014).

What glues and tools she used for preparation is largely unknown, though among the last of her fossils donated to the RTMP was a small tool resembling a harness awl (*sensu* Brown, 2012; Figure 158), of the type used by early twentieth century fossil preparators. Some of her plaster infill work is recognized by its rough, unfinished nature and by its brown colouration throughout. She is known to have used brown watercolours to paint white plaster reconstruction (Figure 159). From repreparation of some of her larger projects, there is no indication that she used epoxy glues to stabilize and reassemble bone pieces back together (Tanke, pers. observation; R. Russell, pers. comm., 2015). She does mention the use of white glue to glue dinosaur bone pieces back together in a mid-1970s letter to a friend. An adult hadrosaur ischium specimen (TMP 1979.014.0381), reassembled by her, used plaster of Paris as the bonding agent. The specimen has since been disassembled, cleaned and restored. A *Euoplocephalus* skull (TMP 1979.014.0747) was prepared by her using plaster (R. Russell, pers. comm., 2005). An unidentified, yellowed, clear coat of glue that was soluble in acetone was also seen on this skull. This may be Glyptal™, a product used by PMA/RTMP staff in the 1970s and 1980s. It may have been applied by them after the skull came to the PMA in 1979.

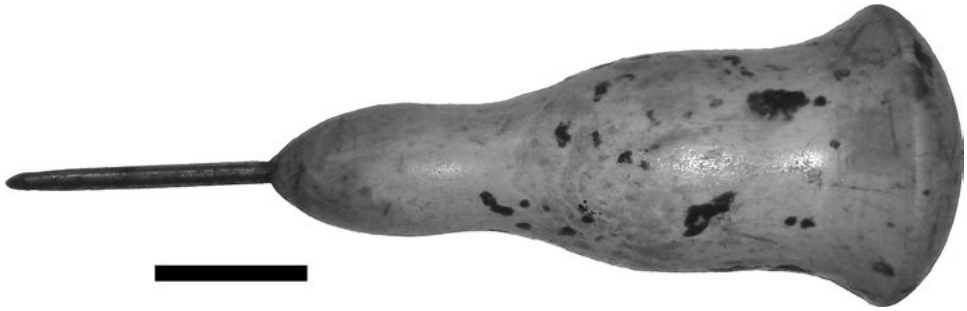


Figure 158. Small harness awl-type tool found among Hope's boxed fossil collection. The tool is 6.2 cm (2.5″) long (scale bar = 1 cm). Photograph by the author.

34. (Page 102) A letter to Dr. Dale A. Russell from Hope (Johnson, 1975a) explains the presence of a hadrosaur tibia in the tyrannosaur section of the DPP fossil exhibit. She writes: "At the moment, the tibia of a hadrosaur is sitting in the tyrannosaur section of the Park's fossil exhibit! It is of a very unusual colour—pale grey + smooth + does look much more like a tyrannosaurid bone than hadrosaur bone but the shape just 'won't do at all.' It weighs so much I can't move it but I shall have to move it, *i.e.* get it removed this year even if the shelf does go empty! Just too bad!" The tibia never was removed, however. It was still there four years later when the author saw and photographed it (Figure 98) and it remained in place a year or so later until the exhibits were finally dismantled.

35. (Page 107) Dr. John E. Storer (1944–) started with the PMA on July 1, 1970. He left that institution on November 1, 1975 and joined the (now Royal) Saskatchewan Museum of Natural History in Regina. His PMA position was filled by a then little-known 27-year-old named Philip J. Currie on July 5, 1976. Storer did have a field program collecting dinosaurs in the Drumheller area from 1970–*c.* 1974, but was more interested in Cenozoic mammal collecting and research.

36. (Page 108) A working title for the book was "A Collector's Guide to Alberta Vertebrate Fossils."

37. (Page 116) Dryden (1981, p. 151) states that the interview is held at the Provincial Archives of Alberta in Edmonton, with catalogue number 4–528/73.209. The author has a digitized version.

38. (Page 119) The names that come up most frequently in Hope's fossil catalogue books are Gareth Jones, Gord Edmund, Charles Heath and his wife Velma, and William and Shirley Lawson. She also did much collecting with Ron and Sharon Stiem and Shane and Vicki Leuck in more recent years.

39. (Page 125) This specimen may be a *Corythosaurus casuarius* skeleton with skull collected by the PMA in 1980 (TMP 1980.040.001; Quarry 150). Hope was aware of the site by no later than 1969: a 35 mm slide with a summer 1969 processing date exists, the image captured by her friends, the Lawsons, to whom she showed the site. Park employee Ron Chamney recalled that she recognized earlier diggings on the site that were done by the early twentieth century dinosaur collectors. The specimen has been on display at RTMP since the museum opened in 1985.

Figure 159. Two fossils: hadrosaur pedal phalanx, Hope's catalogue no. 811; and ceratopsian phalanx, Hope's catalogue no. 705; both in dorsal view showing rough plaster reconstruction and some painting of same. Her numbers were always black India ink written directly on to the bone as seen here or written onto a painted white circle. Scale bar = 1 cm.

40. (Page 125) The author first became aware of Hope Johnson in 1977 from this documentary and was surprised to see her—a woman, and relatively unknown—featured among the male palaeontologists who dominated the science even at that time. The rest of the program is filled with interviews and research work conducted by contemporary leaders in the vertebrate palaeontology field—the likes of Drs. Dale A. Russell, Robert T. Bakker, Farish A. Jenkins, Jim Farlow, Alfred W. Crompton, R. McNeil Alexander, Armand de Ricqlès, and John Ostrom. So how did Hope make an appearance in the film? It is possible there was no provincial palaeontologist at the time and Hope filled the void. John E. Storer had left Alberta in November, 1975; the Park Ranger (Roy Fowler) who had

much experience with fossils was ill and passed away in 1975 (Anonymous, 1975b); and Storer's replacement, Philip J. Currie, did not begin work in Edmonton until July 5, 1976.

41. (Page 125) Alberta dinosaur collecting fieldwork was first professionally filmed during George F. Sternberg's 1921 field season in DPP. C.M. Sternberg's 1925 field season was also filmed by Canadian Government ornithologist Percy A. Taverner (1875–1947). The footage from both of these expeditions has been reused often in more modern documentaries. The oldest effort to film a palaeontological excavation in Alberta and intended for use in a television documentary is the 1974 BBC television production *Before the Ark* (a ten-part series). Filming of the 1972 Ghostpine Creek (near Drumheller) *Edmontosaurus* (TMP 1970.018.0001) quarry took place that summer. British vertebrate palaeontologist Alan J. Charig (1927–1997) accompanied the film crew. It is uncertain if the Alberta footage was actually used in the BBC production.

42. (Page 161) The newspaper article stated she was 92; at the time she was actually 93.

43. (Page 161) Dr. Don Brinkman and the author (with Patty Ralrick) were the RTMP representatives. From the APS were Editor Howard Allen, APS President Dan Quinsey and his wife Valerie. A number of other local amateur fossil collectors also attended.

44. (Page 161) The author recalls seeing a video camera on a tripod recording the events of that afternoon, but has been unable to determine who made the recording.

45. (Page 165) While the Tyrrell Museum did not open its doors until 1985, nearly all palaeontological accessions from the Provincial Museum (Edmonton) went to the Tyrrell. The PMA did not open until December 6, 1967 but there were fossils collected for it several years prior to its opening. Where these were stored, prepared and curated prior to the museum's opening is presently unknown. D.A.E. Spalding (pers. comm., 2016) recalls: "The PMA building was occupied when I joined the staff at the beginning of October 1967. But I don't think they had been there very long. Previously the offices, staff and collections had been based in Beaver House, a historic building in downtown Edmonton. I believe Don [Taylor] had been collecting for the museum for a couple of years or so by the time I arrived."

46. (Page 169) Indeed, this element confused her for quite some time. In one of her fossil catalogue books (Johnson, undated 2, p. 16) under a September 1, 1965 entry she catalogues (her #863) and figures one of these peculiar symmetrical bones.

47. (Page 173) Jane Colwell (1941–) was the first woman with academic training in vertebrate palaeontology to work in DPP, on behalf of the UA (Edmonton) each summer of 1966 through 1968. She worked at the NMC from about 1969 to 1979 and PMA/RTMP from 1979 to 1991. She volunteered at the RTMP from 2008 to 2015, then continued at home until the summer of 2017, picking fossils from Late Cretaceous microvertebrate concentrate she collected in the 1960s. See Tanke (2015a, 2017a and 2017b) for a more detailed accounting of her many Earth science activities.

48. (Page 78, Page 180) Hedwig "Hedy" Hobberlin (1931–2017) was an amateur fossil collector who took some adult courses in geology from the University of Toronto. She and her husband Paul

(1913–1997) met Hope through Hope's fellow collectors, the Lawsons, of Medicine Hat. Hedy collected fossils with Hope in today's DPP several times in the early to mid-1960s and Hope later sent fossils to Hedy from Alberta.

Hedy had a large fossil and geology collection in the basement of her house and school groups would visit (Anonymous, 1971c). Eventually it was suggested that Hedy be hired on contract by the school division in North York, Ontario to teach about the Earth sciences as part of the school curriculum. She began doing this in September, 1975. Hedy's natural history collection was housed in Drewry School (known as the "Hobberlin Museum") in North York and for twenty-seven years many grade six students were taught by her (alone, at first) until 2002 when there was a staff of three and 15,000 students per annum were taught in half-day and full day programs.

The relationship ended suddenly in 2002 with the amalgamation of Toronto with numerous nearby cities, creating a "super city." Much politicking and infrastructure reshuffling ensued. School divisions were amalgamated and Hedy's program was deemed an unneeded "luxury" in a newly expanded city that already had a museum with palaeontology exhibits—the ROM. Ninety-five percent of the Hobberlin Museum collection was put into storage, but some of the fossils Hope and Hedy collected in the Alberta badlands are still used to this day for educational purposes, Hedy's work carried on at a much smaller scale by her daughter Christine Hobberlin-Hynson.

49. (Page 189) As a couple of examples of Hope probably being undercredited for all her finds, RTMP has several trays of bulk-catalogued ornithomimid pedal phalanges from DPP. TMP 1979.014.0490 includes forty-eight phalanges and one distal metatarsal (though some bones may be small tyrannosaur or maniraptoran). Of these, seven bear Hope's handwriting or reconstruction work. TMP 1979.014.0865 comprises a tray of fifteen maniraptoran and ornithomimid phalanges. Of these, seven bear Hope's handwriting and/or plaster reconstruction. These are but two examples of fossils bulk-catalogued under the TMP 1979.014 series, which catalogued the original DPP collection.

Addendum:
Hope Johnson's art in colour

Over the next 28 pages, more of Hope Johnson's art of various styles and media are presented in colour, to give the reader a fuller sense of her diversity of styles and subject matter. Known best in the public and arts community for her drawings, acrylic and oil painting and watercolours, Hope was equally adept at calligraphy, needlepoint and batik wall hangings, though apparently she did not often work in these media. Information on the art pieces is given where known. Some of these images are copied from originals, her 35 mm slides, scans, or photographs. Some bore no information. Hope sold many of her works, and the final repositories of some these pieces are unknown—most are probably in the homes of anonymous private citizens.

After an artist dies the value of their work often increases, sometimes dramatically. During the dozen or so years the author worked on piecing Hope's story together, he checked on the Internet regularly for any sales of her work. Other than a couple of benefit auctions, he did not see or hear of any individuals or art galleries selling Hope Johnson art, which suggests that current owners are happy with her work and are keeping it displayed in their offices and homes.

Figure 160. Watercolour illustration of prairie parsley. Undated, unpublished image courtesy of the Johnson family.

Figure 161. "Laddie" the family horse, Hope's earliest surviving artwork, painted, *c.* 1932, when Hope was 16 or 17. Image courtesy of the Johnson family.

Figure 162. Pencil crayon study of a cat done by Hope at age 90. Image courtesy of the Johnson family.

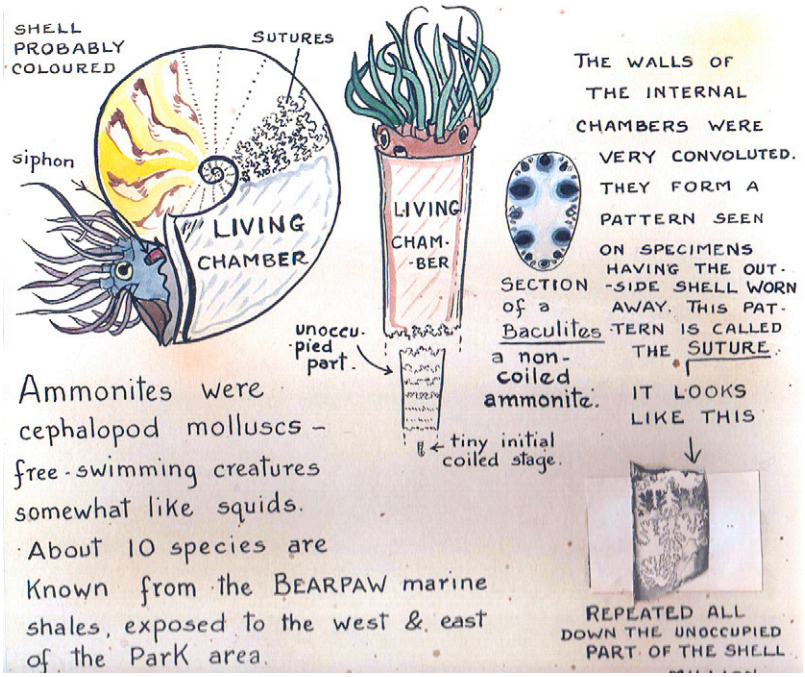


Figure 163. Explanatory illustration of Late Cretaceous ammonites and baculites; undated inked drawing and watercolour on illustration board by Hope Johnson. Original 25.5 × 23 cm. Former DPP fossil exhibit building signage, *c.* 1969–1980. Collection of Clive Coy, Edmonton.

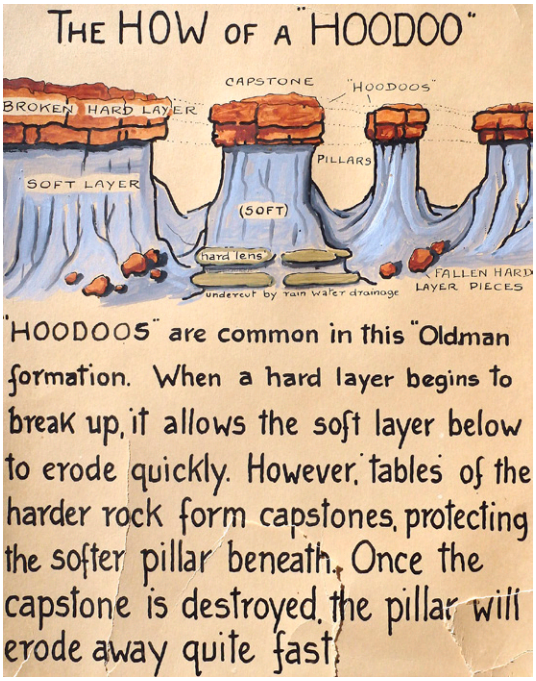


Figure 164. Watercolour on paper illustration explaining the formation of badlands hoodoo structures. Watercolour and ink on paper. Former DPP fossil exhibit building signage, *c.* 1969–1980. Author's collection.

Figure 165. Mountain scene painted by Hope. It is undated, but its simpler, abstracted style (compared to her later works) suggests this piece was painted in the 1960s. Collection of Rachel Noone, Medicine Hat.

Figure 166. An uncharacteristic painting by Hope done in 1964. Image courtesy of the Johnson family.

Figure 167. Hope's palaeontology mural in the Graham Community Library, painted in 1959 (not 1954 as the caption at top states). See discussion, Pages 62–64.

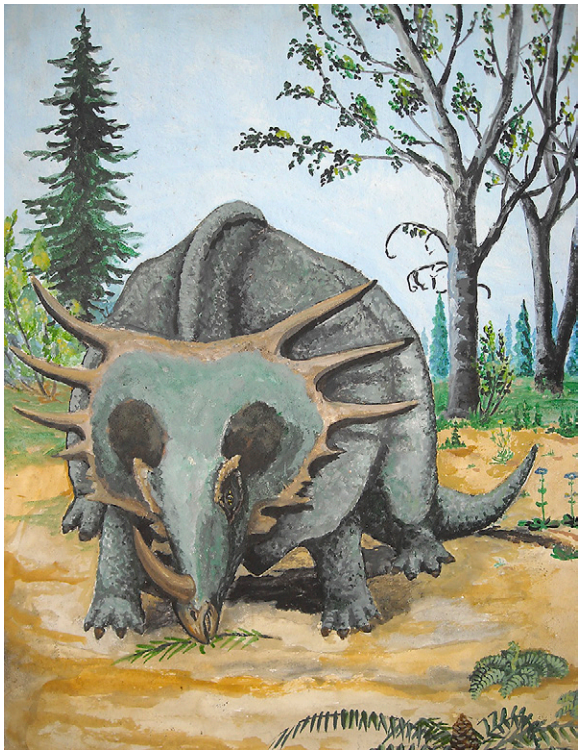


Figure 168. *Styracosaurus* life reconstruction; undated painting by Hope Johnson, adapted from Scheele (1954, p. 115). Acrylic on medium-density particle board (possibly recycled ceiling tile), 61 × 45.5 cm. Former DPP fossil exhibit building signage, *c.* 1969–1980. Author's collection.

Figure 169. *Centrosaurus* life reconstructions; signed and undated painting by Hope Johnson, adapted from Zdeněk Burian *Triceratops* (1972). Acrylic on canvas-covered heavy card, 61 × 45.5 cm. Former DPP fossil exhibit building signage, *c.* 1969–1980. Author's collection.

Figure 170. Armored dinosaur life reconstruction; undated painting by Hope Johnson, adapted from a drawing of *Paleoscincus* by Scheele (1954: 82). Acrylic on medium-density particle board (possibly recycled ceiling tile), 61 X 45.5 cm. Former DPP fossil exhibit building signage, *c.* 1969–1980. Author's collection.

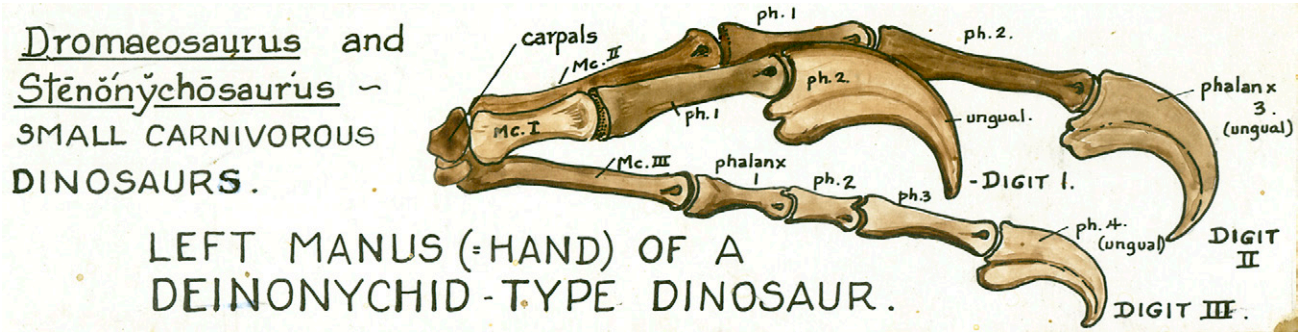


Figure 171. Left manus of maniraptoran theropod, internal view; undated painting by Hope Johnson possibly adapted from Ostrom (1969). Watercolour on card, 30 X 7.5 cm. Former DPP fossil exhibit building signage, *c.* 1969–1980. Author's collection.

Figures 172, 173. Mid-1960s paintings by Hope of Dinosaur Park Formation outcrops in the Denhart Coulee area, within today's DPP. Images courtesy of the Johnson family.

Figure 174. Batik wall hanging, 34.75" × 23.75", 1980. The dragon head motif is based on the early Viking "Oseberg ship." The use of Nordic runes on the border is random and decorative only; they do not spell anything. Batik is not an art medium usually associated with Hope; it is unknown how many, if any, other batik pieces she made. Collection of S. Leuck, Medicine Hat.

Figure 175. These fantastic, Peter Max-influenced psychedelic pop art portraits of the British band The Beatles were inspired by their *Sgt. Pepper's Lonely Hearts Club Band* album released in the early summer of 1967. They were painted by Hope that year and she also painted a chest of drawers in a similar psychedelic style (R. Noone, pers., comm., 2015). They are a total departure from her usual style of work—so much so that the author at first believed they were painted by someone else. Each piece measures about 16 in × 20 in. Clockwise from top left: John, Paul, Ringo, George. Collection of Rachel Noone, Medicine Hat.

Figure 176. One of Hope's watercolours of local wild plants. Image courtesy of the Johnson family.

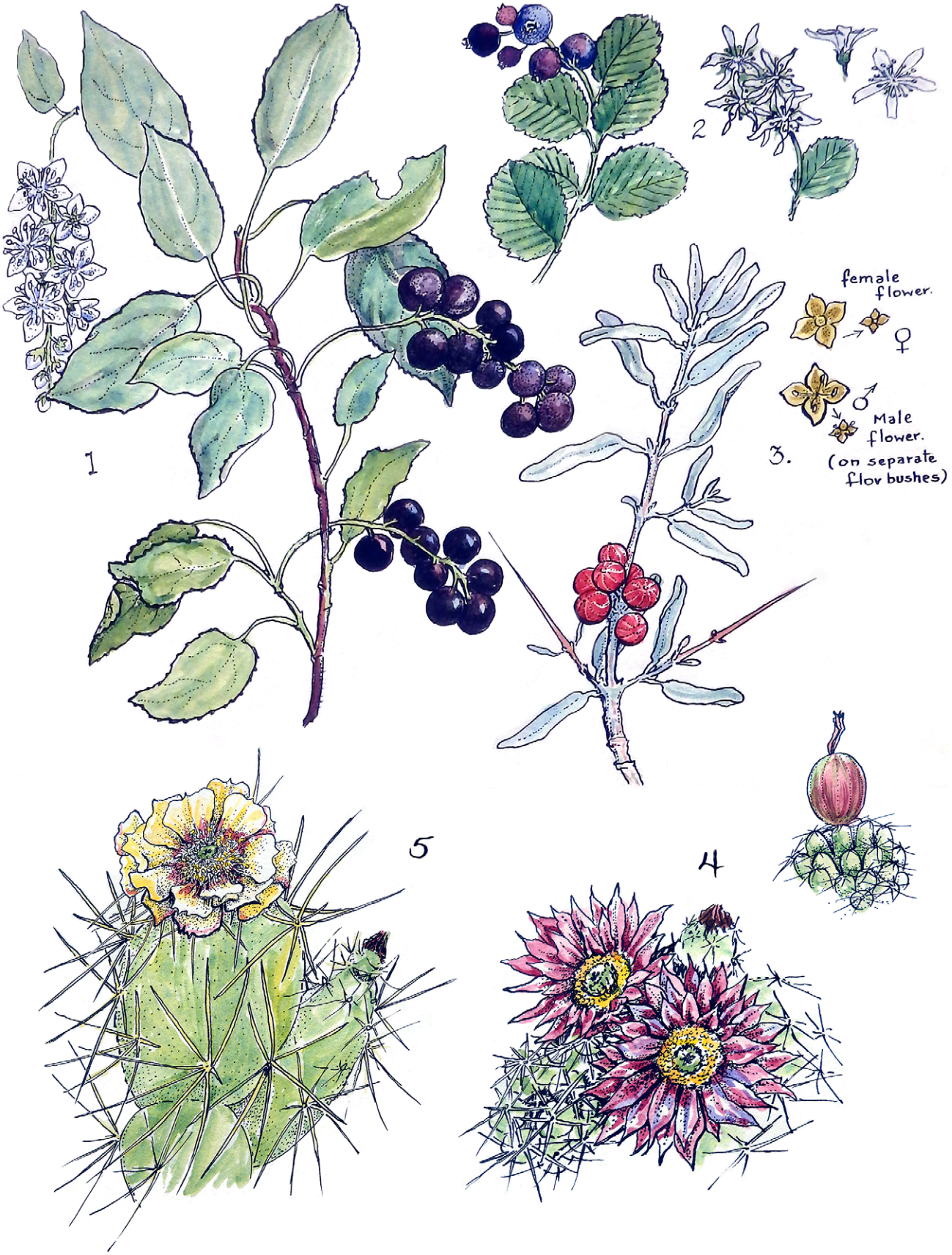


Figure 177. Some of Hope's wonderful renditions of local flora done in watercolours. From Johnson (1975d–i), selected images cropped and rearranged here. 1. Choke cherry (*Prunus virginiana*); 2. Saskatoon berry (*Amelanchier alnifolia*); 3. Thorny buffalo berry (*Shepherdia argentea*); 4. Cushion or ball cactus, flowers and fruit (*Mamillaria vivipara*); 5. Prickly pear cactus (*Opuntia polyacantha*). Images courtesy of Graham Public Library, Ralston, Alberta.

Figure 178. Left: Hope's portrait of Weasel Tail, of the Blood First Nation (?*c.* 1859–1950). Right: another portrait, based on work by German-born artist Winhold Reiss (1886–1953), a prolific portraitist of Blackfoot First Nations people, this one being *Julia Wades in the Water* of the Peigan First Nation. Images courtesy of the Johnson family.

Figure 179. A 1973 oil painting by Hope: "Columbian Mammoths and Sabre-toothed Cats," copied from artwork by Zdeněk Burian. This painting was originally used in the mammoth display case at the Medicine Hat Museum and Art Gallery. Image courtesy of Valerie Martins, Medicine Hat.

Figure 180. *Windy Sky, Red Rock Coulee*, 1989. Watercolour. Image courtesy of the Johnson family.

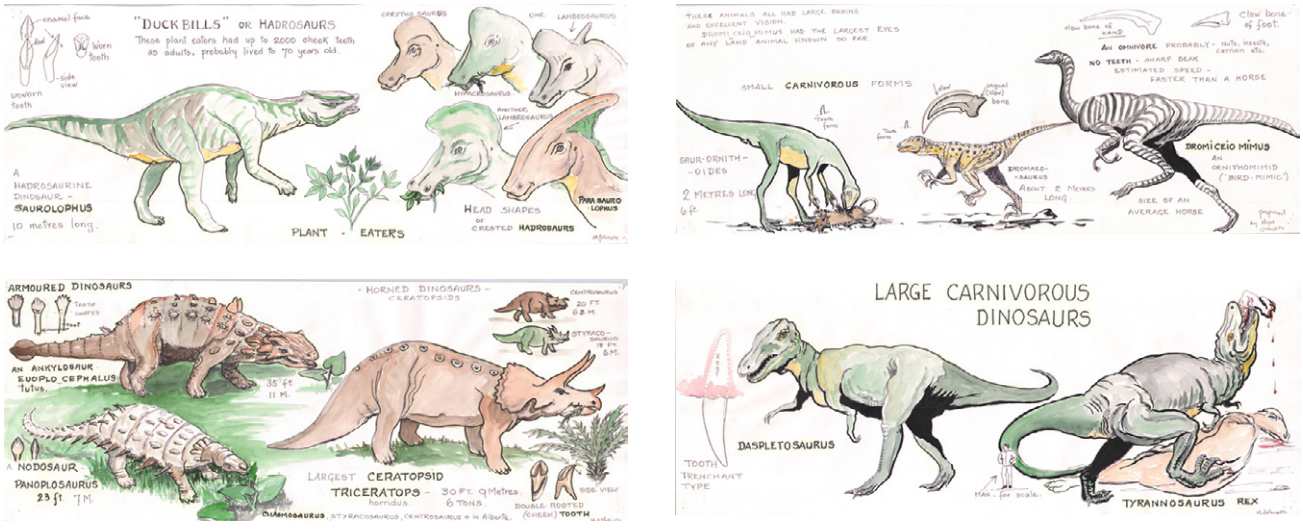


Figure 181. Four undated watercolour dinosaur illustration plates found at the Redcliff Museum. It is not known when these were drafted. The *Daspletosaurus* is copied from a figure in D.A. Russell (1977), so they were made no earlier than that.

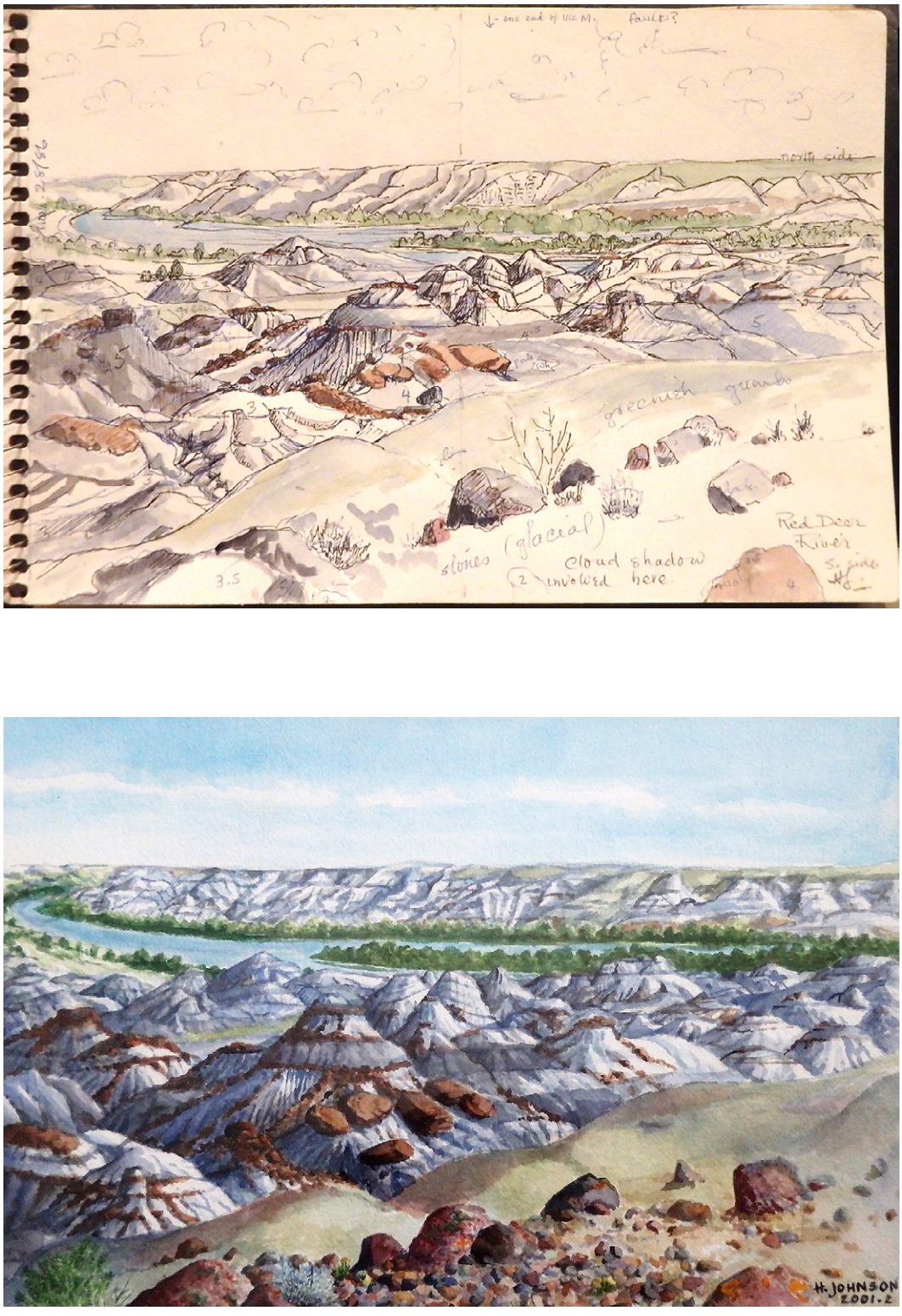


Figure 182. (Top) Small reference watercolour with notes, of DPP badlands, summer 1986. Author's collection, gift of Bill and Karen Spencer. (Bottom) The same scene as a final watercolour, 2001–2002. Esplanade Arts and Heritage Centre Archives, Medicine Hat.

Figures 183 (top) and 184. Hope Johnson paintings with glacial till pebbles as the main subject matter. Figure 184 courtesy of Dr. René W. Barendregt.

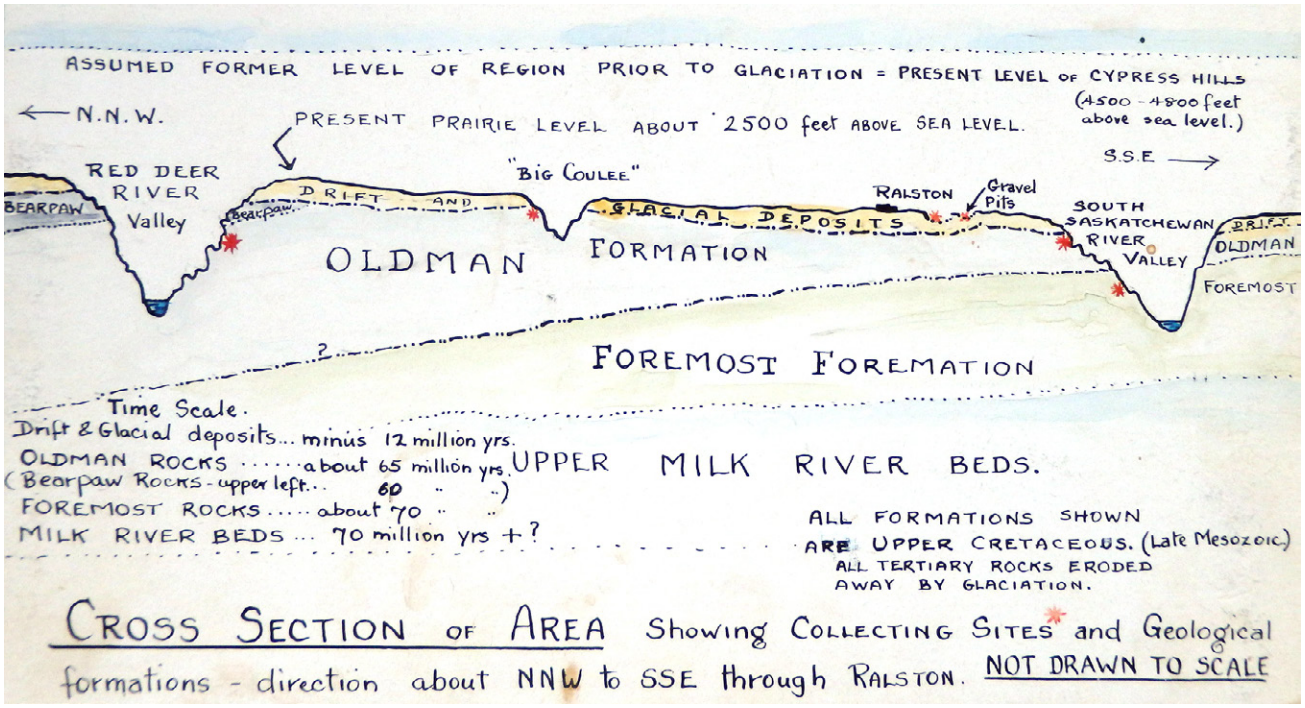


Figure 185. Geological cross-section illustration of the Ralston, Alberta area. The author encountered a number of these and geological stratigraphic column figures during his research. Hope was clearly trying to inform the public about the geological context of their areas and what was going on underfoot. Image courtesy of the Graham Public library, Ralston.

Figure 186. Badlands scenery at the Jenner Crossing (now Jenner Bridge) area, 1965. A note on the 35 mm slide states that the original was sold to a member of the British military. Image courtesy of the Johnson family.

Figure 187. Hope called this painting "Hoodoo, boulders. Badlands of the Red Deer River. March, 1968." The piece was originally a wedding gift, rushed to Gilles and Jane (Colwell) Danis, who were married March 15, 1969. Collection of the author.

Figure 188. Hope's painting *Burmis Tree*. Image courtesy of Shane Leuck, Medicine Hat. The original is now in the University of Lethbridge art collection.

Figure 189 a-b. a) Great horned owl, *Bubo virginianus*. Unknown medium, painted around the mid-1960s, 19.7 cm × 26.7 cm. Author's collection. b) An assortment of owls, undated. The outer circle of owls are tentatively identified, clockwise from upper left: Great horned owl; great grey owl, *Strix nebulosa*; snowy owl, *Bubo scandiacus*; perhaps a variant great horned owl; probably a screech-owl, *Megascops* sp. Left centre: northern saw-whet owl, *Aegolius acadicus;* right centre: boreal owl, *Aegolius funereus*. Collection of Barbara Preston.

Figure 190. Hope's watercolour renditions of old stone buildings in the countryside of Wales. Images courtesy of the Johnson family.

Figure 191. A mixed medium 1997 Christmas card sent to the Leuck family in Medicine Hat. Hope did not want to paint each picket of the fence seen here, so took a photograph of the fence, had that printed, cut it out and affixed it to a card and then painted the family's house and trees. Image courtesy of Shane Leuck, Medicine Hat.

Figure 192. *The Coven's Convention of 2005*, a watercolour painted by Hope in that year. Image courtesy of Sheila Schwab, Valleyview Retirement Community, Medicine Hat.

Figure 193. Another undated painting with a picket fence, this time each board carefully hand painted. Image courtesy of Rachel Noone, Medicine Hat.

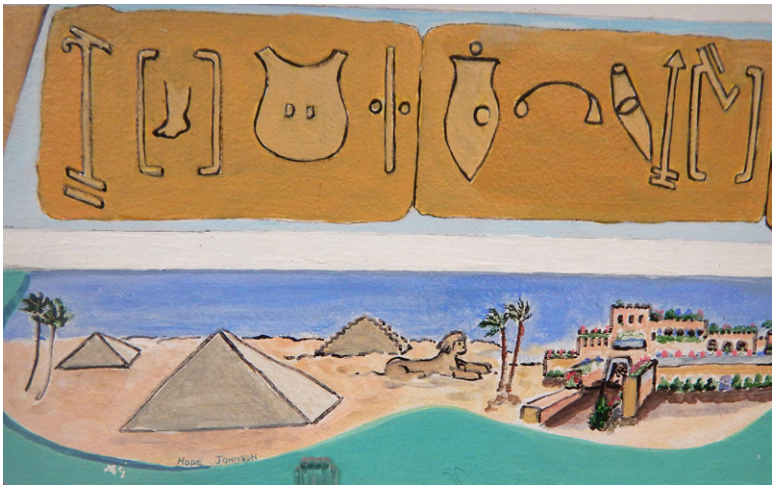


Figure 194. Egyptian-themed detail signed by Hope in the 1983 *Odyssey* mural. Photograph by the author.

Figure 195. Detail of the palaeontology-themed portion of the 1983 *Odyssey* mural in the Medicine Hat Public Library. Photograph by the author.

Figure 196. Seahorses painting, undated. Courtesy of the Johnson family.

Figure 197. An example of Hope's calligraphy work, undated. Image courtesy of the Johnson family.

Figure 198. "Scatha the Worm" dragon sketch, blue and red ballpoint pen on graph paper, undated. Image courtesy of Hope Johnson.

Figure 199. Crewel-style type needlepoint work by Hope, design based on a decorative motif on a silver arm bracelet. Image courtesy of the Johnson family.

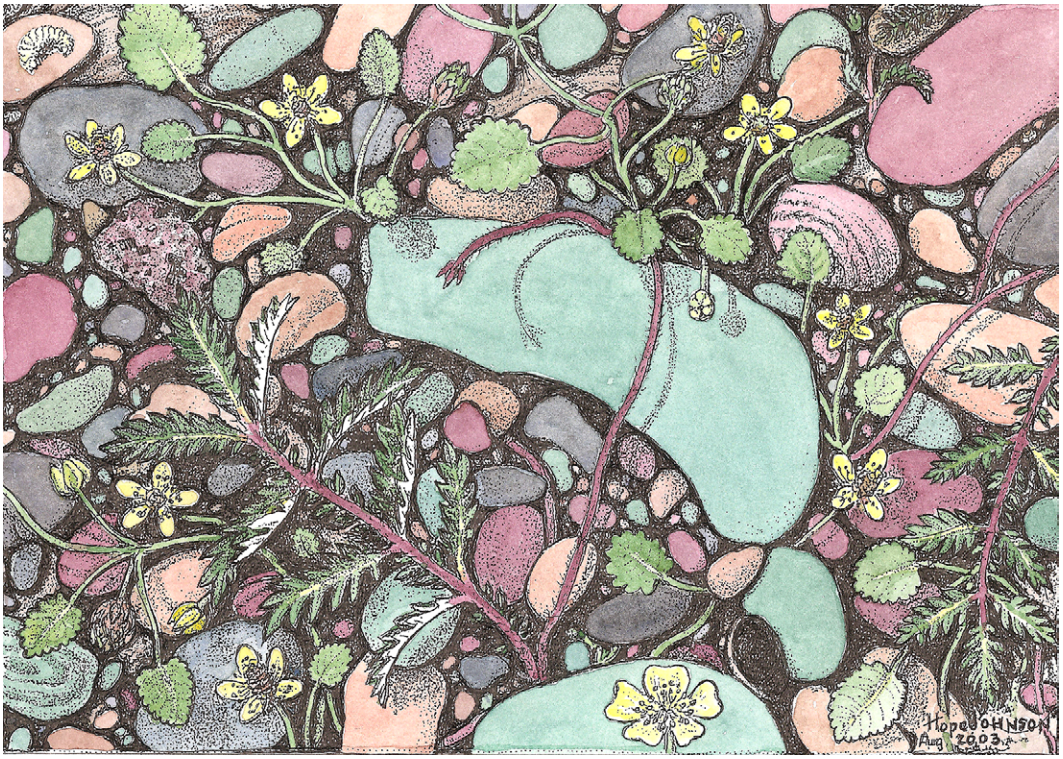


Figure 200. Hope's notes written on the back of this pretty 2003 pen-and-ink and watercolour piece identify these flowers as the buttercup *Ranuculus cymbalaria* from the Waterton, Alberta/Flathead Lake, Montana region. In another art print of this piece, Hope further identified the plant with the red stems and toothed leaflets as silverweed, *Potentilla anserina*. The colourful background stones are a piece of igneous rock and water worn pebbles of Precambrian argillite. Image courtesy of Shane Leuck, Medicine Hat.

Figure 201. Badlands painting, undated, but was acquired between 1962 and 1975. Collection of Duncan and Pat Currie, Victoria, British Columbia.

Figure 202. Painting of assorted badlands flowers. Collection of Margaret Noone, Medicine Hat.

Figure 203. Dead Cottonwood trees and sage brush at Police Point, Medicine Hat, 1989. Image courtesy of Hope Johnson.

Figure 204. Fall colours in the Cypress Hills. Watercolour, painted in 1997. Figure 62 shows Hope painting this piece in the field. Image courtesy of the Johnson family.

Figure 205. Watercolour detail of lichens on fragmented Late Cretaceous concretions at one of Hope's favourite places, Red Rock Coulee, Alberta. From an undated and unsigned work, courtesy of Allan C.J. Jensen, Medicine Hat College, where this piece is curated.

Figure 206. Oil painting of badlands on the South Saskatchewan River. Collection of Al Lawson, Redcliff.

Figure 207. Oil painting. Pinnacles of Oldman Formation in White Rock Coulee above the South Saskatchewan River. Collection of Bill Lawson, Redcliff.

Figure 208. South Saskatchewan River, south of Bowell, Alberta. 20 in × 18 in. Photograph by the author.

Figure 209. Oil painting of White Rock Coulee badlands on the South Saskatchewan River. Collection of Bill Lawson, Redcliff.

Figure 210. Mill Creek, southwestern Alberta, 1989. Image courtesy of Allan C.J. Jensen, Medicine Hat.

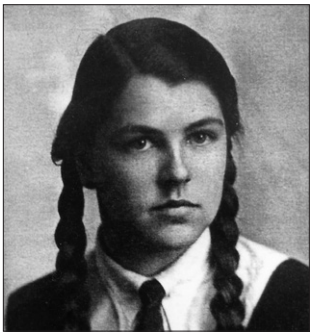
Figure 211. South Saskatchewan River and First Nations teepee ring, 1989.

Portraits of Hope through the years

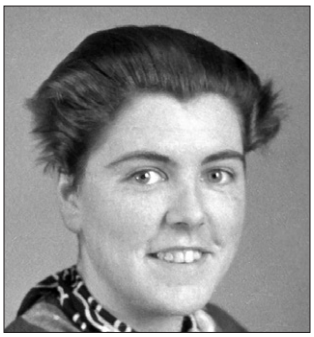
1917—18 months

1924—8 years

1929—13 years



1932—16 years



1936—20 years

1937—20 years

1939—23 years

1942—26 years

1943—27 years

1944—28 years

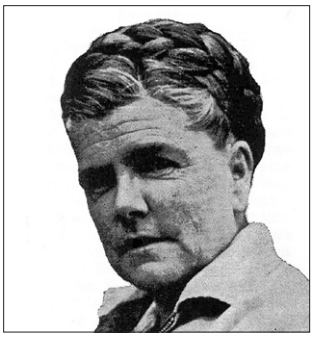
1945—29 years

1946—30 years

1953—37 years

1963—47 years

1966—50 years



1969—53 years

1972—56 years

1975—59 years

1981—65 years

1982—66 years

1984—68 years

1985—69 years

1988—72 years

1990—74 years

1992—76 years

1994—78 years

1999—83 years

2003—87 years

2005—89 years

2006—90 years

2009—93 years

2010—93 years

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Index

Page numbers in **bold type** refer to illustrations.

A

Ablett, Thomas L. (artist) 3, 29
Actinosepia 52, 192
Adler, Leslie 109
agoraphobia 197
Alberta
  dinosaurs in Ottawa 42
  government support for arts 49
  Hope moves to 34, 43
Alberta Archaeological Society 107, 112
Alberta Foundation for the Arts 142
Alberta Historic Resources Foundation 171
Alberta Palaeontological Advisory Committee (APAC) 146
Alberta Palaeontological Society 75, 109, 143, **147**, 148, **149**, 151, 152, 155, 156, 160, 161, 163, 166, 189, 190, 192, 195
  Life Membership 152, **153**, 192, 194
Alberta, Provincial Museum of (PMA) 1, 5, 58, 71, 74, 75, 77–79, 81, **82**, 86, 89, 91, 99, 103, 107, 108, 110, 113, 116, 117, 119, 120, 127, 128, 130, 131, 133, 137, 140, 152, 166, 169, 175, 189, 190, 201, 202, 204, 206
Albertosaurus 152, 193
Allan, John A. 53, 58, 169, 198, 201
Alzheimer's disease, Society 137, 144, 149, 152, 168, 185
amateur roles in palaeontology 2, 4, 6, 12, 74, 99, 108, 114, 147, 168, 180, 181, 195, 205
American Museum of Natural History, NY 87
American Quaternary Association (AMQUA) 129
ammonites 53, **63**, **66**, 74, 111, 117, 123, 142, 146, **172**, **174**, 175, **179**, **186**, 198, 201, **209**
Anderson, Leroy and Rena 129
Andrews, Roy Chapman 137
animals, pets **31**, 173, 177, 182, 184, **209**
ankylosaurs (unidentified) 106, 203, **213**
archaeology 2, 81, 86, 93, 94, 101, 106, 113, 121, 132, 140, 196, 199
Arizona, USA 120, 167
Arnes, MB 45, 164
art exhibitions 4, 8, 45, 54, 67, 85, 94, 110, 112–114, 118, 120, 122, 127, 128, 130, 138, 141–143, 145, 147, 149, 150, 153, 156, 159, 164, 168, 171, 187
art, performance 49, 171
art, visual 2, 3, 5, 38, 40, 45, 49, 53–55, 58, **66**, 78, 83, 102, **103**, 120, 147, 159, 163, 166
  batik **215**
  botanical **51**, 62, 68, 120, 122, 130, 141, 142, 145, 160, **165**, **188**, **208**, **217**, **218**, **230**, **231**
  calligraphy **229**
  clay sculpture 93, **94**
  etchings 166, **168**
  greeting cards **69**, 148, 157, **158**, **226**
  in childhood 3, 27, 29, 30, **31**, 196, **209**
  instruction, by Hope 3, 4, 54, 82, 85, 94, 110, 127, 128, 130, 141, 143, 148–150, 152, 154, 171, 187
  logo, crest design 67, 85, **86**
  mountain scenery **210**, **224**, **225**, **235**
  murals 62, **63**, 140, 158, **159**, 183, 188, **211**, **227**, **228**
  needlepoint **229**
  pen-and-ink 4, **51**, 57, **66**, 83, **87**, **90**, 103, **104**, **105**, 108, 120, 130, 142, 148, **149**, 153, **156**, **157**, **162**, **165**, **170**, **172**, **178**, **179**, **186**, **188**
  portraits 8, **130**, 166, 177, **216**, **219**
  puppets 140
  sales, by Hope 45, 54, 138, 142, 145, **155**, 156, 160, 164, 166, 208
  technique 3, 68, 157, **162**, 166
  unconventional subjects 79, 140, **178**, **210**, **215**, **216**, **226**, **227**, **228**
Aspideretoides **78**, 104, 193
astronomy 32
awards 46, 136, 141, 147, 150, 152, **153**, 171, 194

B

Babine Lake, BC 127
Baculites 53
Baird, Donald 191
Banff, AB 54
Barbados 19
Bardack, David 60, **73**, 88, 191
Bare Creek, AB 121
Barendregt, René W. 129, 130, 157, **158**
Baresco, Dennis 145
Barrie, John "Jack" 89, 126
Bassano, AB 66
bats (animals) 103
Bayrock, Lou A. 74
BBC television documentary 125, 194, 206
Beare, James "Jim" 182, **183**–185
Bearpaw Formation 51, 111, 112, 114, 115, 120, **122**, 123, 128, **129**, 142, 148, **149**, **156**, 165, **179**, 190
bentonite clay 157
Berton, Pierre 30
Beta Sigma Phi (club) 120
Bickerton, Florence "Flo" 177
"Big Coulee", DPP 67
"Big Coulee", not DPP 88
biological weapons testing 44
bird fossils 149
birdwatching 78
births
  Barbara (Hope's daughter) 49
  Herbert (Hope's father) 18
  Hope 17, 196
  Lawrence (Hope's son) 49
  Margaret (Hope's daughter) 49
  Olafur Johnson 45
  Rachel (Hope's mother) 20

Robert (Hope's son) 49
Bison 74, 88, 93, 94, **95**, **98**, 108, 128, 129
Bonn, Germany 147
books, papers written or co-written by Hope 171
Alberta Palaeontological Society, 2009 (vertebrate fossils) 1, 133, 152, 153, 155, 156, 160, **161**, **162**, 192, 195
Droppo & Johnson, 1988 (berries) 145, 171, 188, 189
Johnson, 1963 (geology, fossils) 75
Johnson, 1968a (dinosaur fossils) 84, 89, **90**, 192
Johnson, 1977b (dinosaurs, fossils) **128**
Johnson, 1983 (prairie plants) 141, 142, 147, 189
Johnson & Hallworth, 1975 (botany) 122, 188, 189
Johnson & Storer, 1974 (vert. fossils) 1, 8, 69, 83, 101, 107, 108, 110, 115, 120, **121**, 130, 147, 152, 153, 155, 156, 159, 161, 190, 192, 204
Michael & Johnson, 1981 (Elkwater) 138, 171, 194
Storer & Johnson, 1974 (sawfish) 113, 120, 192
boots, Hope's hobnail **97**, 175
botany 2, 3, 5, 7, 45, 50, 62, 96, 118, 121, 122, 131, 135, 141, 145, 151, 158, 168, 178, 188
childhood interest 27, 28, 43
Bothriolepis 140
Bow River, AB, badlands 114, 123, 128, 131
brachiopod fossils 52
Bradley, Cheryl 191
Braunberger, Wayne **160**
Brinkman, Donald B. 104, 191, 206
British Army Training Unit Suffield (BATUS) 44, 53
British Columbia Women's Service Corps (BCWSC) 34, 35, **36**, 41, 187, 196
British Guiana (Guyana) 19
Broadstairs, Kent, UK 146
Brockville, ON 43, 185
Brontops 64
Brooks, AB 5, 54, 127, 128, 137
Brown, Barnum 58, 137
Budd, Archibald 51
Burdett, AB 93
Burian, Zdeněk 102, 117, **212**, **219**
Burnaby, BC 26, 33
Burquitlam, BC 33
Burton, William "Bill" **84**

C

cactus 175, 176, **218**
Cadboro Bay, BC 33
caenagnathid dinosaurs 133, 192, 193
Calf Creek, SK 91, 114, 115, 117
Calgary, AB 21, 58, 71, 72, 74, 82, 95, 112, 113, 116, 123, 143, 166, 188, 191, 194, 199
Calgary Zoo 5, 58, 95, **116**
"Camel", DPP 67
Canadian Museum of Nature, Ottawa 2, 42, **43**, 59, 62, 120, 166, 189, 190, 203
Canadian Nature Federation 3, 7, 112, 114, 120, **158**
Canadian Quaternary Association (CANQUA) 143, 201
Canadian Shield 51
Canadian Women's Army Corps (CWAC) 3, 35–**39**, **40**, **41**, **42**, 43, 44, 46, 47, 62, **176**, 185, 187, 196, 198
Hope joined 32
Hope retired from 46
Canal Creek, AB 129
Canal Flats, BC 47
Cardiff, Wales, UK 21
"Castle Rock", DPP 67
"Cathedral Ridge", DPP 67
Centrosaurus 67, 99, **100**, 139, **140**, **212**
cephalopod fossils 52, 53, **63**
ceratopsians (unidentified) 63, 78, 87, 100, **104**, 129, 153, 156, 192, 193, 195, 201
Chamney, Ronald "Ron" 54, 119, 205
Chamney, Thomas Potter 54, 60, **73**, 82, 191, 200
Champsosaurus 62, 63
charity 35, 55, 84, 138, 142, 143, 154, 162, 164, 197
Chasmosaurus 60, 67
chemical weapons testing 44, 46
Chilliwack, BC 17
Chinook, MT 129
Churcher, Charles S. "Rufus" 60, **73**, 76, 84, 93, 94, 106, 109, 112, 119, 124, 133, 142, 191
"Citadel", DPP 67
Claresholm, AB 107
Clemens, William A. 71, **73**
Clementi, Kenneth "Ken" R. 53, 75
Colbert, Edwin H. 6, **73**, 74, 87
Cold War 55, 80
colour theory 54
Colwell, Jane 83, 86, 89, 91, 92, 101, 173, 191, 206. *See also* Danis, Jane and Gilles
concretions, fossils in 52, 183
"Conference Room", DPP 67
conferences attended by Hope 168, 191, 192
AMQUA, 1978, Lethbridge 129, 130, 192
ASPG 16th annual, 1965 81
CANQUA, 1985, Lethbridge 143, 192
Community Art Class, Edmonton 67, 110, 187
Dinosaur Provincial Park, 2005 9, 159, 171, 192
International Geological Congress, 1972 113
SVP, 1974, Flagstaff, AZ 120, 191, 202
SVP, 1977, Los Angeles 129, 191, 202
SVP, 1978, Toronto 130, 191, 202
SVP, 1988, Drumheller 146, 191, 202
Vertebrate Paleontology, 1963 6, 71, **72**, **73**, 74, 76, 123, 169, 191
Coqualeetza School, Sardis, BC 18, 22
coral fossils 52
Corythosaurus **56**, 57, 58, 63, 75, 81, 101, **103**, 203, 205
CPR pumphouse locality, AB 67, 70, 75
Craigdarroch Castle, Victoria, BC **30**, **31**, 197
crinoid fossils 52
crocodile fossils 62, 63, **105**, 107
Crossman, Edwin J. 60, 191
"Crown", DPP 67
Crowsnest Pass, AB 139
Cryptantha 118, 188
Cupressinocortex 65, 192
Currie, Philip J. 1, 2, 8, 58, 60, 100, **124**, 129–131, 133, 134, 136, 154, 155, 191, 204, 206
Cutler, William E. 58
Cypress Hills, AB and SK 64, **70**, 75, 81, 84, 93, 128, **131**, 141, 199, **232**
Cypress Hills Formation 51, 91

D

Danis, Jane and Gilles 1, 99, **133**, 137. *See also* Colwell, Jane
Darwin, Charles 19
Dawe, Robert "Bob" 132
Dean, Solomon 20
deaths
Charles M. Sternberg 138
Herbert Large (Hope's father) 20, 27
Hope Johnson 10, 163, 184
Olafur Johnson 144
Rachel Large (Hope's mother) 27
Debert, NS 45, 46, 198
Denhart Coulee, DPP 67, 70, 77, 79, 81, 83, **214**
Depression era (1920s–1930s) 30–32, 39, 40, 58
Devillers, Charles **73**, 74, **122**, 123, 166, 191, 201
Devonian fossils 52, 140
dinosaur babies, nests 139
dinosaur footprints, tracks 120, 133
"Dinosauroid" (D.A. Russell) 122
Dinosaur Park Formation (outside DPP)

131, 136, 139, 165
Dinosaur Provincial Park (DPP), AB 1, 5, 67, **69**–72, 74, 75, 77–79, 81, 83, 86, 87, 89, 91, 92, 94, 95, **99**–**104**, 106, 107, 109, 110, 115–121, 123–125, **128**, 129, 131, **133**, **134**–137, 139, 140, 144, 149, 163, 165, 166, 169, 178, 192, 195, 204
creation of 54, 169
exhibit buildings ("chicken coops") 1, **101**–**104**, 107, 117, 131, 135, 144, 190, 203
Field Station 131, 144, 171
in situ fossil displays 5, 79, 81, 100, 101, 107, 203
politics 119, 124, 173
restricted areas 109, 110, 127, **133**
staff accommodation 97, 202
UNESCO World Heritage Site 131, 135, 169
vehicles, tours 97–**99**, 104, 106, 118, 127, 183, 190
Warden, Head Ranger **77**, 87, 92, 96, 97, 99, 104, 106, 119, 121, 181, 190, 205
displays. *See* fossil displays; art exhibitions
Dodson, Peter 1, 91, 92, 101, 112, 119, 136, 140, 155, 191
Dromaeosaurus 88, 125, 166, **213**
Dromiceiomimus 117, 125
Drumheller, AB 5, 64, 74, 75, 77, 110, 120, 125, 146, 153, 168, 169, 178, 187, 204, 206
Drumheller fossil museum (not RTMP) 5, 64, 71, 72, 74, 76, **77**, 91, **92**, 189, 190, 196
Dry Island Buffalo Jump Provincial Park, AB 152
Dunbar, J.S., Brigadier-General 36
Dunsmuir, Robert 30
dust 1, 48

E

Eagle Butte, AB 114, 141
East Coulee, AB 148
Eastend, SK 65, 72, 112, 117, 150, 151
Edmonton, AB 52, 57, 58, 66, 67, 71, 74, 85, 86, 92, 93, 110, 112–114, 116, 119, 124, 125, 127, 129, 133
Edmontosaurus 76, **77**, 206
Edmund, A. Gordon 1, **57**, 60–62, 71, **73**, 83, 87, 91, 119, 191, 196, 201, 205
eggs, dinosaur 135
Elderhostel (senior travel group) 145
employment (Hope)
art instructor 143
Dinosaur Prov. Park 1, 96–104, 106, 107, 113, 115, 116, 166, 189, 191, 202
fruit picker **30**
illustrator 62, 68, 76, 93, **103**, **132**, 138
military service **35**–**47**, 185, 187, 196
odd jobs 31, 32, 196
office clerk 32
stenographer 32, 33, 41
teacher 33
teaching assistant 26, 32
typist 32
Empress, AB 120
English grammar and spelling 99, 118, 153, 167
Eocene fossils, rocks 51
Equus 124
Erickson, John Uno **96**
Esplanade Arts and Heritage Centre, Medicine Hat, AB 17, 113, 160, **161**, 166, 171, 182, 188, 189, 196
Esquimalt military base, BC 36, 185
Estes, Richard 93
Etzikom Coulee, AB 112
Euoplocephalus 99, 193, 203, 204
excavation of fossils 64, 77, 90, **91**, **98**, 99, **100**, 104, 106, 115, 119, 133, **140**, 142, 168
permit requirement 124, 146
exhibits. *See* fossil displays; art exhibitions
explosions (military testing at Suffield) 44, 46, 47, 80, 84, 199
explosives for fossil extraction 64

F

farm, Large family, Sardis, BC 20, **23**, 24, 25
Federation of Alberta Naturalists 7, 118, 171, 191, 194
field jackets for fossils 64, 77, 89, 104
field preparation of fossils 64, 77, 90, 99, **204**
field trips
attended by Hope 71–**73**, 82, 120, 129, 143, 152, **158**, 169
led by Hope 66, 70, 71, 79, 81, 91, 112, 118, 119, 127–130, 132, 139, 143, 145, **147**
Filanti, Amerigo (Rick) 85
First Aid training 29, 41, 185
First Nations Peoples 22, 81, 86, 101, 104, 130, 132, 140, 166, 177, **219**, **235**
fish, fossil 60, 62, 63, 88, 107, 109, 111–113, 120, 140, 193
Forbis, Richard G. 94
Foremost Formation 57, 63, 112, 113, 120, 129, 165, 192, 195
Forey, Peter 112
Forsen, Carolyn (Virovec) 119
Forsen, Gerhard 104, 119
Fort McLeod, AB 109
Fort McMurray, AB
Oilsands Interpretive Centre 82
"fossil boxes" 87, **89**, 95, 100, 112
fossil collecting
abusive, irresponsible 6, 116, 146, 178, 180, 181
Hope begins collecting 50
Hope's collection sent to RTMP 153
Hope's first catalogue 60, 189
Hope's last catalogue entry 147, 189
invertebrate fossils 51–53, 75, 117, 121, 123, 127, 146, **156**, **174**, 192
law/registration/disposition 12, 13, 146
techniques, tools **98**, 175, **204**
fossil displays, exhibits 130, 190
absence of, in Medicine Hat 181, 182
Dinosaur Provincial Park 96, 190
Medicine Hat Museum 5, 89, 90, 115, 117, 124, 128, **129**, **132**, **156**, 190
personal 51, 112, 153, 154, **161**, 162
Police Point Interpretive Centre **143**
Ralston Library 88
Suffield Officers' Club 47, 71, 81, 129, 190, 196
Fowler, Roy **77**, 96, 102, 205
Fox, Richard C. 5, 60, 83, 86, 92, 93, 99, 102, 106–109, 173, 191
French language education 21, 27, 28, 37, 74, 173
Frisch (Maser-Frisch) site, AB 129, 132, 133
frogs and toads, fossil 106
frogs and toads, living 171, 188

G

Galt Island Bluff, AB 114, 118
gardening 25, 27, 28, 33, 48, 127, 171
gender inequity in palaeontology 72, 83, 119, 125, 169, 173, 195, 202, 205
Geological Survey of Canada 53, 58, 60, 84, 87, 201
geology
Hope's interest and activities 50, 51, **139**, **222**, **223**, **230**
"German Helmet", DPP 67
German language education 29, 33, 37, 41, 160, 173, 185
Gilmore, Charles W. 58
Gimli, MB 45
glacial landforms, deposits 75, 120, **222**
glacial till 51, 52, **222**
Golden Valley Bluff, AB 129
Gorgosaurus 57, **58**, 64, 87, 90, **91**, **92**, 192
Grasslands Naturalists 142, 150, 152, 188, 196
gravel pits, fossils in **51**–53, 56, 66, 93, 112, 120, 183
Graves, Warren 124
gravestones
Hope and Olafur Johnson **163**, 164
Hope's parents **27**

"Great Dinosaur Rush" 58
Greek language education 30, 52
Guides, Girl 28, 35, 75, 79, 171, 185
Guild of Friendship (church group) 35, 197
Gunson
  Barbara Avice (Stacey) **22**
  Edward "Teddy" Richard **20**, 21, 197
  Rachel "Dollie" Emily 20, **21**
"Gypsona" plaster bandages 64

H

hadrosaurs (unidentified) 57, **61**–64, 76, 78, **79**, **98**, 99, **102**, **103**, 116, 123, 125, 131, 139, 141, 166, 169, 193, 203, **205**
Hage, Keith 191
Halloween 65, **159**, 178, **226**
Hallworth, Beryl 122, 191
Halmrast, Lawrence D. 69
Hand Coulee, AB 65
Harington, C.R. "Dick" 109, 124, 132, 191
Hat Art Club, Medicine Hat 4, 130, 138, 142, 148, 164, 167, 187
Hatley Memorial Gardens, Colwood, BC 20
Hays, AB 123
headstones
  Hope and Olafur Johnson **163**, 164
  Hope's parents **27**
health (Hope's) 42, 44, 47, 114, 122, 125, 138, 145, 146, 148, 152, 159, 160, 162, 163, 167
Heath, Charles 5, 84, 85, 91, 119, 146, 191, 202, 205
Heath, Deanna 84, **85**, 146
Heath, Velma (Burton) 5, **84**, **85**, 89, 119, 146, 205
Heather Valley Site, AB 142
Helmcken, Dr. John S. 20
"Herkimer diamonds" 51
High School Correspondence Branch, Victoria, BC 32
Hilda, AB 44, 138
Hills, Len V. 5, 60, 191
Historical Resources Act (Alberta) 12, 13, 124, 146
Hobberlin, Hedwig "Hedy" 78, 81, 180, 189, 190, 206, 207
Hobberlin-Hynson, Christine 207
honorary degree. *See* LL.D. honorary degree
"Hopeless" (nickname) 29, 92
Hornell, Ralph R. 71, 201
Horner, John Robert "Jack" 135, 147, 155, 191
horses, as draft animals 117
horses, fossil 62, 108, 117, 124, 128, **132**, 190, 201
Horseshoe Canyon, AB 75
Horseshoe Canyon Formation 66
Hot-Blooded Dinosaurs, The (BBC TV) 125, 194
housekeeping 175, 176
Hrapko, Julie 7, 191
Hudson, Sir Peter 129
humour (writing, anecdotes) 96, 150, **176**, 177, 181, 182, 184
Hunt, Caroline Ada **21**
"Hunter Quarry", SK 91, 93
Huxley, Thomas H. 19

I

Ice Age deposits, fossils 64, 75, 76, 84, 87, 93, 106, 120, 128, 143, 171, 201
Iceland (Olafur's ancestors) 45, 148
Iddesleigh, AB 79
Iguanodon 122
Inoceramus 52, 53, **66**
International Geological Congress, 1972 113
Irvine, AB 75, 79, 86, 88, 93, 106, 107, 127, 128, 129, 131, 133, 140–142, 148, 149, 165, 171
Ischyrhiza 62, 113, 120, 192

J

Jackson, "Happy Jack" 116
Jackson, J.-P. Christopher 4, **123**, 136
James, Frederick 91
Jenner, AB 43, 56
Jenner (Bridge, Crossing, Ferry), AB 75, 77, 109, 114, 115, **223**
Jensen, Allan C.J. 3, 4, 53, 77, 166
Johnson, Barbara (Preston; Hope's daughter) 47, **48**, **49**, 86, 112, 137, 138, 196
Johnson, Lawrence (Hope's son) 47, **49**, 58, 59, 123, 124, 138, 159, 163, 174, 178, 196
Johnson, Margaret (Noone; Hope's daughter) **48**, **49**, 196
Johnson, Olafur "Ollie" 3, **45**, **46**–**50**, 51, 52, 55, 56, 62, 70, 74, 112, 127, 129, 131, 137–139, 142, 144, 145, 148, 164, 174–178, 198, 202
Johnson, Robert (Hope's son) **48**, **49**, 50, **53**, 196
Jones, Gareth H.S. 54, 56, **57**, 60–62, 64, 66, **67**, 71, 178, 191, 198, 205
Jónsson, Jon (Olafur's father) 45
Jurassic rocks, fossils 146

K

Keating, Brian 94, 95
Kennedy, Joan 37, 39, 46
Kielan-Jaworowska, Zofia 83
Killdeer, SK 83, 114, 115
Kingsley, Rev. Charles 19
Kitchener, Gordon H. and Elsie F. 110
Kitchener, ON 41, 185
Kleis, Evelyn 54, 70
"Knife Ridge", DPP 67
Knight, Charles R. 102
Kootenay Lake, BC 21
Krause, David (and Krause Ranch, AB) 138
Kroetsch, Robert 54

L

Lambe, Lawrence M. 58, 59
Langston, Wann Jr. 57, 60, 64, **73**, 77, 110, 191
lapidary, commercial, hobby 116, 123, 147, 180
Lapp, George, Ranch 57
Large
  Ada Elisabeth 20
  Barbara Elizabeth Hope
    birth, name **17**
  Caroline **19**
  Herbert "Bert" Ernest Evylyn **17**, 18, **19**, 20, **23**, **24**, 25–27, 33, **47**, 197
  James Joseph **18**
  Rachel Emily (née Gunson) 19, 20, **21**, **22**, **23**, **24**, 25–27, 32–35, 46, **47**, 197
Latin language education 27, 29, 30, 52
Lawson, Shirley, William "Bill", Al **97**, 115, 133, 139, 173, 191, 205, 207
lectures, talks by Hope 70, 78, 81, 84, 93, 107, 118, 120, 130, 137, 141, 144, 146, 150, 157, 168, 190, 191, 194
Lemna, Deanna 147, 187
Lemon, Roy Richard Henry **57**, 60, 191, 201
Lepidosteus 63
Lethbridge, AB 96, 107, 112, 118, 123, 136, 143, 153, 187, 188, 194
Leuck, Shane and Vicki **iv**, 151, 166, 173, 205
Lindoe, Allan 99, 173
Lindoe, Luke 173
Littlehampton, West Sussex, UK 21
Liverpool, UK 21, 25
lizard, eastern short-horned 66, 171, 188, 194
lizards, fossil 107, 193
LL.D. honorary degree 2, 4, 124, **135**–137, 141–143, 167, 168, 194
London, UK 21, 146
Longfellow, W.W. (poet) 28
Los Angeles, CA 120, 129, 191
Lost River badlands, AB 83, 88, 129, 135
Lull, Richard S. 58

M

Makela, Robert "Bob" 135
malaria, Hope's father 19, 20, 24
mammals, fossil 64, 71, 72, 83, 84, 86, **87**, 88, 91, 93, 94, 98, 106, 108, 109, 112–114, 117, **118**, 119, 120, 126–128, 131–133, 142, 152, 171, 190, 192, 193, 195, 201, 204
mammoth, fossil 120, 126, 127, 128, 131, 133, 190, 195
Manyberries, AB 112, 115, 122, 123, 142, 147, 165
Marsovsky, Vaclav **153**, 162
Maryańska, Teresa 83
"Massif", DPP 67
Mattson Ranch, AB 131
McCorquodale, Bruce A. 6, **73**, 81, 191
McCracken's Mine, AB 75, 121, 129
McDougall Ranch, AB (= Pat C Ranch) 112. *See also* Pat C Ranch
McGowan, Christopher 191
McKenzie, Edward 84
McKinley Ranch, AB 112, 114, 115, 131, 142
Medicine Hat, AB 2, 3, 44, 47, 48, 53–55, 62, 70, 76, 77, 81, 90, 93, 94, 106, 109, 112–114, 117–119, 122, 126, 129, 130, 132, 137, 138–143, 145, 147–149, 151, 152, 157, 159, 161, 163, 164, 167, 171, 181, 184, 187, 193, 194, 199
  newspaper, Hope's writing **66**, 68, 75
Medicine Hat and District, Historical Society of 84, 131, 138, 140, 185, 193
Medicine Hat College 112, 115, 139, 145, 153
Medicine Hat Hospital 145, 163, 187
Medicine Hat Museum and Art Gallery 4, 5, 8, 84, 86, 88, 89, 91, 93, 94, 106, 109, 114, 115, 124, 126–128, **129**, 131, **132**, 133, 139, 142, 147, 156, 185, 189, 190, **219**
Medicine Hat Public Library 140
Medicine Hat Rock and Lapidary Club 147
Medicine Hat University Women's Club 70, 84, 109, 185
Meniscoessus 193
Merychippus 117, **118**
meteorology (Olafur) 45, 46, 48, 49, 52, 70
Michael, Hope 138, 194
microfossils (microvertebrate fossils) 57, 59, 62, 83, **84**, 86, 93, 100, 107, 117, 119, 120, 133, 136, 166, **170**, 192, 206
military service **35–47**, 185, 196
Milk River, AB 65, 69, 72, 75, 87, 88
Milne, Fiona 84
Milne, Lucy 84
Miocene fossils, beds 51, 83, 114, 117, 165
Mitchell Bluff, AB 93, 117
Mongolian dinosaurs 94
Monoclonius 63
Montana, USA 51, 165
Montreal, QC 25, 40, 113
Morgan Creek, SK 114
Morris, William J. 129, 191
mosasaurs 63, 110, 115, 122, 123, 129, 141, 142, 201
Mosasaurus 111, 141
moulding & casting of fossils 117
Mountains of the Moon Coulee, AB 66, **67**
mud 44, 48, 97, **176**
Murray Lake, AB 128
Museum of the Rockies, Bozeman, MT 147
mushrooms 92
music 29, 32, 41, 160, 167
Myers, Memory 88
Myledaphus 62, 63, 109, 119, **170**

N

Nanaimo, BC 50
Nanton Rock and Mineral Society 107
National Museum of Canada, Ottawa 3, 4, 8, 53, 57, 60, 76, 86, 92, 109, 110, 120, 124, 132, 201, 206
Natural History Museum, London, UK **146**
nautiloids 63
Naylor, Evelyn 74
New Westminster, BC 20, 23, 26, 33
"Nick Point Cave", DPP 67
nicknames, geographical 67
Noone, Rachel 166
NOVA (PBS television show) 125
Nursall, John R. 123

O

Oakalta (oil company) 62, 65
Oenothera **131**, **165**
Oldman Formation 63, **233**
Oligocene fossils, beds 72, 91, 114, 117, 119, 165
Ordovician fossils 52
Ornatotholus 100, 203
ornithomimids (unidentified) 70, **71**, **82**, 87, **90**, 92, 93, **104**, 117, 152, 192, 193, 202, 207
Osmólska, Halszka 83
osteology 59
Ottawa, ON 41, 42, 45, 100, 102, 109, 112, 120, 137, 145, 185, 187
Oviraptor 125
owls, paintings 118, 127, 166, **167**, **225**

P

pachycephalosaurs (unidentified) 57, 100, 203
palaeobotany 65, 93, 94
Palaeozoic fossils **52**
Paleocene fossils 74
Parasaniwa 193
Parasaurolophus 63
parenting 175, 176, 177
park development, Hope's activism 64, 122
Parks, William A. 58
Pat C Ranch, AB (= McDougall Ranch) 66, 70, 75, 79, 81
pathology (of fossil bones) 71, 87, 90, 99, 115, 123, 153, 166, 192, 203
Patricia, AB 155
Peace River, AB 92
performing arts 23, 32, 35, 49
Peterborough, ON 43, 185
Petrified Coulee, AB **50**, 54, 57, 61, 62, 64–67, 70, 75, 76, 79, 81, 83, 87, 93, 112, 113, 115, 123, 128, 131, 133, 139, 146, 147–149, 165, 200
"Petrified Teepee", DPP 67
"Phred the Camel", DPP 67
Phrynosoma brevirostris 66, 171, 188, 194
Pitman College, Vancouver, BC 32
Placenticeras 53, 117, 123, **174**, **186**
plaster, for fossil preparation 64, 77, 89, 101, 104, 119, 126, 204, **205**
Pleistocene fossils, geology 53, 75, 76, 87, 106, 109, 117, 119, 120, 129, 133, 142, 143, 152, 190, 201
Plesiobaena 193
plesiosaurs 52, 56, **111**, 114, 192
Plioplatecarpus 115, 141
poaching (of fossils) 109, 110, 116, 146, 147, 178, 181
poetry 144
  childhood 28
Poikans, John 82, 99, 201
Police Point Interpretive Centre 139, 141, **143**, 145, 151, 184, 185, 187, 189, 190, 195, 196
"Pond Mesa", DPP 67
Poquelin, J.-B. (French playwright) 21
porcupine 70
Porsild, Alf E. 62
poverty 23, 30, 31, 32, 39, 45
"Praying Lady", DPP 67
preparation of fossils 64, 66, 97, **98**, 102, 114, 119
Preston, Edward 84, 112, 191
Princess, AB 62, 81, 91, 115, 146
prisoners of war camp, Alberta 43
Procleptes hopejohnsonae 192, **193**, 195
Prognathodon 111
Pteraspis 140

pterosaurs 81, 99, 100, 110, 149, 192, 193, 203
Puckett, Jerrold Alair "Jerry" 116, 194

Q

quarries, "lost", locality unknown 91, 117, 149
Quarry 42 (Bonebed 42), DPP **82**, 91, 137
Quarry 78, DPP 67
Quarry 79, DPP 70
Quarry 119, DPP 109
Quarry 128, DPP 75
Quarry 135, DPP 99, **100**
Quarry 139, DPP 99, 149
Quarry 143, DPP 67, 139, **140**
Quarry 150, DPP 205
Quaternary deposits, fossils 74, 87, 106, 119, 120, 129, 143, 190
Quebec (military service) 40
Quinsey, Dan 75, 156, **160**, 206

R

Ralrick, Patricia (Patty) 31
Ralston, AB 3, 31, 38, 44, 47, **48**, 54, 68, 74, 96, 112, 114, 127, 129, 133, 177, 185, 187, 192, **223**
  Art Club 3, 53, 54, 65, 67, **68**, 75, 79, 80, 82, 93, 94, 120, 122, 164, 167, 187, 188
  Drama Club 3, 49, 185
  fossil collecting near 50, **51**, 52, **53**, 56
  gravel pits **51**, 52, 53, 56, 93, 112, 183
  newspaper, Hope's writing 65, 83, 85, 109, 120, 121, 122
  Public Library (Graham Community) 62, **63**, 88, 122, 153, 183, 196, **211**
  School 49, 65, 70, 120, 183, 185
Ralston, J.L. 38
Ramanujam, Chakravartula G.K. 191
Ramchandra, Reddy P. 65, 191
rattlesnakes 59, 91, 107, 175, 182, 183
Raymond, Mrs. C. Hugo 36
Read, Wesley R. **73**, 74, 76
Red Cross 35
Red Deer River, AB 67, 71, 79, **224**
Red Rock Coulee, AB 54, 114, 115, 142, 143, 144, 145, 147, 148, 149, 152, 165, 171, 194, **220**, **232**
Redcliff, AB 8, 65, 93, 122, **127**, 128, 130, 131, 133, 141, 167, 171, 192, 196
  Johnsons move to **127**
  Museum 189, 190, 196, **220**
Regina, SK 45
religion 32, 37, 174, 197
Research Council of Alberta 74
Reservoir Gully (Saamis Gully), AB 93
residential schools 22
road-kill 59
Robinson, Caroline (Hope's paternal grandmother) 18
Robinson, Dorothy A. 43
Romer, Alfred Sherwood 6, 58, **73**, 74, 154
"Rose's Ranch", AB 67, 79, 131
Rothwell, Nick 86–87
Royal Alberta Museum, Edmonton 1, 5, 71, 156, 169, 188, 189
Royal Canadian Astronomical Society 32
Royal Ontario Museum, Toronto 6, 8, 53, 56, 57, 60, 65, 67, 71, 76, 78, 83, 84, 87, 91, 117, 124, 166, 189, 190, 196, 200, 201
Royal Saskatchewan Museum, Regina 6, 68, 83, 124, 151, 166, 187, 204
Royal Tyrrell Museum of Palaeontology 5, 52, 64, 74, 83, 87, 91, 108, 111, 117, 119, 128, 131, 134, 135, 141–144, 146, 148, 151, 153, 157, 159, 165, 168, 169, 171, 180, 189, 190, 194, 195, 201, 205, 206, 207
Rudyard, MT 135
Russell, Dale A. 1, 60, **86**, 87, 91, 92, 94, 96, 98, 100, 102, 106, 108, 110, 112–115, 117, 121–125, 136, 166, 191, 203
Russell, Loris Shano 6, 8, 53, 57, 58, 60, 62, 66, **73**, 74, 76, 112, 123, 191
Rustington, West Sussex, UK 21, **25**

S

Saanich, BC 20, 33
Saint John, NB 25
Sainte-Anne-de-Bellevue, QC **39**, 40, 185
Salvation Army (charity) 138
Sand (= Little Sandhill) Creek 65, 86
Sandy Point, AB 44, 129, 131, 133
Sardis, BC 17, 18, 20, 22–25, 33
Saskatchewan, fossil collecting in 83, 87, 91, 93, 112, 114, 115, 119, 165, 166
Saurolophus **61**
Saurornithoides 94
Saurornitholestes 115
sawfish, fossil 62, 113, 120, 192
Scabby Butte, AB 110
Scapherpeton 107
Schaeffer, Robert "Bob" 191
Scheele, William E. 102
Schowalter, Tim 86
science education (Hope's early) 29, 30
science fair, judging 139
Scollard, AB 71
Scollard Formation 136
scorpions 66, 188
Scouts, Boy, Cub 29, 31, 32, 35, 41, 54, 57, 65, 66, 68, 70, 71, 75, 79, 82, 84, 109, 171, 177, 185
seahorse paintings 166, **228**
Seven Persons, AB 128
Shaman's Head 65
shark, fossil 62, 66, 70, 114, 120
Shearman, Hank 77
Sherbrooke, QC 41
Sheridan College, ON 78
sinkholes, fossils in 88, **98**
skin impressions, dinosaur **79**, 101, 142
smallpox epidemic (1892) 20
Society of Vertebrate Paleontology 120, 129, 130, 146, 191, 192
  Hope nominated to 62, 83, 202
Sons of England Benefit Society 20
South Sandhill Coulee, AB 109
South Saskatchewan River, AB 62, 64, 65, 67, 70, 112, 113, 122, 133, 158, 182, **233**, **234**, **235**
Southeast Alberta Archaeological Society 157
Spalding, David A.E. 5–9, 89, 90, 114, 124, 128, 130, 191, 206
Spencer's Wood, Reading, Berkshire, UK 25
sports (Hope's involvement in) 173
SS Lake Manitoba (ship) 21
SS Megantic (ship) 25
St. Agatha's School 12, 25, **26**, 33, 197
St. Mary River, AB 110, 111, 123
St. Swithin's School, Winchester, England 18
Stacey, Barbara Avice (Gunson) **22**
Stacey, Christopher Cyril **22**, **28**
Stacey, Wilson 22
Stalker, Archibald MacSween 60, 75, **76**, 84, 93, 94, 106, 109, 112–114, 117–119, 126, 129, 131–133, 142, 145, 150, 152, 191, 201
Stegoceras 57, 102, 106, 107, 193, 203
Stelck, Charles R. 5, 53, **73**, 191
Sternberg, Charles M. 6, 58, 59, 60, 71, **73**, 74, 76, 81, 92, 96, 102, 137, 191, 200, 206
Sternberg, George F. 58, 206
Sternberg, Levi 60, 173
Stevenson, William L. 82
Steveville, AB 54, 93, 99, 107, 114, 169
Stewart, William C. 53
Stewart, Wilson N. 60, 92, 93, 94, 191
Stiem, Sharon and Ronald "Ron" 159, 173, 205
Storer, John E. 1, 7, 8, 58, 60, 99, 107, 108, 110, 113–118, 120, 124, 178, 191, 204, 205
Struthiomimus 64
Styracosaurus 63, **82**, 83, 91, 137, 140, **212**
"Submarine", DPP 67
Suffield, AB (military base) 3, 43, **44**, 45–49, 51–53, 56, 62, 70, 79, 84, 85, 88, 93, 94, 112, 118, 119, 129, **176**, 177, 185, 191, 196

Sweetgrass, MT, USA 65
Switzerland 74, 159

T

Taber, AB 143
Taylor, Donald "Don" A. 5, **73**, **75**, 82, 89, **91**, 99, 191, 201, 206
Taylor, Jack B. 94
teaching (by Hope)
  art. *See* art instruction
  geology, palaeontology 112, 114, 115, 145
television, Hope's appearances on 66, 116, 124, 125, 148, 150, 194
tennis (Olafur) 48
Texas 19
Thain, Joan 53
theropods (unidentified) 79, **80**, **90**–92, **102**, **104**, 115–**117**, 120, 123, 125, 129, 133, 152, 166, 193, **213**
Thorvaldsdottir, Guthrun (Olafur's mother) 45
Tingley, "Ting" (CWAC Captain) 43
Tolkien, J.R.R. **178**
Toronto, ON 45, 78
Trans-Canada highway 44, 48, 86
travel (by Hope)
  Eastern Canada 39, 78, 109, 112, 137, 145
  Europe 122, 127, 159, 178
  Great Britain 25, 122, **146**
  Manitoba 46, 137
  United States 120, 167
Tremblay, T.L., Major-General 41
Triassic rocks, formations 120
Triceratops 63, **146**
Trinidad 19
Troodon 166, 193, 200
turtles, fossil 57, 59, 63, 77, **78**, 104, 107, 128, 192, 193
Tylosaurus 63
Tyrannosaurus 64, 139, 151

U

UFO sighting 121
United Way (charity) 138, 185
University of Alberta (Edmonton) 3, 5, 52, 53, **56**, 58, 65, 66, 74, 82, 86, 87, 89, 91–93, 99, 100, 102, 106–108, 110, 112, 123, 149, 166, 169, 173, 189–191, 196, 198, 201
  palaeontology museum 57, **58**
University of British Columbia 30
University of Calgary 82, 94, 95, 113, 121, 122, 128, 131, 188, 199
University of Lethbridge 2, 4, 123, 136–138, 141–143, 146, 149, 154, 167, 168, 188, 194
University of Manitoba (Winnipeg) 45
University of Paris (France) 166, 180, 189, 190, 201

V

Vagaceratops 60
"Valley of the Castles", DPP 67
"Valley of the Gold", DPP 67
Valley of the Moon(s), DPP 67
Valleyview Retirement Community, Medicine Hat 153, 154, 158, **159**, 162, 163, 167, 177, 178, 184, 187, 196
  Hope moves in 153
Vancouver, BC 3, 17, 18, 25, 32, 40
Vanderloh, Irene 81, 92, 99, 115, 116, 173, 195
vehicles, Johnson family 79, **126**, 183
Victoria, BC 20, 25–27, 29–33, 35–37, 40, 42, 46, 47, 50, 185
Victoria College, Victoria, BC **30**
Victoria High School, Victoria, BC **29**
Victoria Memorial Museum, Ottawa 42
Victoria Women's Service Club 36
Virovec, Carolyn (née Forsen) 119
Visser, John 82
volunteerism by Hope and family 20, 29, 31, 35, 36, 45, 49, 68, 85, 88, 93, 137–142, 144, 148, 166, 173, 185, 194

W

Waage, Karl M. 142, 144, 191
Wales, UK 54, 178, 198, **225**
Walthamstow, London, UK 21
Warbeck (Warbrook) house, Spencer's Wood, Reading, Berkshire, UK 25, **26**
Wardlow, AB 81
Ware, John (cabin, DPP) 104, 118
Warner, AB 69
Warren, Percival S. 52, **57**, 60, 191, 198
Waterton, AB 109, 143
wedding
  Hope and Olafur 46, **47**
  Hope's parents 23, **24**
Welling, AB 110, 111
Westoll, T. Stanley 6, 71
White Rock Coulee, AB 44, 65, 79, 83, 87, 88, 90, **91**, 93, 127, 128, 131, 132, 153, 165, 192, **233**, **234**
Willock, Tom A. 131, 191
Willow Bunch, SK 114
Wilson, Michael C. 60, **74**, 87, 107, 120, 128, 146, 191
Winnipeg, MB 46
Withyham, East Sussex, UK 18
Wohlfarth, Harry 3, 53, 54, 65, 67, 80, 113
Wolf (Wolfe) Coulee 62, 65
women's roles in palaeontology. *See* gender inequity in palaeontology
wood, fossil 65, **85**, 93, 101, 107, 116, 183
Wood Mountain Beds, SK 83
World War I 17, 25
World War II 3, 34, 35, 58, 197, 198
Worthing, West Sussex, UK 22
Wright, Nelda E. 58
writing, by Hope 159, 168, 171
  botany 95, 96, 120, 122, 147, 189
  geology, palaeontology **66**, 83, 91, 99, 116, 118, 128, 135, 148, 151, 152, 192. *See also* books, papers written or co-written by Hope
  poetry 144
Writing-On-Stone Provincial Park, AB 140, 149, 157

X

Xiphactinus 112

Y

Young, Robert "Bob" 92, 96, **99**

Z

Zandmer, Ursula 94
Zybryski, Frederick "Fred" **159**

About the author

Caricature of the author drawn by a street artist in Aix-en-Provence, France, August 2001.

Darren H. Tanke is the longest serving employee at the Royal Tyrrell Museum of Palaeontology in Drumheller, where he is a Senior Technician II in the field and fossil preparation lab. Born in Calgary in 1960, his interest in Earth sciences began by the age of five. He planned on being a geologist but at the age of 10 he found his first fossil and palaeontology took over as a career choice. A visit to the Drumheller badlands with future ammolite promoter René Vandervelde (1935–2006) hooked him on dinosaurs, especially ceratopsians (horned dinosaurs). While in high school he volunteered with Michael C. Wilson at the University of Calgary, where he worked on bones of American bison. At the same time he took a Special Projects course, studying and writing about Alberta dinosaurs—one report each week—eventually amassing a 600-page final product and a high mark. In 1979 he sent the reports to Philip J. Currie and asked about a job: he was offered a summer volunteer opportunity. Hope Johnson's lavishly illustrated fossil book (Johnson and Storer, 1974) was crucial to his education in dinosaur and contemporaneous animal skeletal anatomy. Straight out of high school he joined the Paleontology Department at the Provincial Museum of Alberta as a volunteer under Philip J. Currie. With Hope's book committed to memory, he hit the ground running and was able to identify nearly every fossil he saw on his first day in the field at Dinosaur Provincial Park and Sandy Point, Alberta. He left home and moved to Edmonton with $200 in his pocket, dreaming of a career working on dinosaurs.

In the past forty years Darren has worked all over Alberta prospecting for fossils, collecting skeletons, working in ceratopsian bone beds, finding lost dinosaur quarries and investigating known but unidentified quarries. In the lab he prepares a diverse variety of mostly vertebrate fossils for research and display. His most technically challenging project was the type-specimen skull of the ceratopsian *Regaliceratops peterhewsi*. Though holding no university degrees, he is active in solo and collaborative research projects on diverse topics, including quarry staking and site documentation, dinosaur pathology, fossil preparation, lab and field safety, ceratopsian ontogeny, taphonomy, and human history related to early dinosaur collecting in Alberta. He has published or is writing a series of biographies on lesser-known "blue collar-type" Albertans involved in vertebrate palaeontology who have made significant contributions to the science but are overshadowed by others, or whose contributions faded with the passage of time. He created and administrates three Facebook groups: Ceratopsian dinosaurs, Women in vertebrate paleontology, and 1950–1960 Buick cars. When he has spare time, he reads about natural and human history, 1950s American cars, gardening, and aquaculture.

The August 24, 2010 passing of Hope Johnson marked the end of an era for Alberta's amateur and professional vertebrate palaeontology communities. Her death affected other disciplines, too, as she travelled in many circles within the province for 65 years. How many among us can truly say they never knew her personally, or of her art work, or learned to identify Alberta prairie plants, or Late Cretaceous bones and teeth through her excellent vertebrate fossil identification books? A relative late-comer to palaeontology, during much of her middle and later life, and especially during the late 1950s to 1980s, Hope was a well-known and respected powerhouse in the Albertan amateur and professional vertebrate palaeontological communities. She was also heavily involved in the naturalist and visual arts communities as well as charitable organizations, but this book focuses more on her extensive activities mostly in Alberta vertebrate palaeontology; as well as providing examples of some of her fossil and botanical drawings and paintings.

—From the Introduction

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